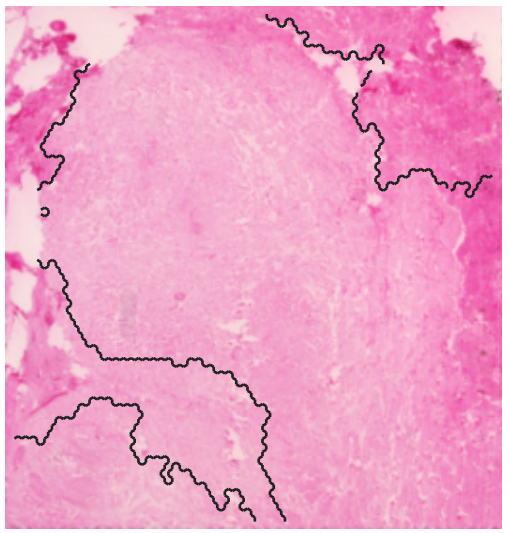
CanDis_2026_Sample1 | Tissue context

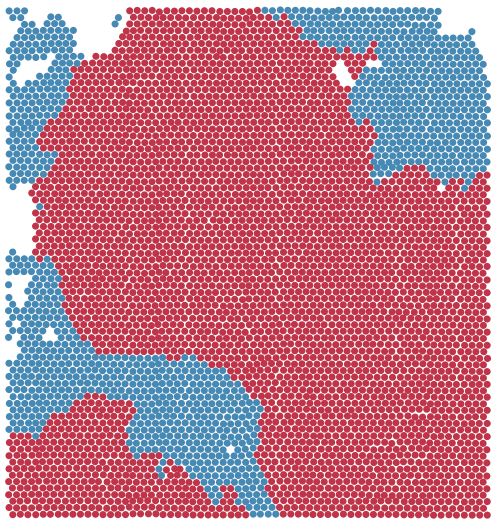
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



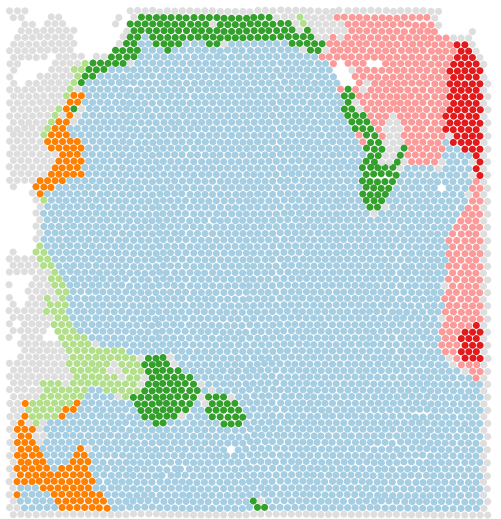
Black line: observed candidate outer edge

Final candidate location



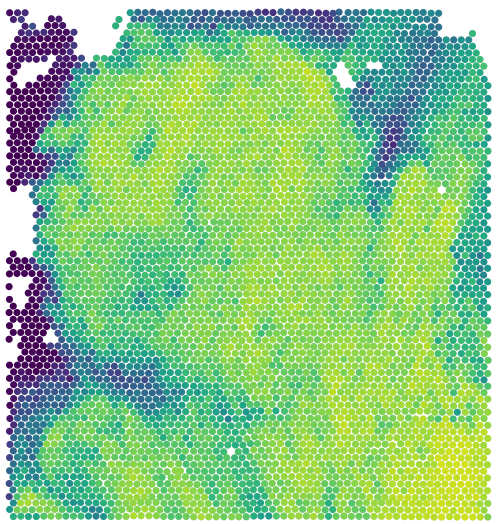
inside outside

Accepted R12 (original 0–8)



0 3 5 6 7 2

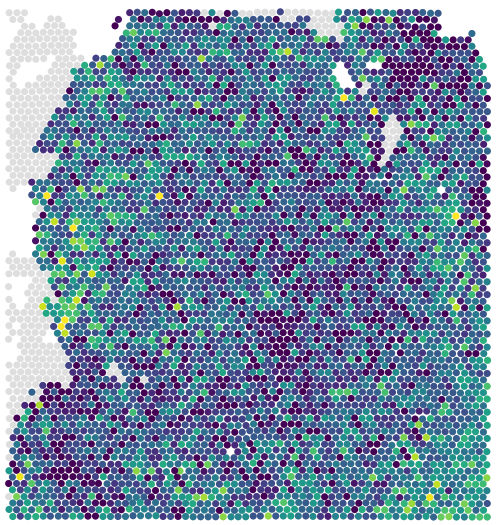
Epithelial support



0 1

Fraction of all 30

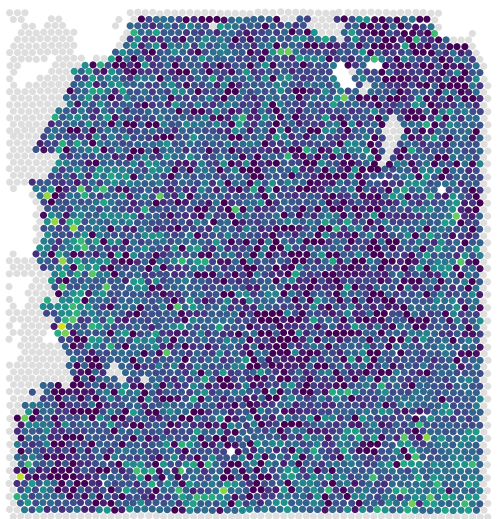
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



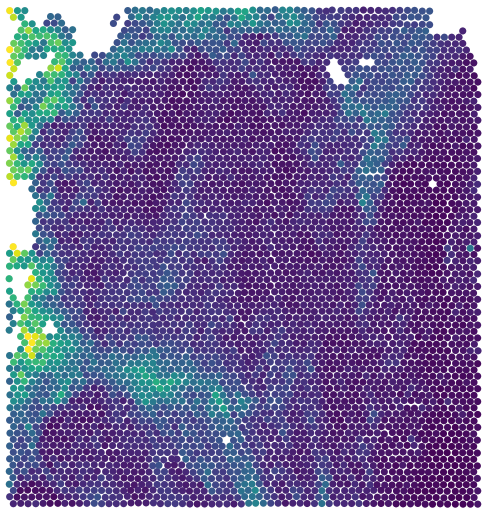
0 0.14

Matched-model AUCell

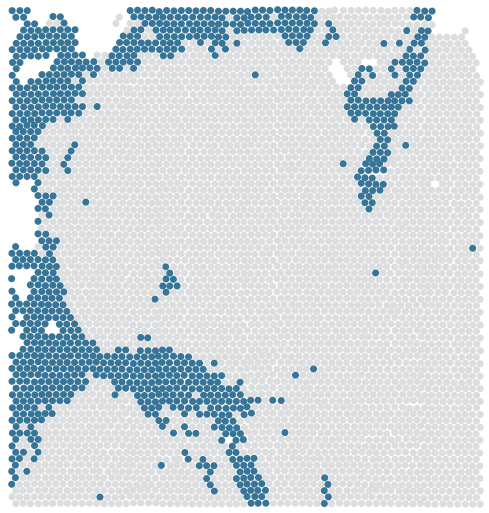
CanDis_2026_Sample1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

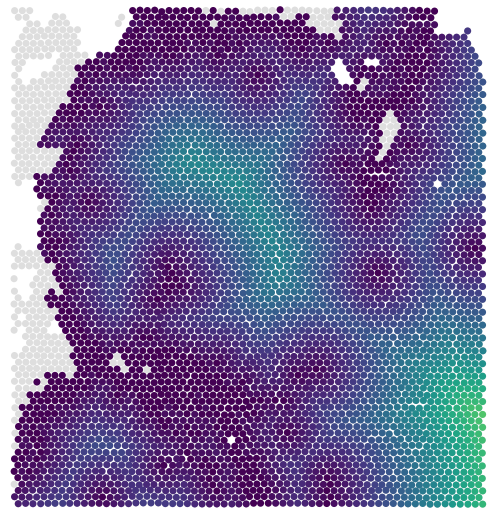
CAF composition



CAF P80 targets



CAF distance

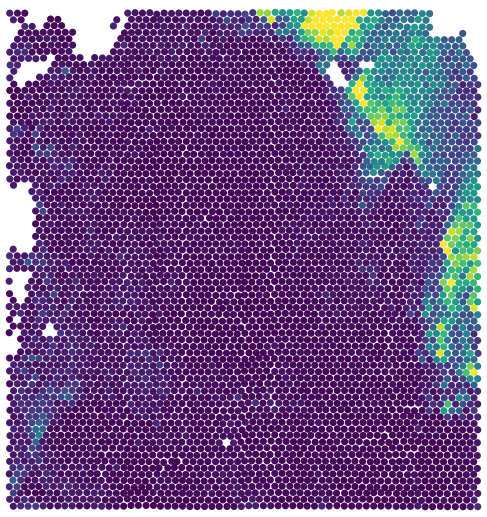


0 0.418
Fraction of all 30

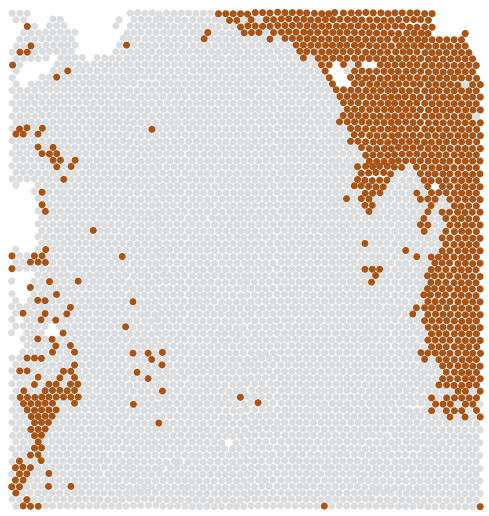
False True

0 3.1e+03
Nearest spot center (μm)

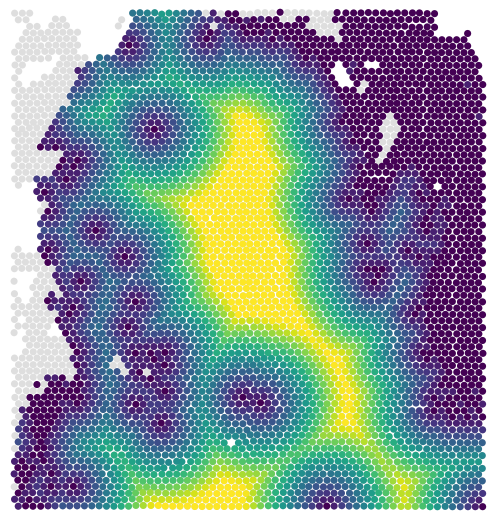
TAM composition



TAM P80 targets



TAM distance

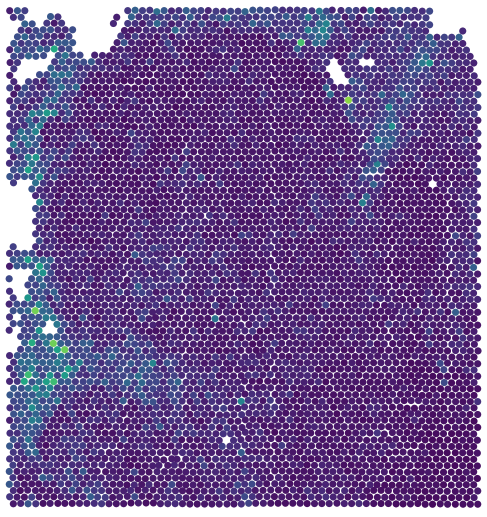


0 0.291
Fraction of all 30

False True

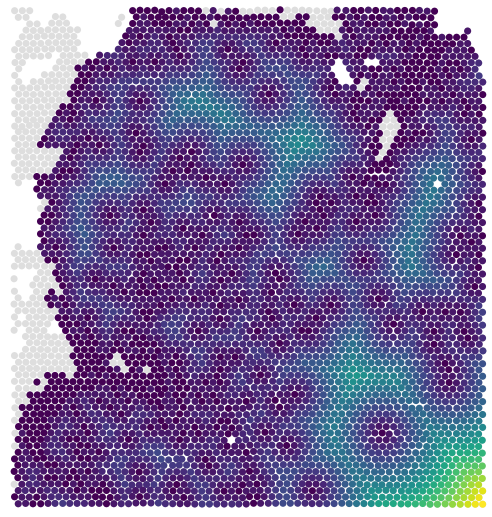
0 1.13e+03
Nearest spot center (μm)

Treg composition



Treg P80 targets

Treg distance



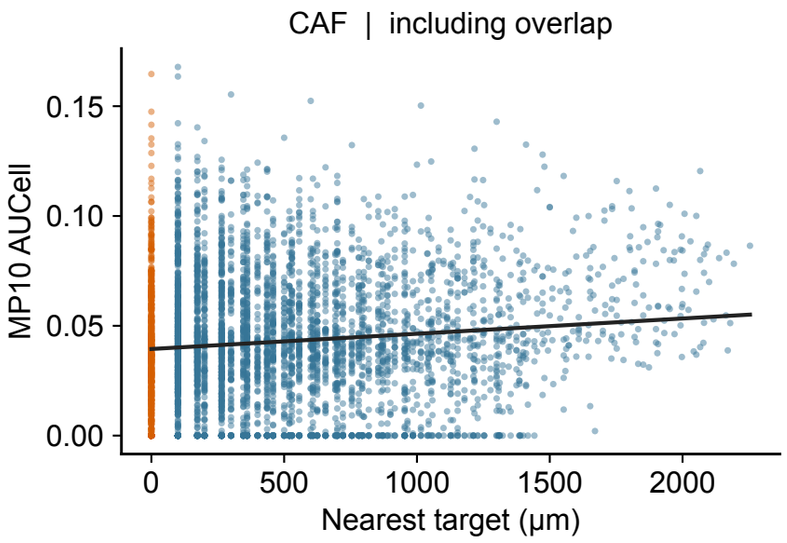
0 0.102
Fraction of all 30

False True

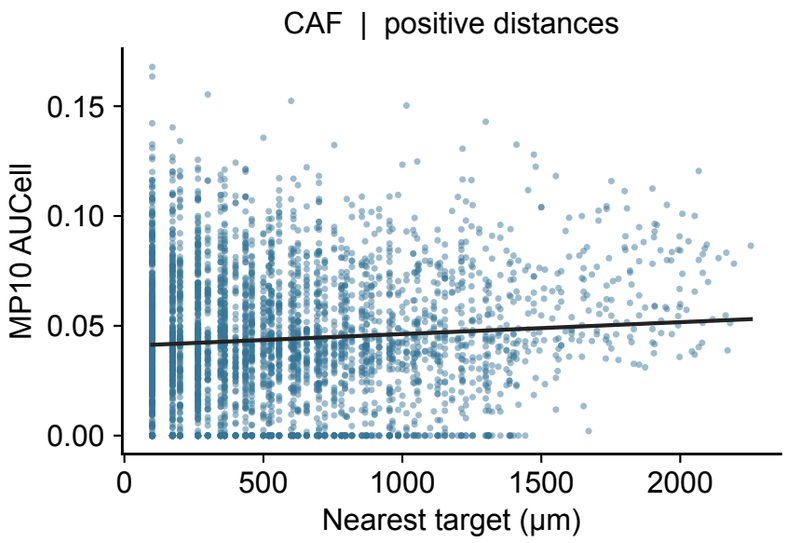
0 1.61e+03
Nearest spot center (μm)

CanDis_2026_Sample1 | MP10 versus nearest target

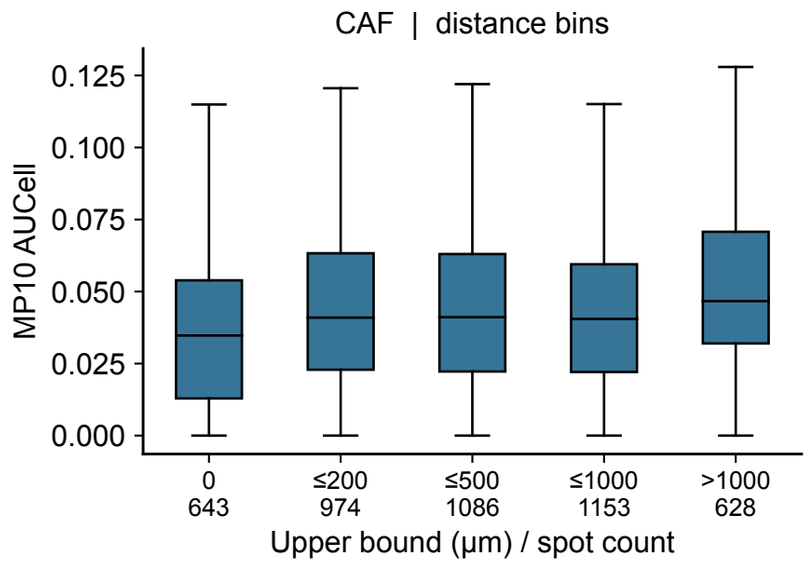
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



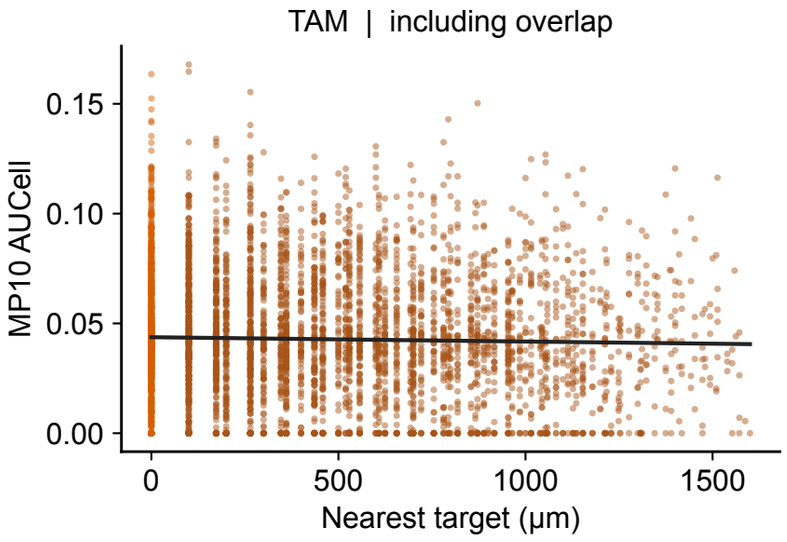
r +0.11 ρ +0.09
n 4484/4486 queries • 965 targets



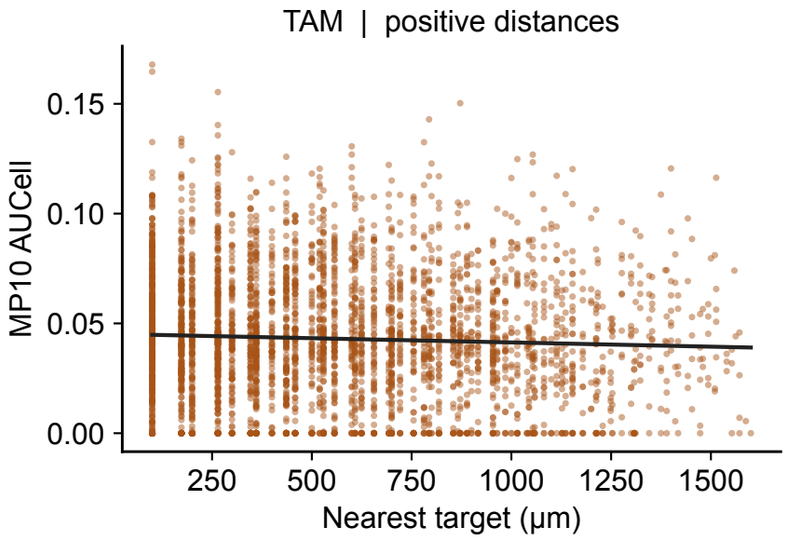
r +0.08 ρ +0.05
n 3841/4486 queries • 965 targets



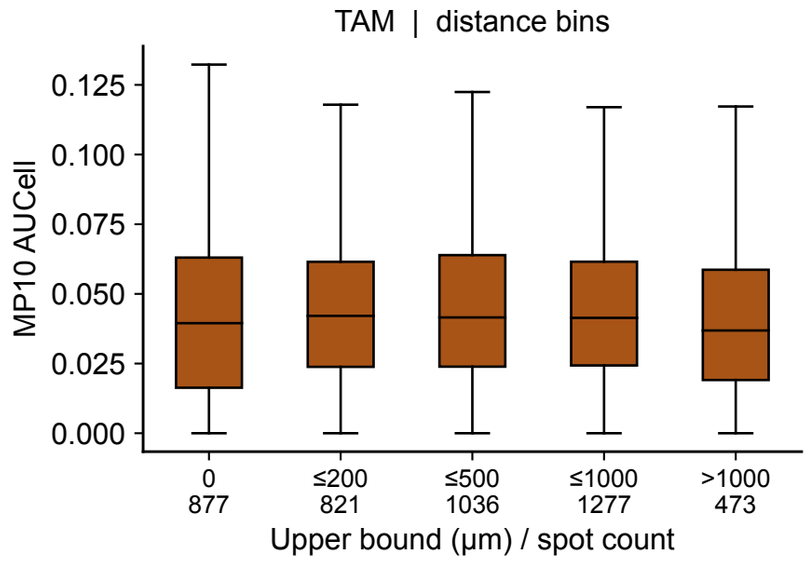
Orange = zero distance
Line = within-section linear fit



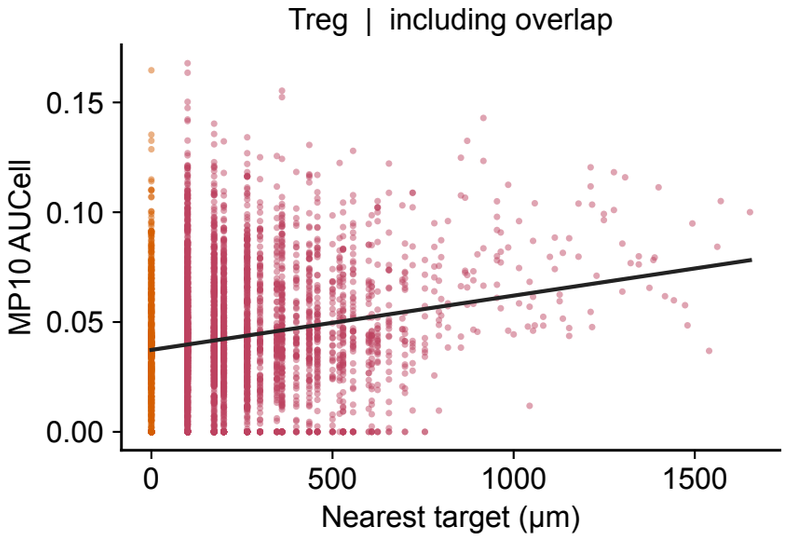
r -0.03 ρ -0.01
n 4484/4486 queries • 965 targets



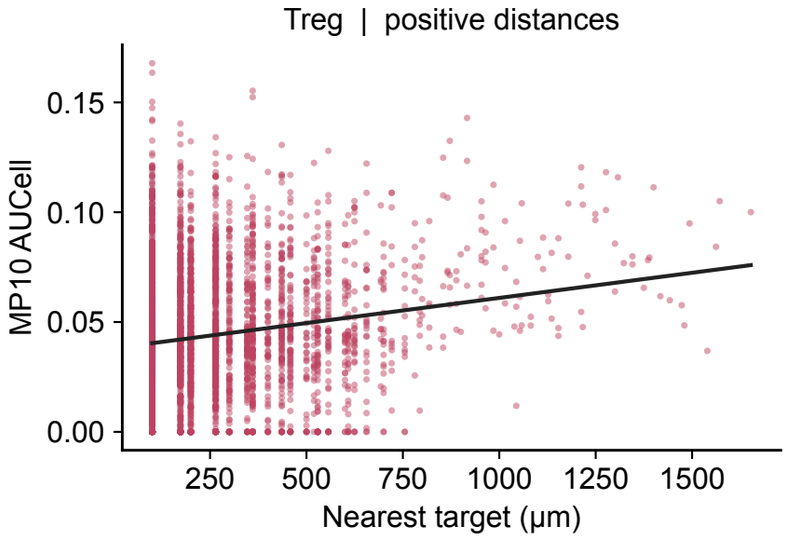
r -0.05 ρ -0.04
n 3607/4486 queries • 965 targets



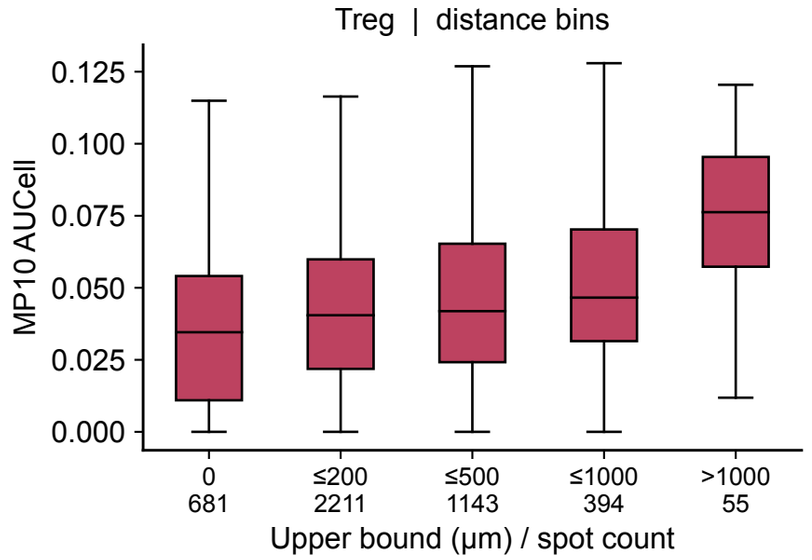
Orange = zero distance
Line = within-section linear fit



r +0.18 ρ +0.15
n 4484/4486 queries • 965 targets



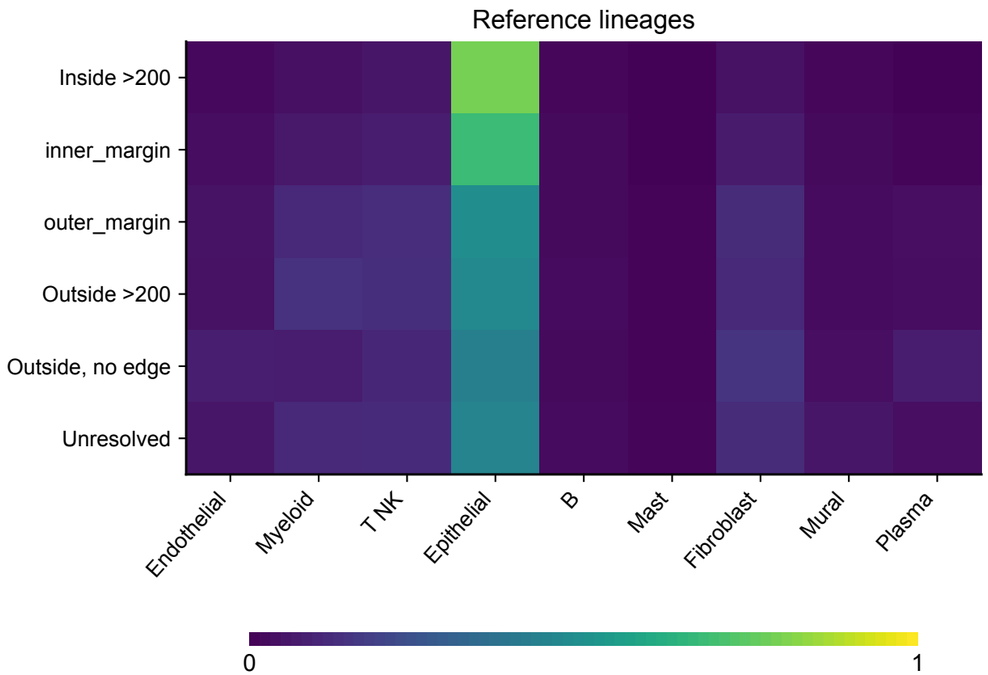
r +0.16 ρ +0.11
n 3803/4486 queries • 965 targets

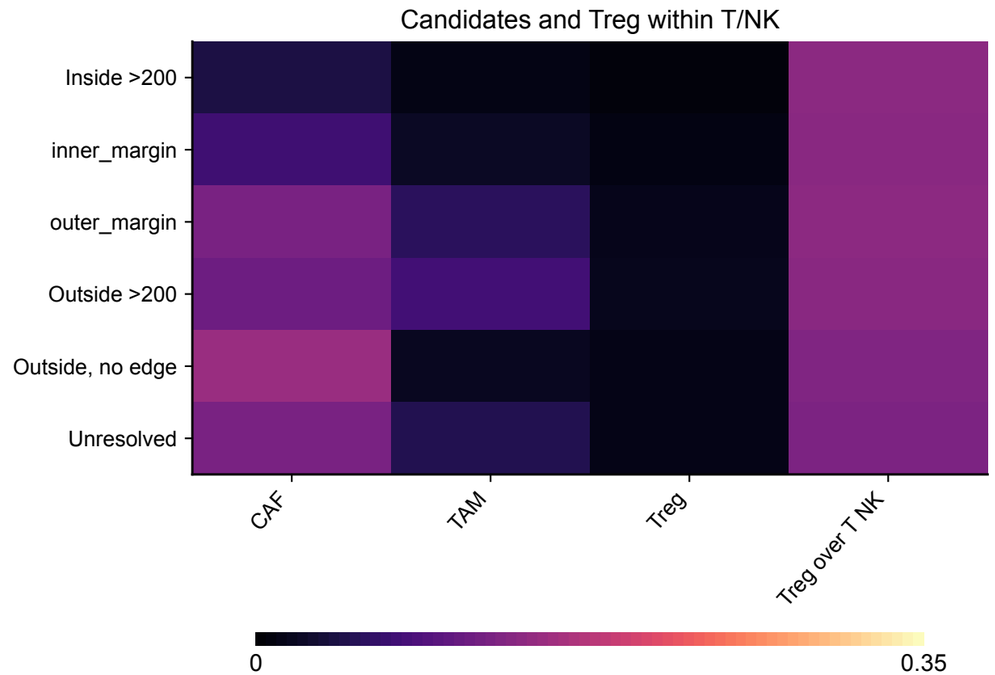


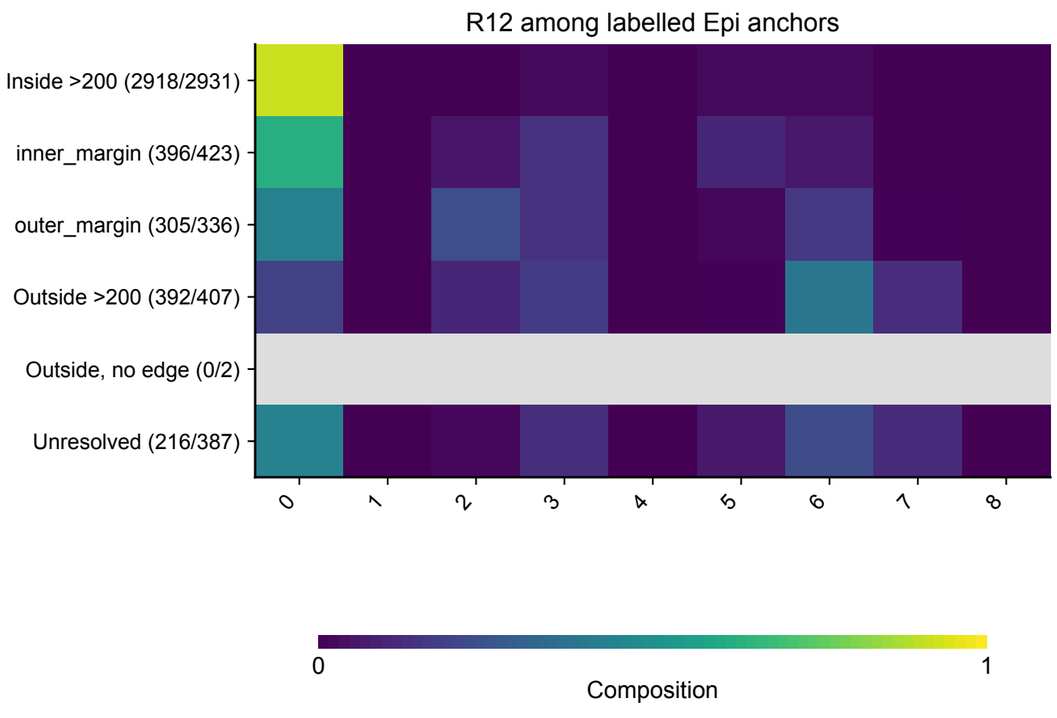
Orange = zero distance
Line = within-section linear fit

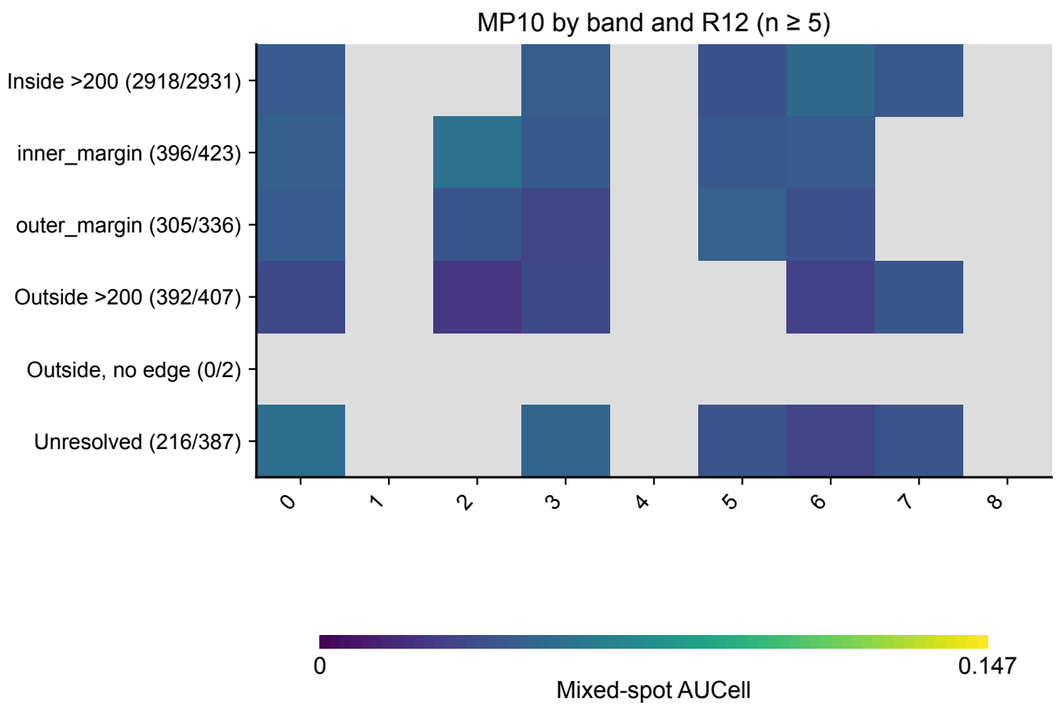
CanDis_2026_Sample1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





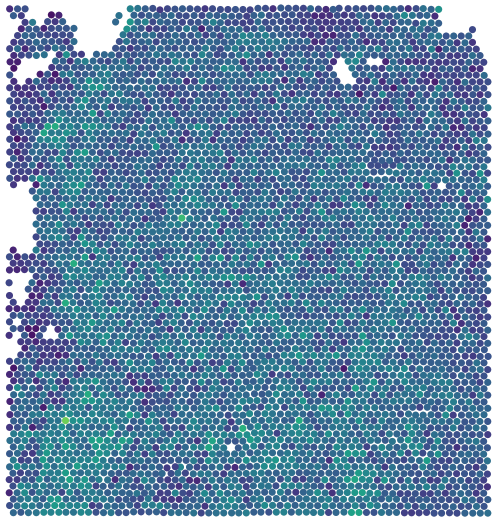




CanDis_2026_Sample1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

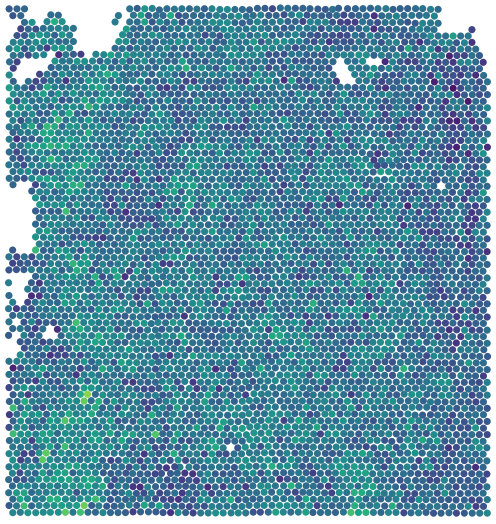
E2F Targets



0 0.152

UCell score

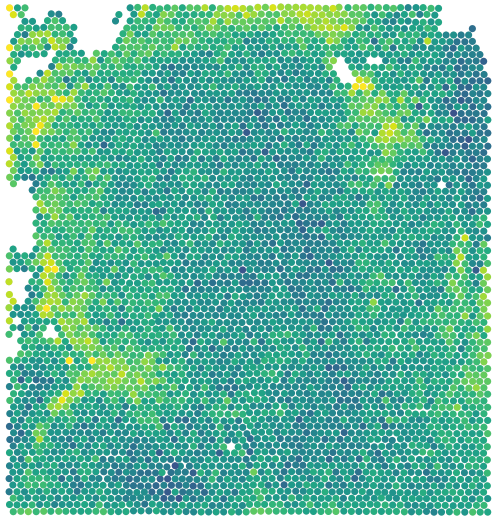
G2M Checkpoint



0 0.134

UCell score

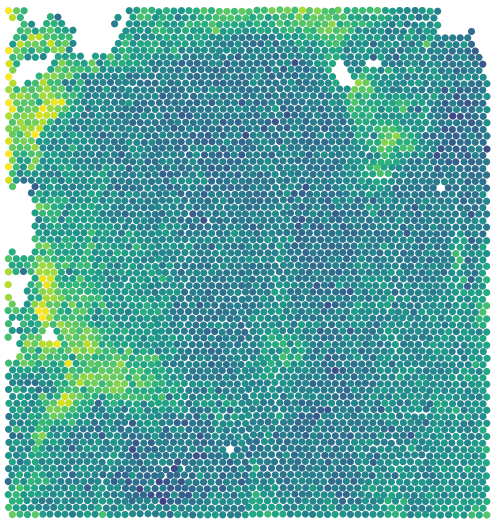
Hypoxia



0 0.211

UCell score

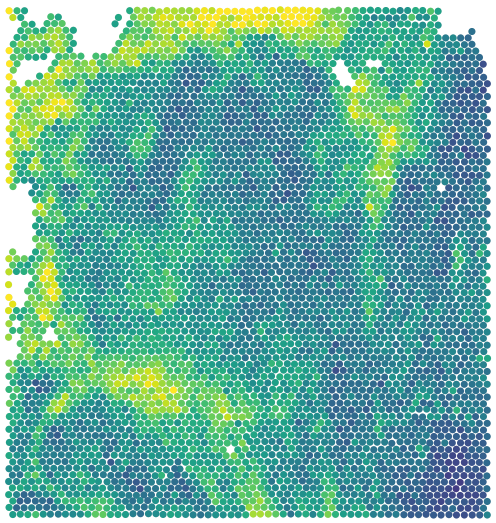
Tnfa Signaling Via Nfkb



0 0.275

UCell score

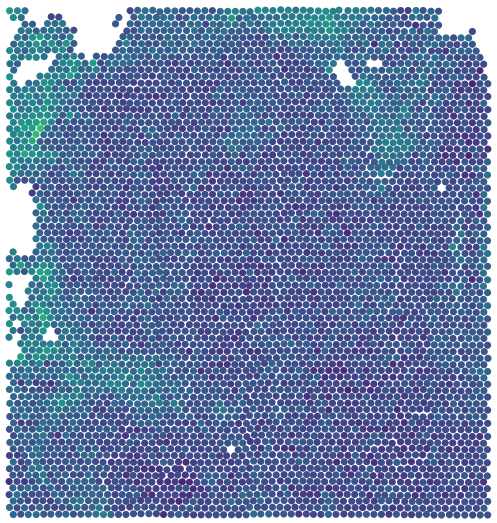
Emt



0 0.386

UCell score

Interferon Gamma Response



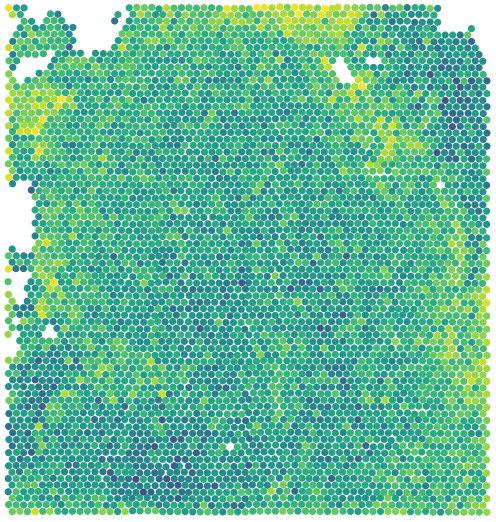
0 0.275

UCell score

CanDis_2026_Sample1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

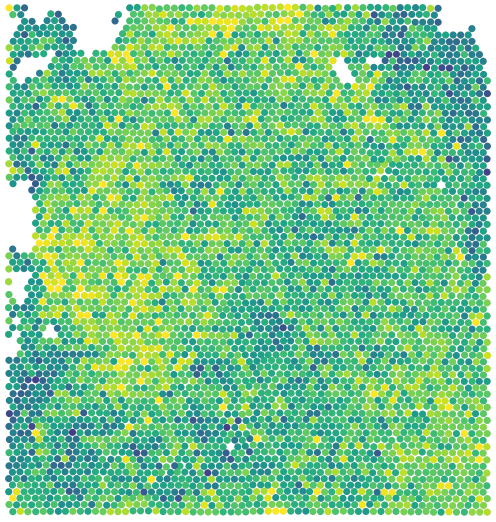
Il2 Stat5 Signaling



0 0.144

UCell score

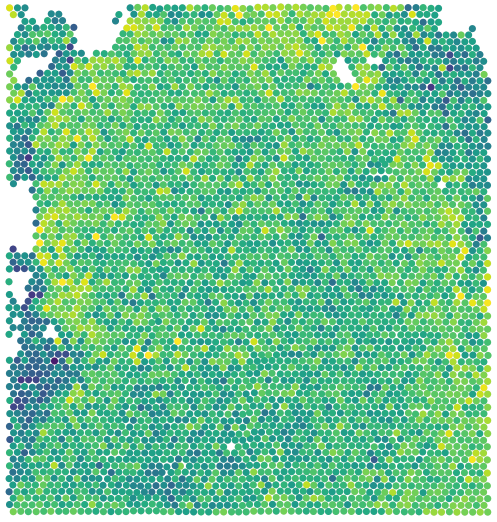
Tgf Beta Signaling



0 0.232

UCell score

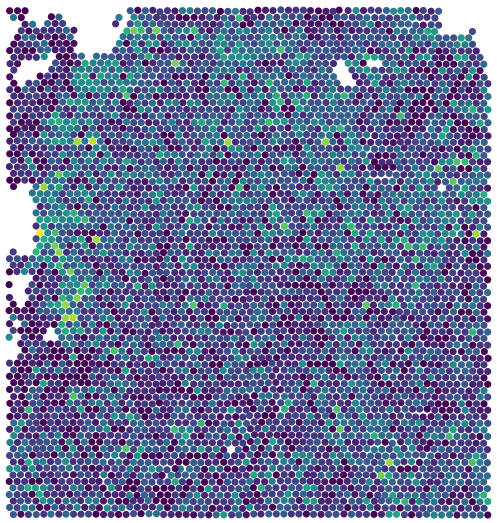
Cholesterol Homeostasis



0 0.21

UCell score

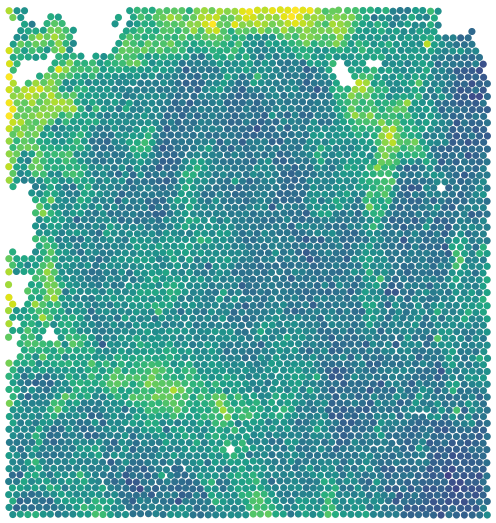
Cholesterol Biosynthesis



0 0.166

UCell score

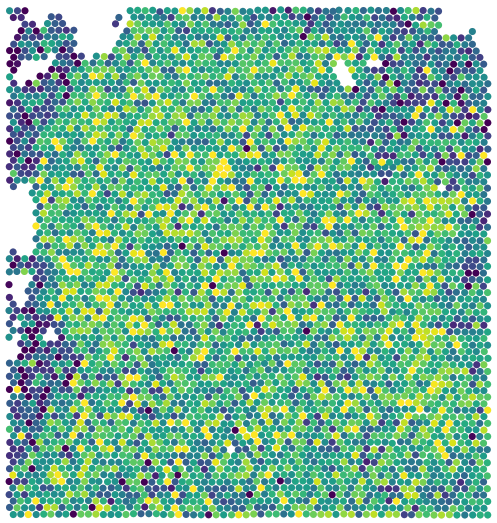
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



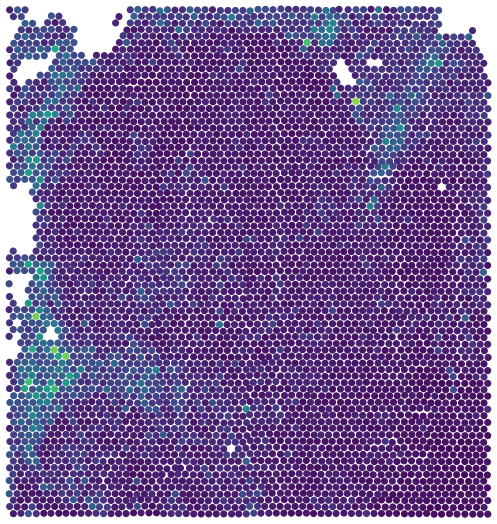
0 0.151

UCell score

CanDis_2026_Sample1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

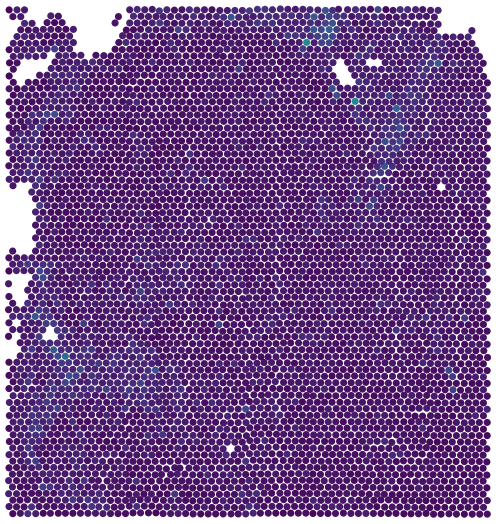
Treg composition



0 0.102

Fraction of all 30

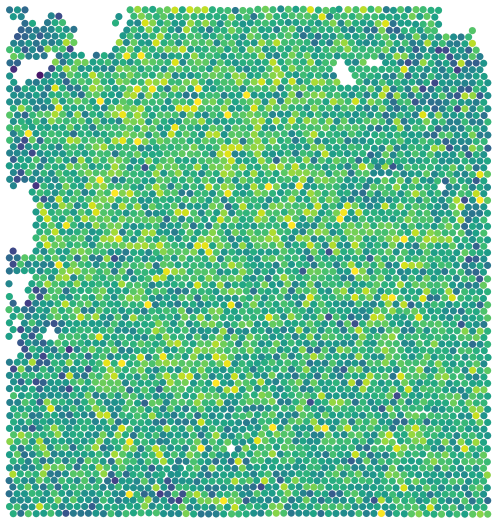
Treg estimated abundance



0 2.85

Posterior mean estimate

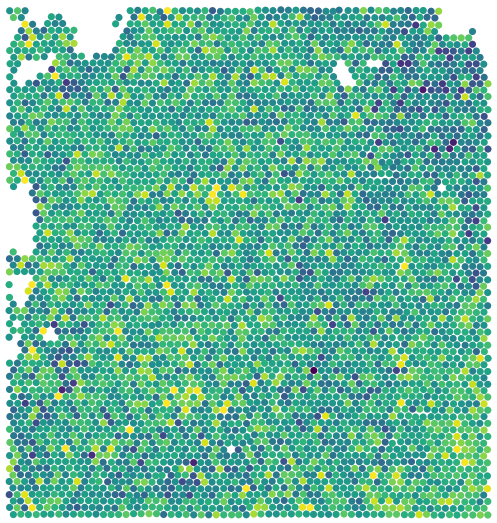
Treg Resting Vs Activated Dn



0 0.0778

UCell score

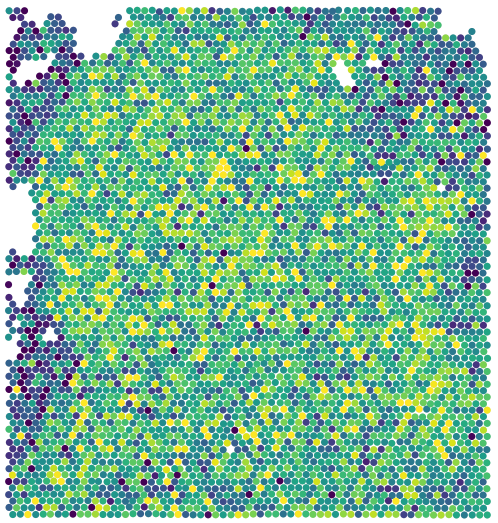
Treg Resting Vs Activated Up



0 0.0609

UCell score

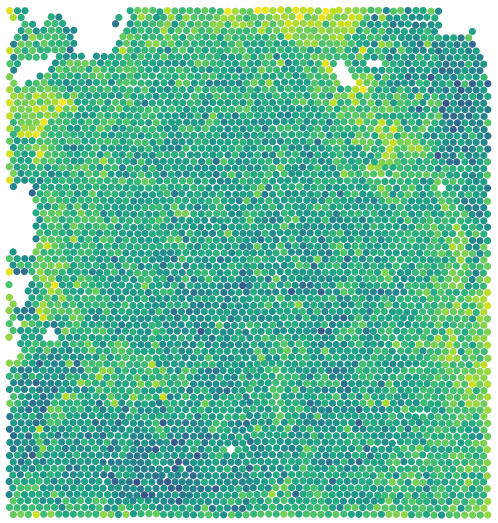
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

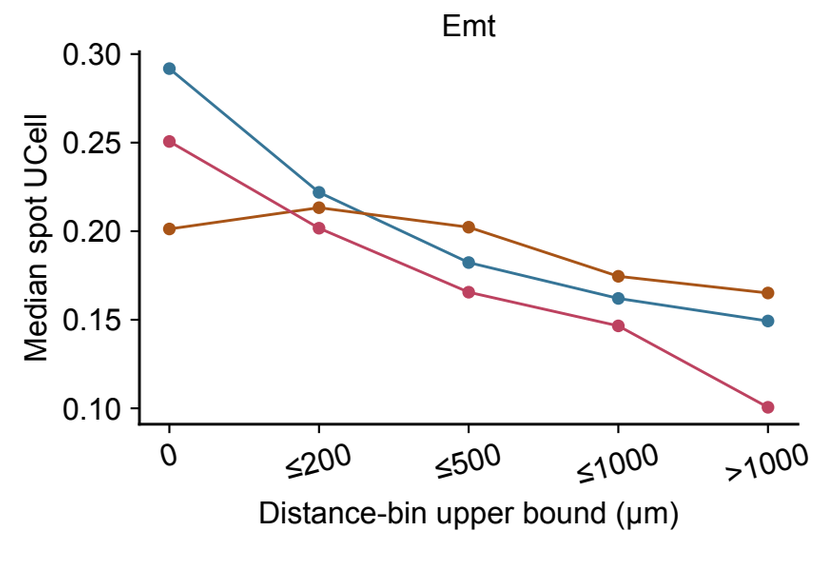


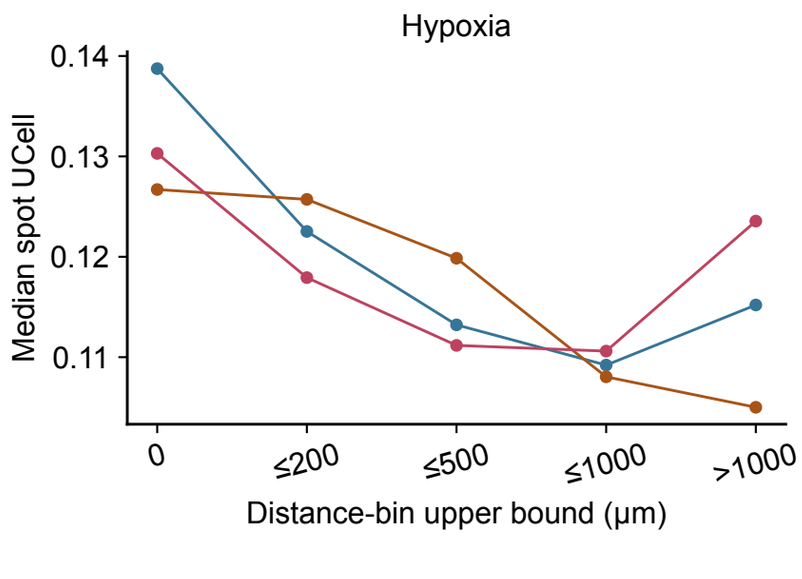
0 0.144

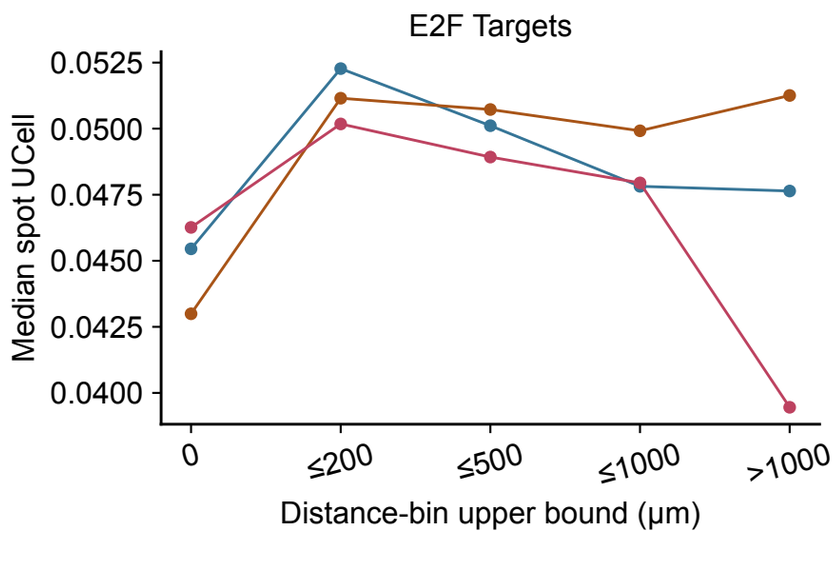
UCell score

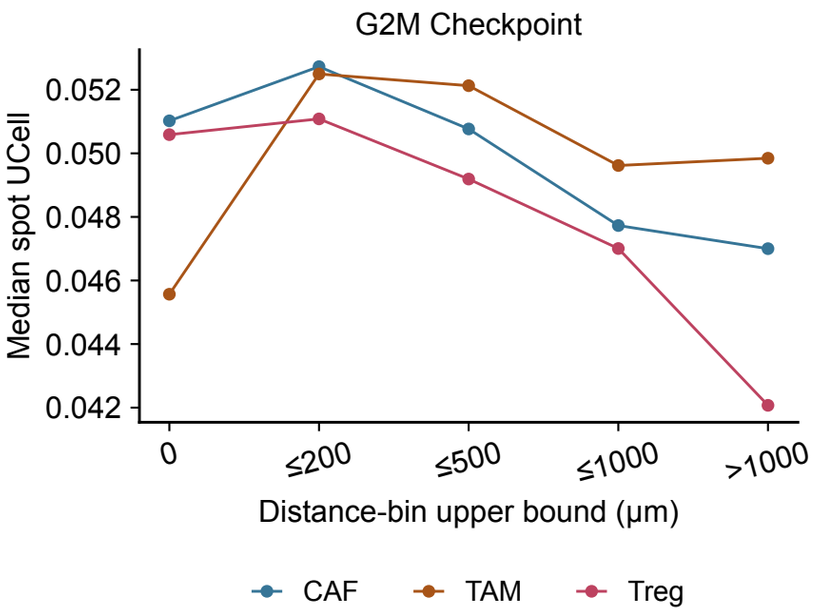
CanDis_2026_Sample1 | Programs by target distance

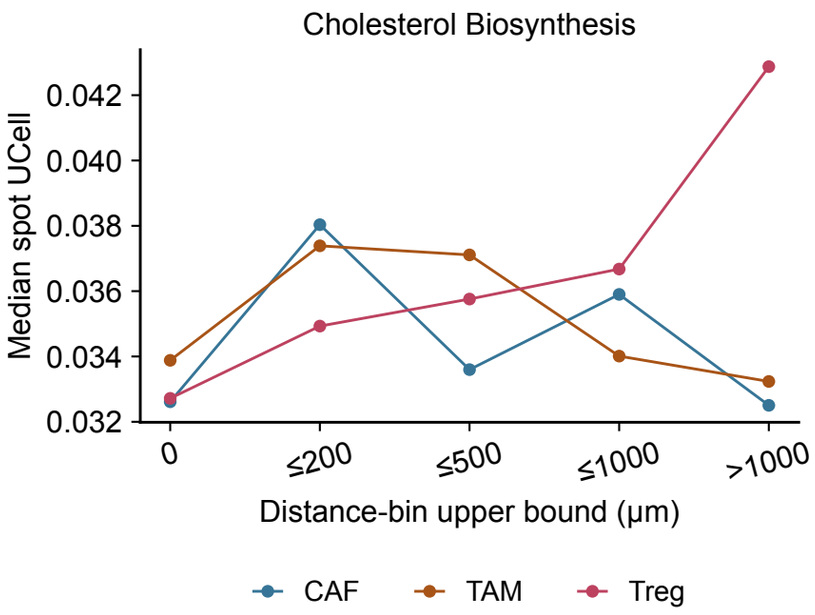
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

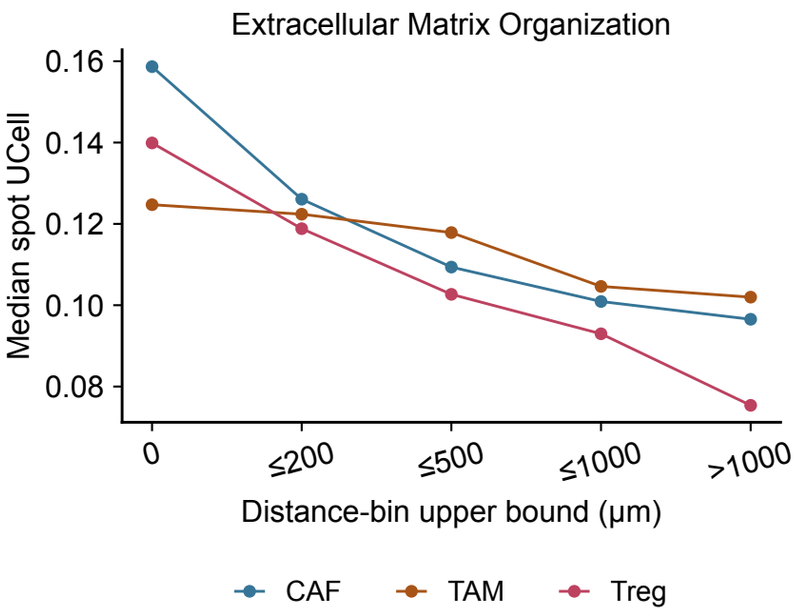






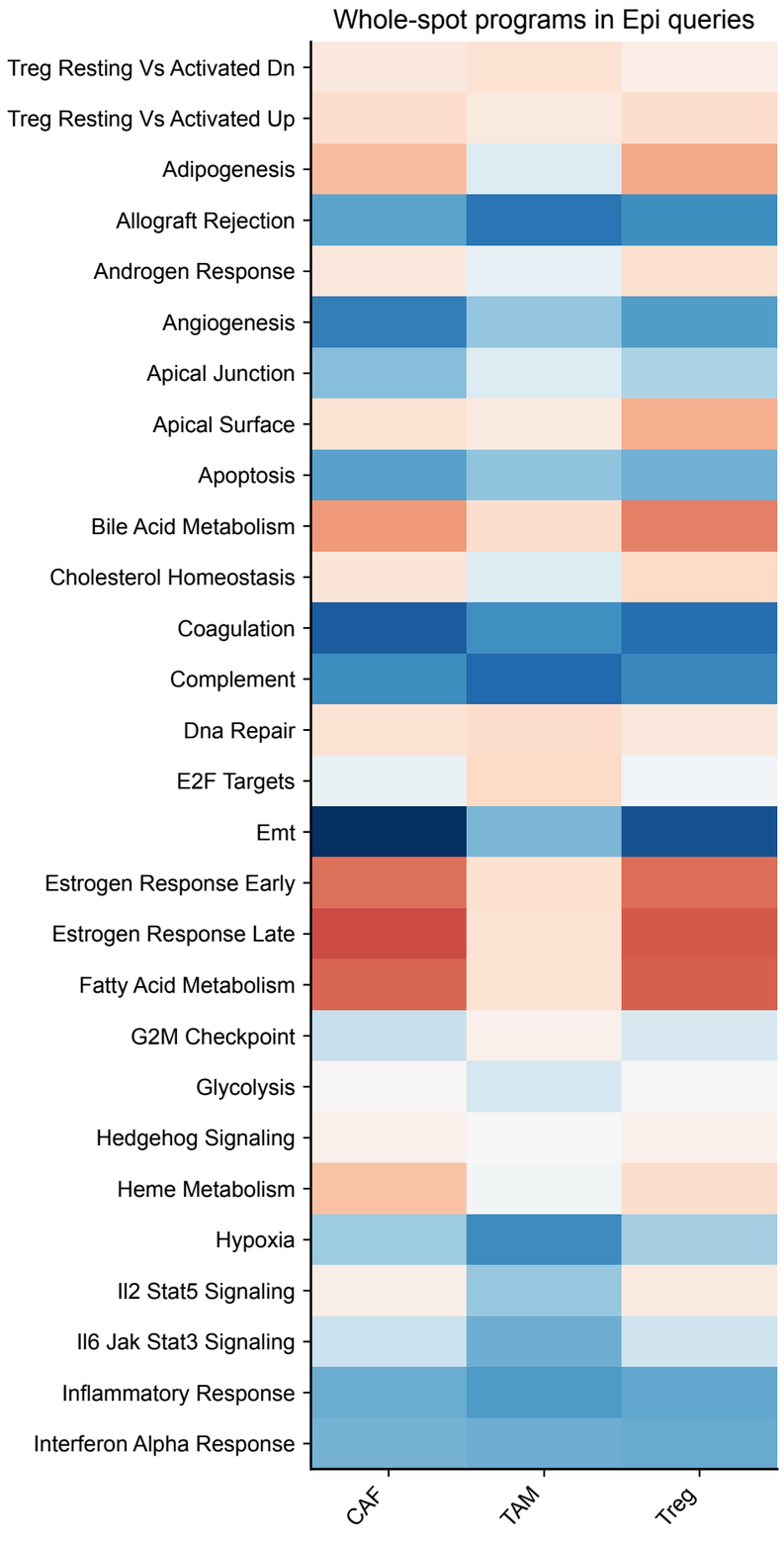


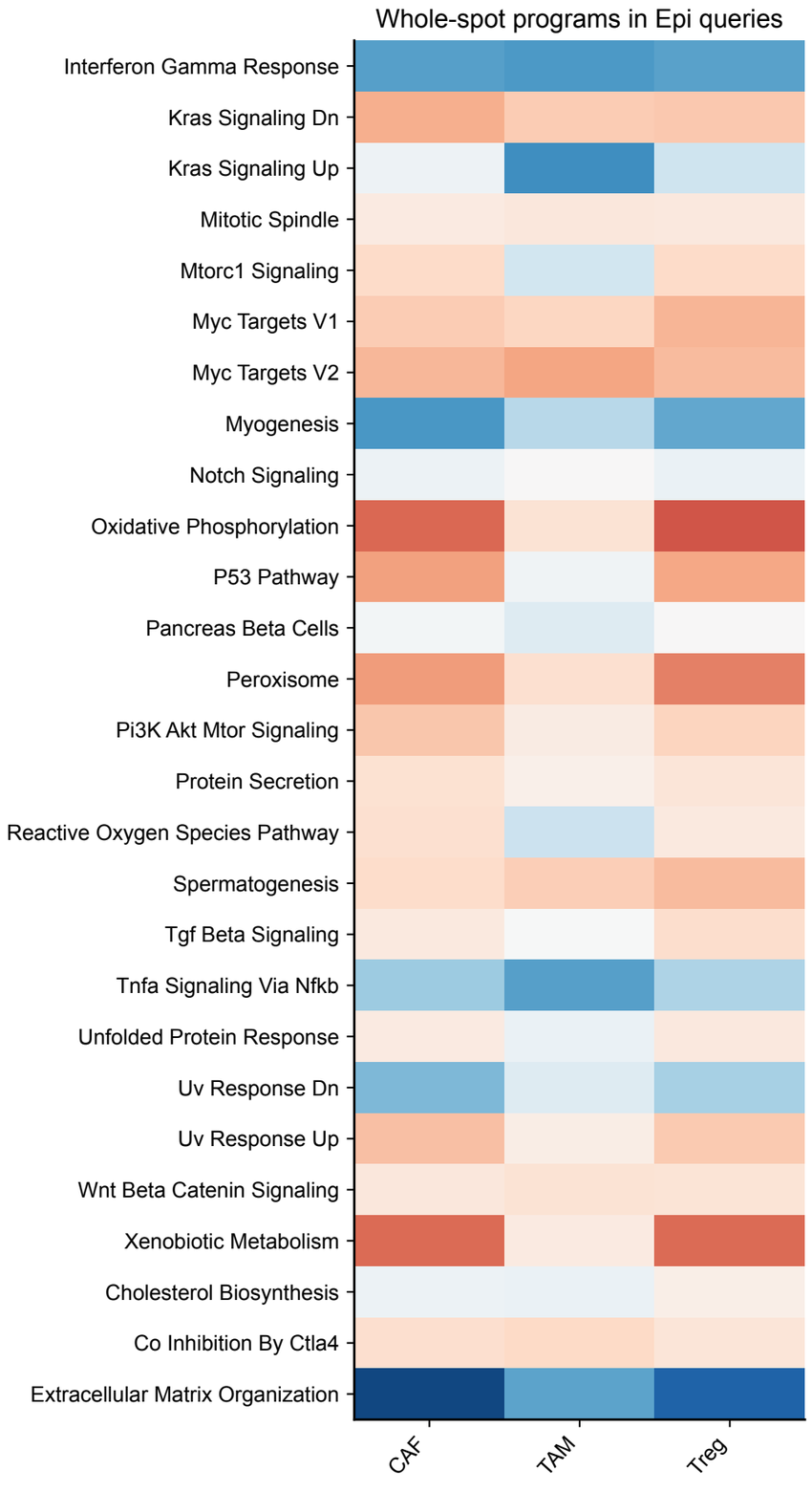




CanDis_2026_Sample1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

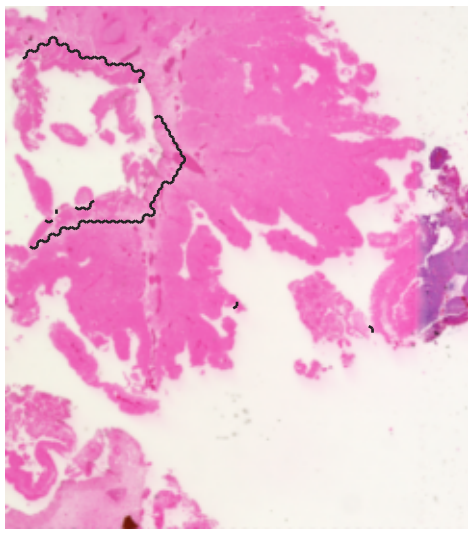




CanDis_2026_Sample10 | Tissue context

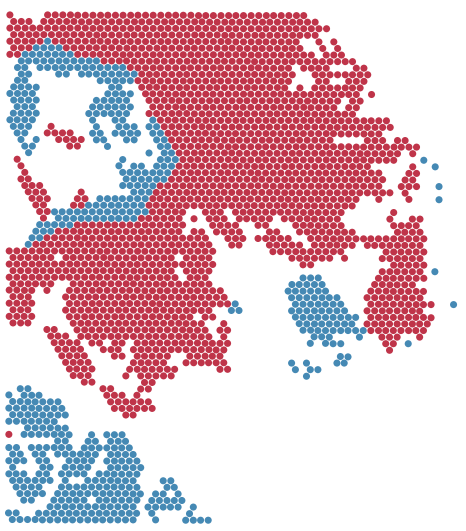
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



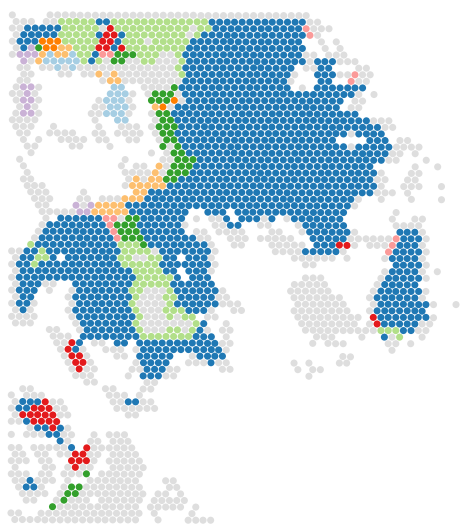
Black line: observed candidate outer edge

Final candidate location



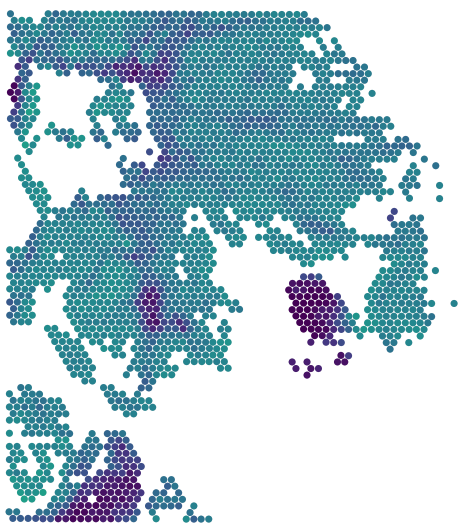
inside outside

Accepted R12 (original 0–8)



0 1 2 3 4 5 6 7 8

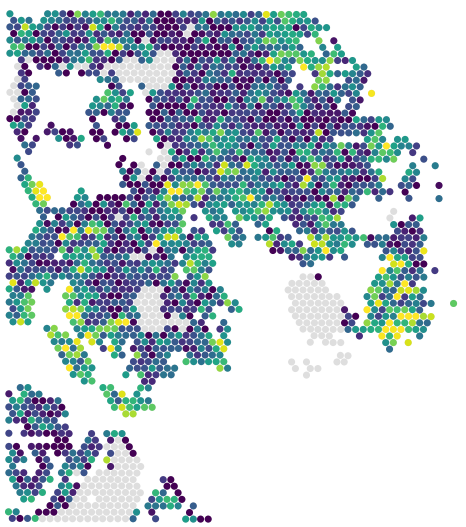
Epithelial support



0 1

Fraction of all 30

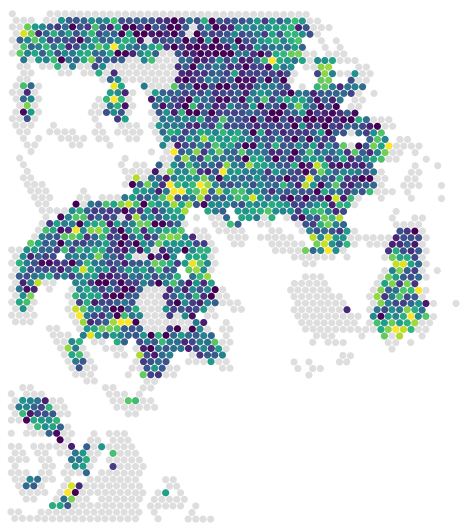
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



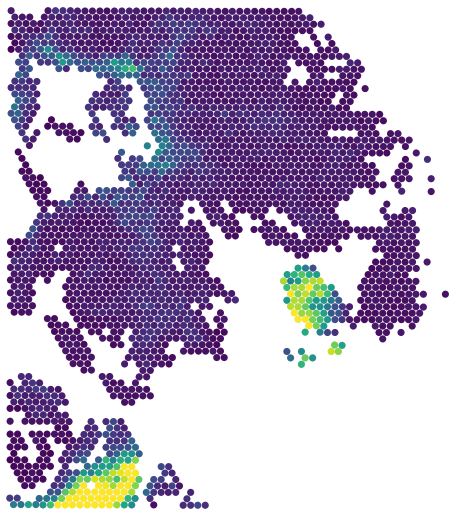
0 0.14

Matched-model AUCell

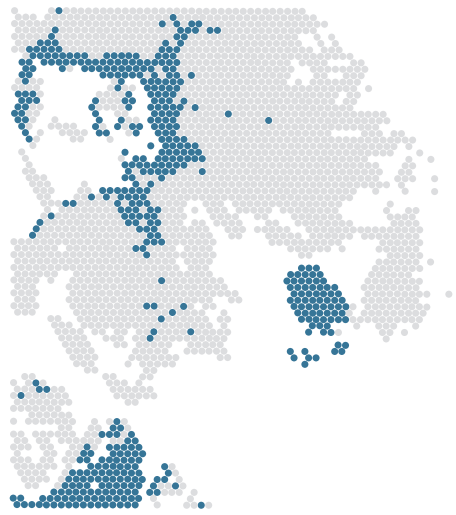
CanDis_2026_Sample10 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

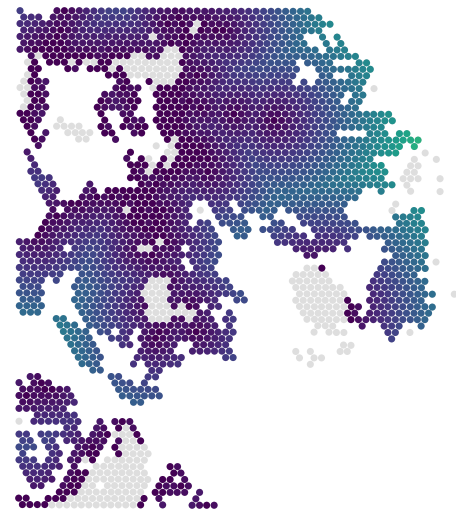
CAF composition



CAF P80 targets



CAF distance

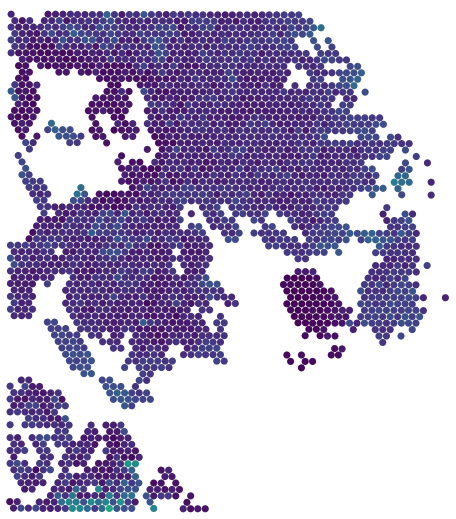


0 0.418
Fraction of all 30

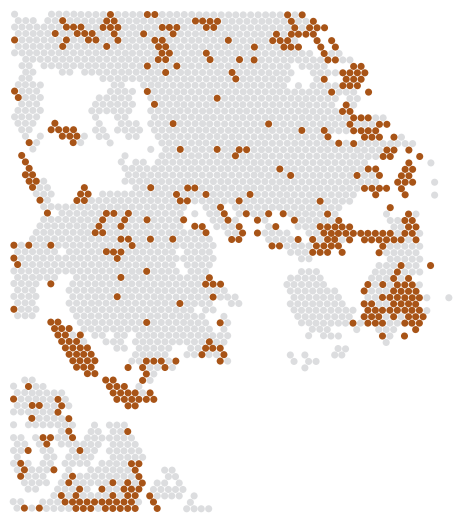
False True

0 3.1e+03
Nearest spot center (µm)

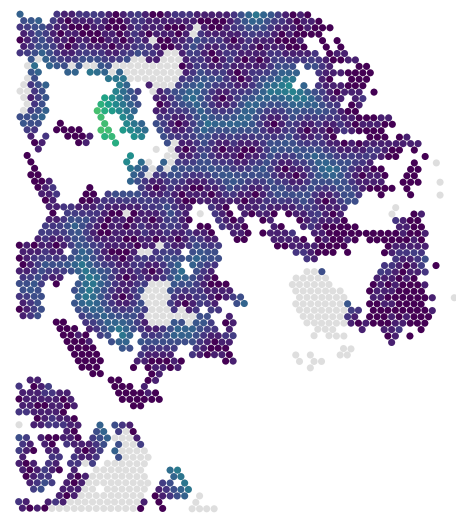
TAM composition



TAM P80 targets



TAM distance

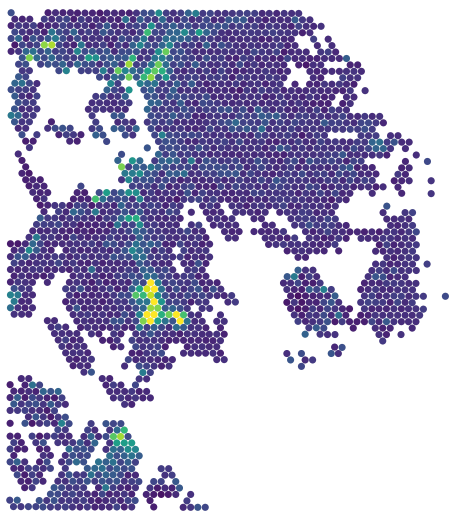


0 0.291
Fraction of all 30

False True

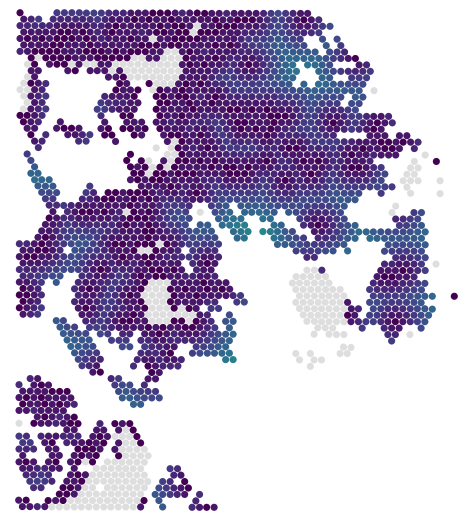
0 1.13e+03
Nearest spot center (µm)

Treg composition



Treg P80 targets

Treg distance



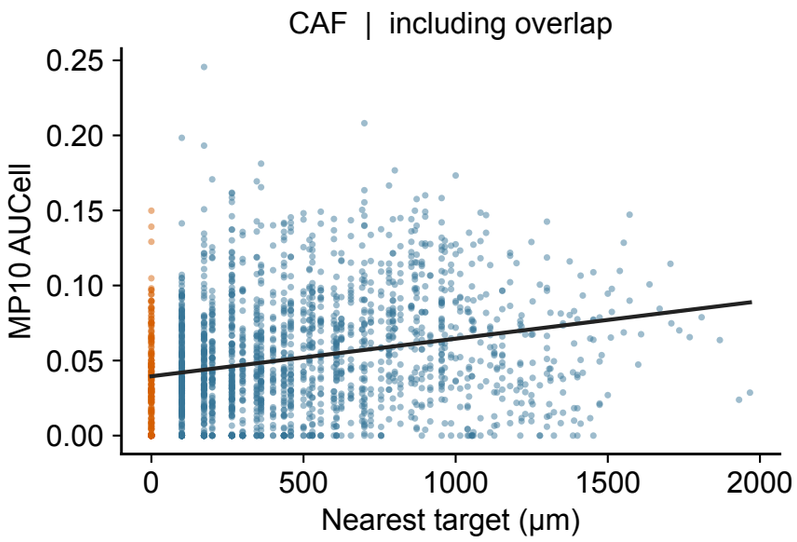
0 0.102
Fraction of all 30

False True

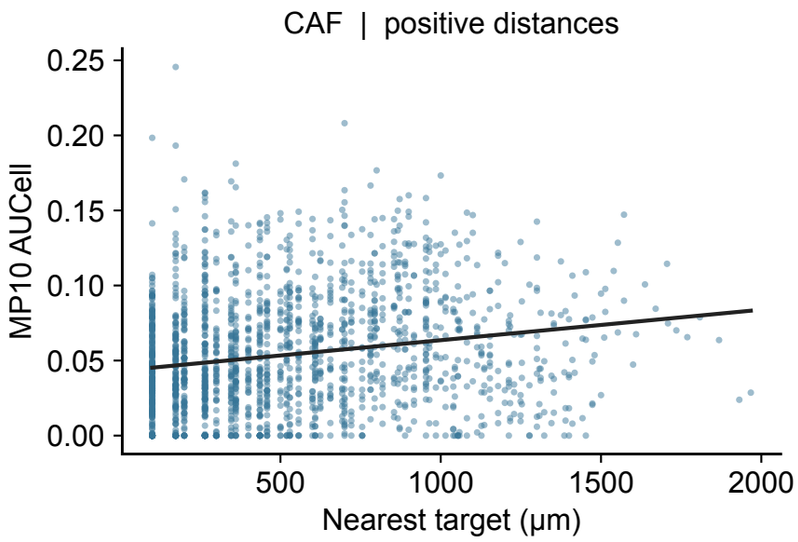
0 1.61e+03
Nearest spot center (µm)

CanDis_2026_Sample10 | MP10 versus nearest target

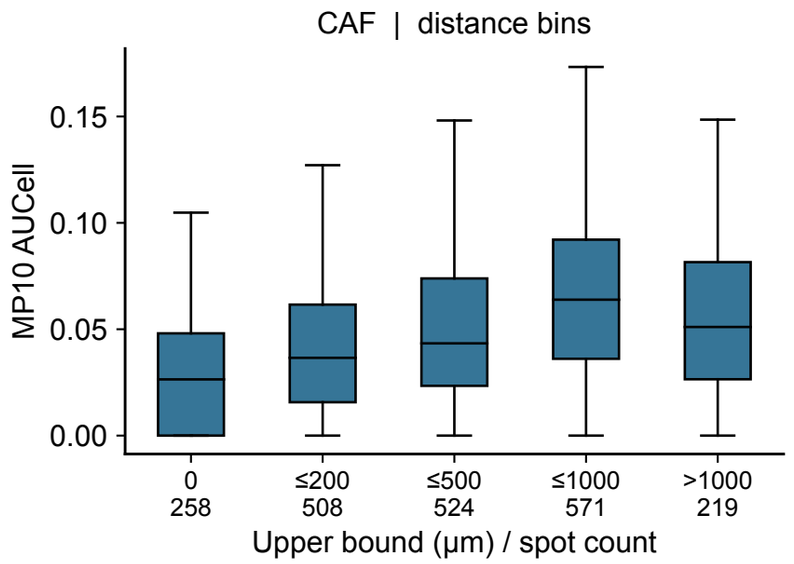
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



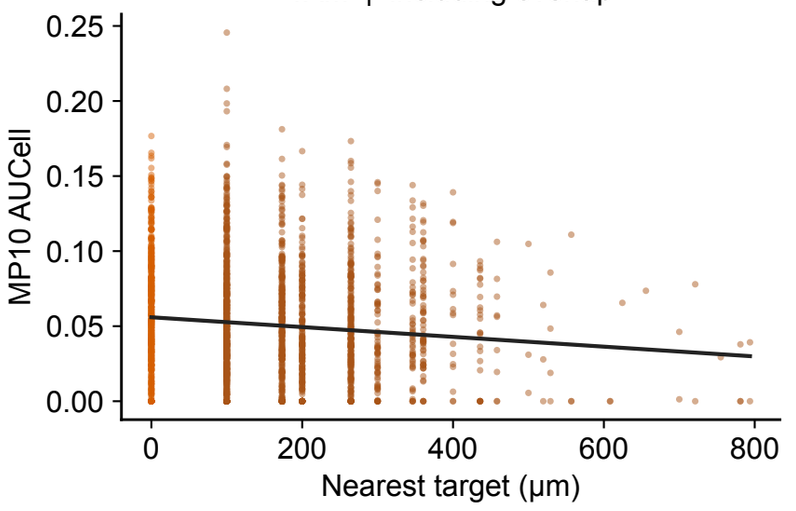
r +0.25 ρ +0.30
n 2080/2111 queries • 472 targets



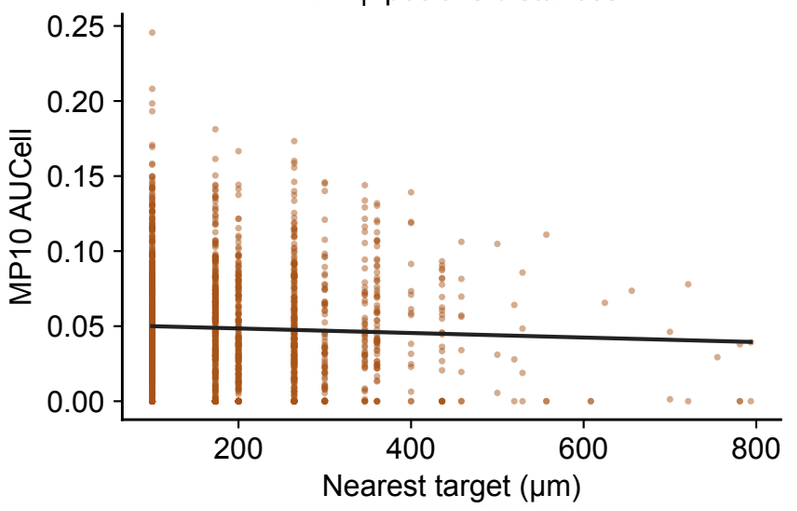
r +0.19 ρ +0.24
n 1822/2111 queries • 472 targets



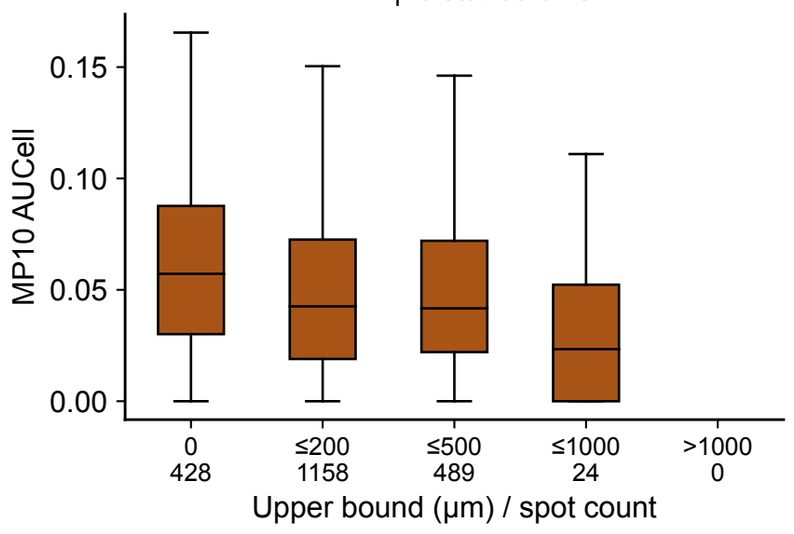
Orange = zero distance
Line = within-section linear fit



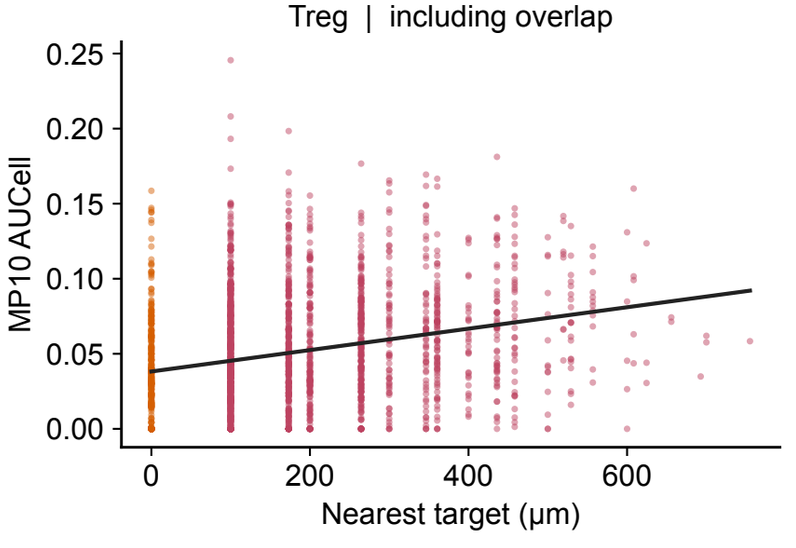
r -0.10 ρ -0.11
n 2099/2111 queries • 472 targets



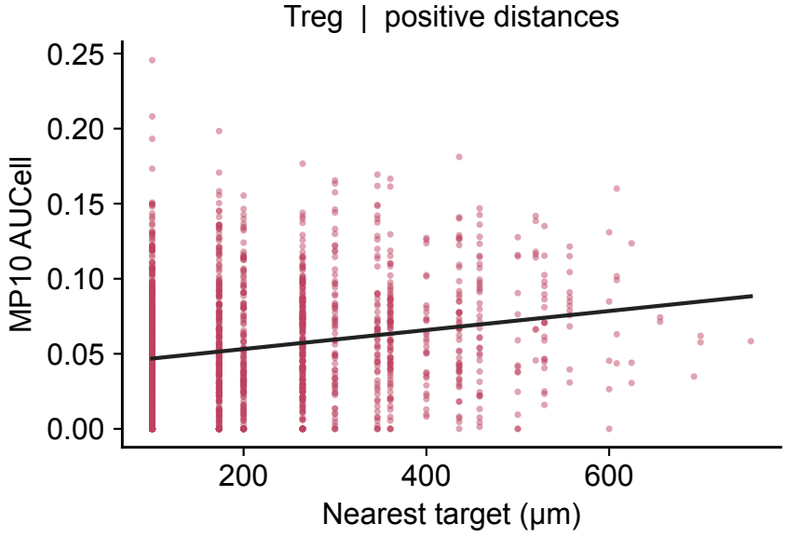
r -0.04 ρ -0.04
n 1671/2111 queries • 472 targets



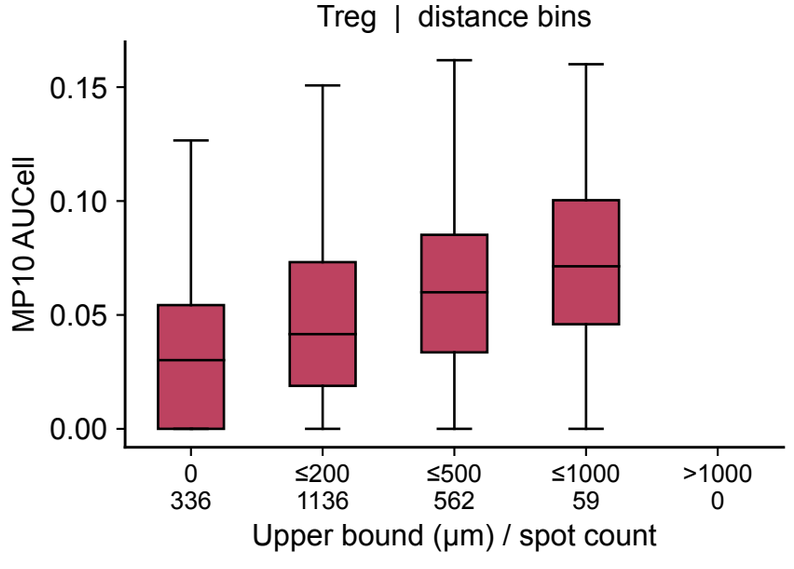
Orange = zero distance
Line = within-section linear fit



r +0.26 ρ +0.27
n 2093/2111 queries • 472 targets



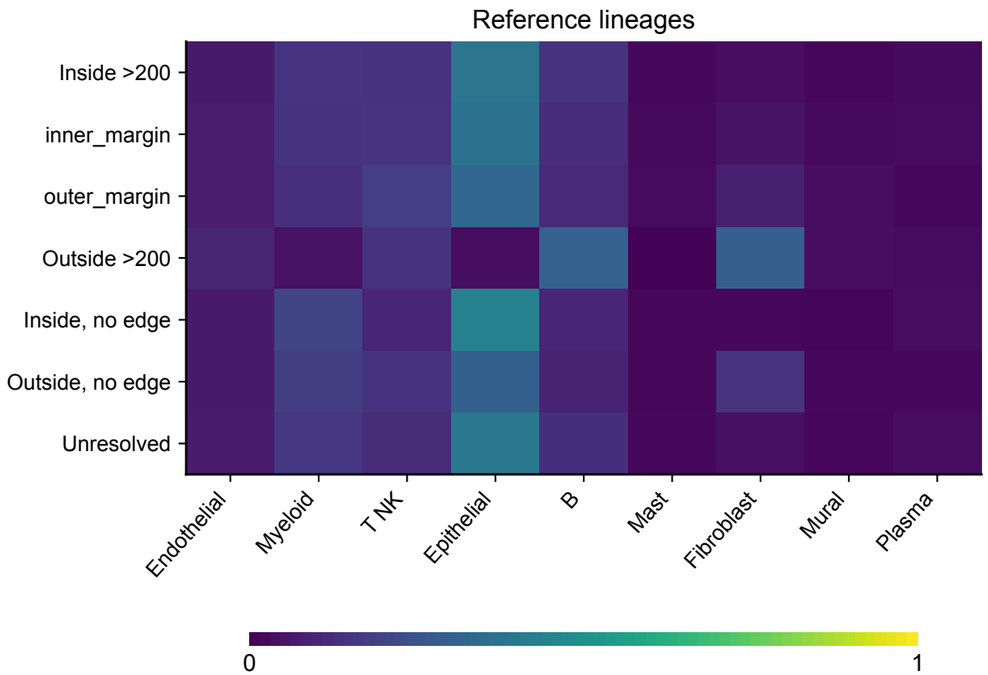
r +0.21 ρ +0.22
n 1757/2111 queries • 472 targets

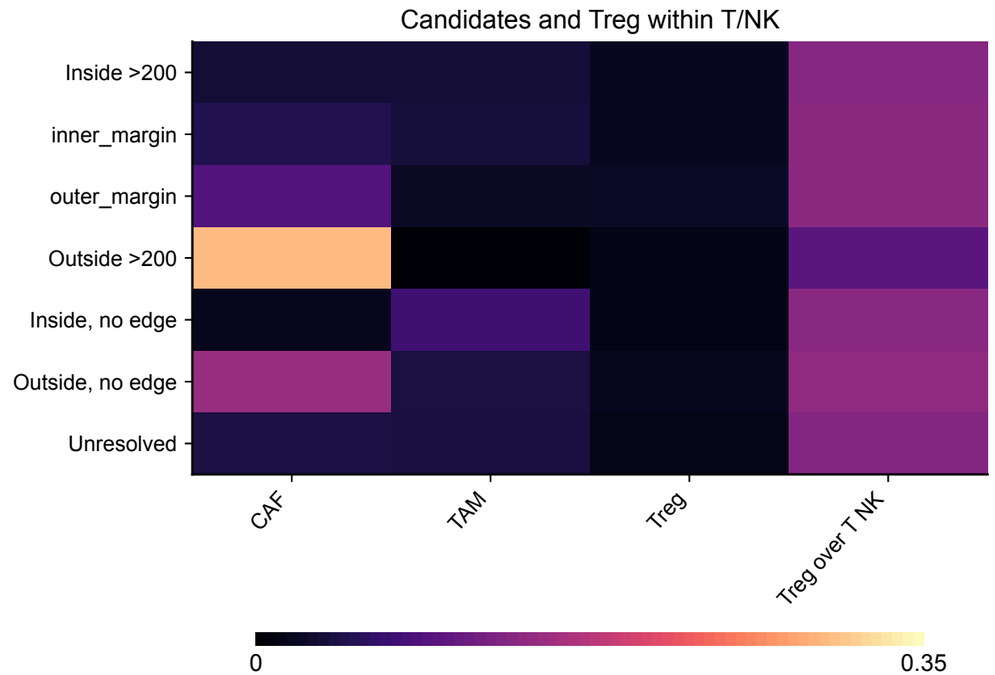


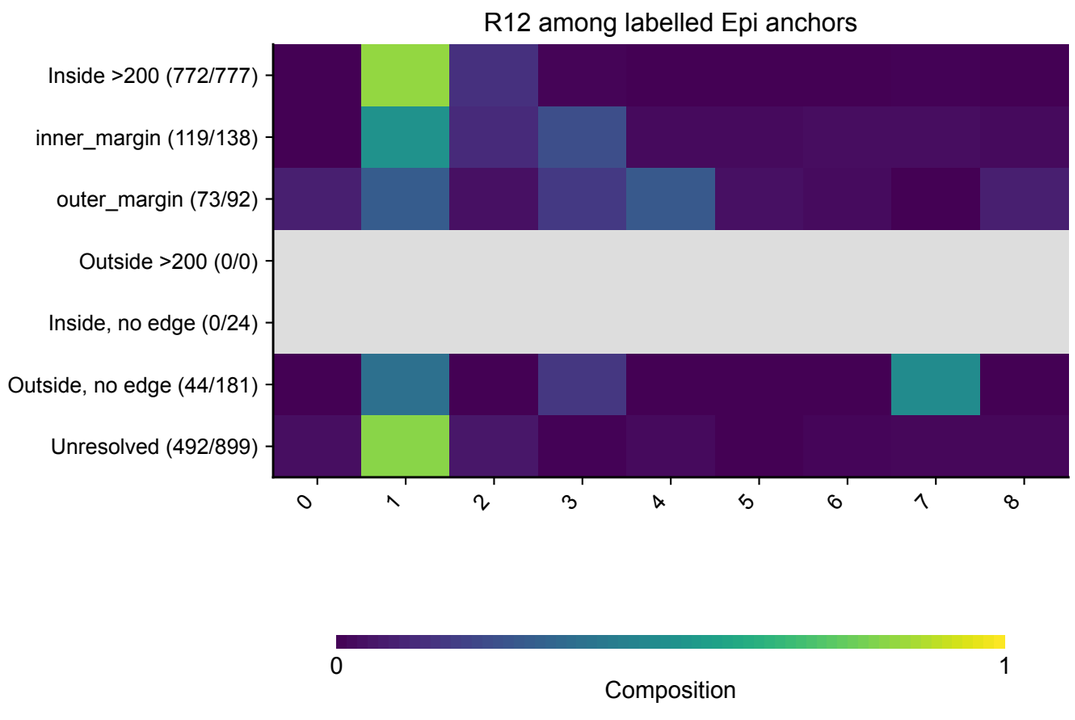
Orange = zero distance
Line = within-section linear fit

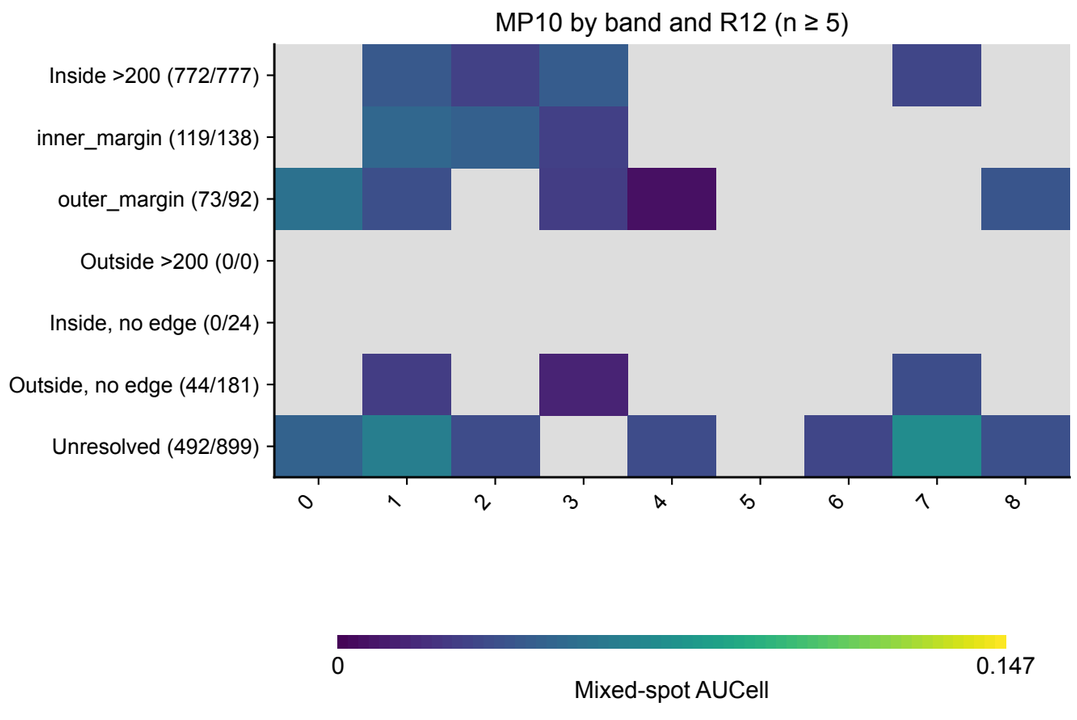
CanDis_2026_Sample10 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable









CanDis_2026_Sample10 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets

0 0.152

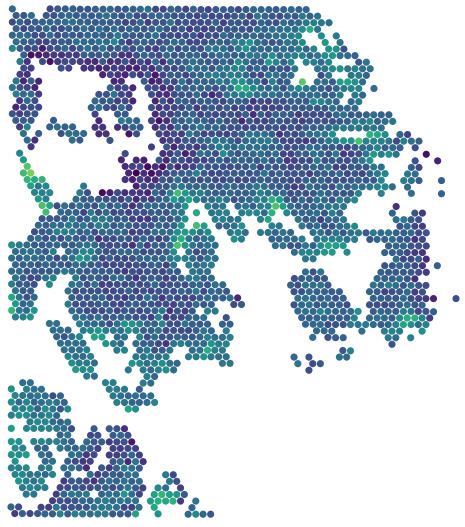
UCell score

G2M Checkpoint

0 0.134

UCell score

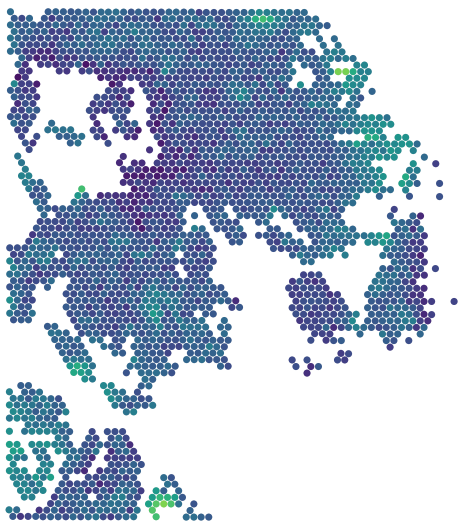
Hypoxia



0 0.211

UCell score

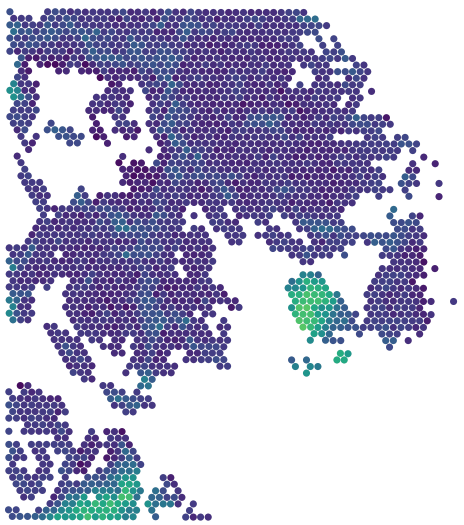
Tnfa Signaling Via Nfkb



0 0.275

UCell score

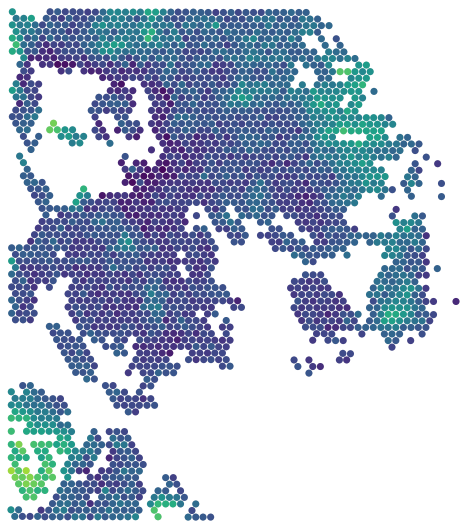
Emt



0 0.386

UCell score

Interferon Gamma Response



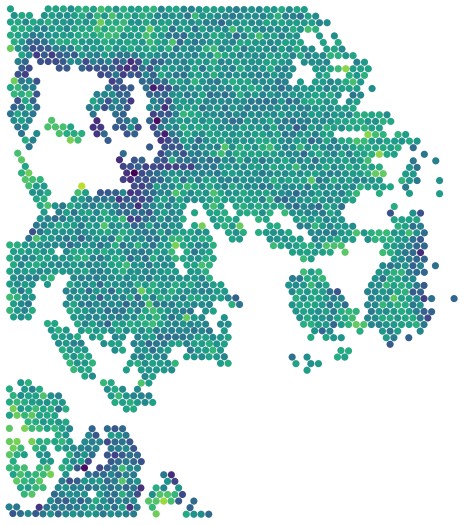
0 0.275

UCell score

CanDis_2026_Sample10 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling



0 0.144

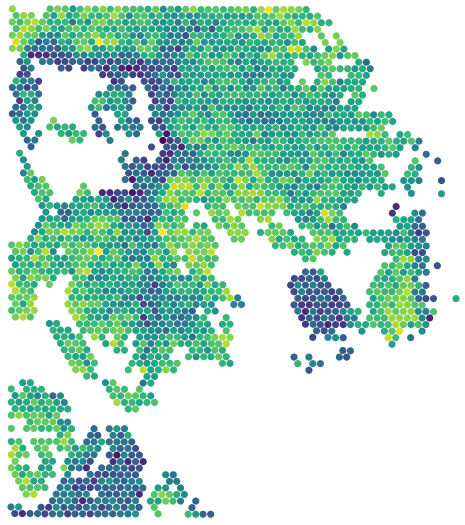
UCell score

Tgf Beta Signaling

0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

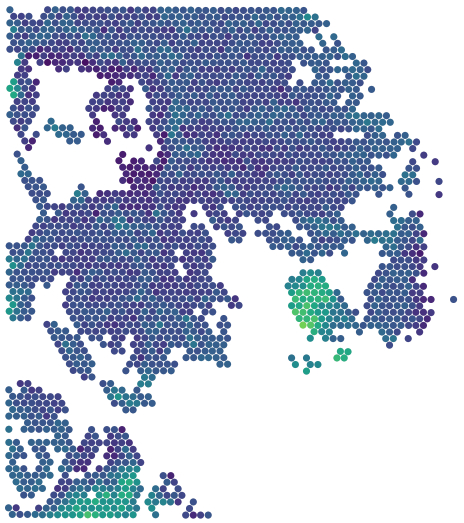
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

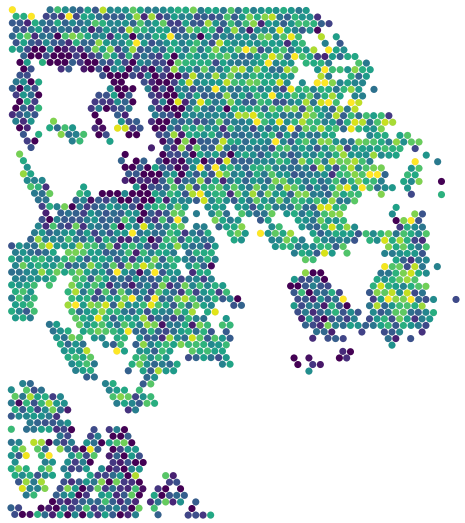
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



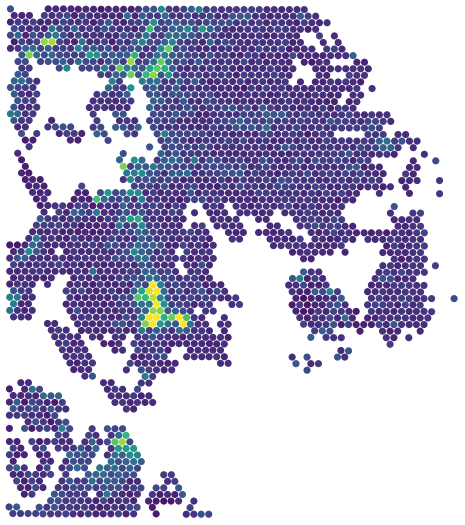
0 0.151

UCell score

CanDis_2026_Sample10 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

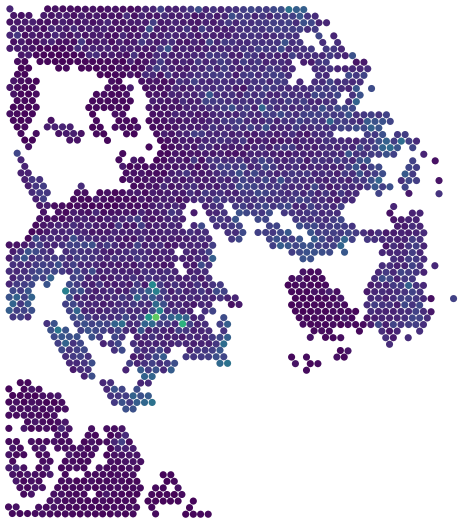
Treg composition



0 — 0.102

Fraction of all 30

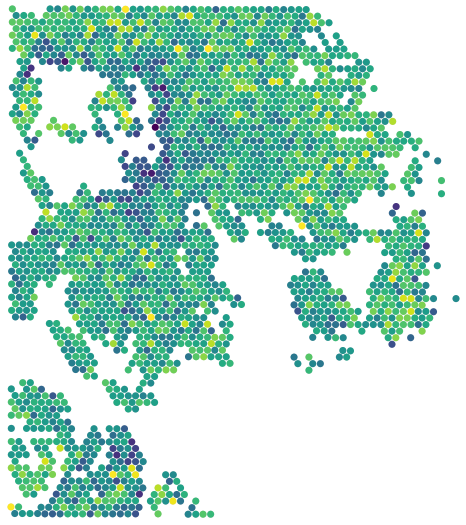
Treg estimated abundance



0 — 2.85

Posterior mean estimate

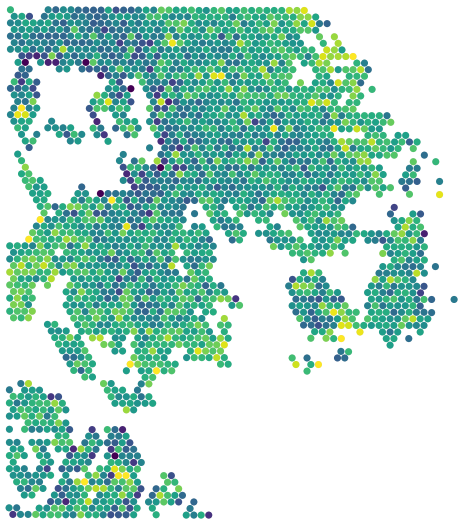
Treg Resting Vs Activated Dn



0 — 0.0778

UCell score

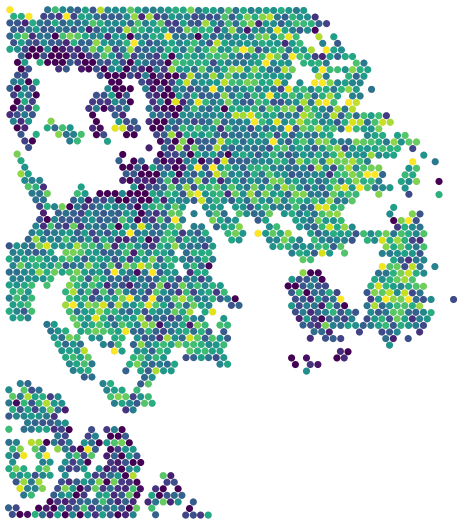
Treg Resting Vs Activated Up



0 — 0.0609

UCell score

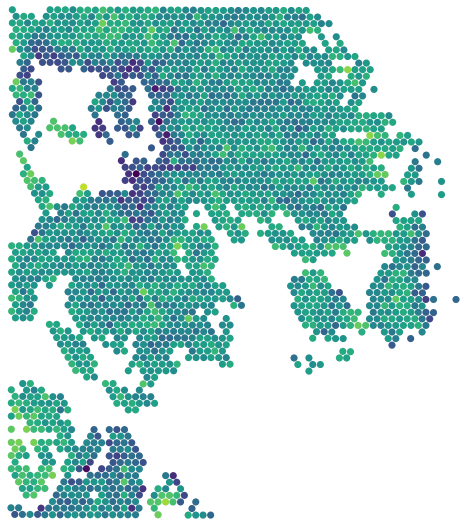
Co Inhibition By Ctla4



0 — 0.151

UCell score

Il2 Stat5 Signaling

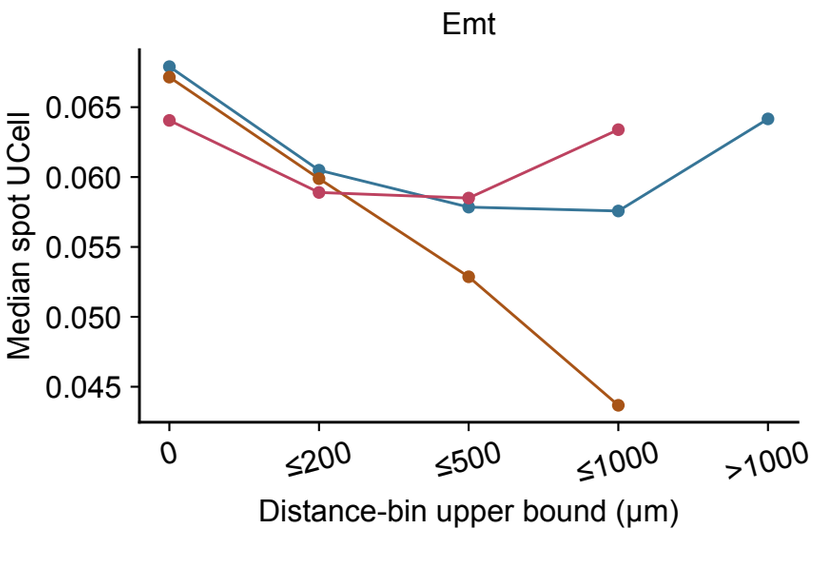


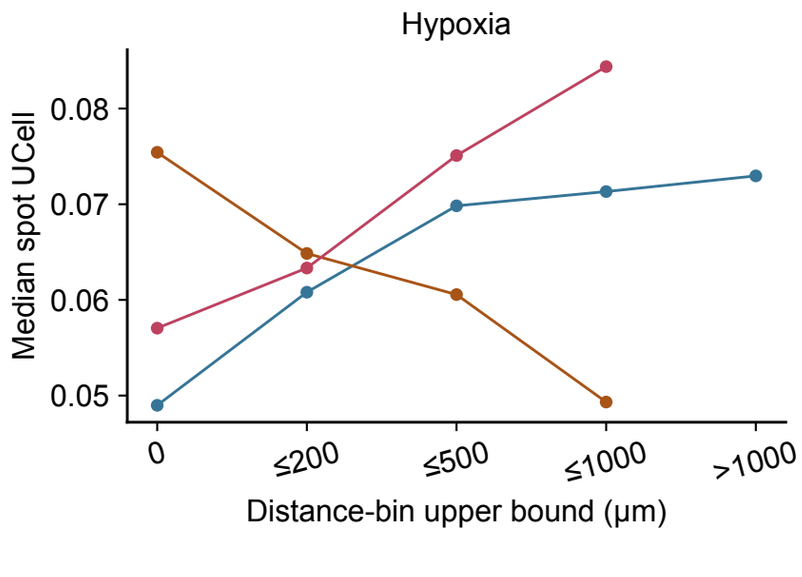
0 — 0.144

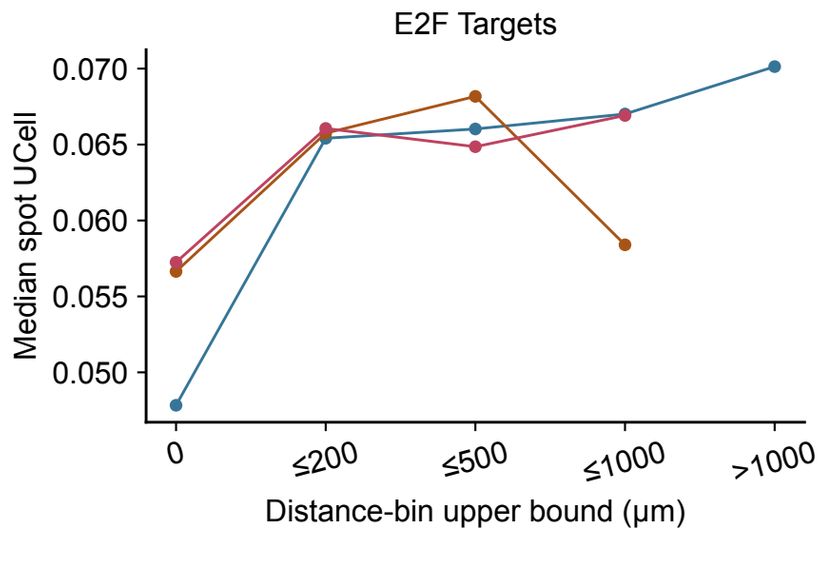
UCell score

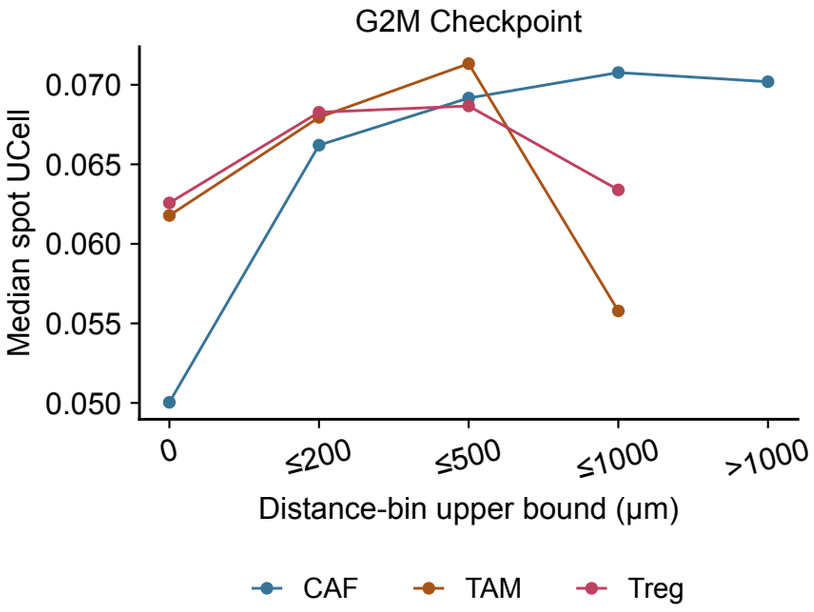
CanDis_2026_Sample10 | Programs by target distance

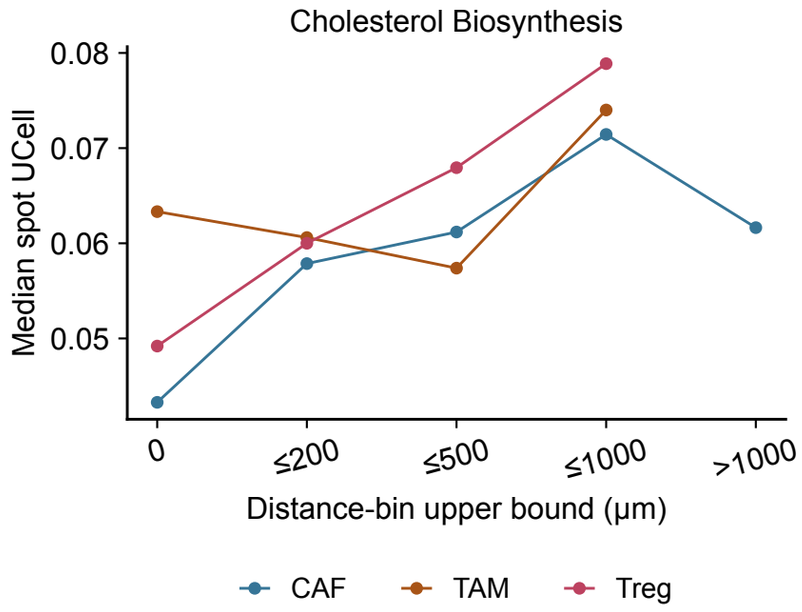
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

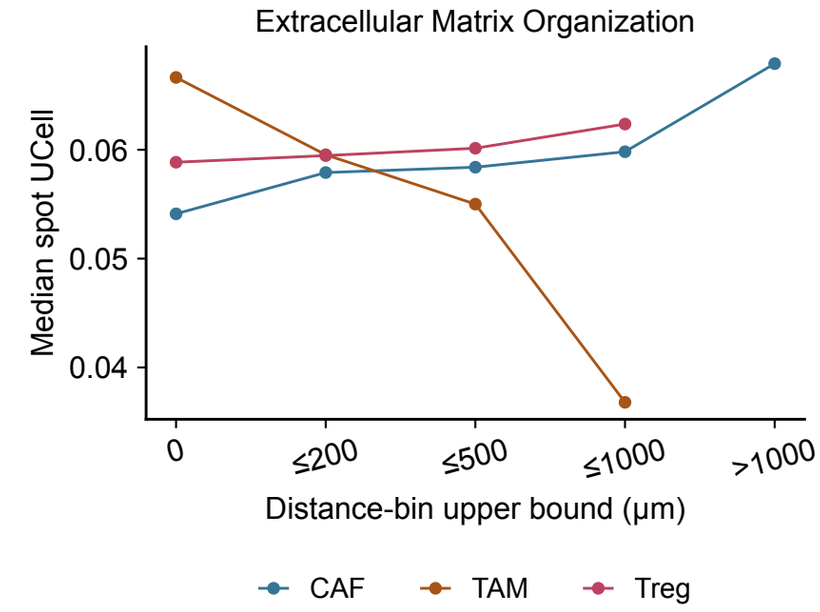






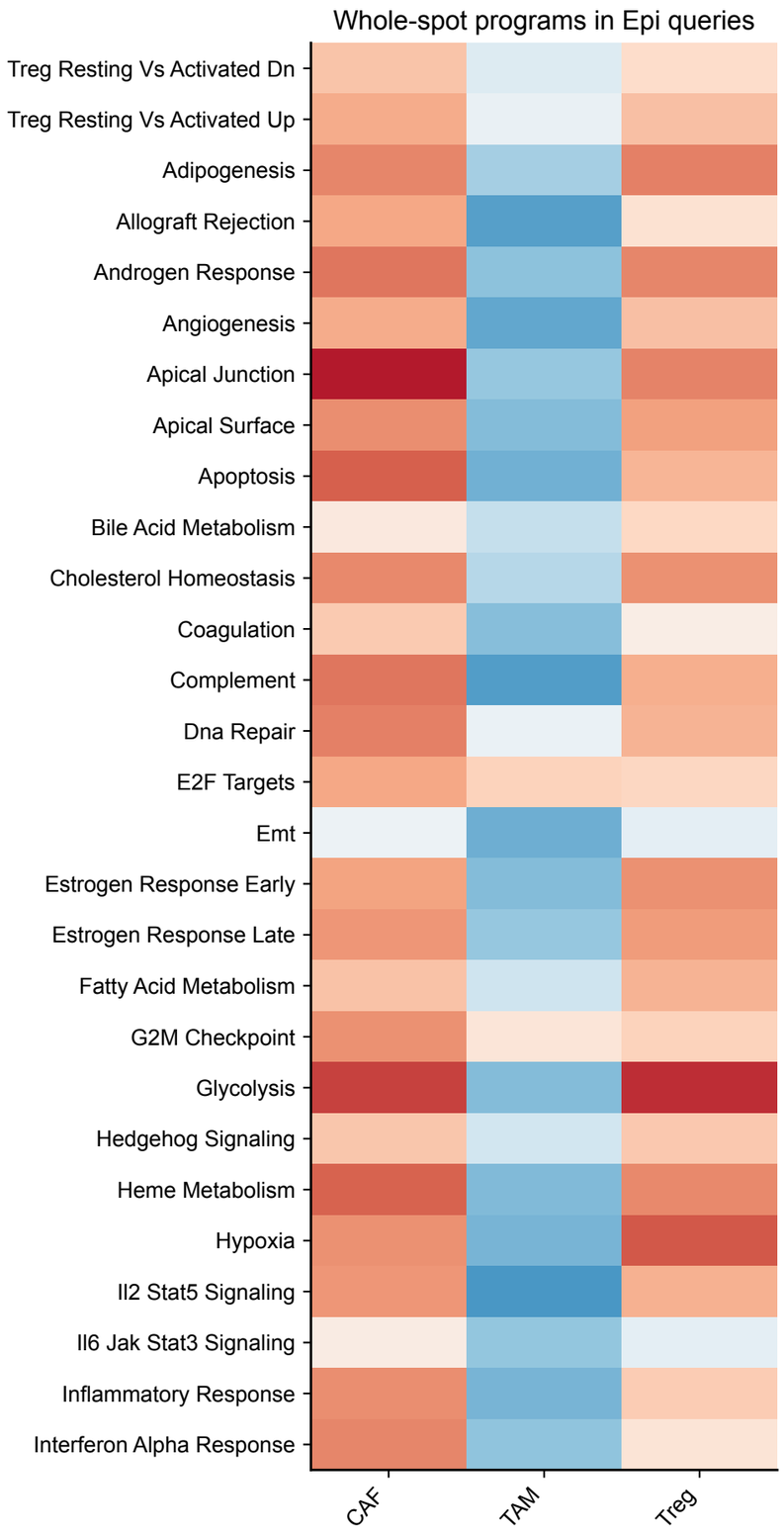


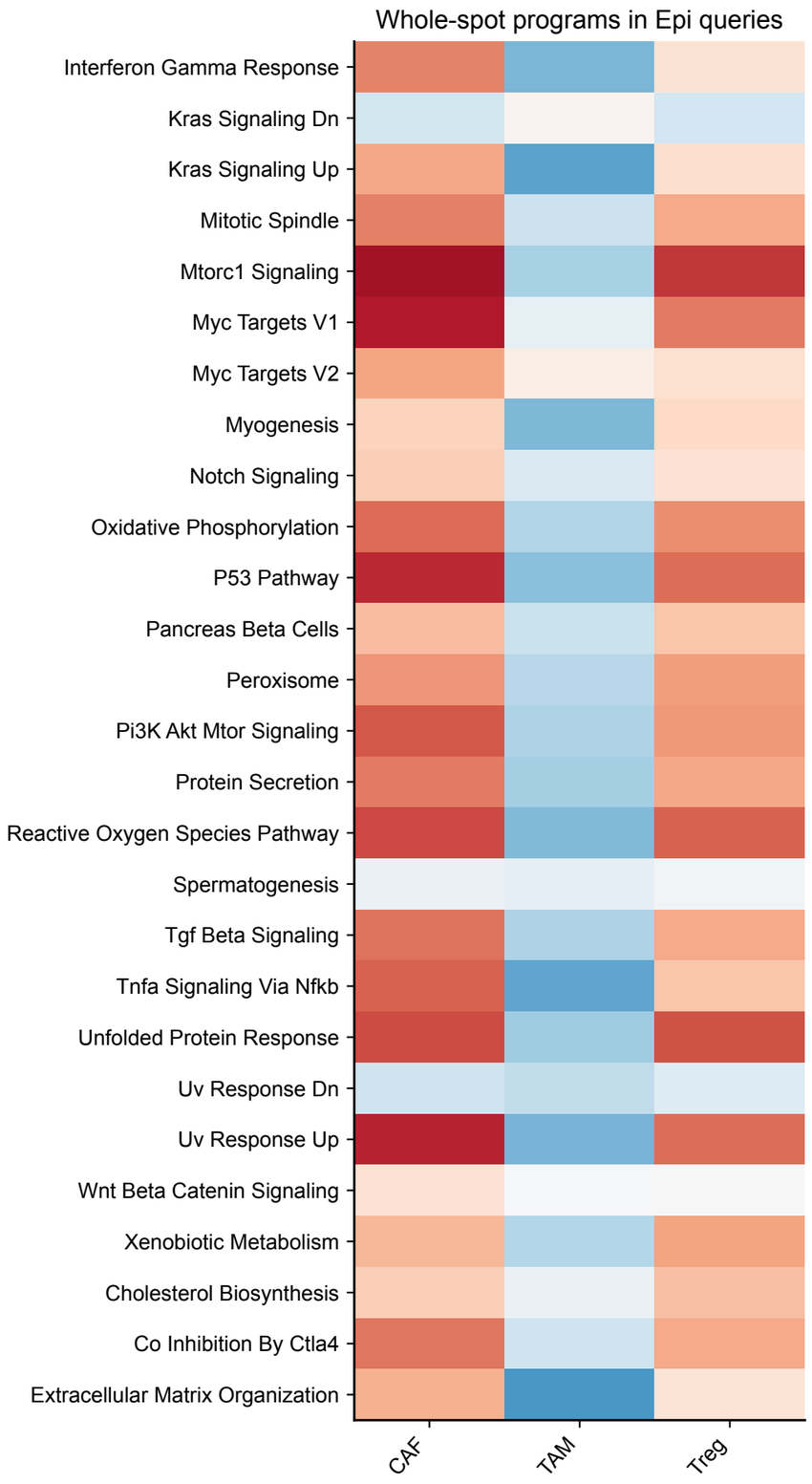




CanDis_2026_Sample10 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

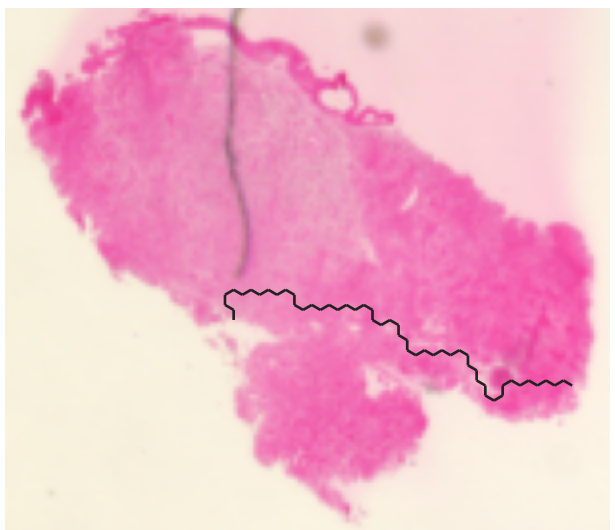




CanDis_2026_Sample11 | Tissue context

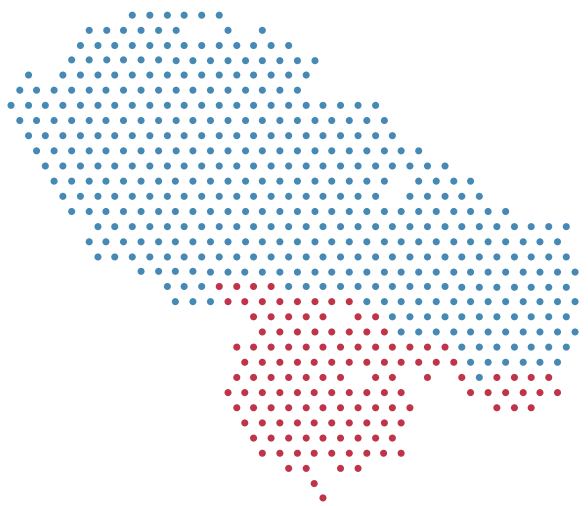
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



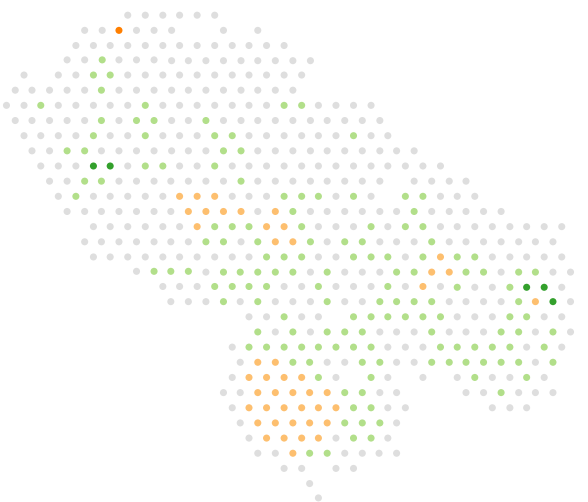
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



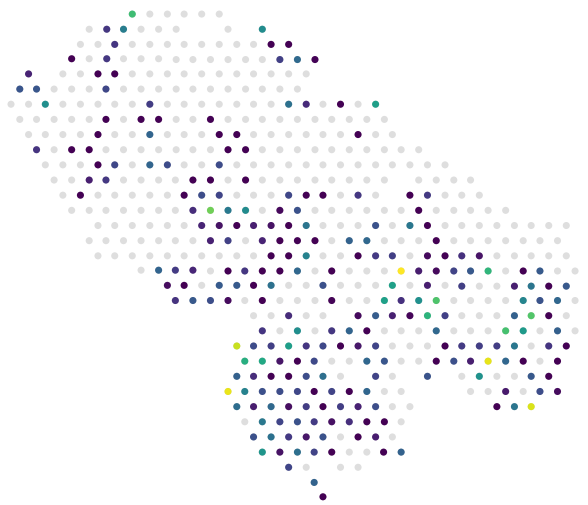
2 3 4 5

Epithelial support

0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

0 0.14

Matched-model AUCell

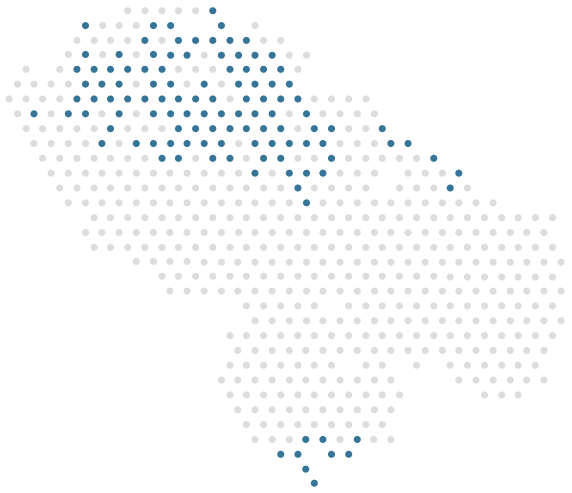
CanDis_2026_Sample11 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

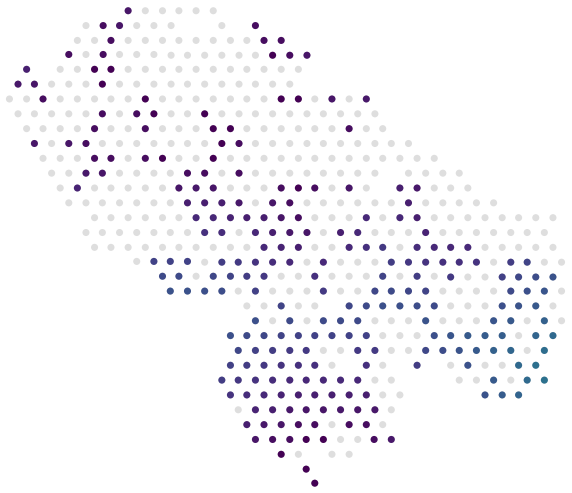
0 0.418
Fraction of all 30

CAF P80 targets



False True

CAF distance



0 3.1e+03
Nearest spot center (μm)

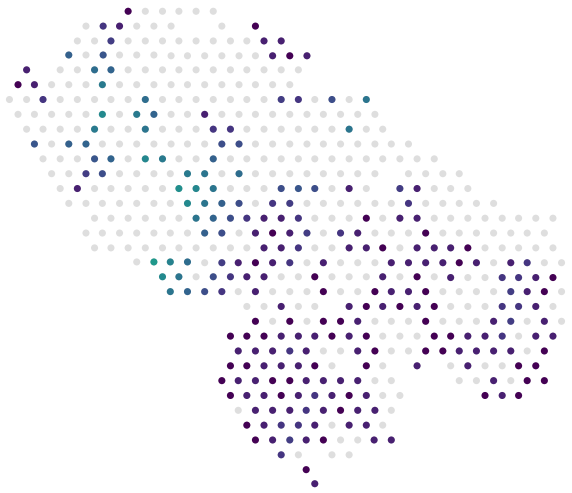
TAM composition

0 0.291
Fraction of all 30

TAM P80 targets

False True

TAM distance



0 1.13e+03
Nearest spot center (μm)

Treg composition

0 0.102
Fraction of all 30

Treg P80 targets

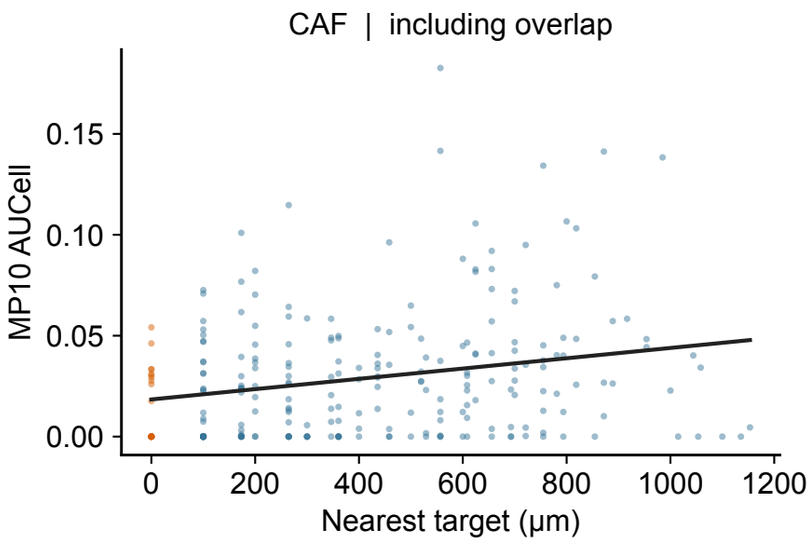
False True

Treg distance

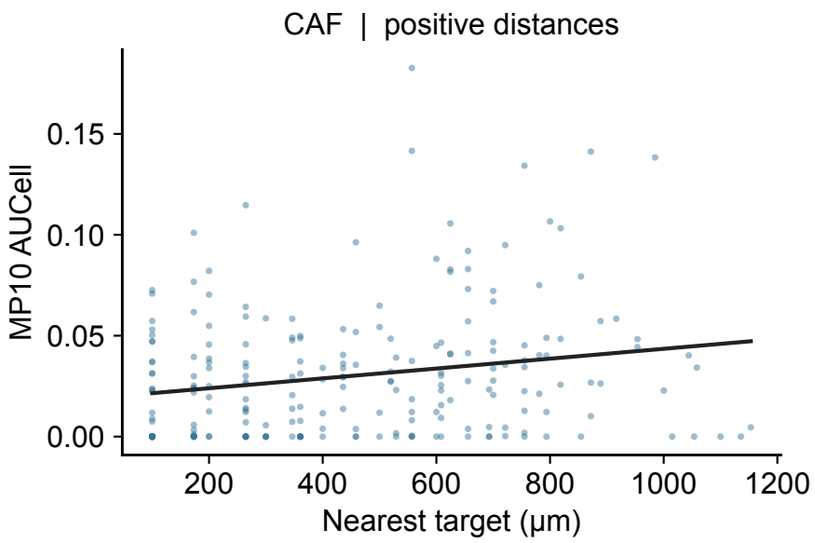
0 1.61e+03
Nearest spot center (μm)

CanDis_2026_Sample11 | MP10 versus nearest target

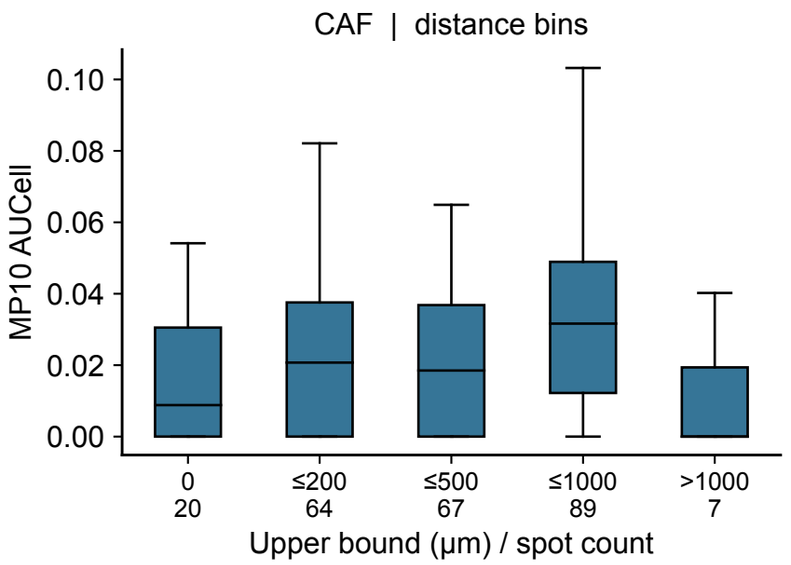
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



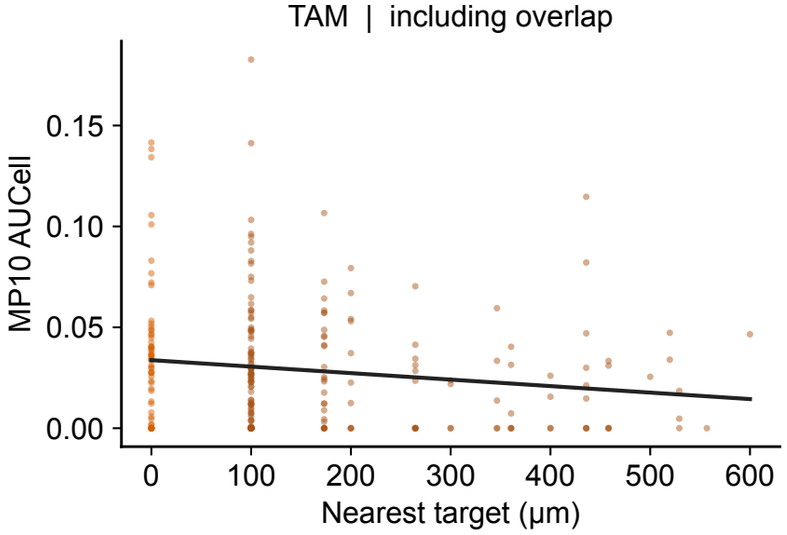
r +0.23 ρ +0.23
n 247/247 queries • 116 targets



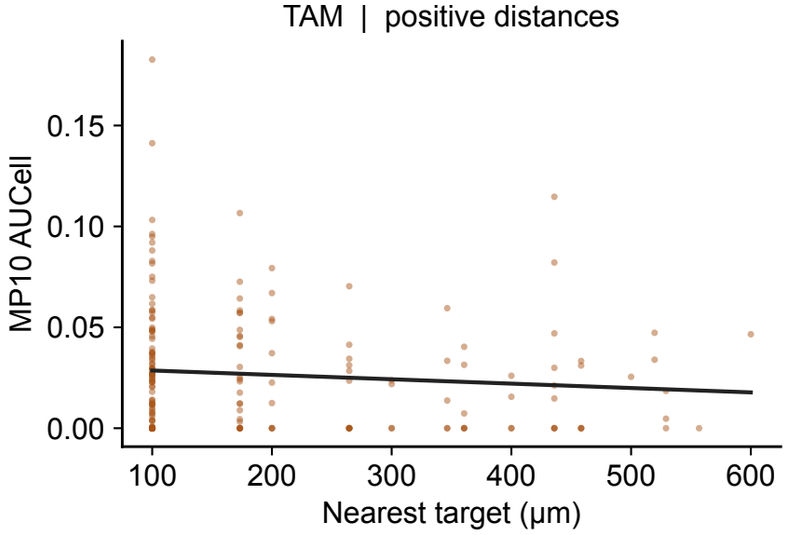
r +0.21 ρ +0.20
n 227/247 queries • 116 targets



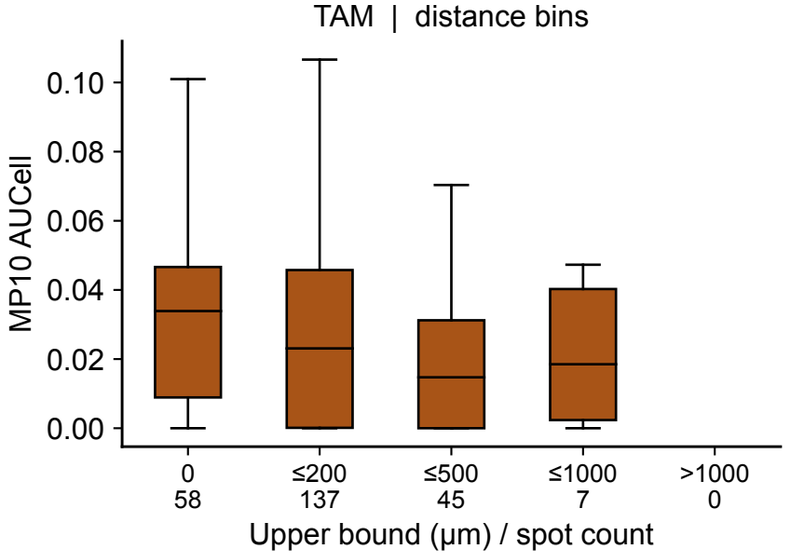
Orange = zero distance
Line = within-section linear fit



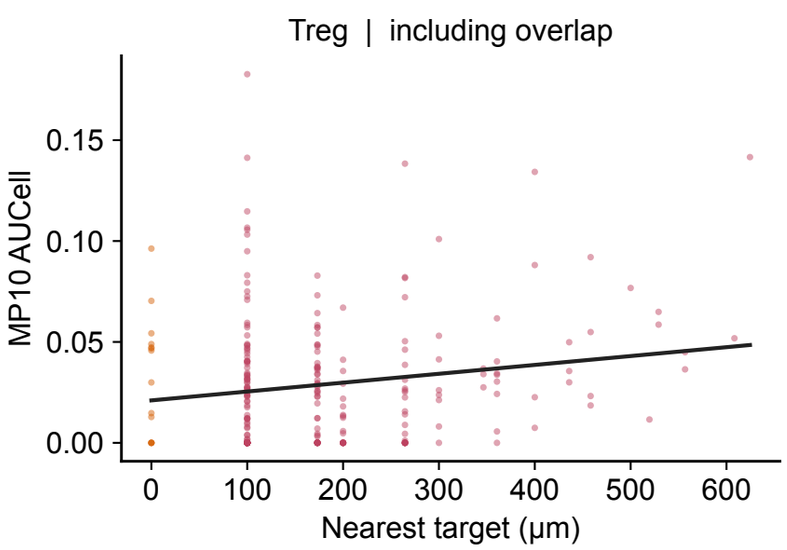
r -0.14 ρ -0.17
n 247/247 queries • 116 targets



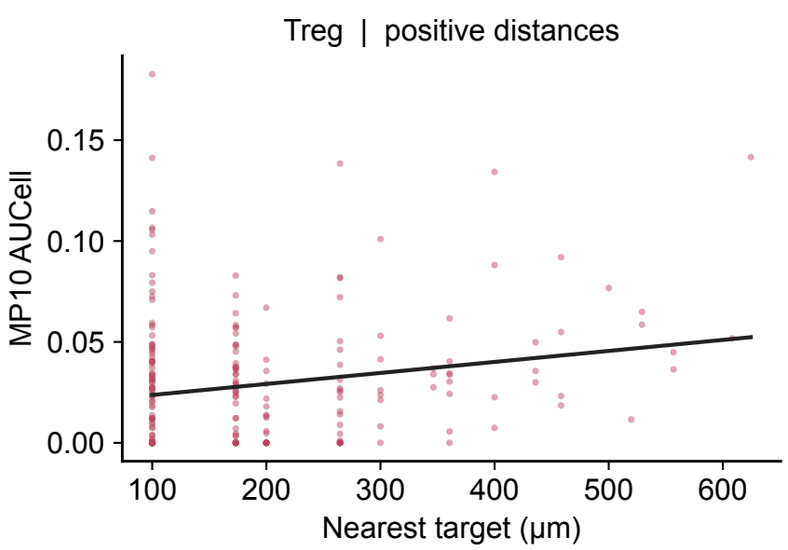
r -0.09 ρ -0.09
n 189/247 queries • 116 targets



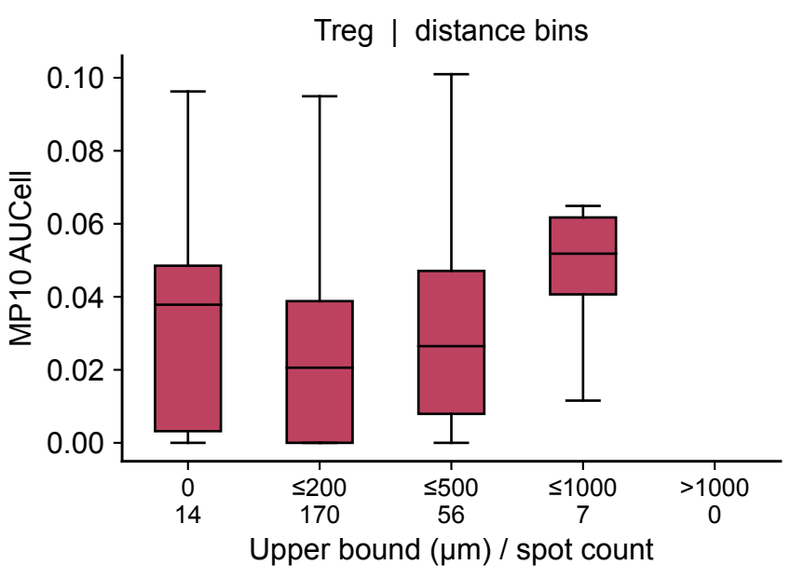
Orange = zero distance
Line = within-section linear fit



r +0.17 ρ +0.09
n 247/247 queries • 116 targets



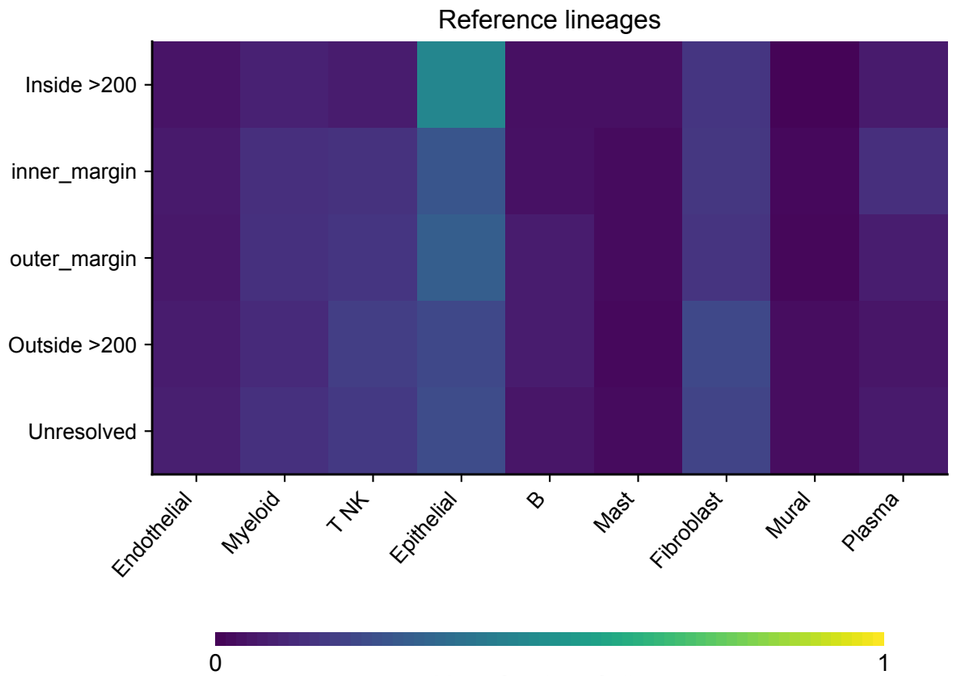
r +0.20 ρ +0.12
n 233/247 queries • 116 targets

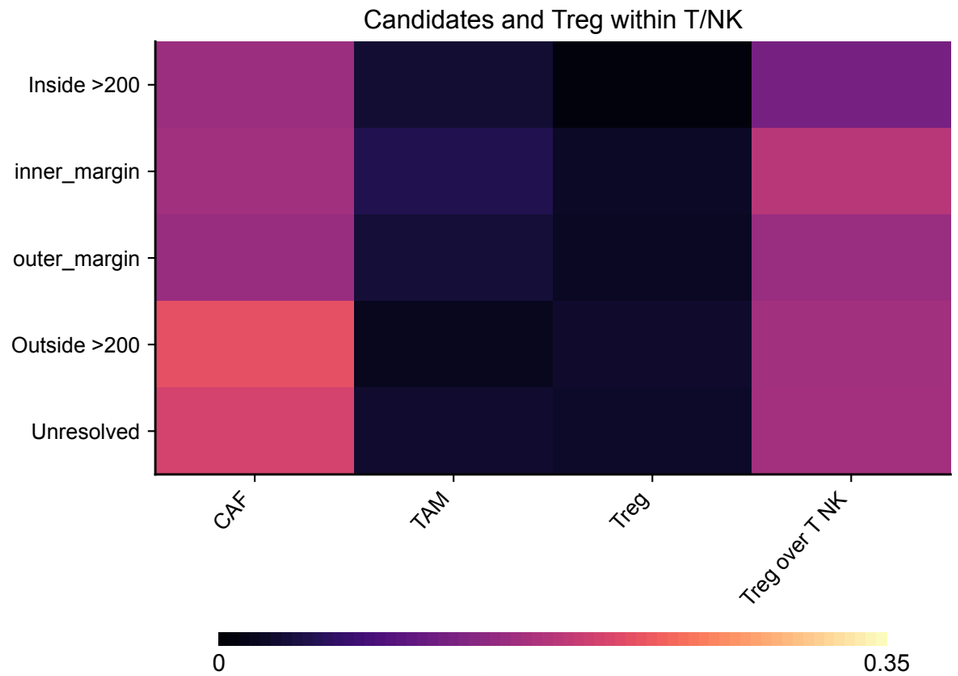


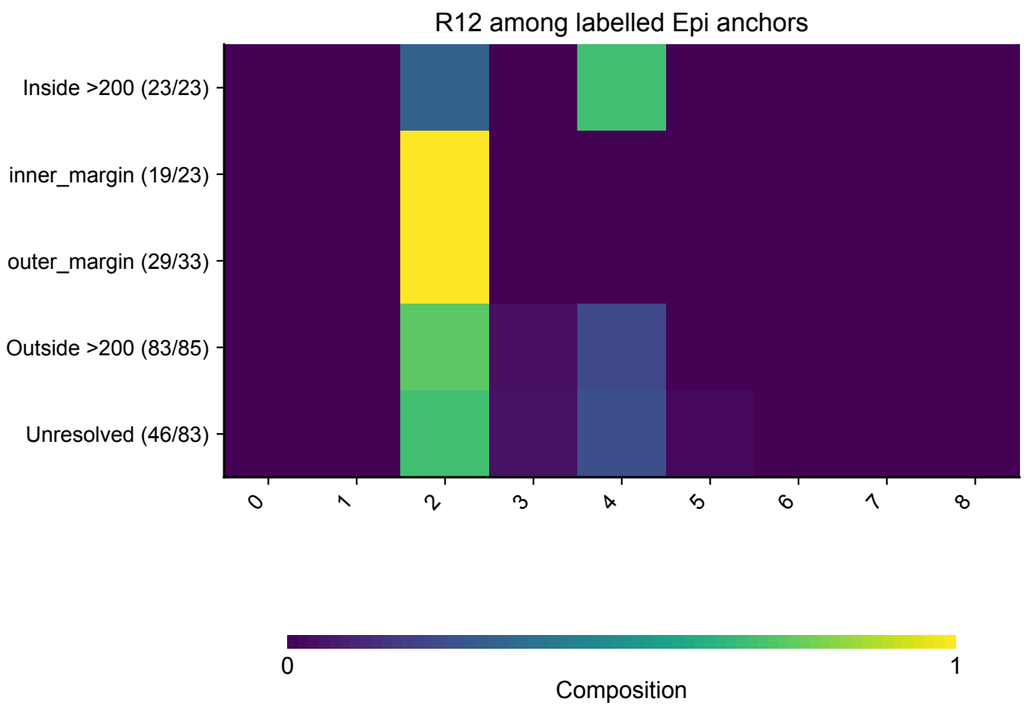
Orange = zero distance
Line = within-section linear fit

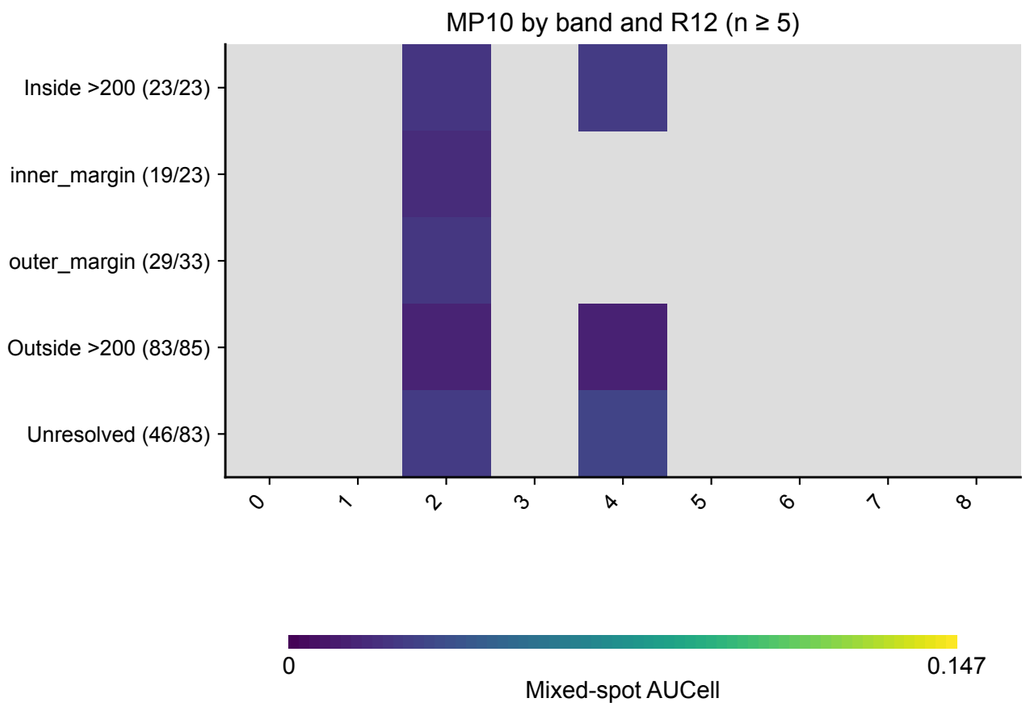
CanDis_2026_Sample11 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable









CanDis_2026_Sample11 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets

0 0.152

UCell score

G2M Checkpoint

0 0.134

UCell score

Hypoxia

0 0.211

UCell score

Tnfa Signaling Via Nfkb

0 0.275

UCell score

Emt

0 0.386

UCell score

Interferon Gamma Response

0 0.275

UCell score

CanDis_2026_Sample11 | Immune, lipid and matrix programs

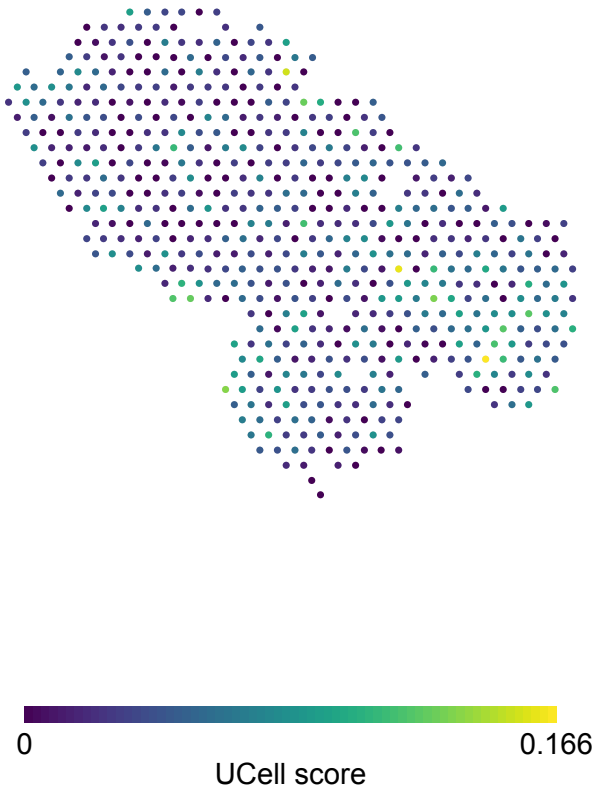
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

Tgf Beta Signaling

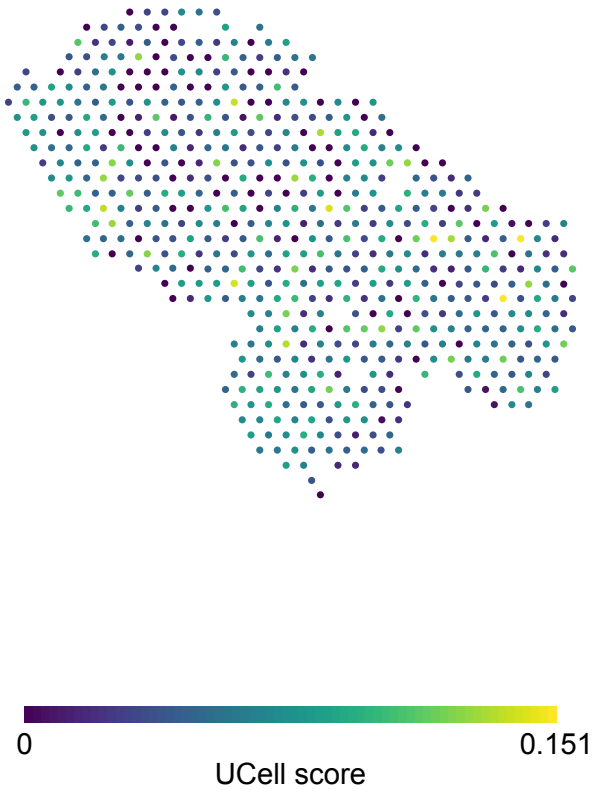
Cholesterol Homeostasis

Cholesterol Biosynthesis



Extracellular Matrix Organization

Co Inhibition By Ctla4



CanDis_2026_Sample11 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

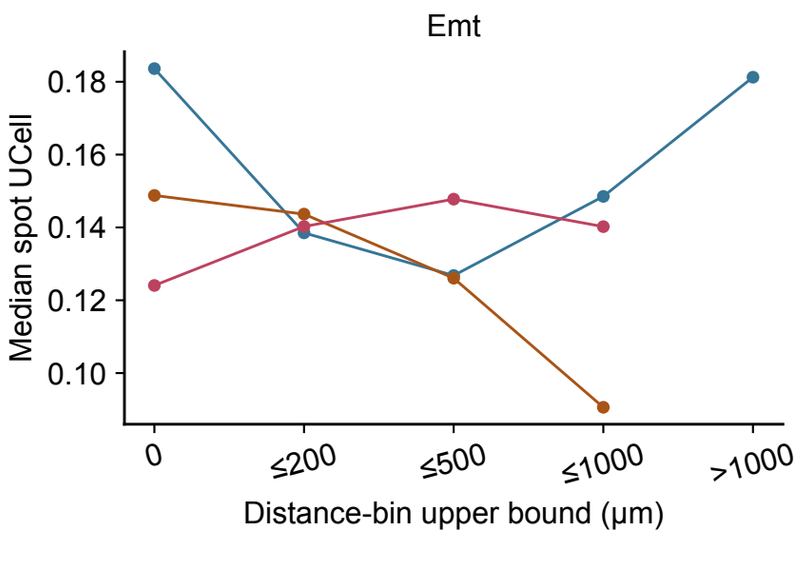
Il2 Stat5 Signaling

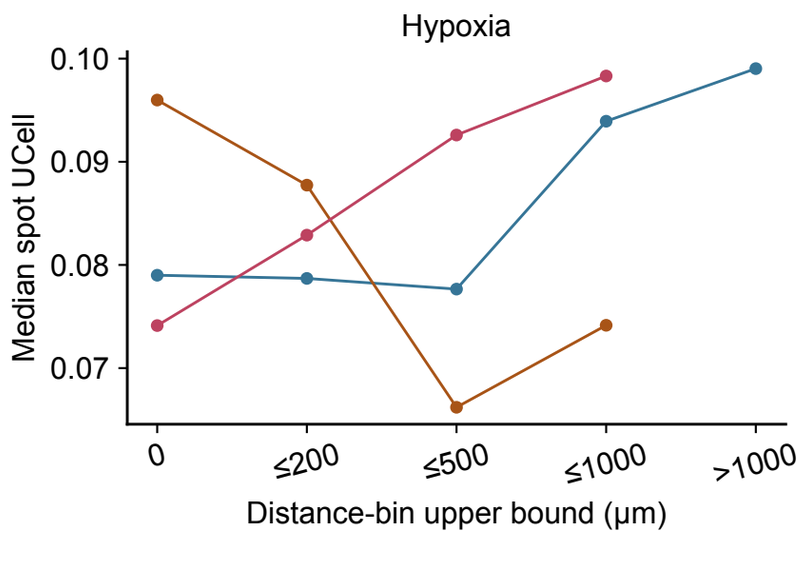
0 0.144

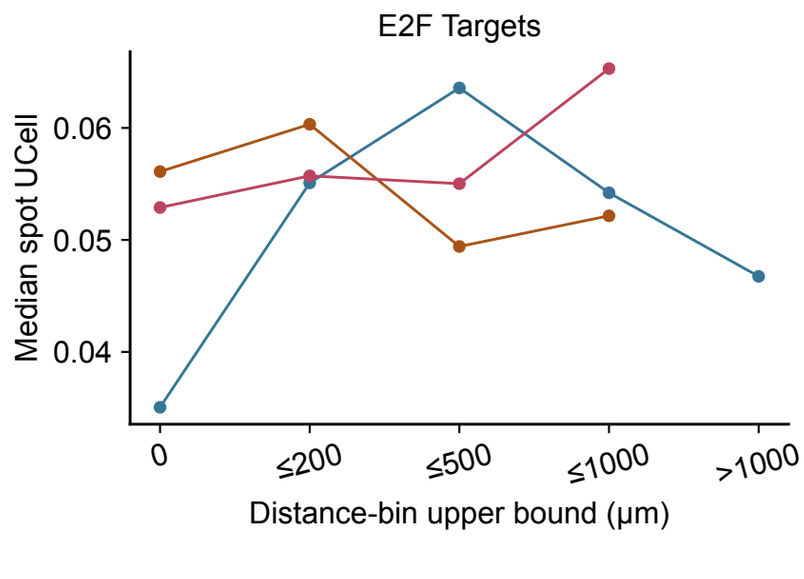
UCell score

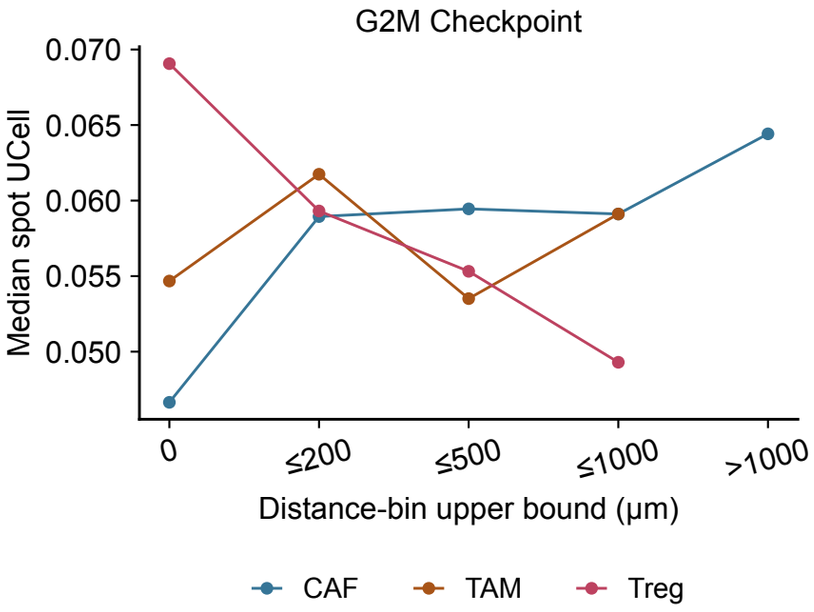
CanDis_2026_Sample11 | Programs by target distance

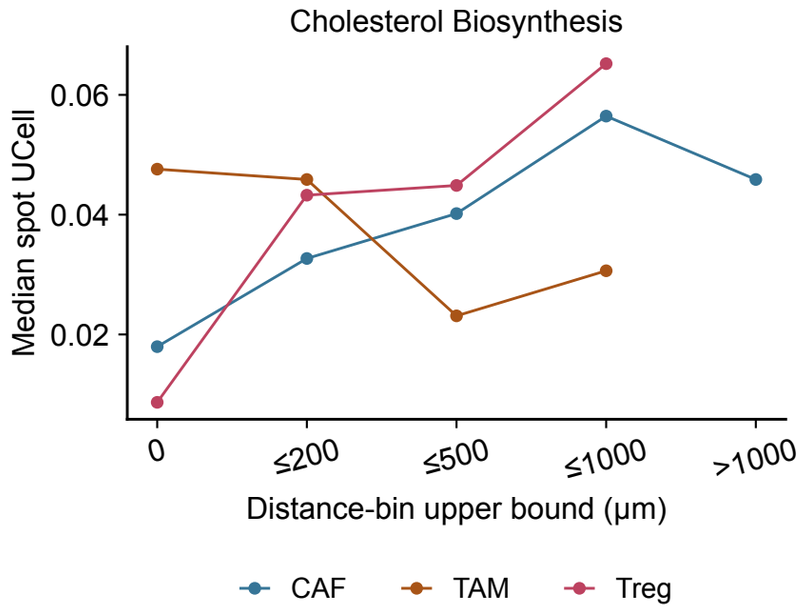
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

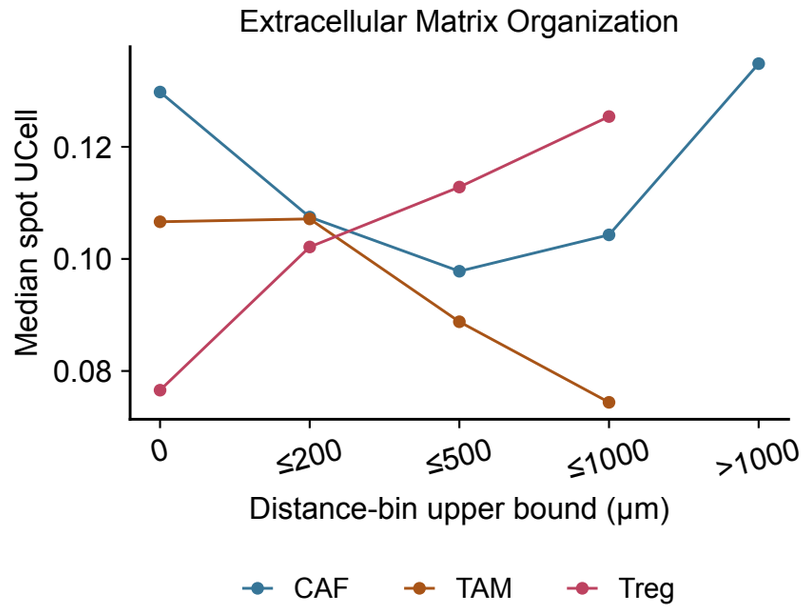






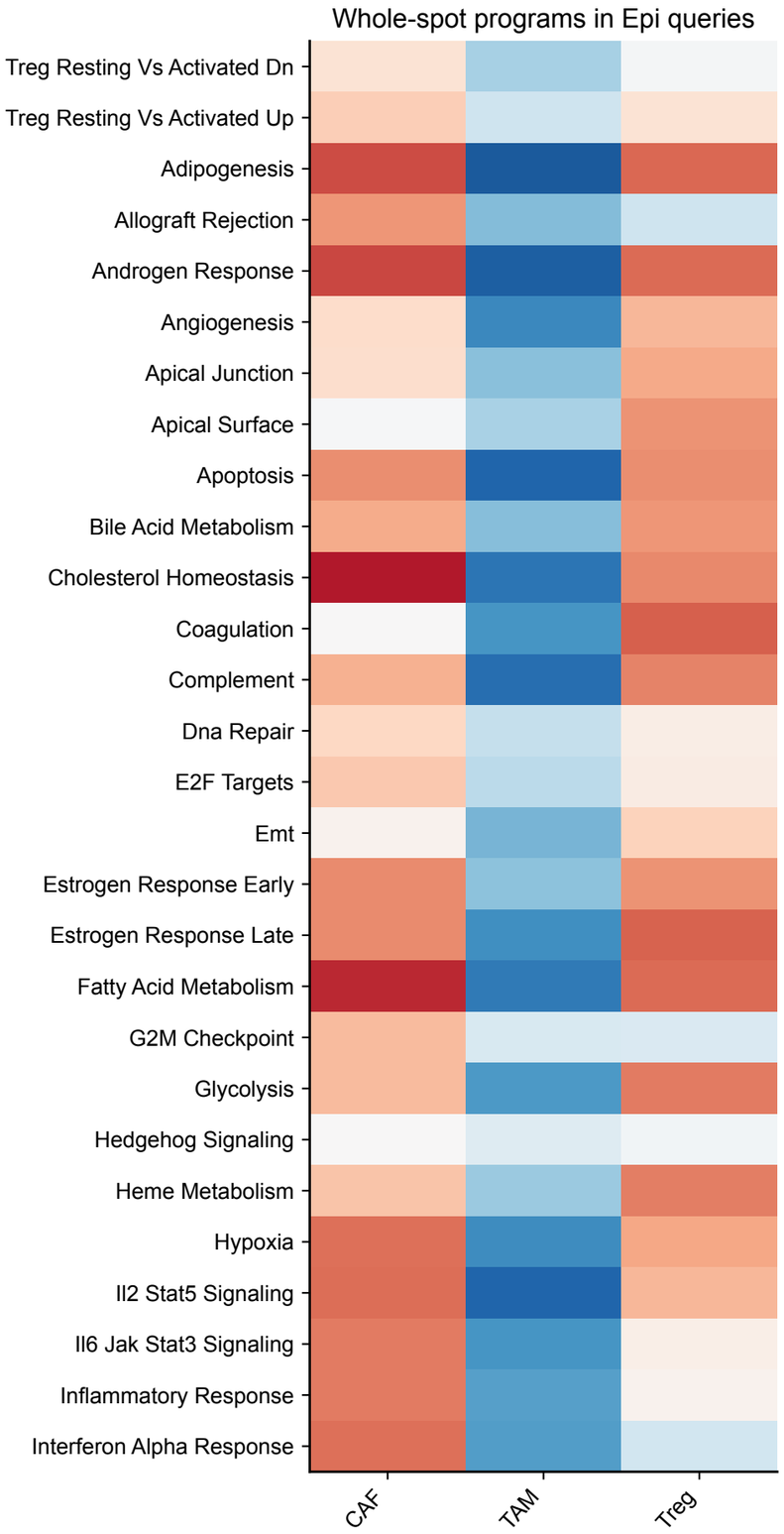


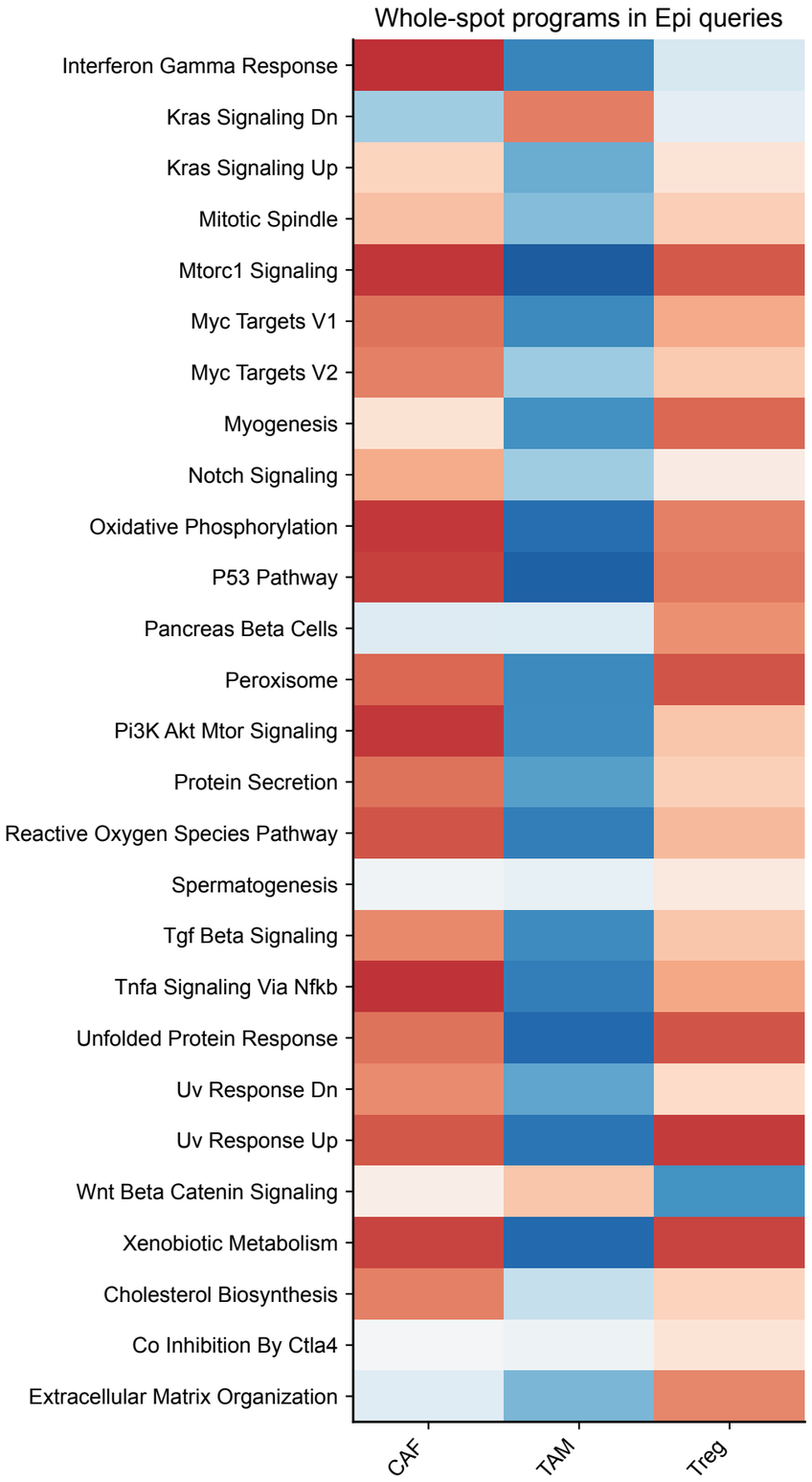




CanDis_2026_Sample11 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

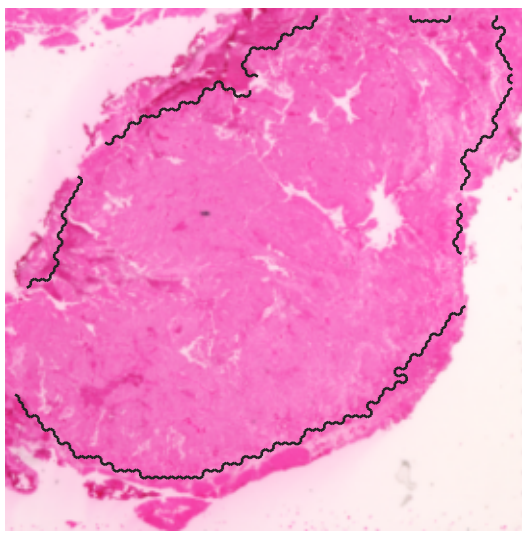




CanDis_2026_Sample12 | Tissue context

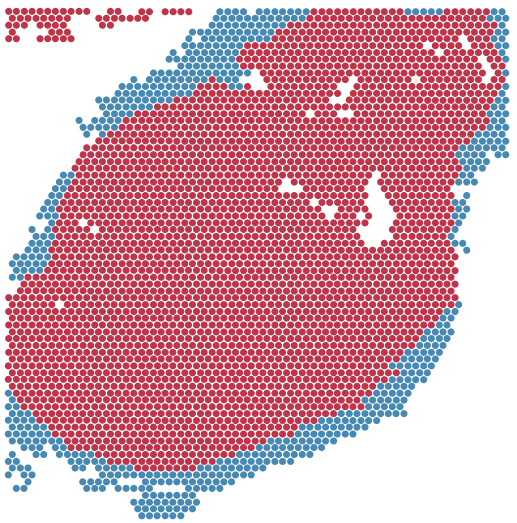
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



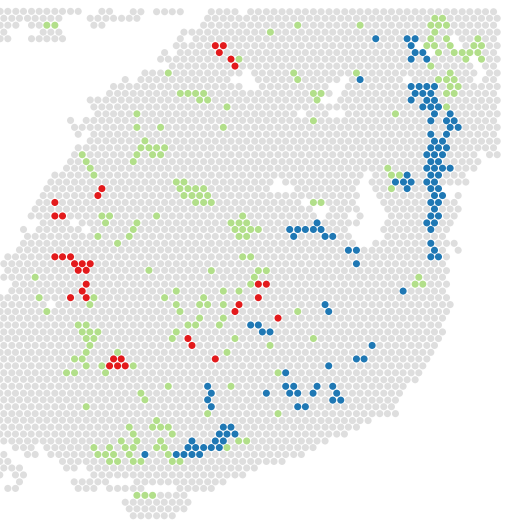
Black line: observed candidate outer edge

Final candidate location



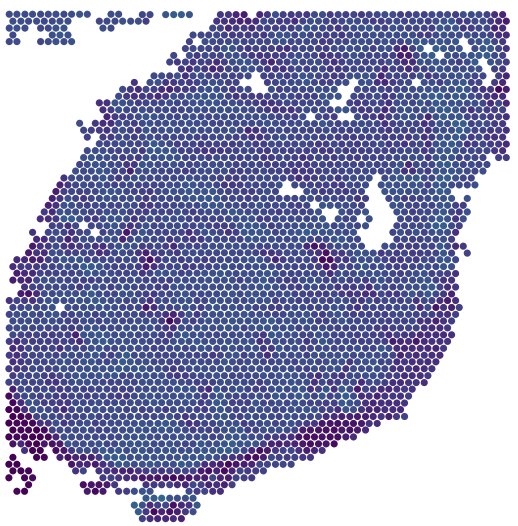
inside outside

Accepted R12 (original 0–8)



1 2 7

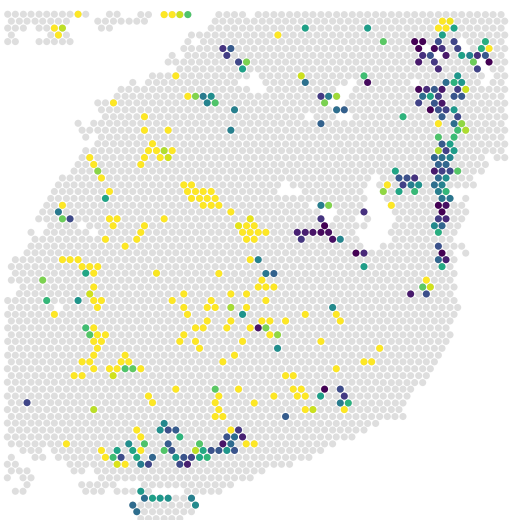
Epithelial support



0 1

Fraction of all 30

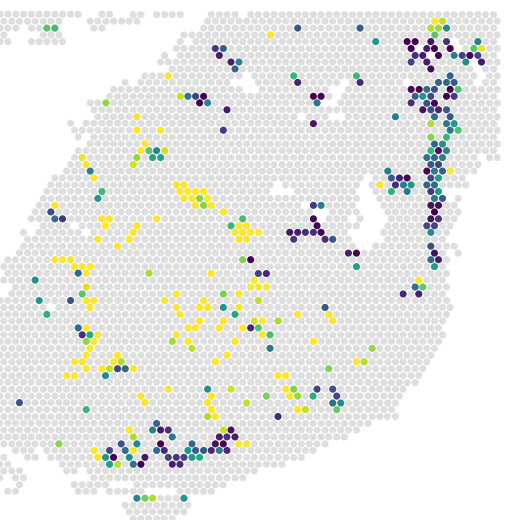
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



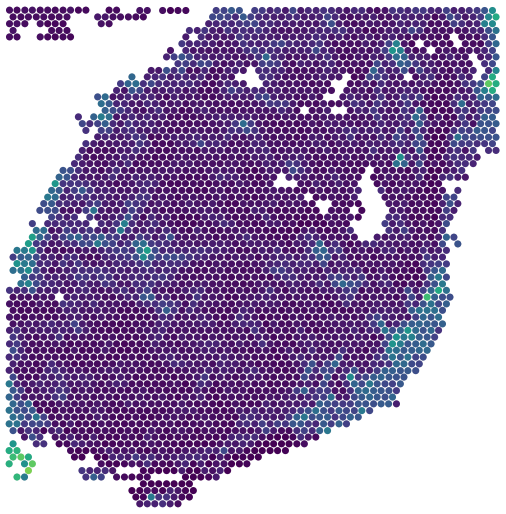
0 0.14

Matched-model AUCell

CanDis_2026_Sample12 | Candidate neighborhoods

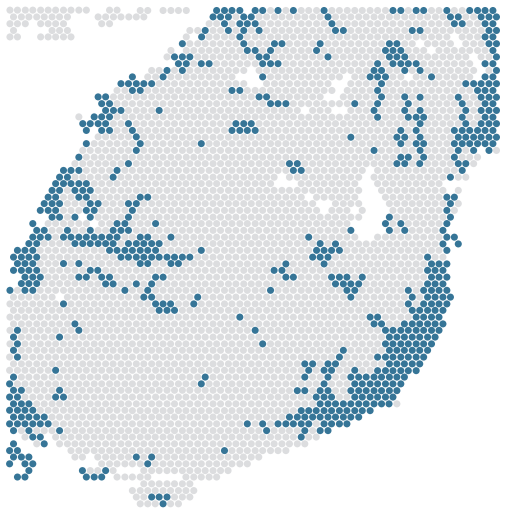
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



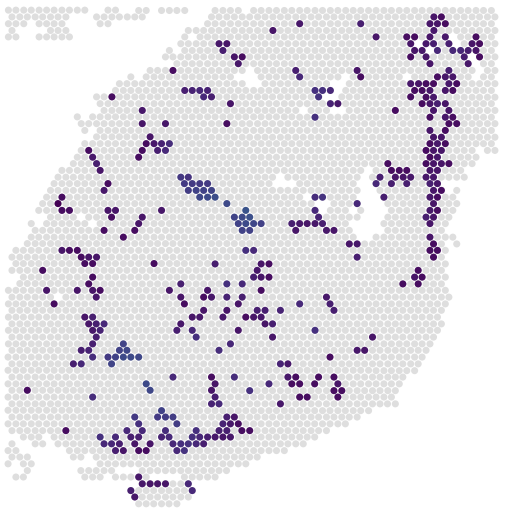
0 0.418
Fraction of all 30

CAF P80 targets



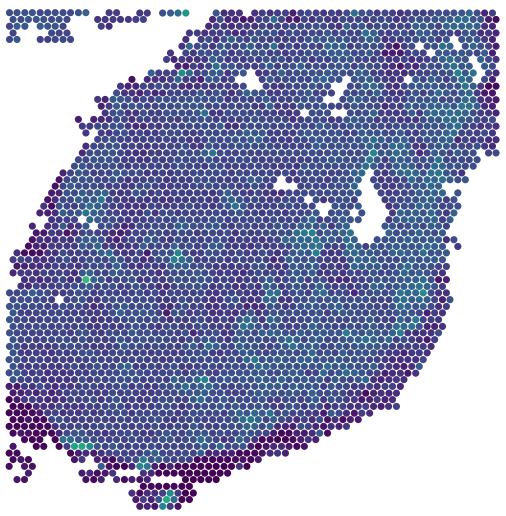
False True

CAF distance



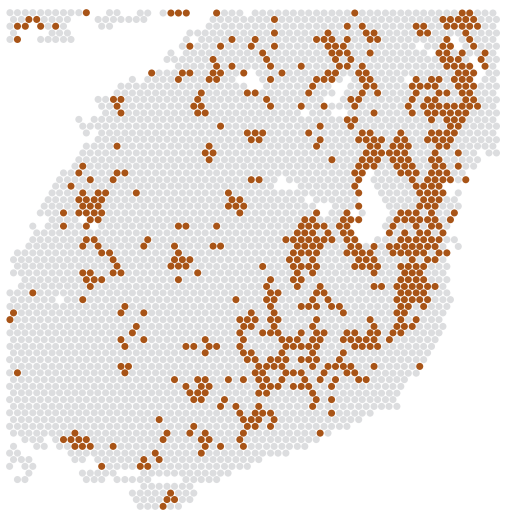
0 3.1e+03
Nearest spot center (μm)

TAM composition



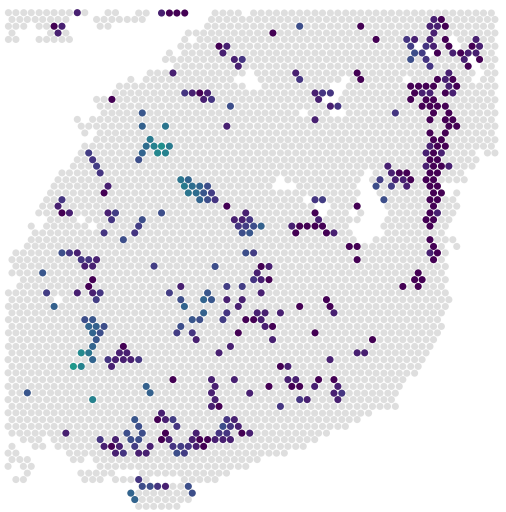
0 0.291
Fraction of all 30

TAM P80 targets



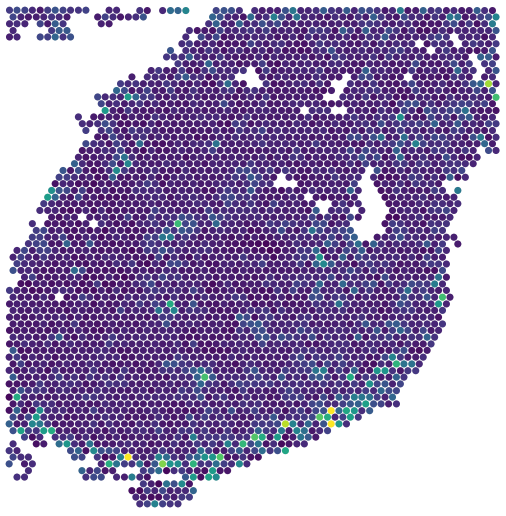
False True

TAM distance



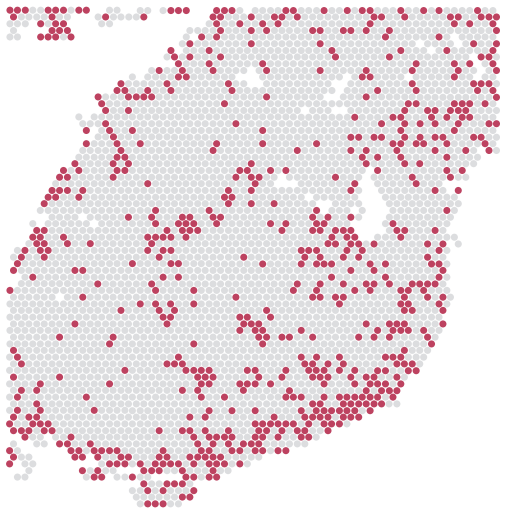
0 1.13e+03
Nearest spot center (μm)

Treg composition



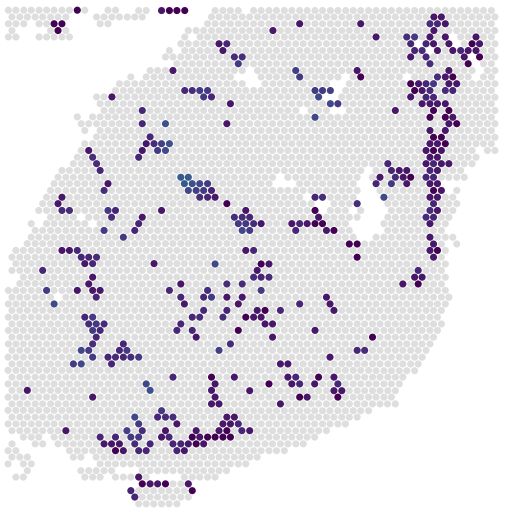
0 0.102
Fraction of all 30

Treg P80 targets



False True

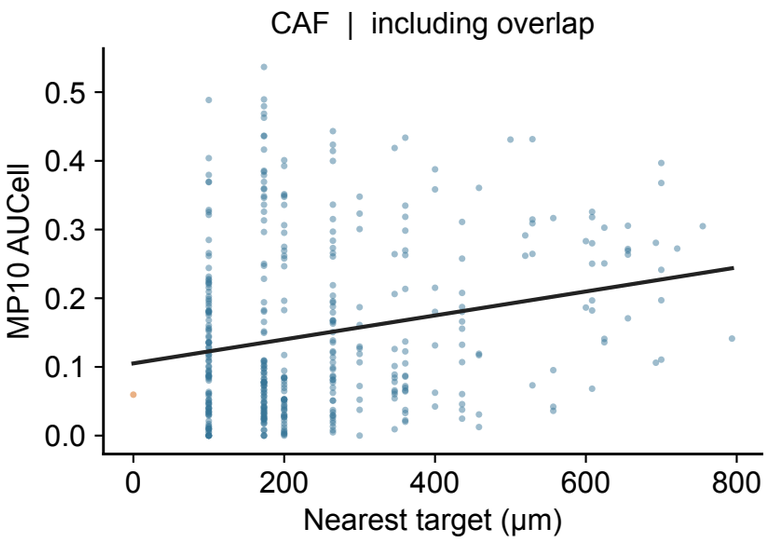
Treg distance



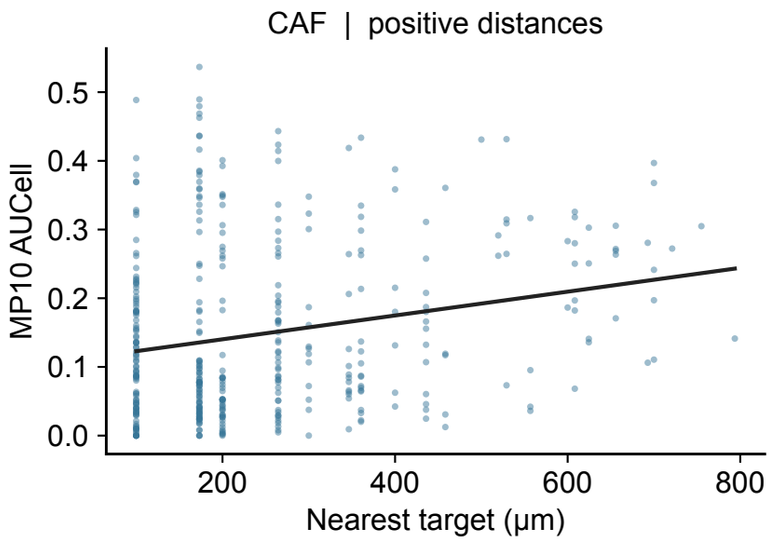
0 1.61e+03
Nearest spot center (μm)

CanDis_2026_Sample12 | MP10 versus nearest target

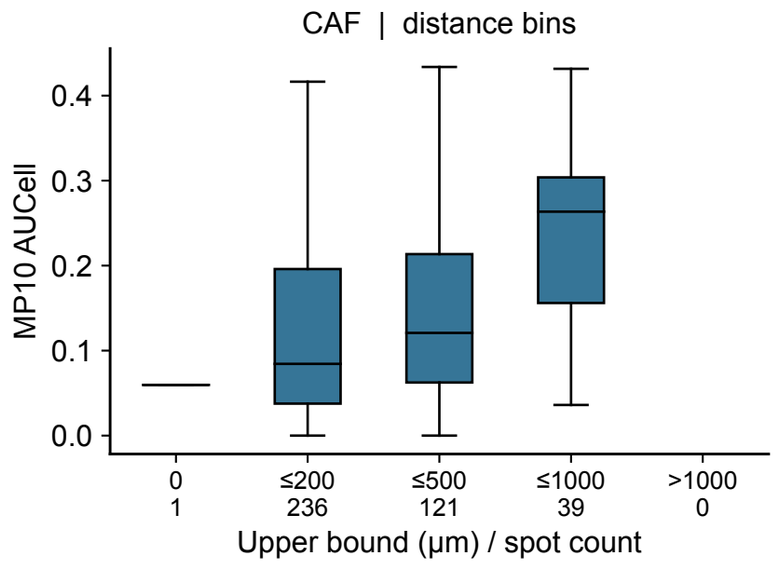
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



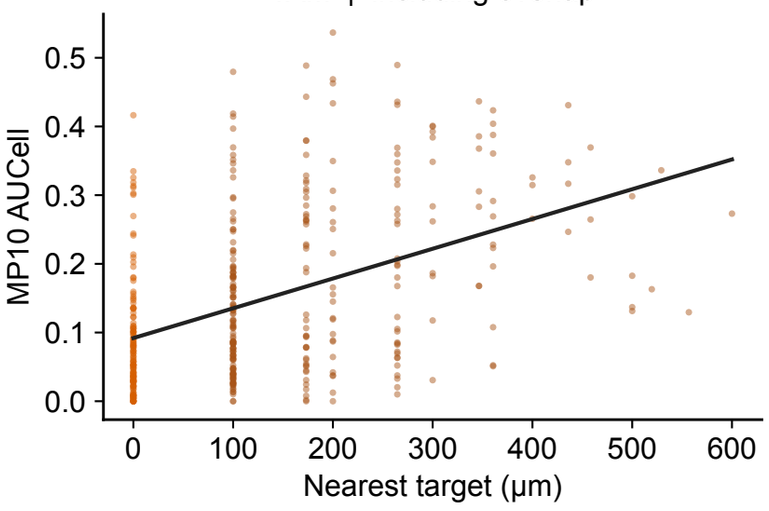
r +0.22 ρ +0.19
n 397/405 queries • 702 targets



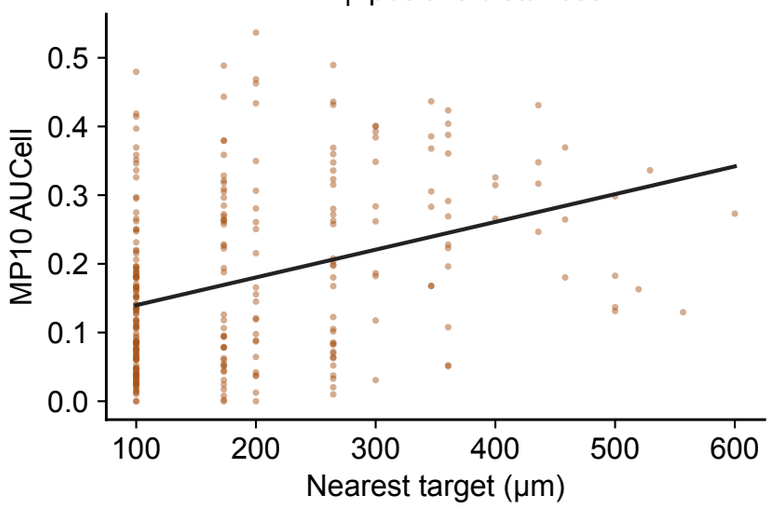
r +0.22 ρ +0.19
n 396/405 queries • 702 targets



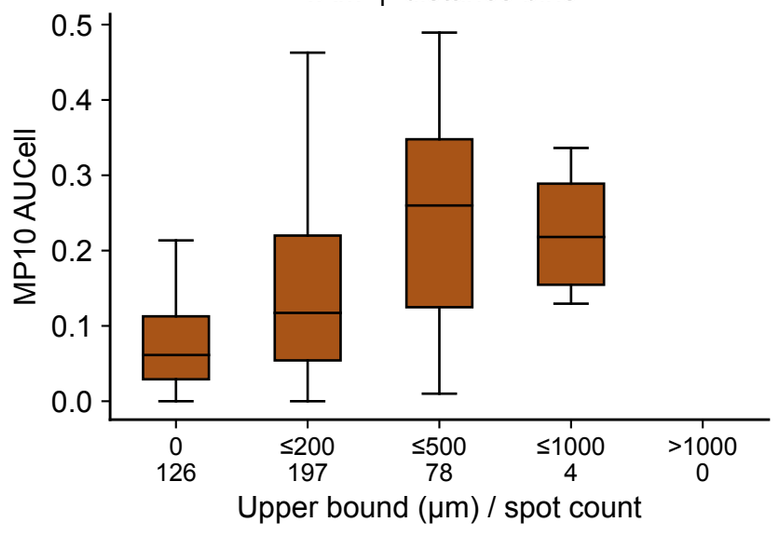
Orange = zero distance
Line = within-section linear fit



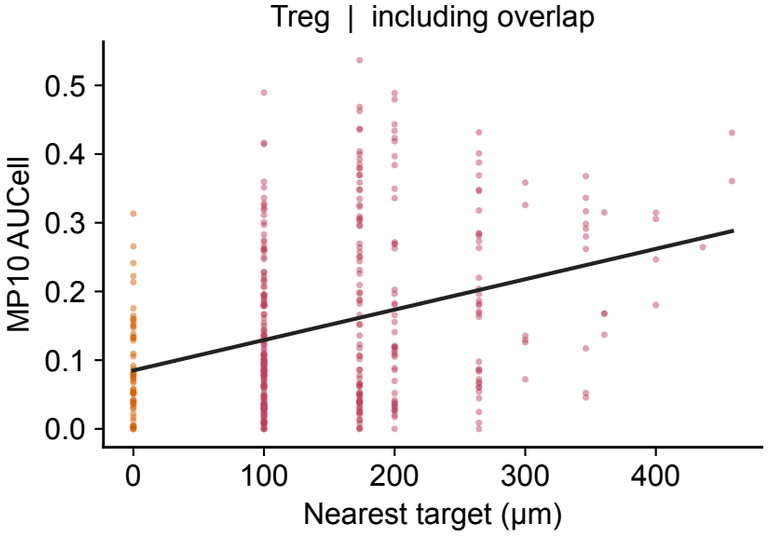
r +0.45 ρ +0.43
n 405/405 queries • 702 targets



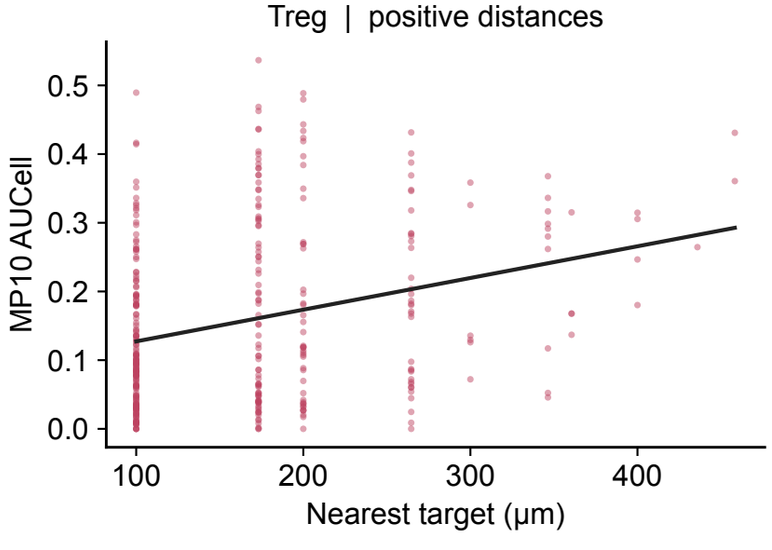
r +0.35 ρ +0.32
n 279/405 queries • 702 targets



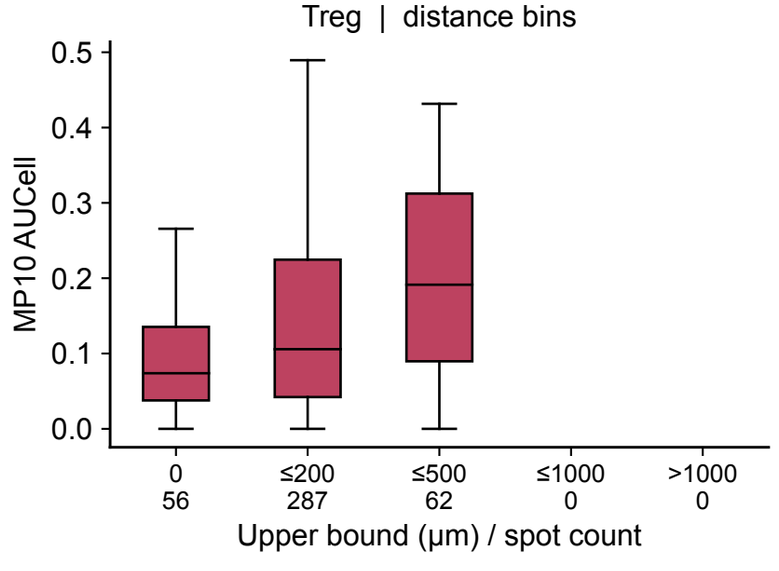
Orange = zero distance
Line = within-section linear fit



r +0.33 ρ +0.28
n 405/405 queries • 702 targets



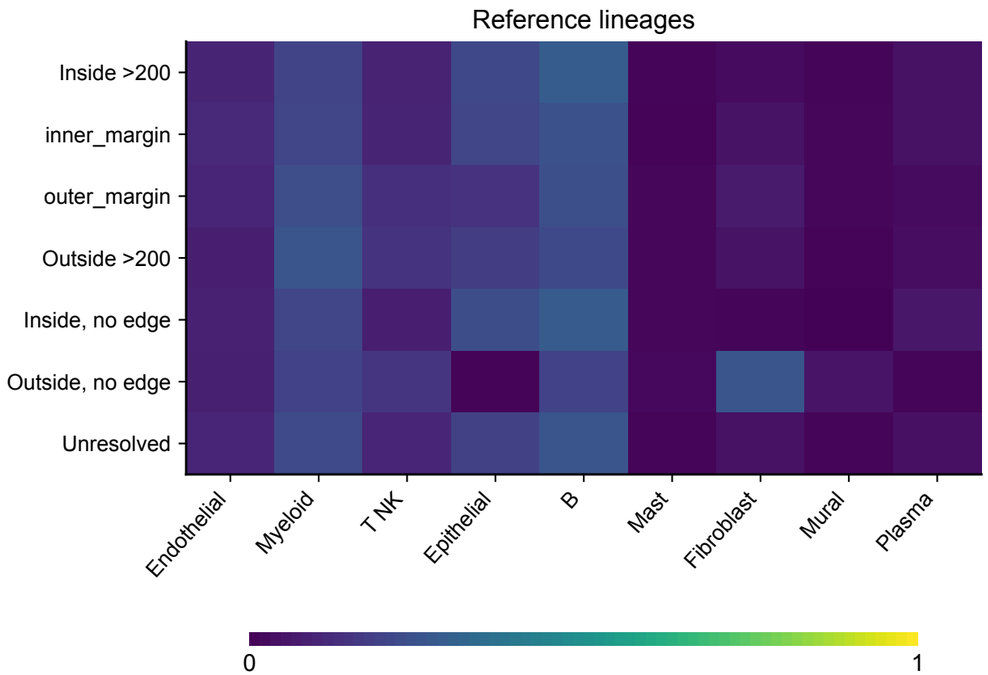
r +0.28 ρ +0.23
n 349/405 queries • 702 targets

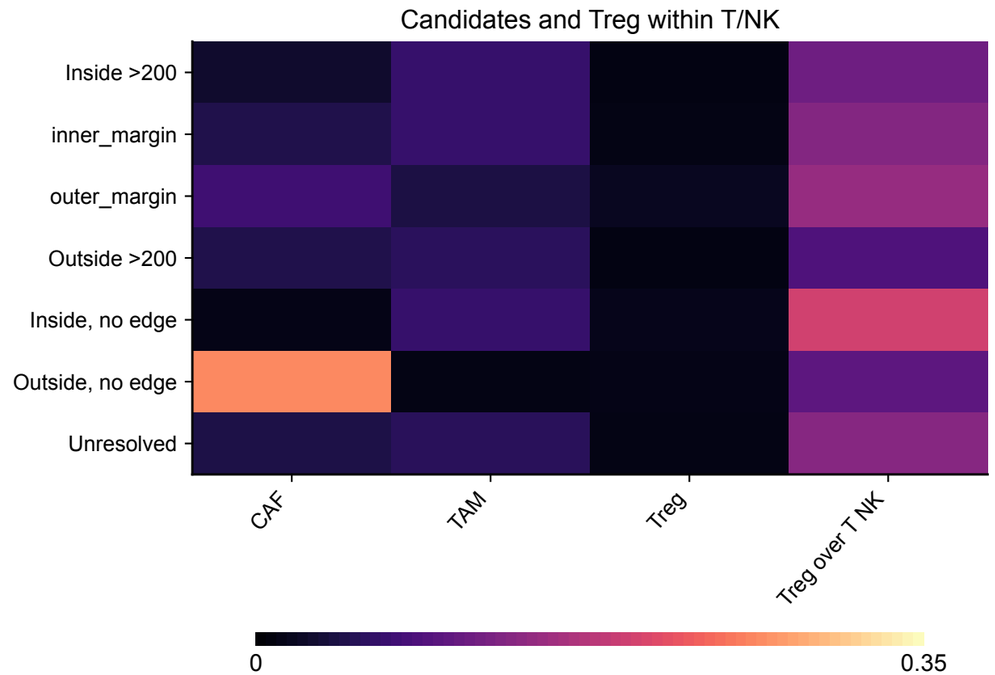


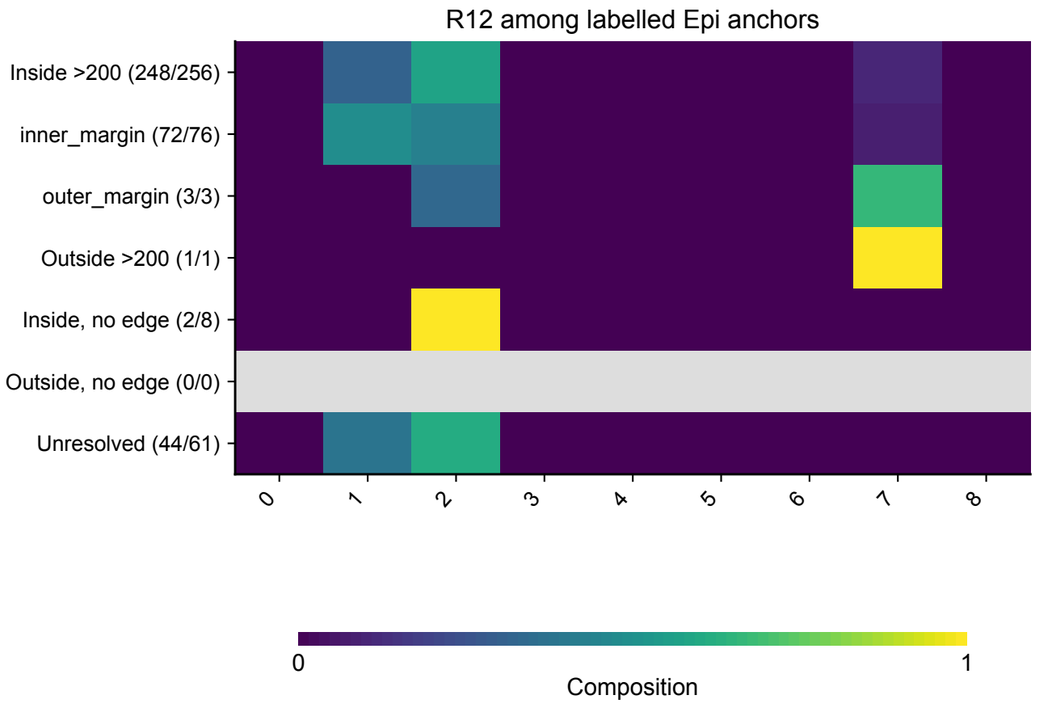
Orange = zero distance
Line = within-section linear fit

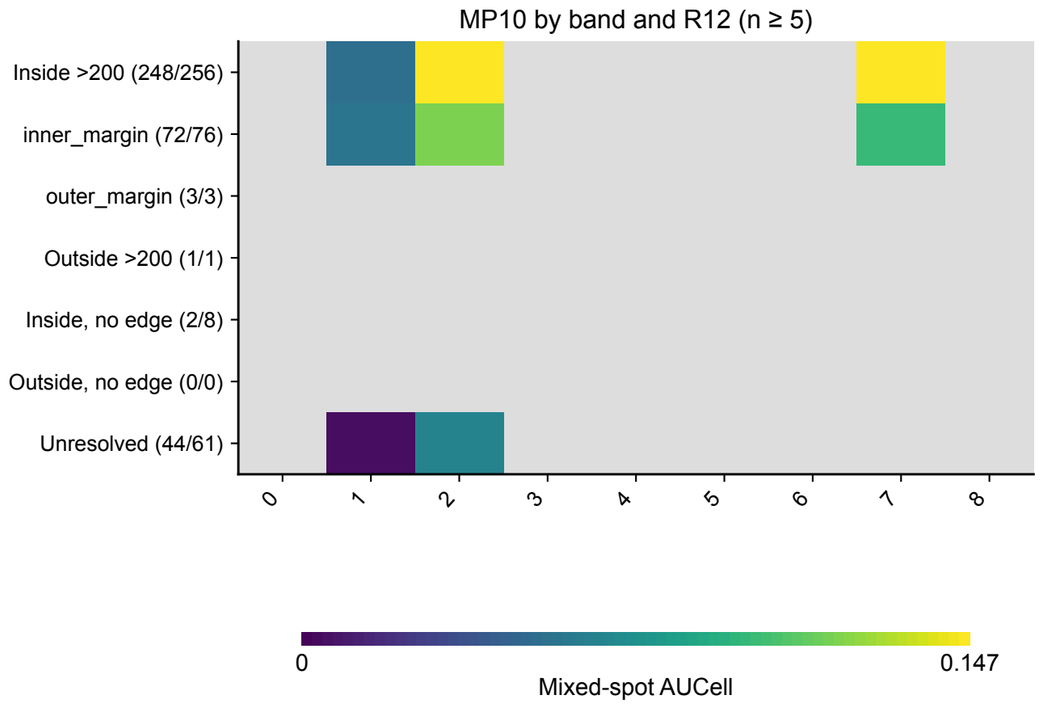
CanDis_2026_Sample12 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





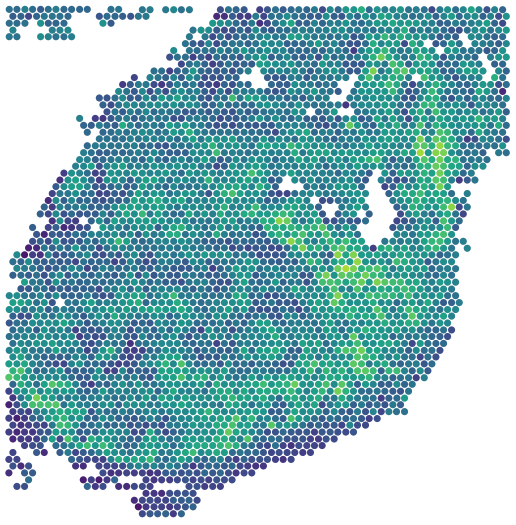




CanDis_2026_Sample12 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

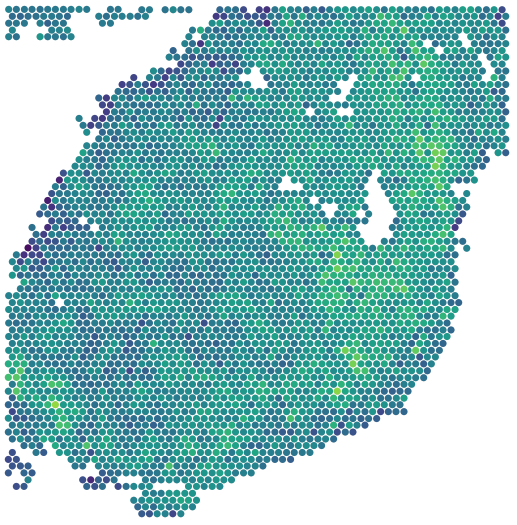
E2F Targets



0 0.152

UCell score

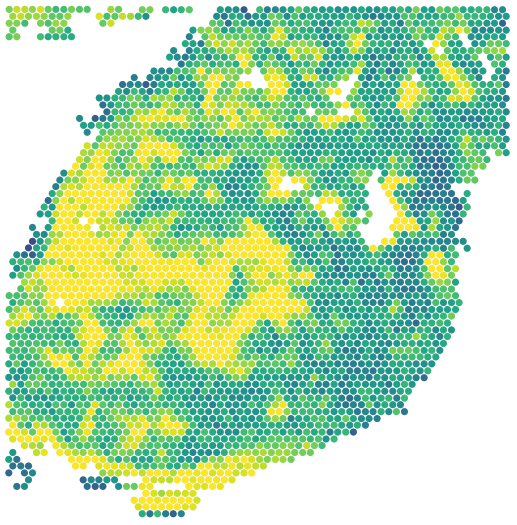
G2M Checkpoint



0 0.134

UCell score

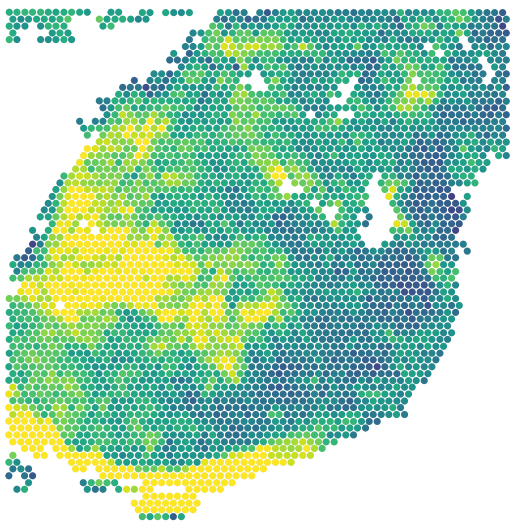
Hypoxia



0 0.211

UCell score

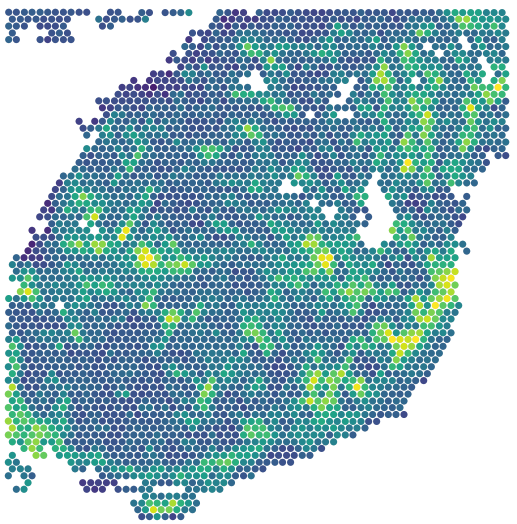
Tnfa Signaling Via Nfkb



0 0.275

UCell score

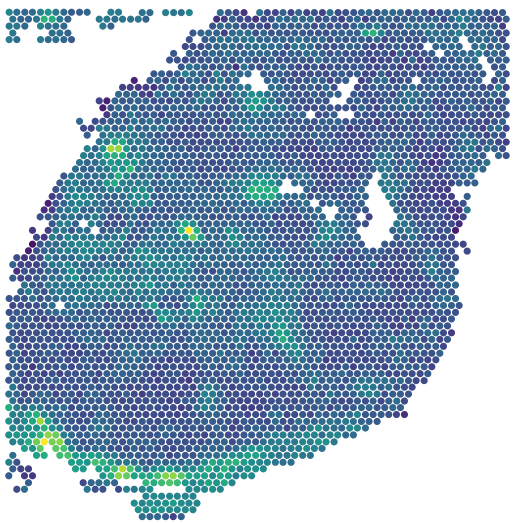
Emt



0 0.386

UCell score

Interferon Gamma Response



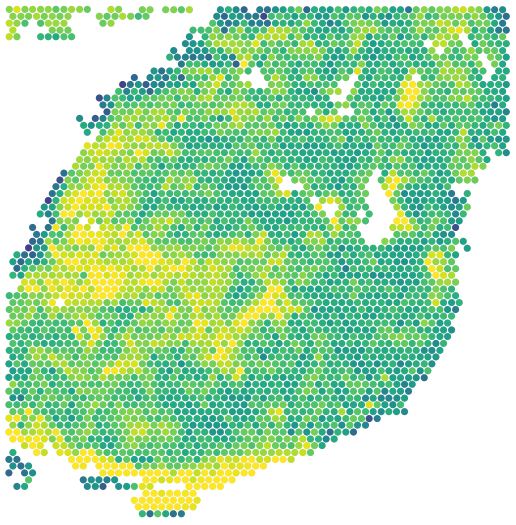
0 0.275

UCell score

CanDis_2026_Sample12 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

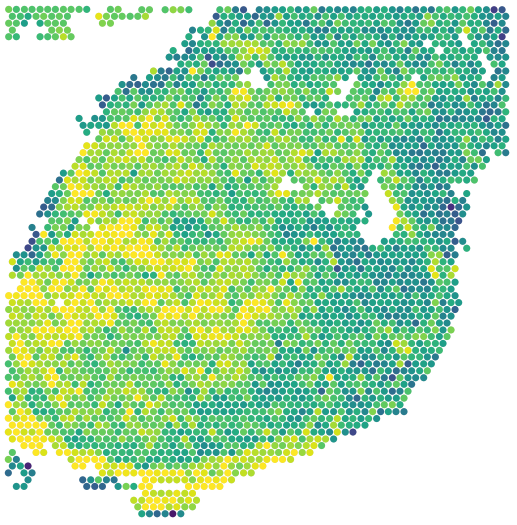
Il2 Stat5 Signaling



0 0.144

UCell score

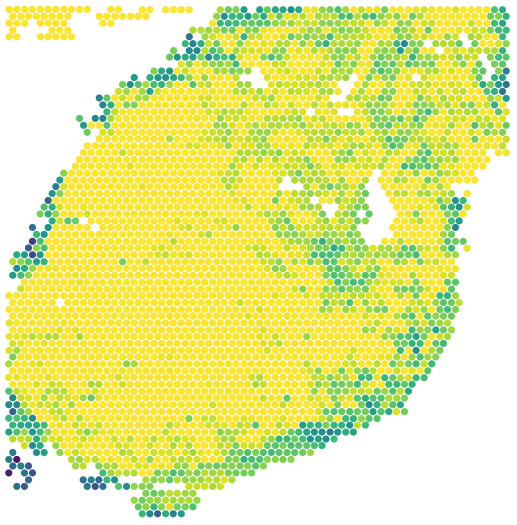
Tgf Beta Signaling



0 0.232

UCell score

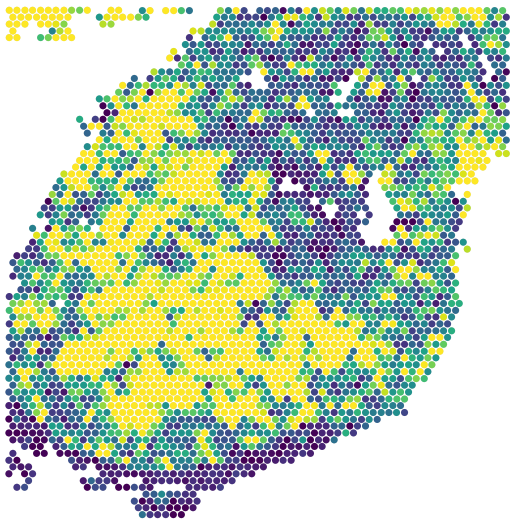
Cholesterol Homeostasis



0 0.21

UCell score

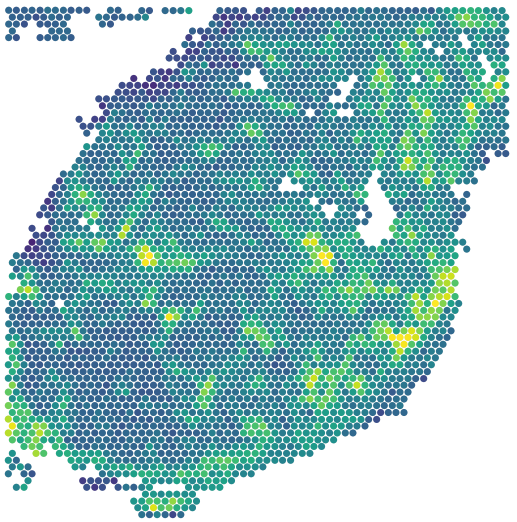
Cholesterol Biosynthesis



0 0.166

UCell score

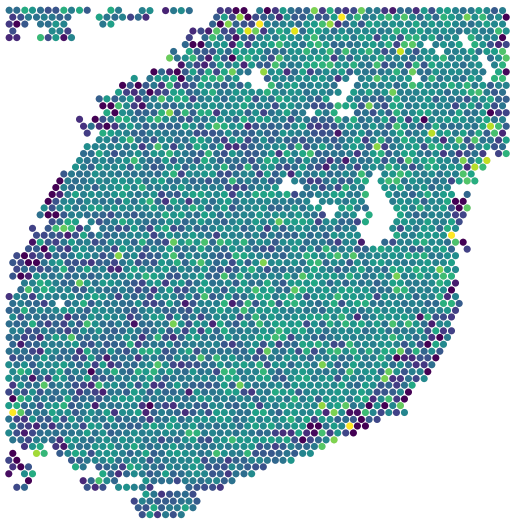
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



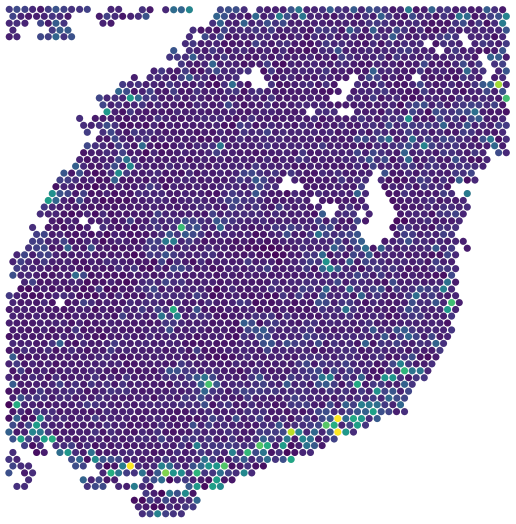
0 0.151

UCell score

CanDis_2026_Sample12 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

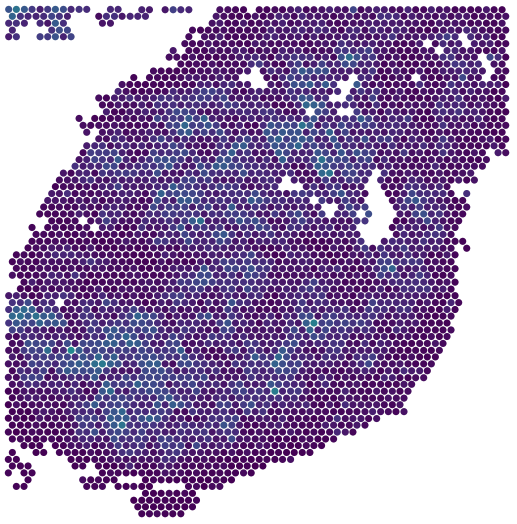
Treg composition



0 0.102

Fraction of all 30

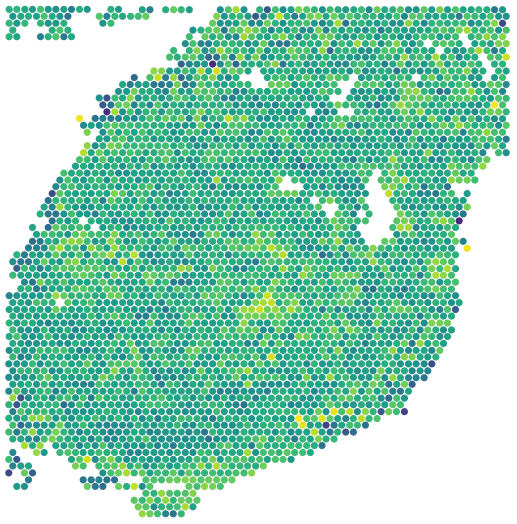
Treg estimated abundance



0 2.85

Posterior mean estimate

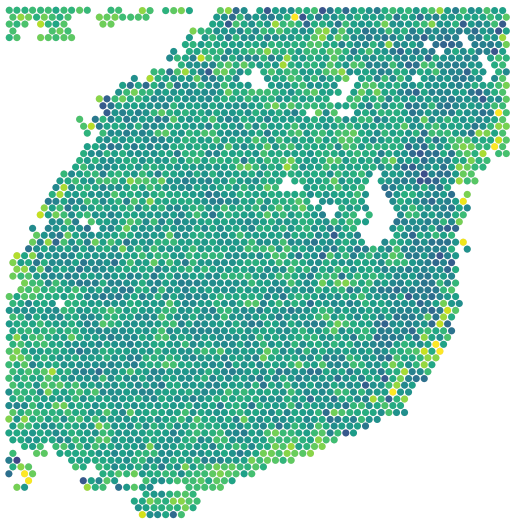
Treg Resting Vs Activated Dn



0 0.0778

UCell score

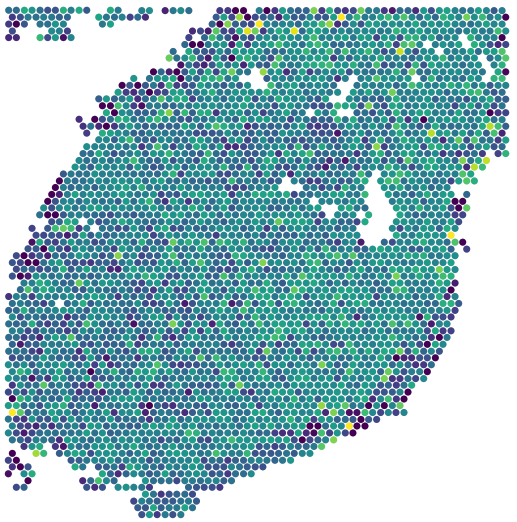
Treg Resting Vs Activated Up



0 0.0609

UCell score

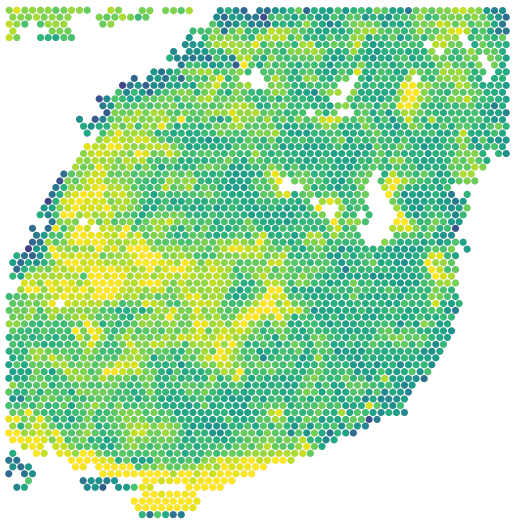
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

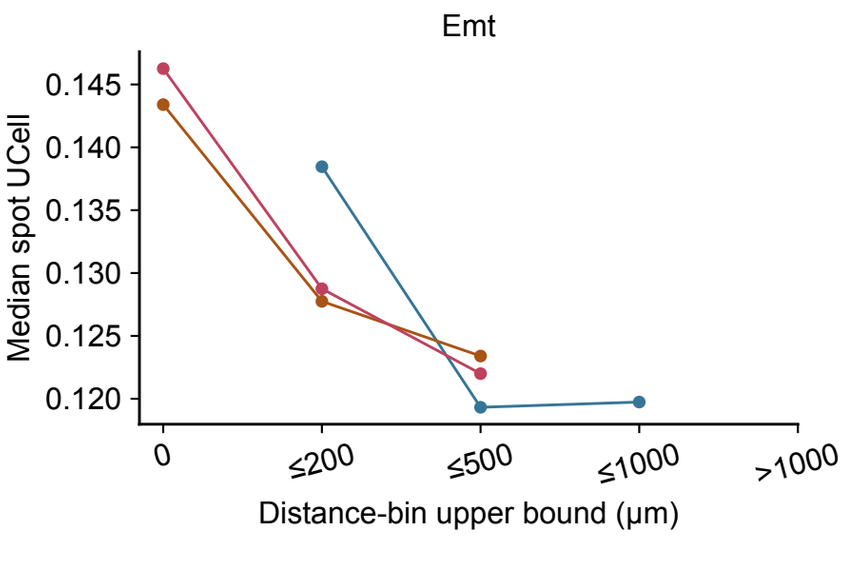


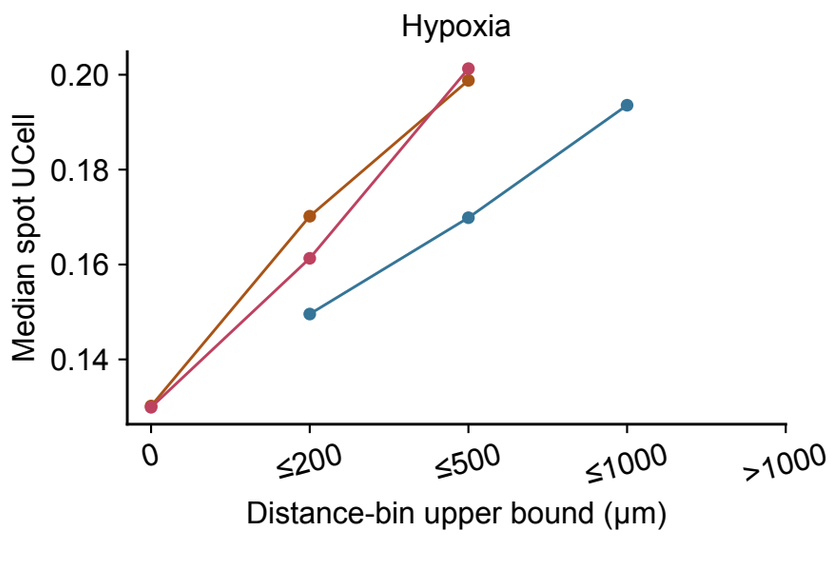
0 0.144

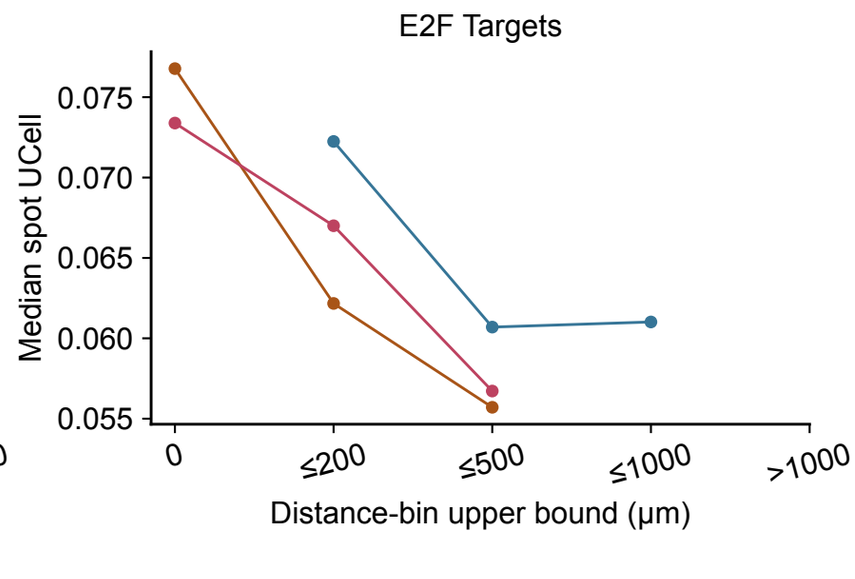
UCell score

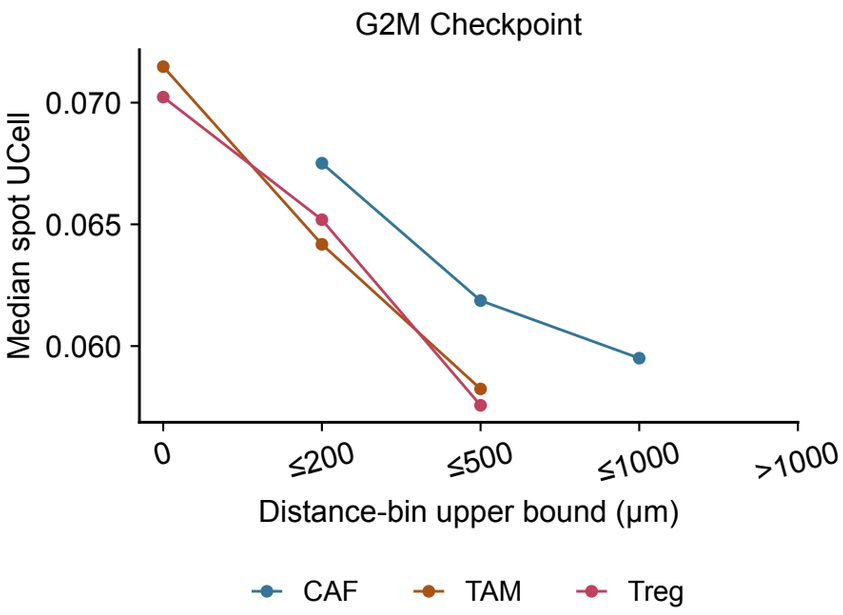
CanDis_2026_Sample12 | Programs by target distance

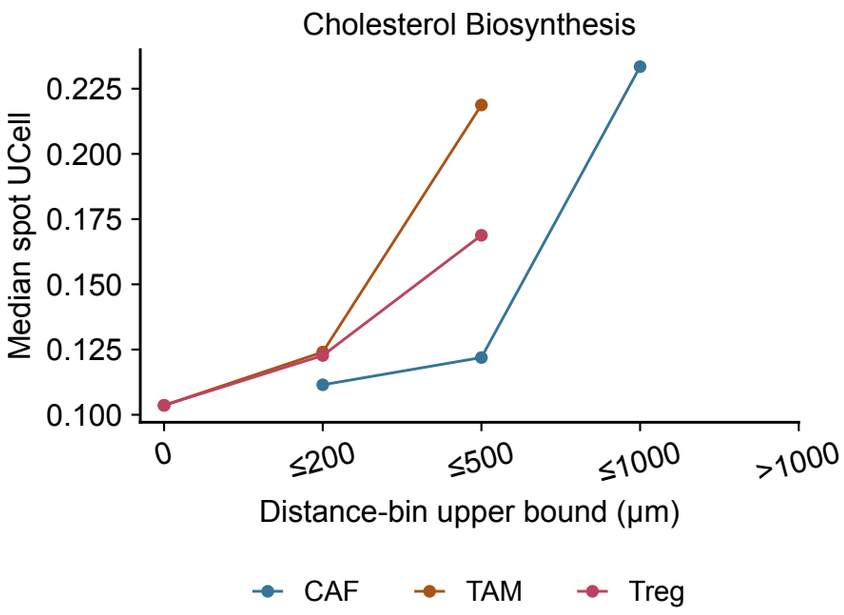
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

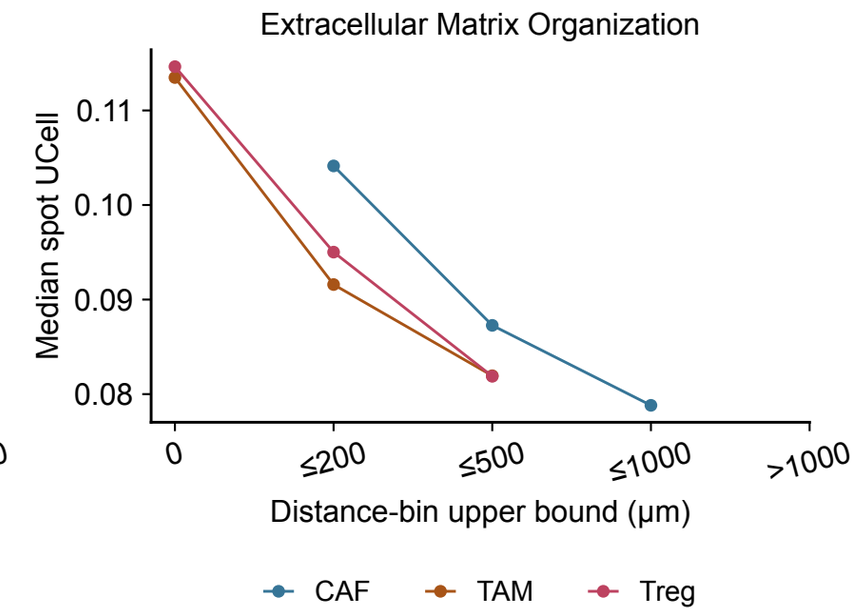






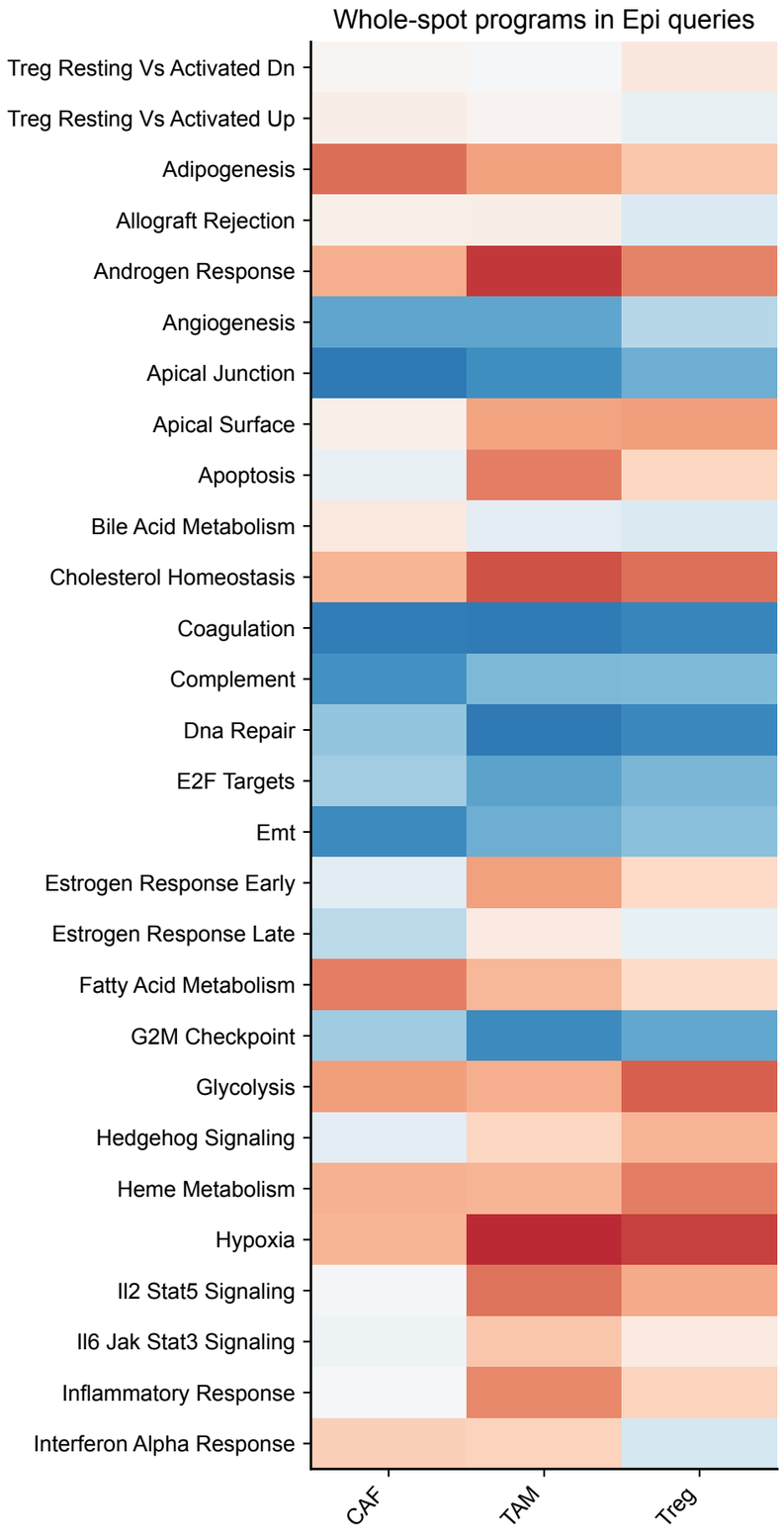


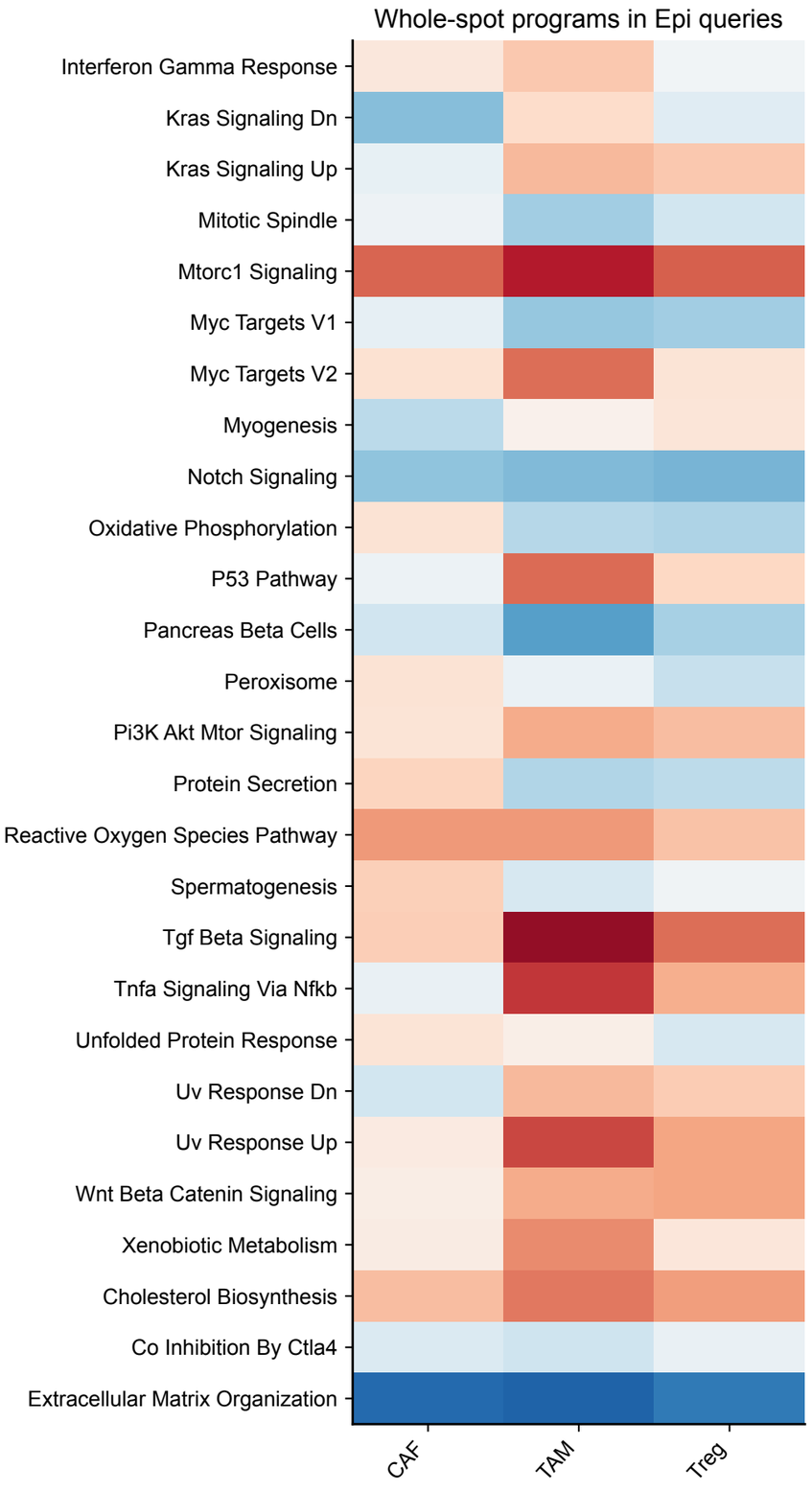




CanDis_2026_Sample12 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

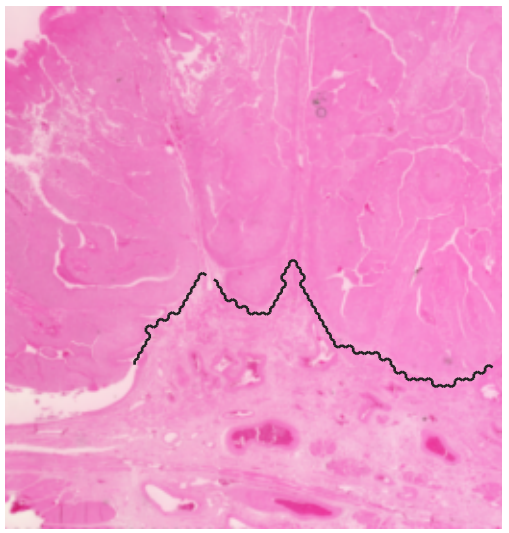




CanDis_2026_Sample13 | Tissue context

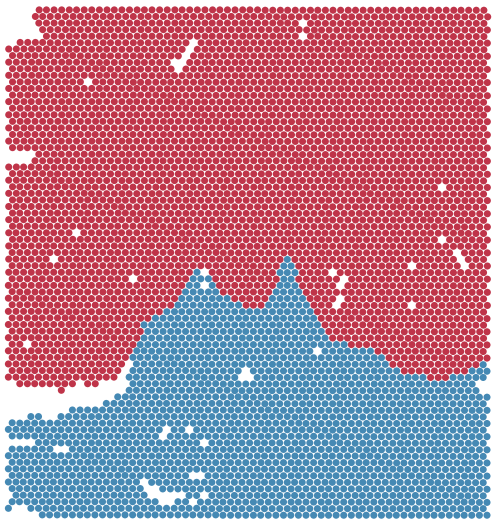
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



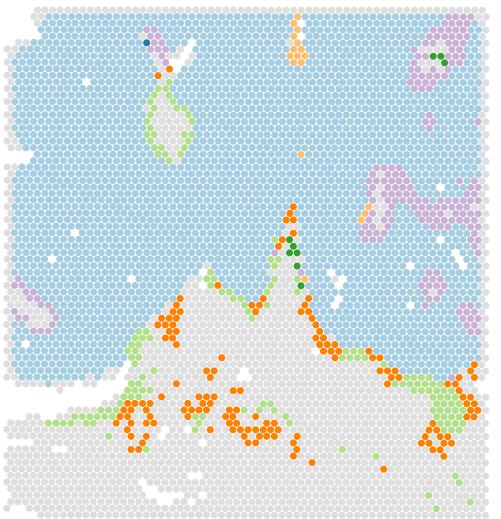
Black line: observed candidate outer edge

Final candidate location



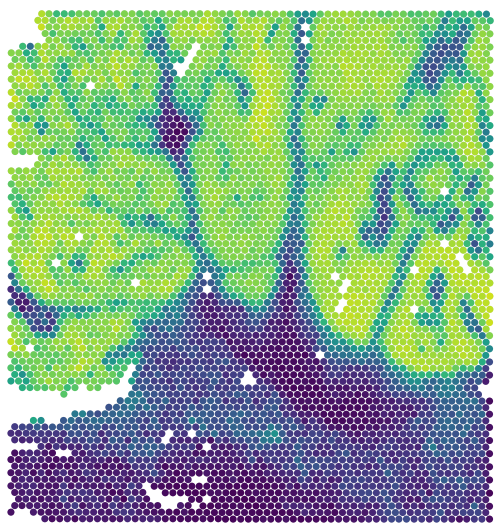
inside outside

Accepted R12 (original 0–8)



0 2 4 5 8
1 3

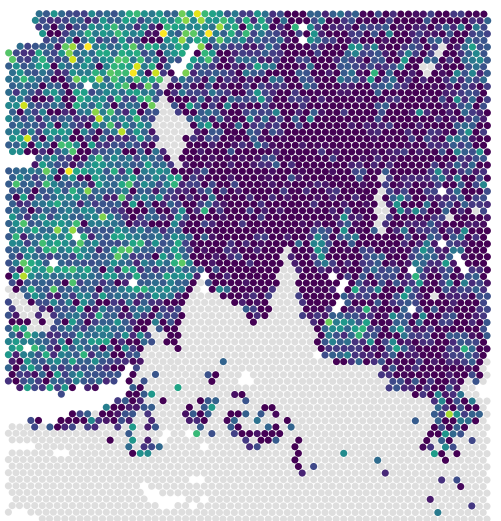
Epithelial support



0 1

Fraction of all 30

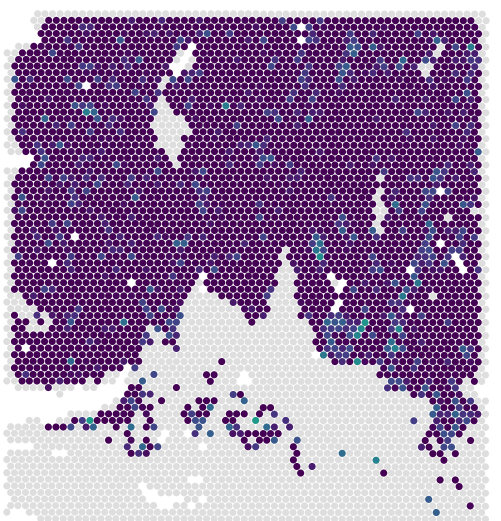
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



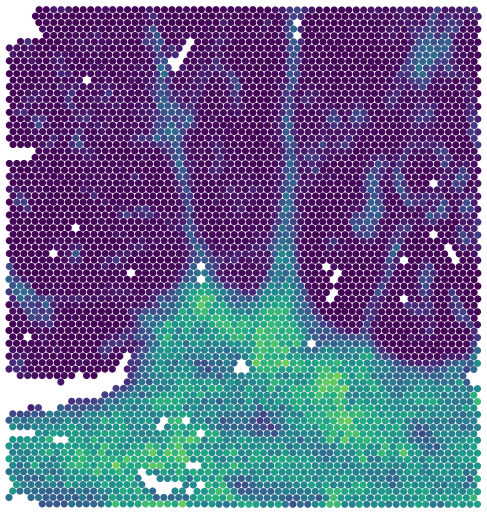
0 0.14

Matched-model AUCell

CanDis_2026_Sample13 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

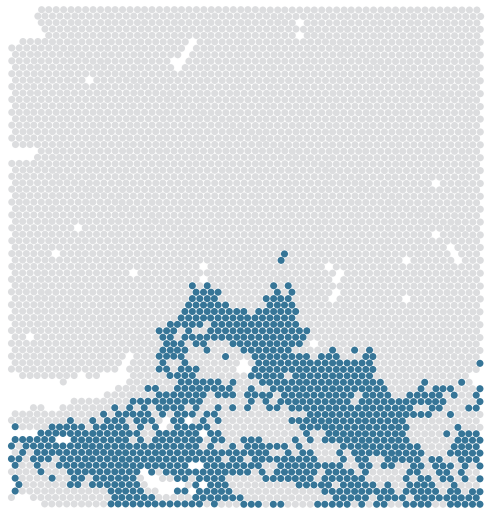
CAF composition



0 0.418

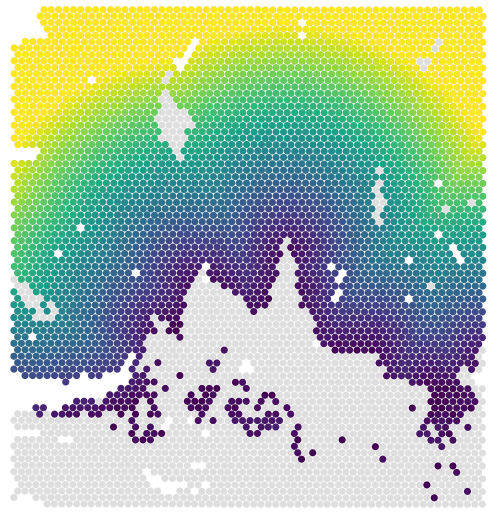
Fraction of all 30

CAF P80 targets



False True

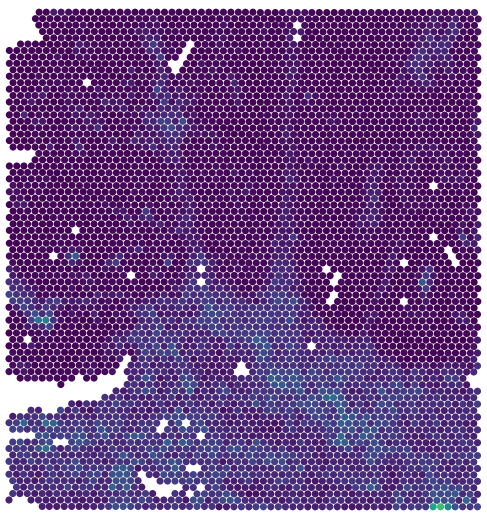
CAF distance



0 3.1e+03

Nearest spot center (μm)

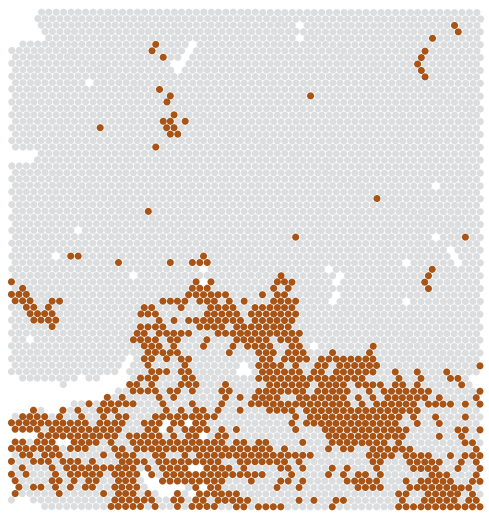
TAM composition



0 0.291

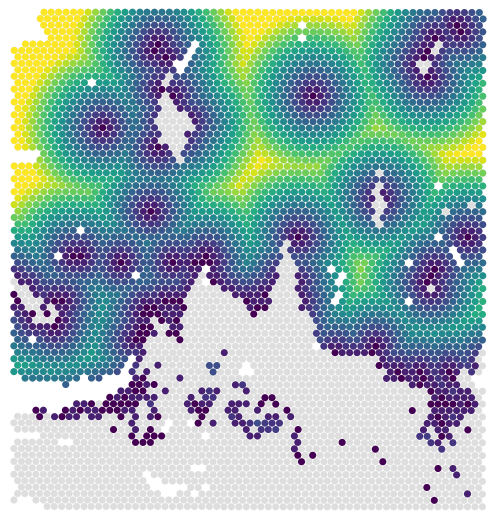
Fraction of all 30

TAM P80 targets



False True

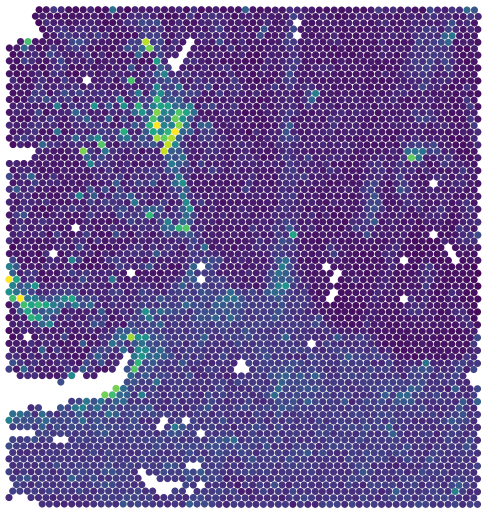
TAM distance



0 1.13e+03

Nearest spot center (μm)

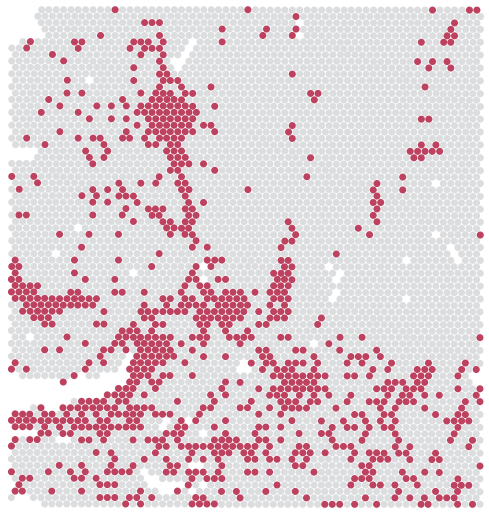
Treg composition



0 0.102

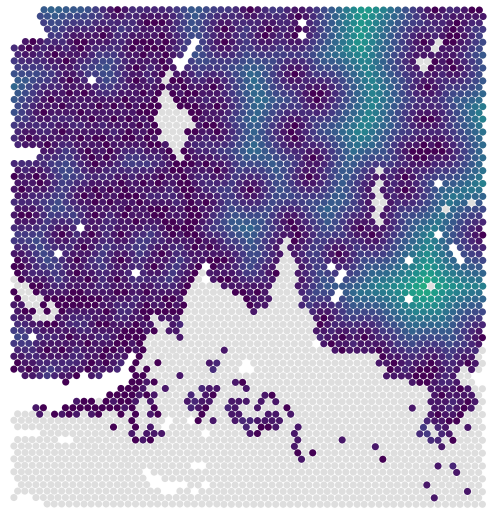
Fraction of all 30

Treg P80 targets



False True

Treg distance

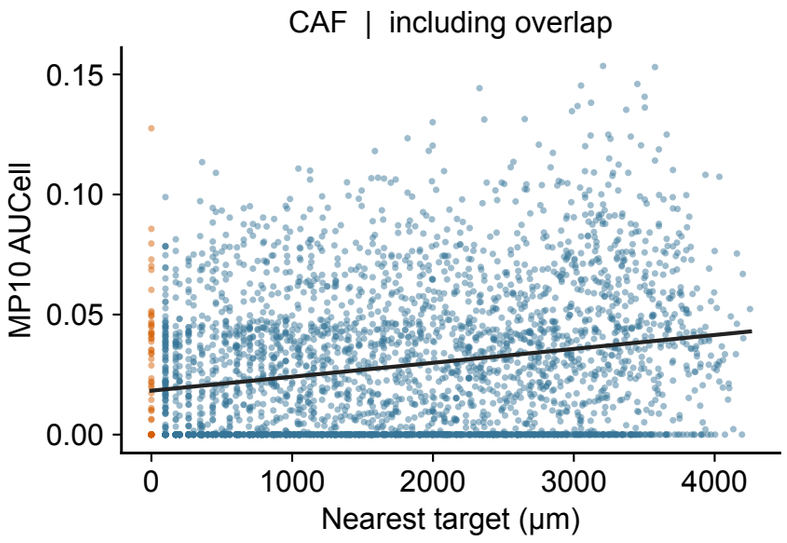


0 1.61e+03

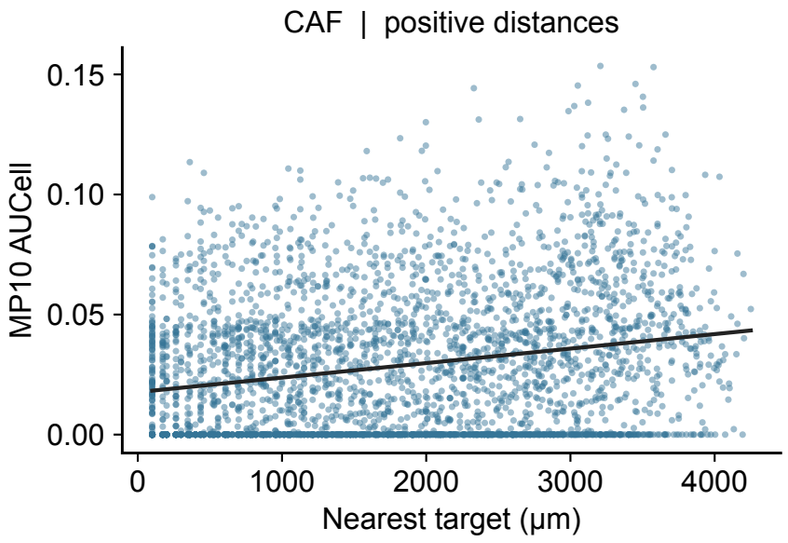
Nearest spot center (μm)

CanDis_2026_Sample13 | MP10 versus nearest target

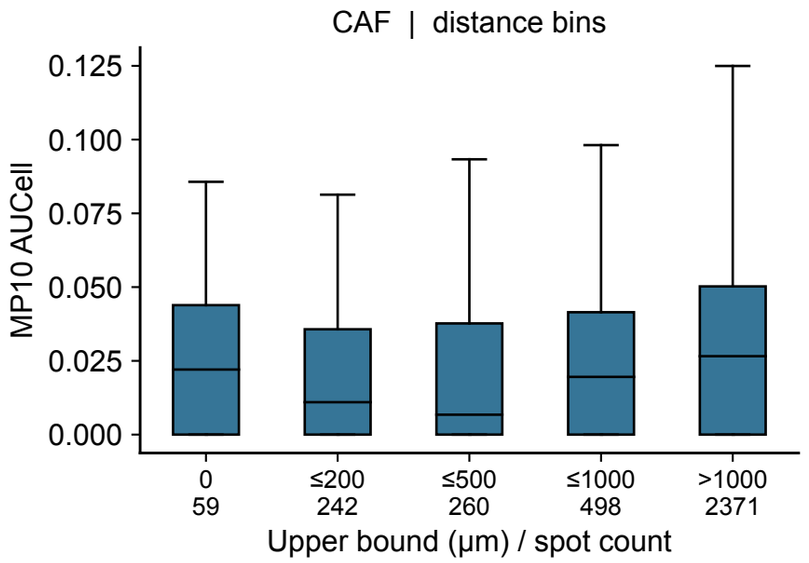
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



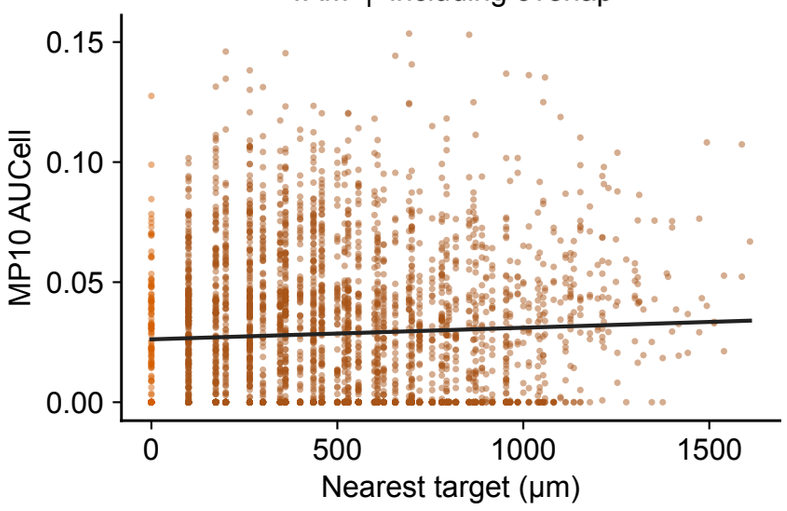
r +0.22 ρ +0.20
n 3430/3430 queries • 967 targets



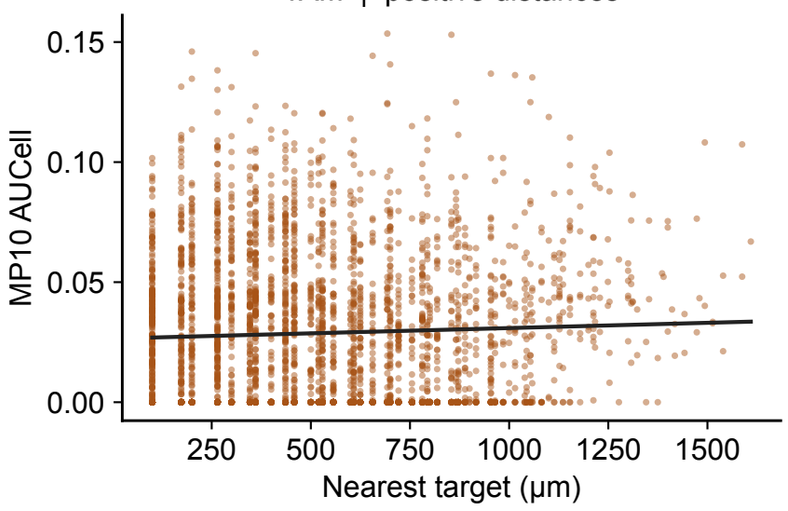
r +0.22 ρ +0.21
n 3371/3430 queries • 967 targets



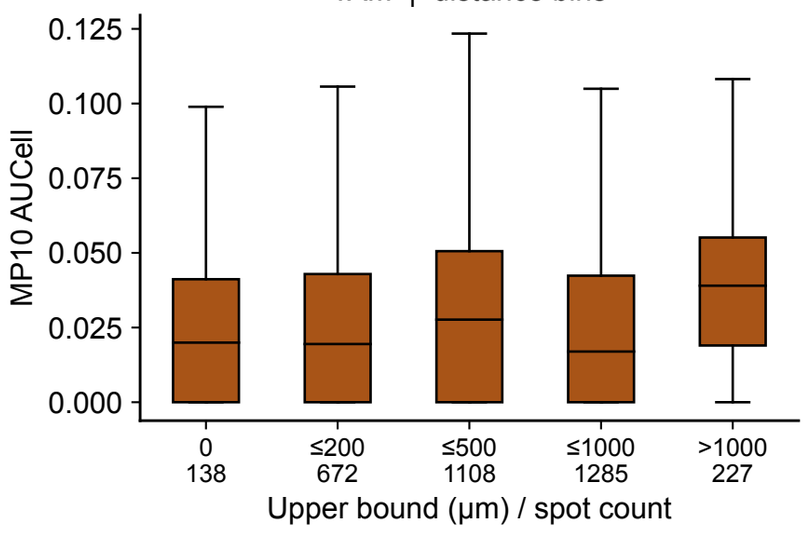
Orange = zero distance
Line = within-section linear fit



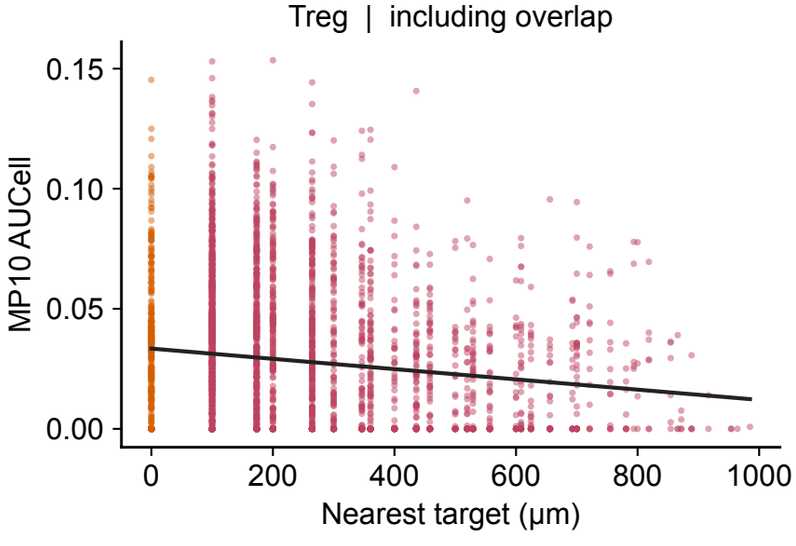
r +0.05 ρ +0.04
n 3430/3430 queries • 967 targets



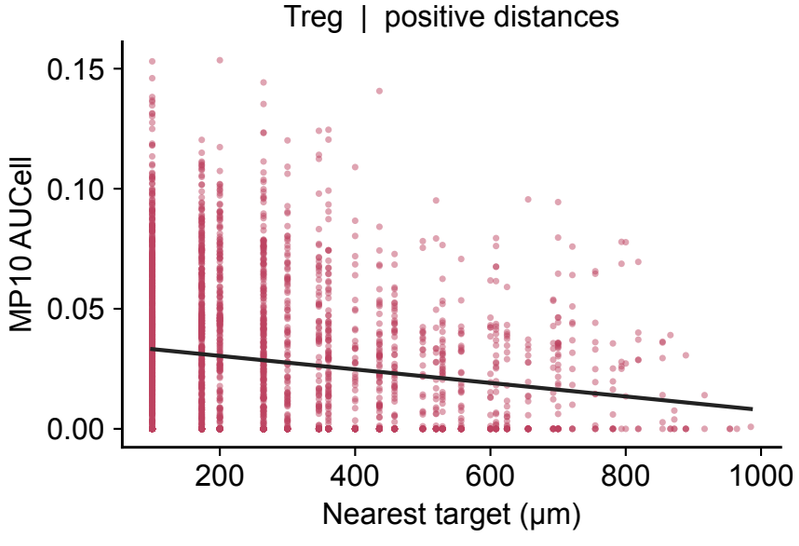
r +0.04 ρ +0.03
n 3292/3430 queries • 967 targets



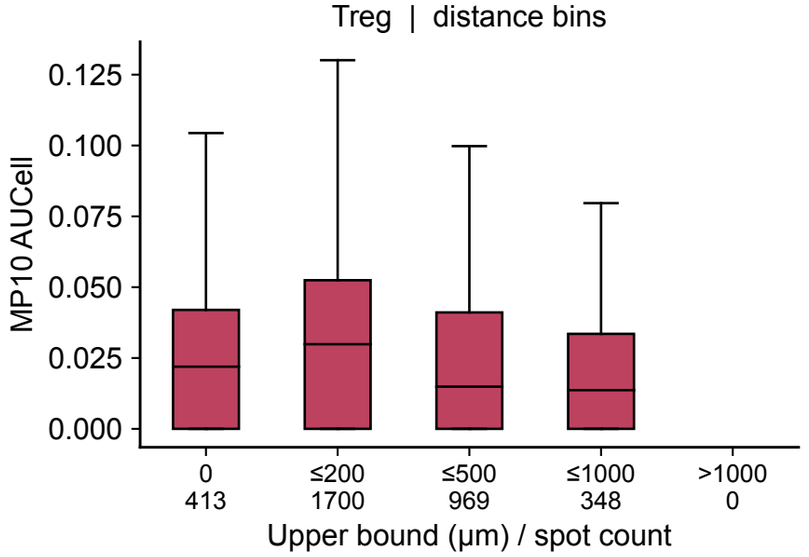
Orange = zero distance
Line = within-section linear fit



r -0.13 ρ -0.10
n 3430/3430 queries • 967 targets



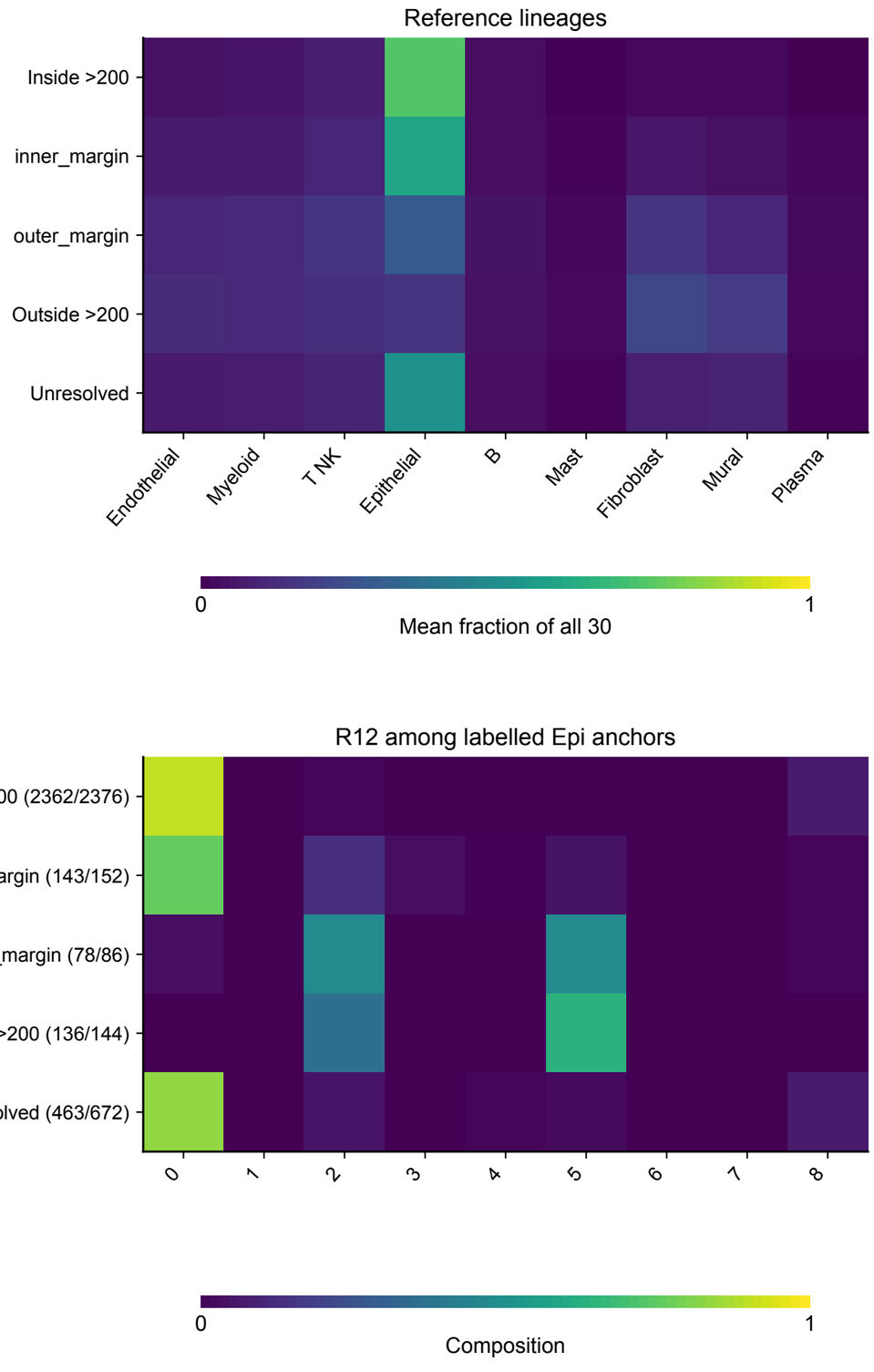
r -0.17 ρ -0.15
n 3017/3430 queries • 967 targets

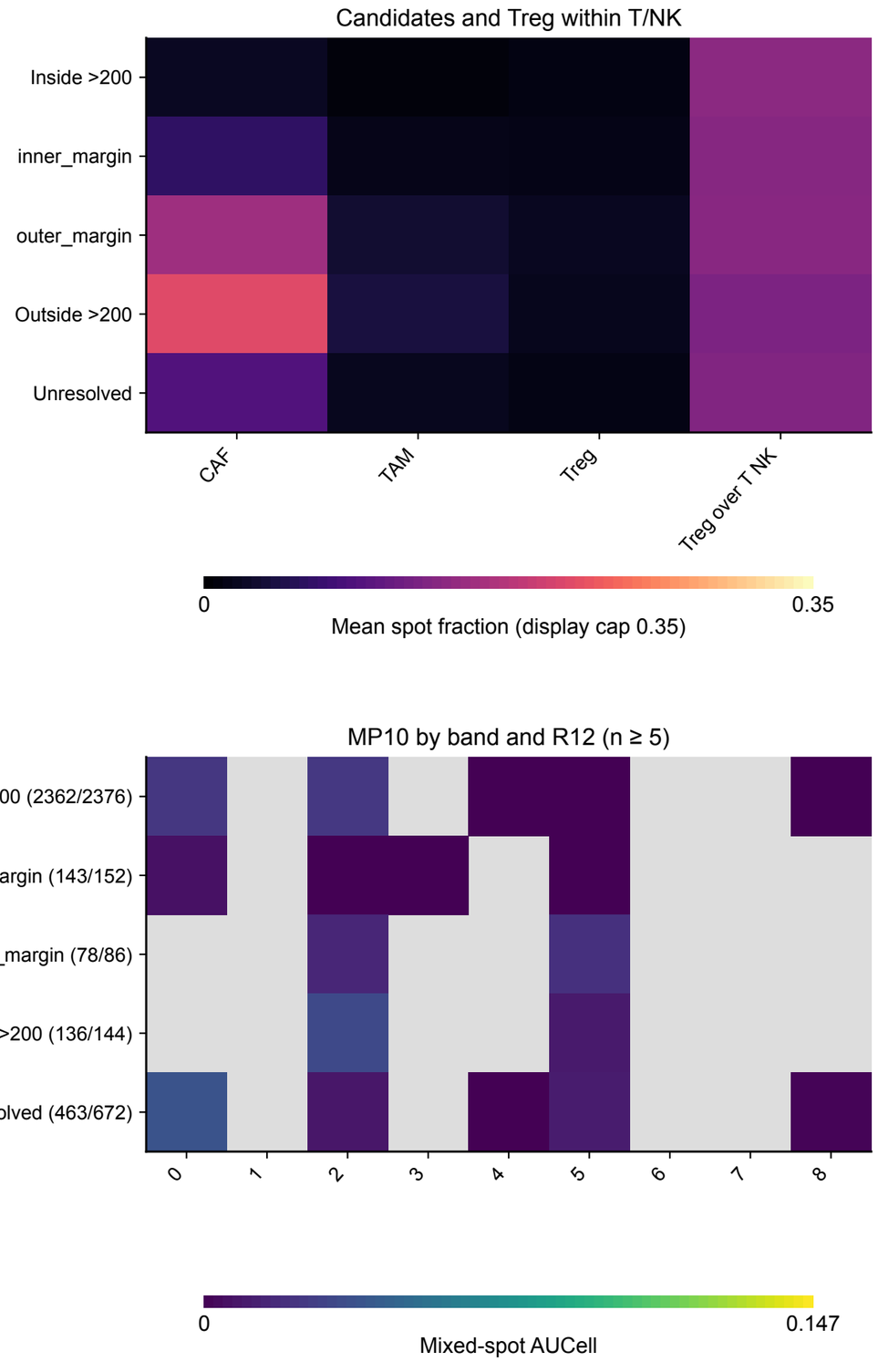


Orange = zero distance
Line = within-section linear fit

CanDis_2026_Sample13 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable

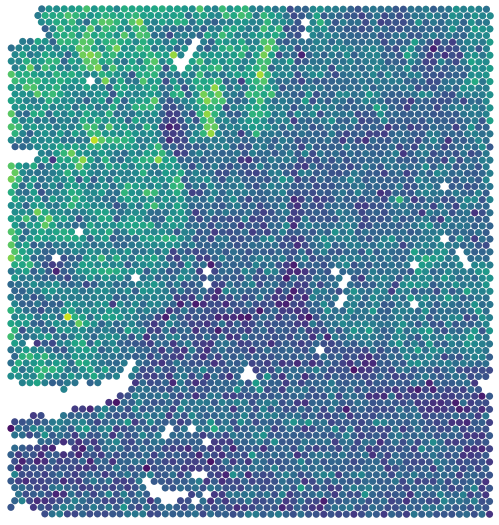




CanDis_2026_Sample13 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets



0 0.152

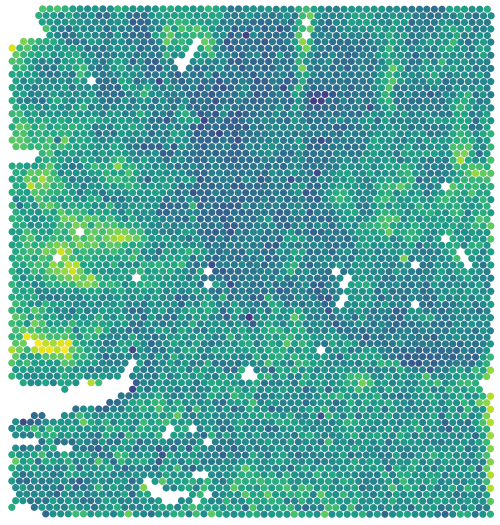
UCell score

G2M Checkpoint

0 0.134

UCell score

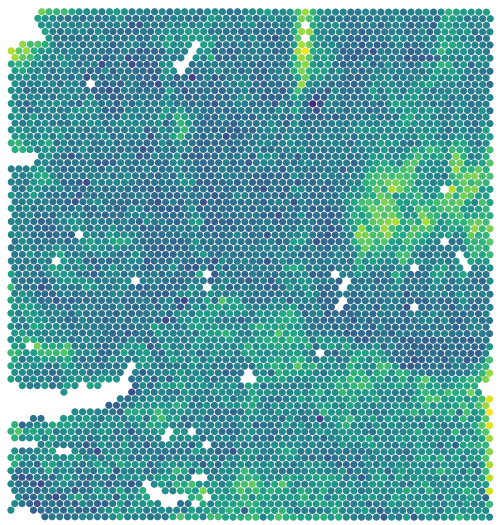
Hypoxia



0 0.211

UCell score

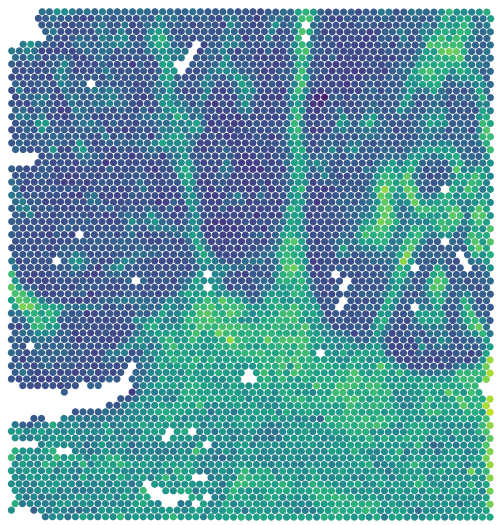
Tnfa Signaling Via Nfkb



0 0.275

UCell score

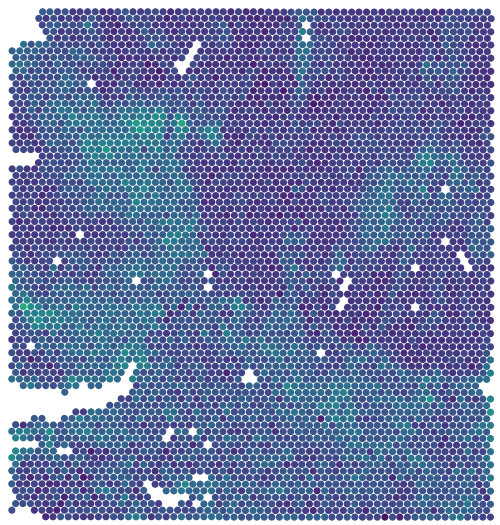
Emt



0 0.386

UCell score

Interferon Gamma Response



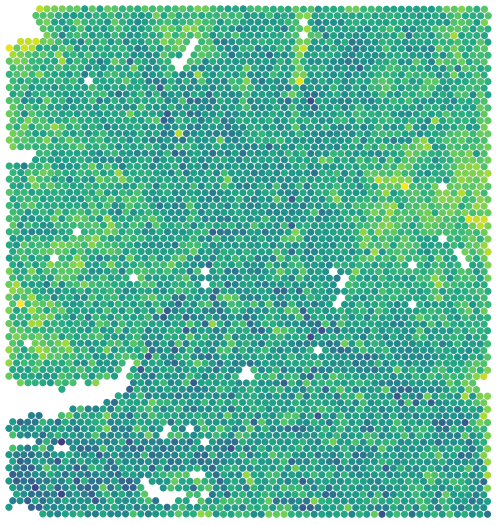
0 0.275

UCell score

CanDis_2026_Sample13 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

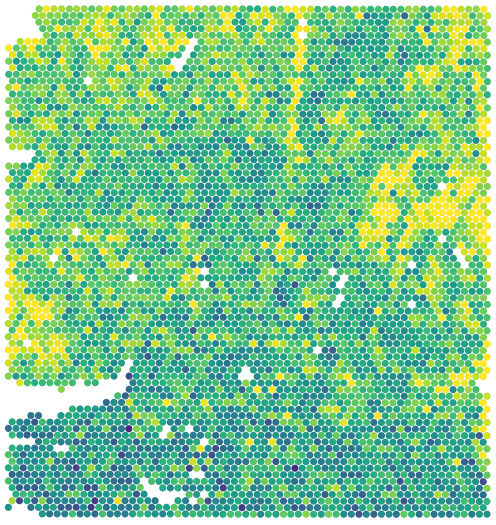
Il2 Stat5 Signaling



0 0.144

UCell score

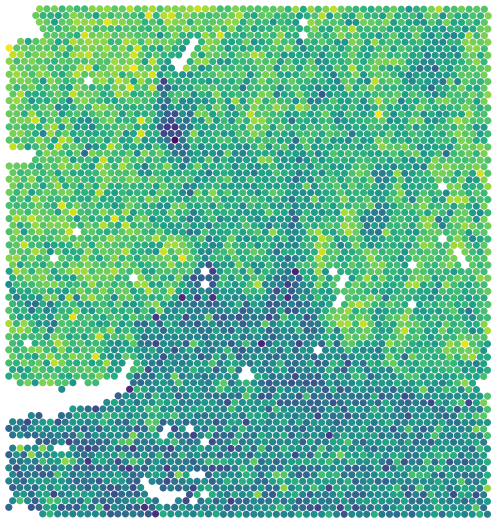
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

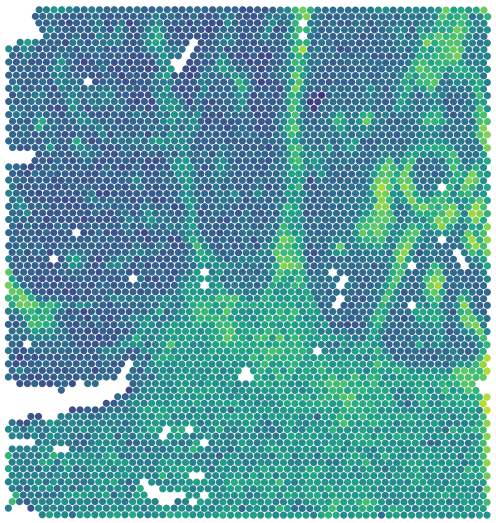
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

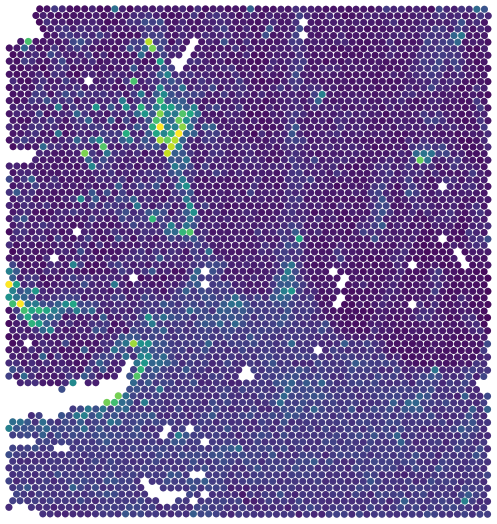
0 0.151

UCell score

CanDis_2026_Sample13 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

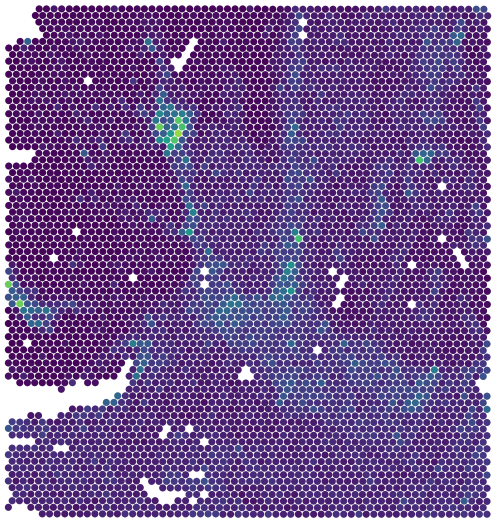
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

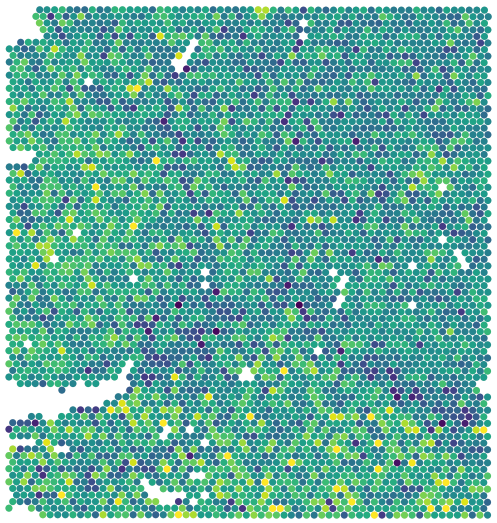
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up



0 0.0609

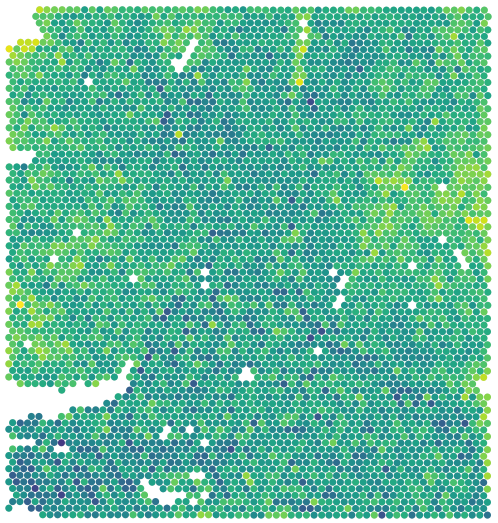
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

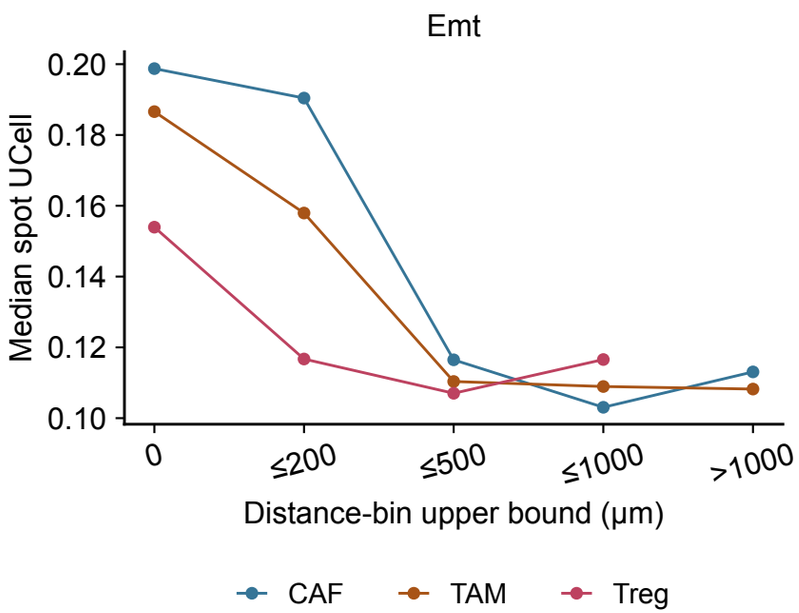


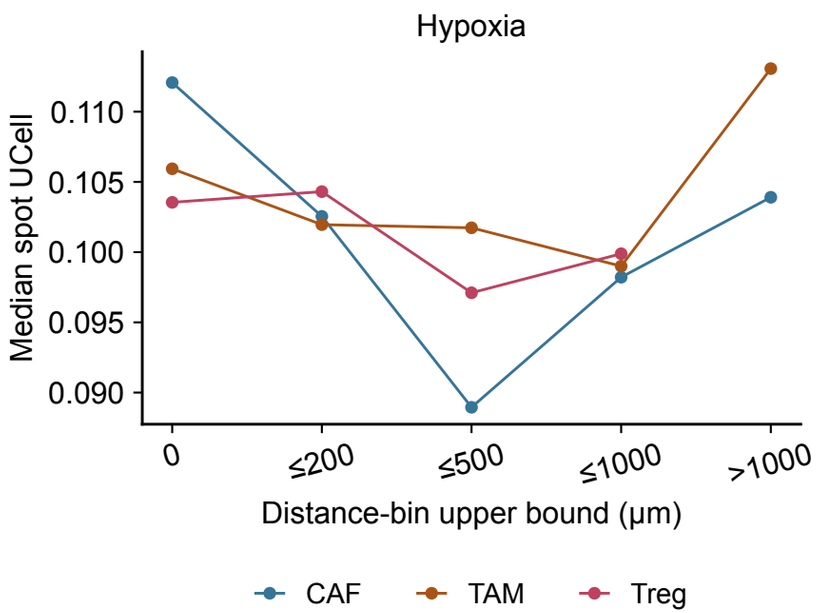
0 0.144

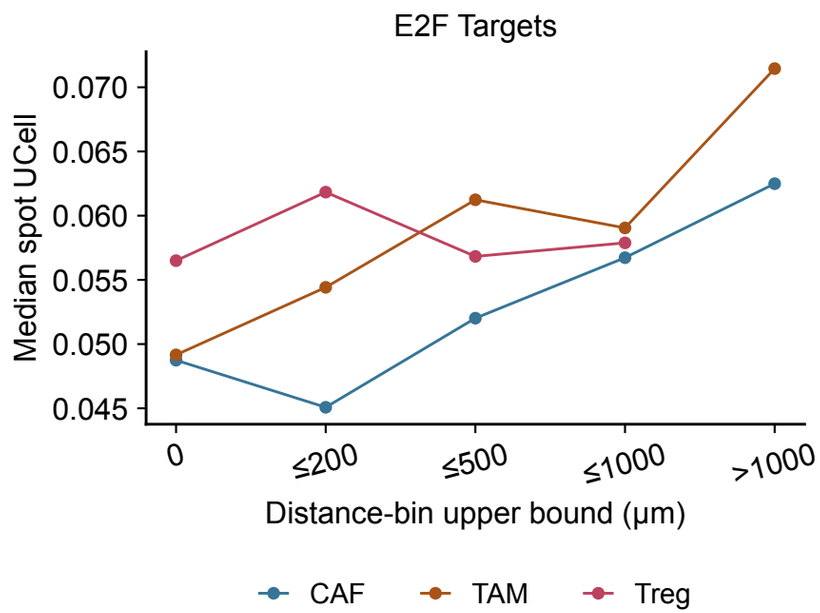
UCell score

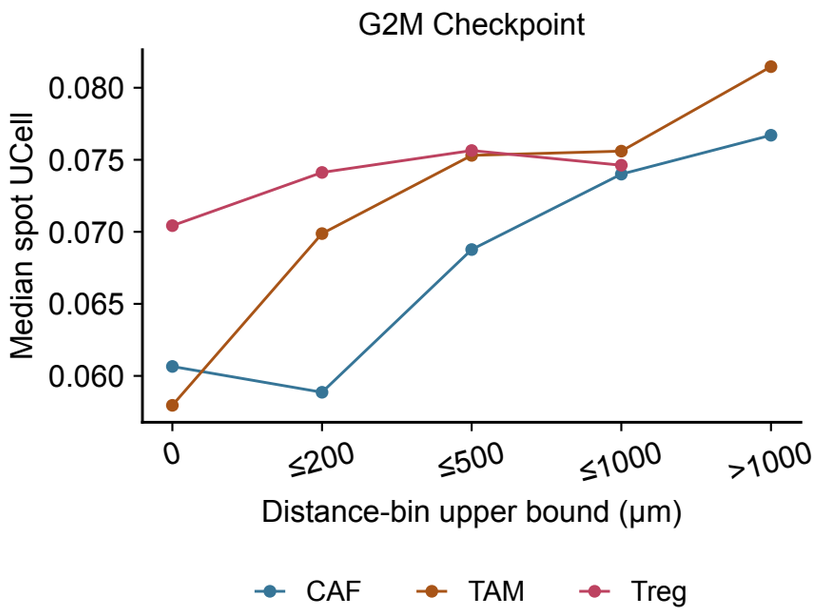
CanDis_2026_Sample13 | Programs by target distance

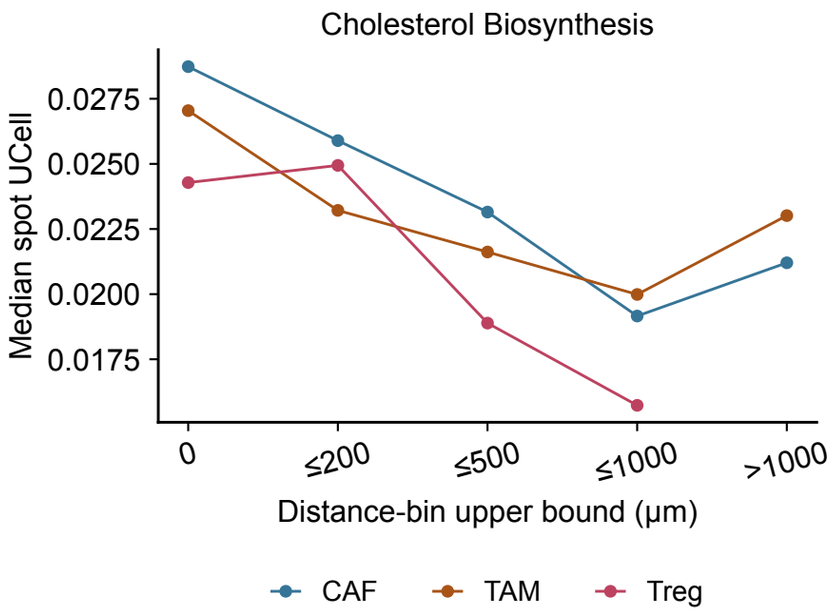
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

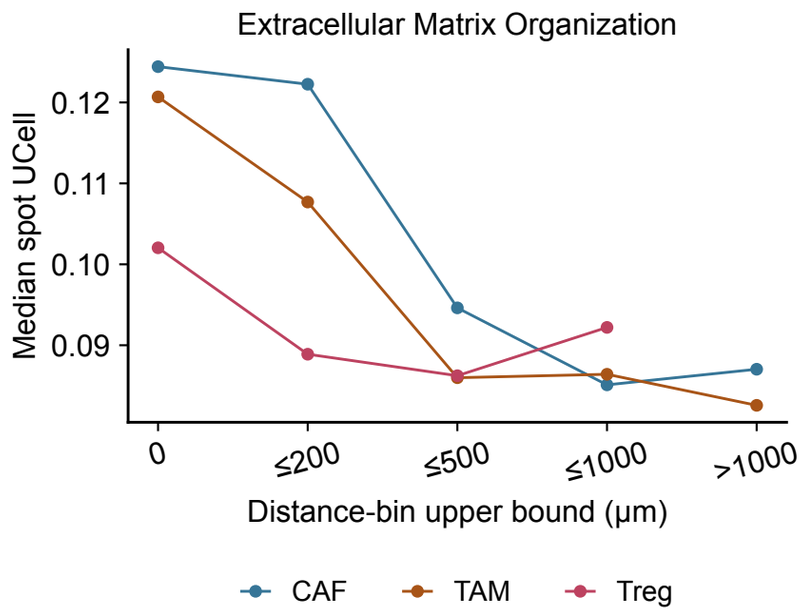






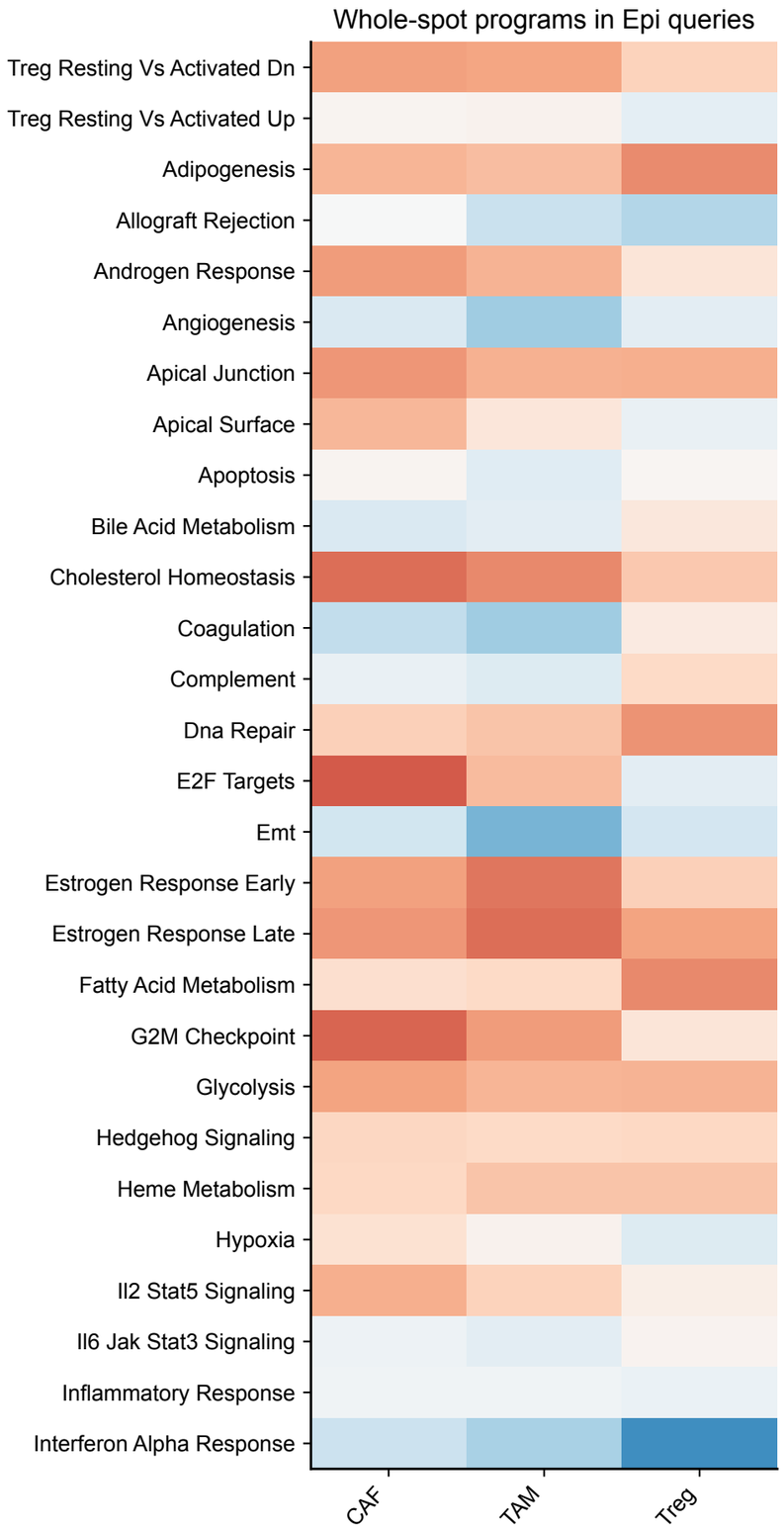


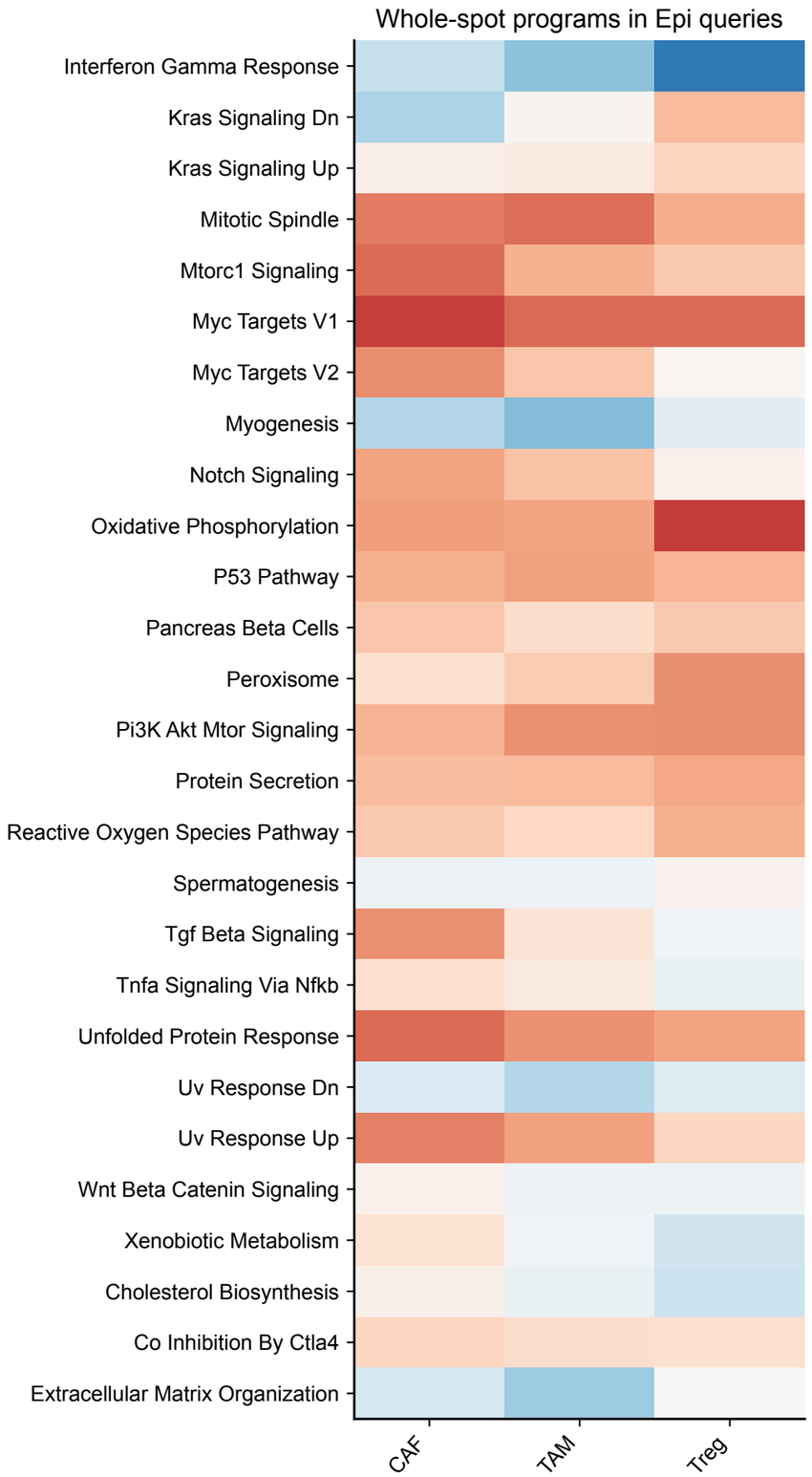




CanDis_2026_Sample13 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

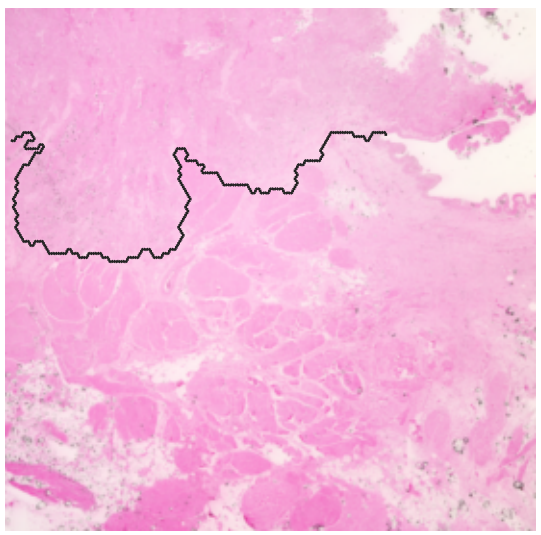




CanDis_2026_Sample2 | Tissue context

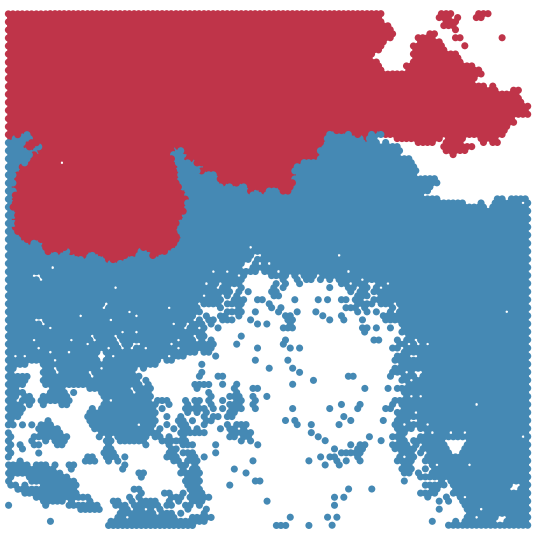
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



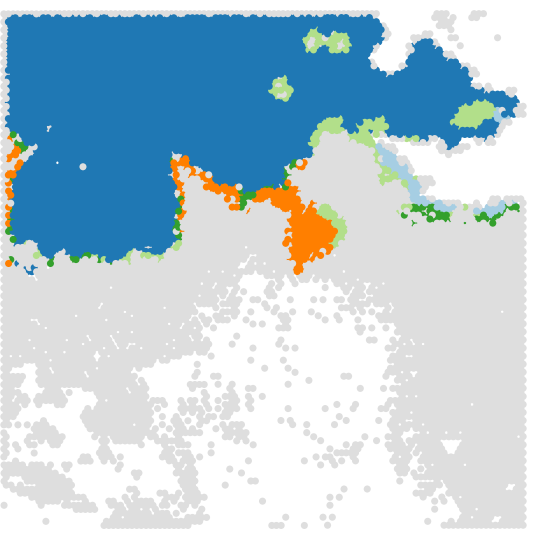
Black line: observed candidate outer edge

Final candidate location



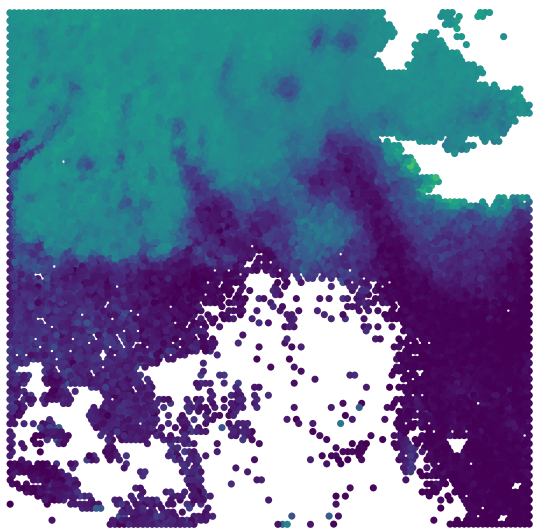
inside outside

Accepted R12 (original 0–8)



0 1 2 3 5

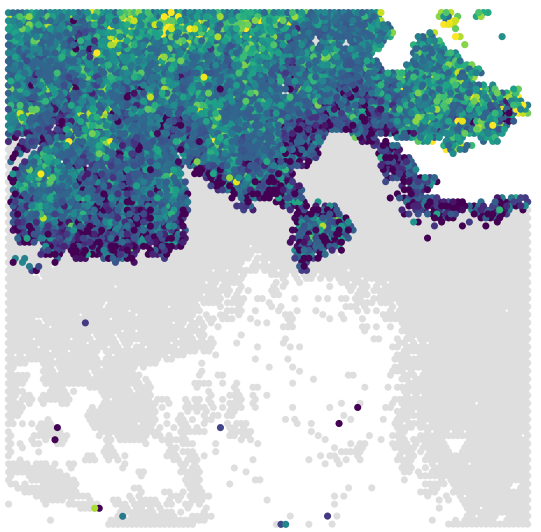
Epithelial support



0 1

Fraction of all 30

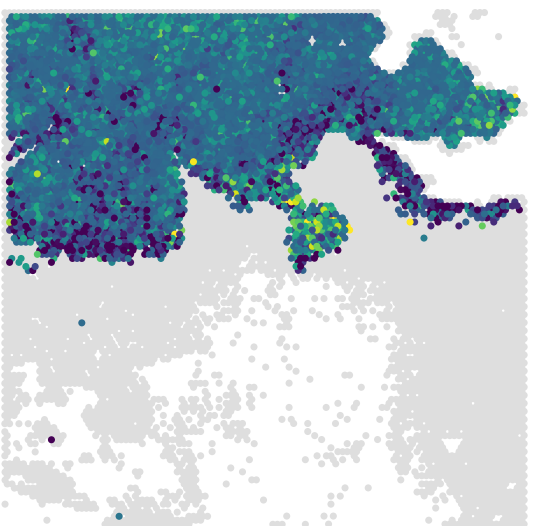
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



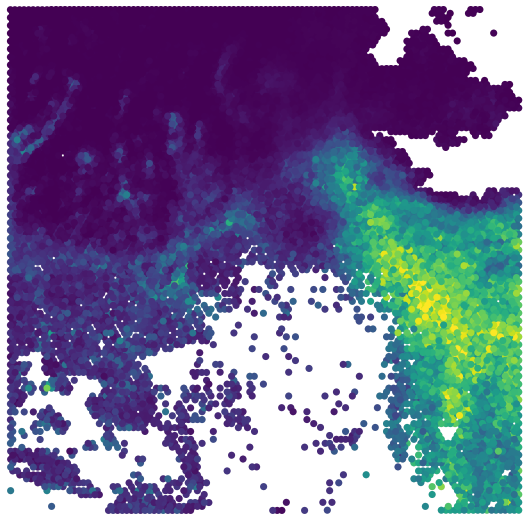
0 0.14

Matched-model AUCell

CanDis_2026_Sample2 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

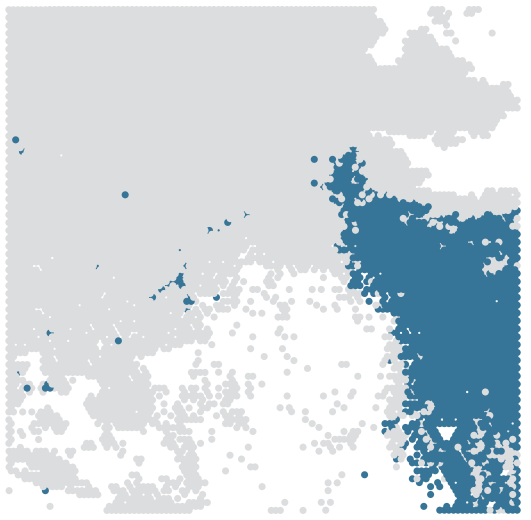
CAF composition



0 0.418

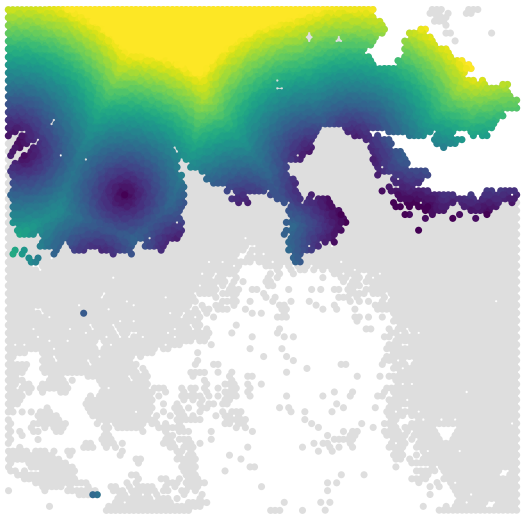
Fraction of all 30

CAF P80 targets



False True

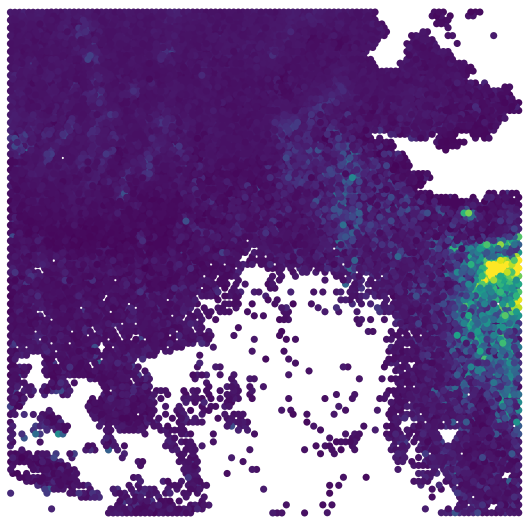
CAF distance



0 3.1e+03

Nearest spot center (µm)

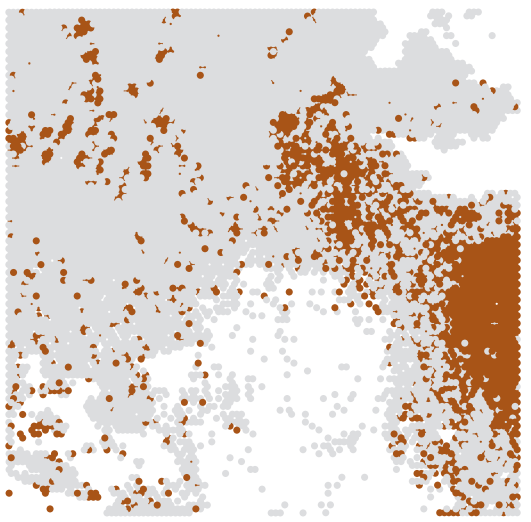
TAM composition



0 0.291

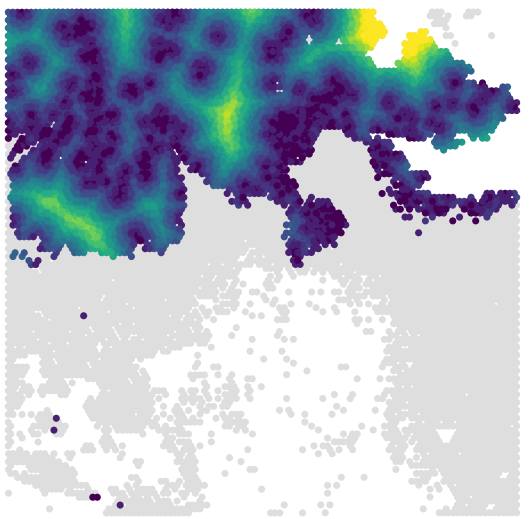
Fraction of all 30

TAM P80 targets



False True

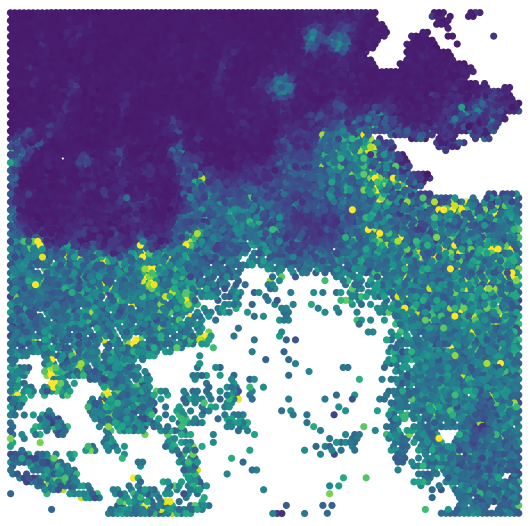
TAM distance



0 1.13e+03

Nearest spot center (µm)

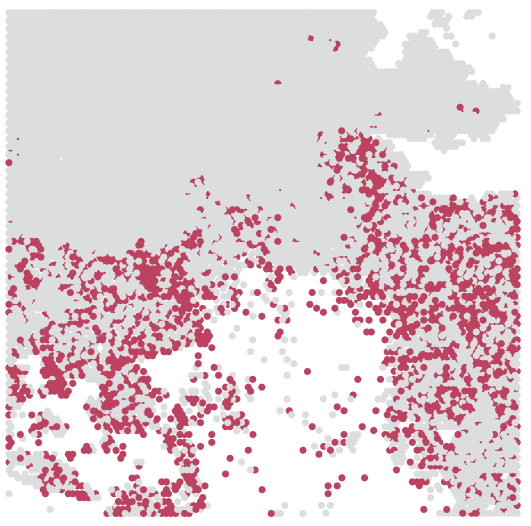
Treg composition



0 0.102

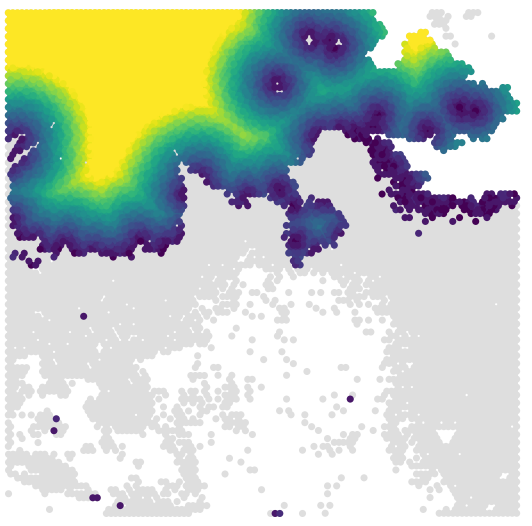
Fraction of all 30

Treg P80 targets



False True

Treg distance

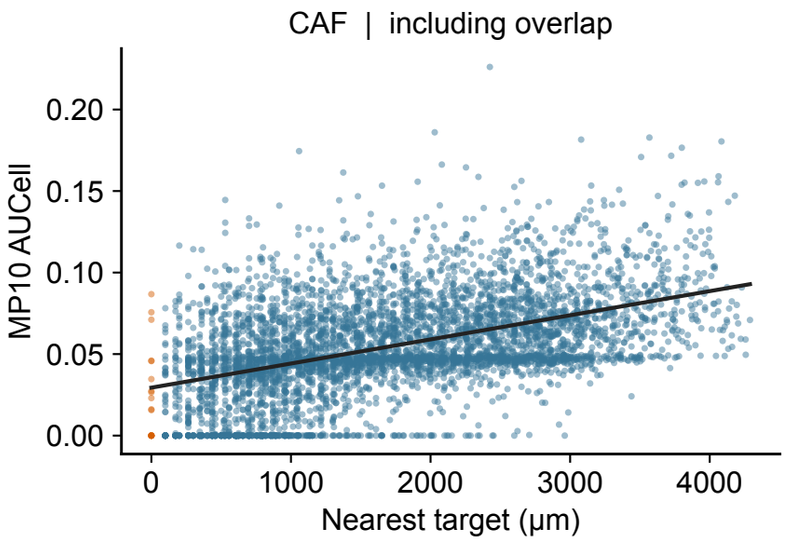


0 1.61e+03

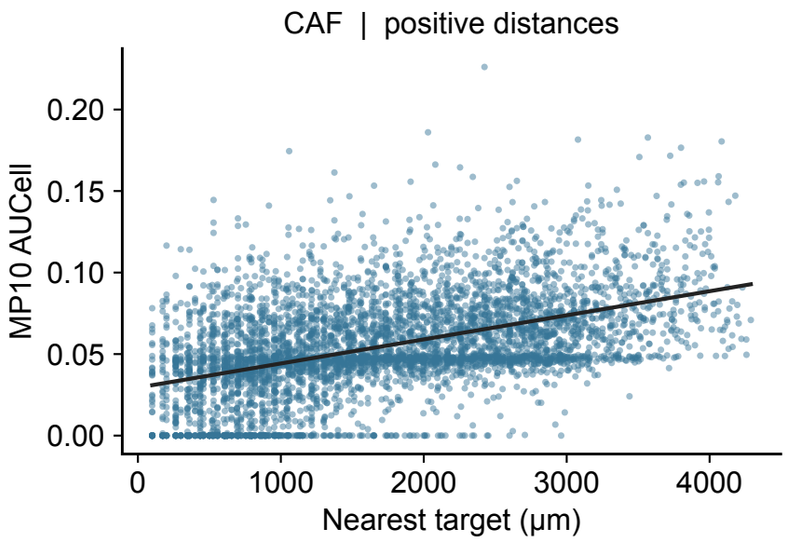
Nearest spot center (µm)

CanDis_2026_Sample2 | MP10 versus nearest target

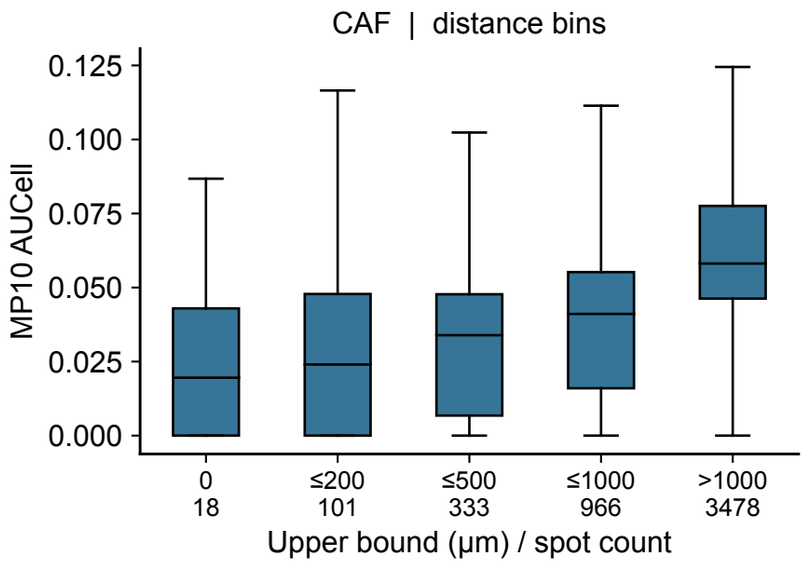
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



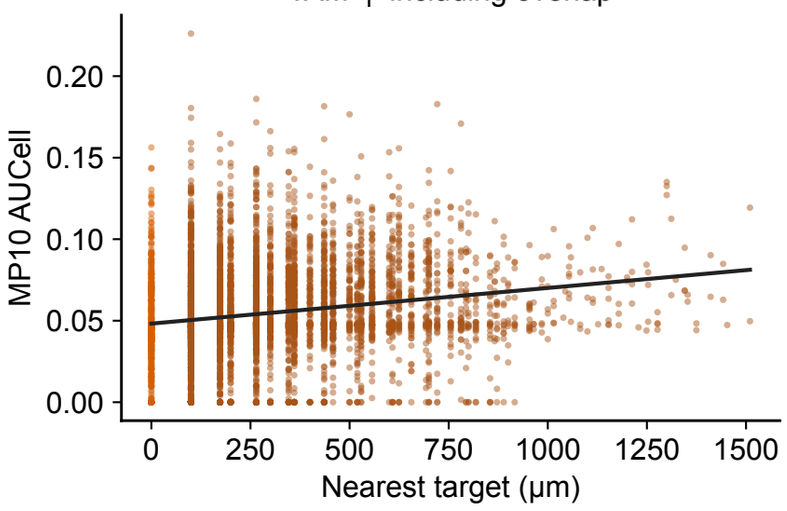
r +0.48 ρ +0.49
n 4896/4928 queries • 2013 targets



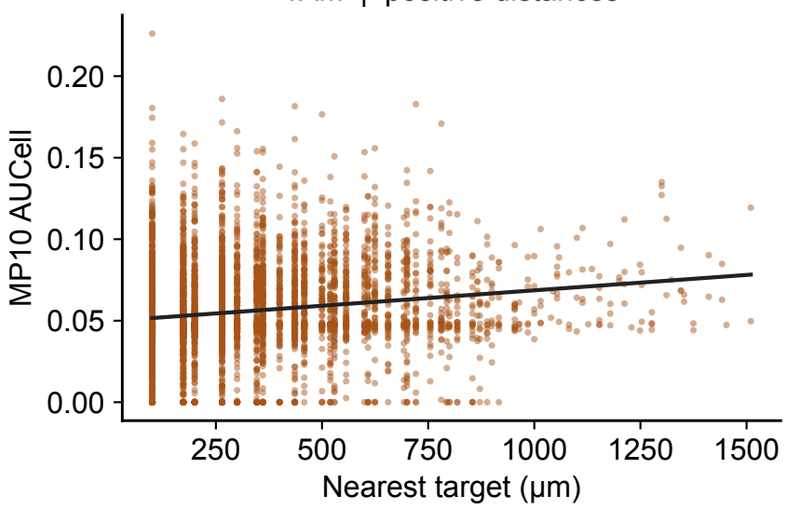
r +0.48 ρ +0.49
n 4878/4928 queries • 2013 targets



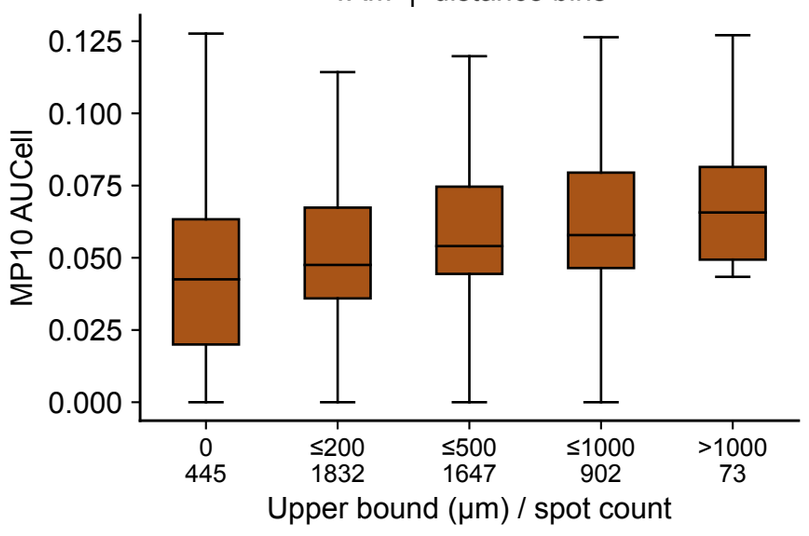
Orange = zero distance
Line = within-section linear fit



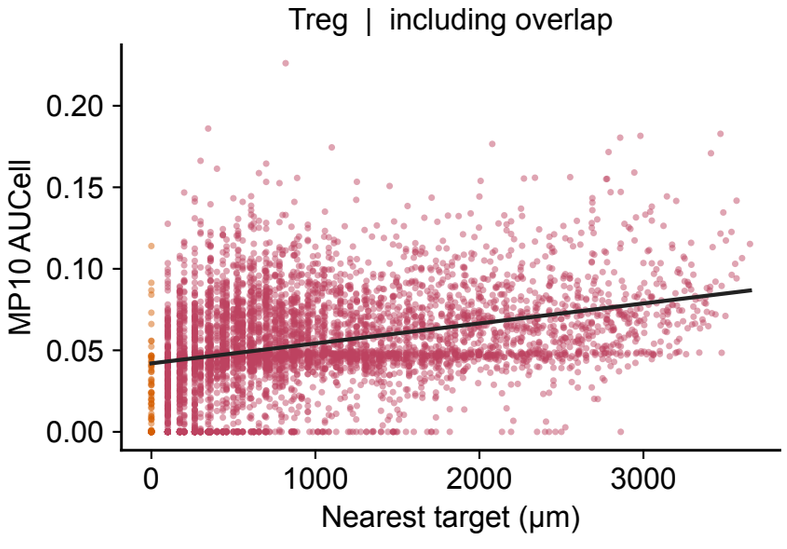
r +0.18 ρ +0.22
n 4899/4928 queries • 2013 targets



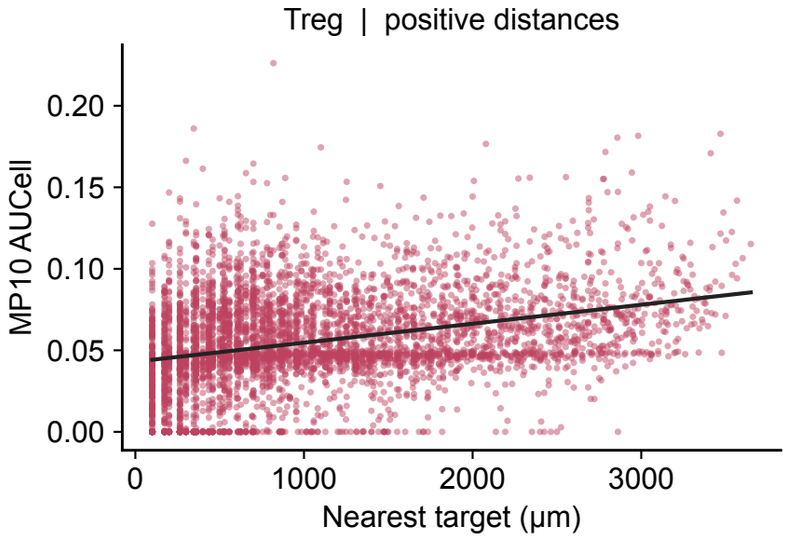
r +0.15 ρ +0.19
n 4454/4928 queries • 2013 targets



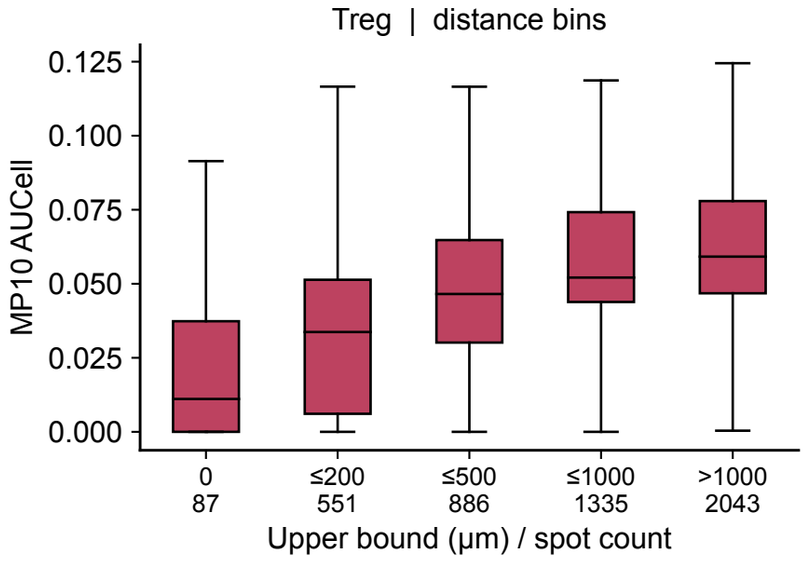
Orange = zero distance
Line = within-section linear fit



r +0.34 ρ +0.36
n 4902/4928 queries • 2013 targets



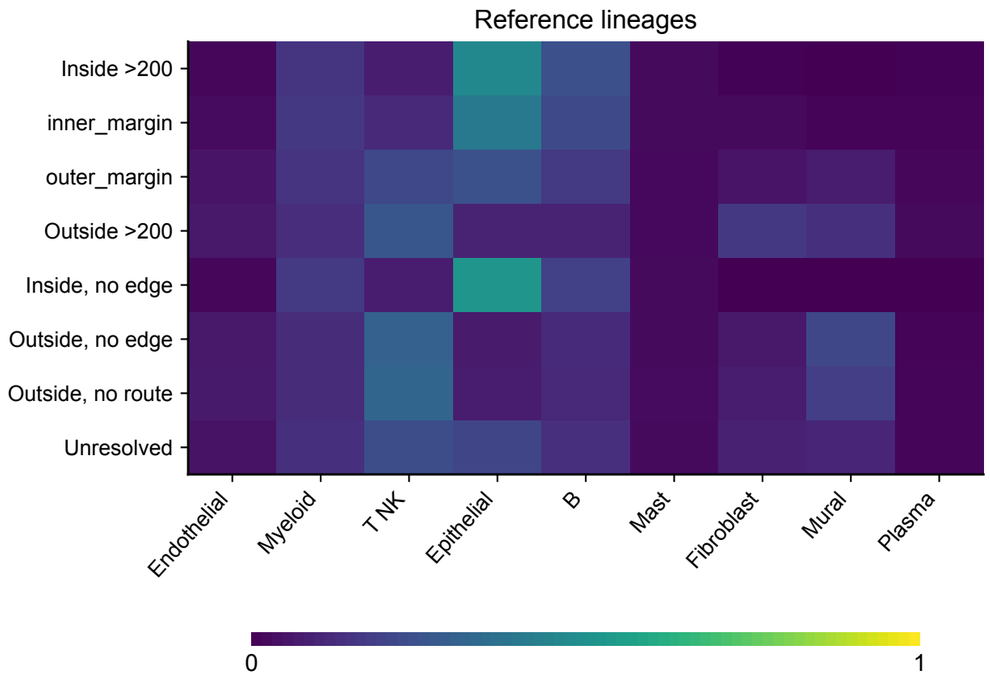
r +0.32 ρ +0.34
n 4815/4928 queries • 2013 targets

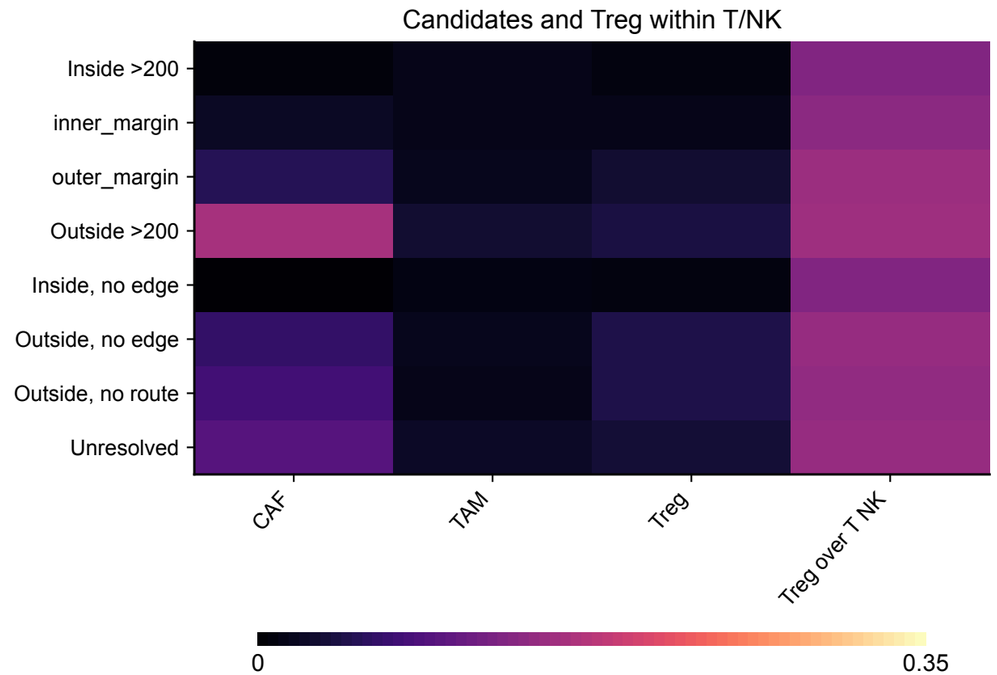


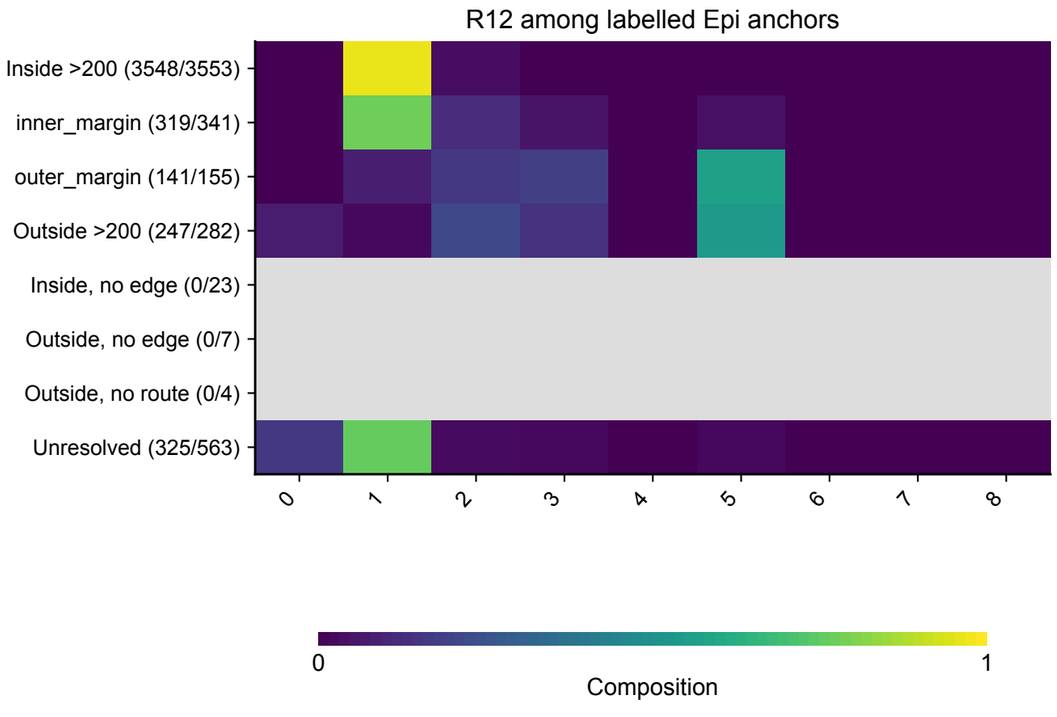
Orange = zero distance
Line = within-section linear fit

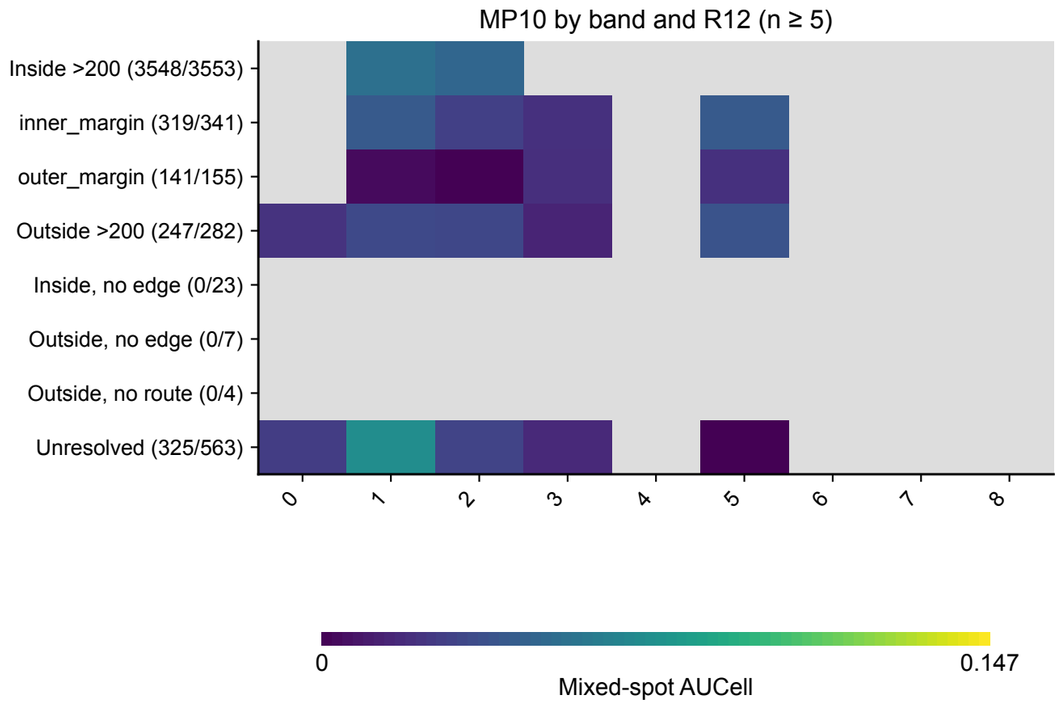
CanDis_2026_Sample2 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





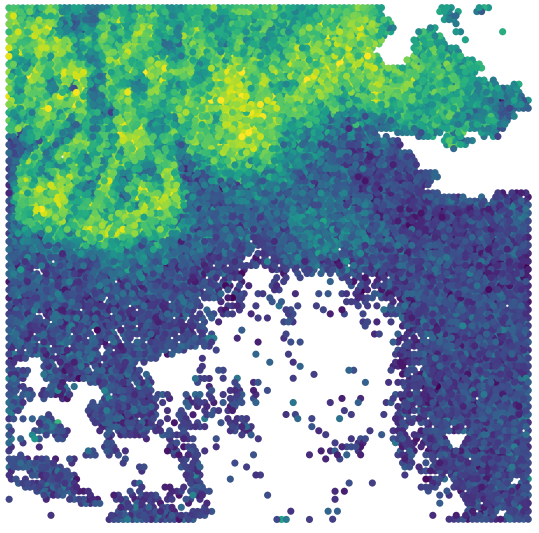




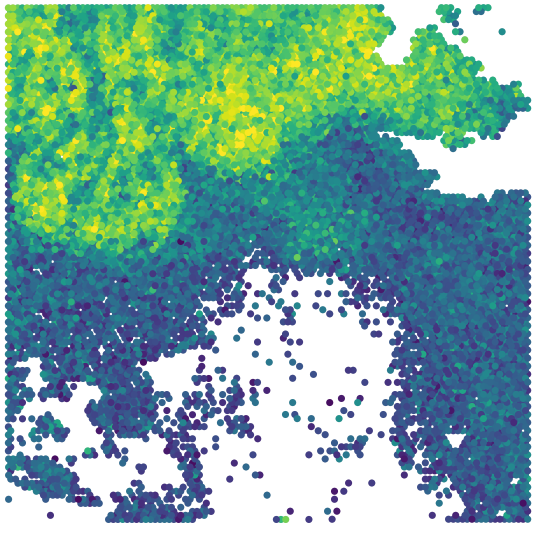
CanDis_2026_Sample2 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

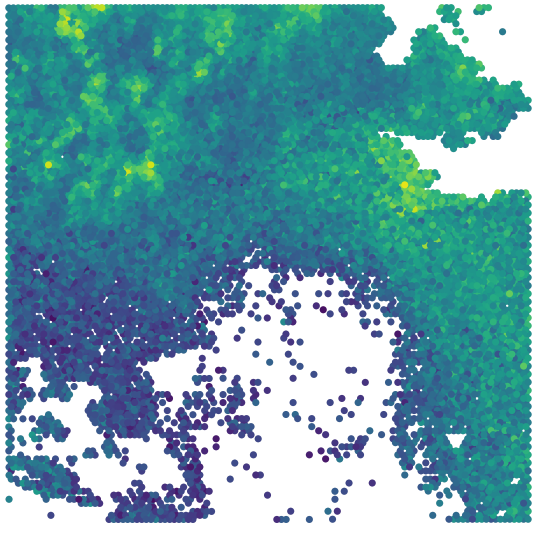
E2F Targets



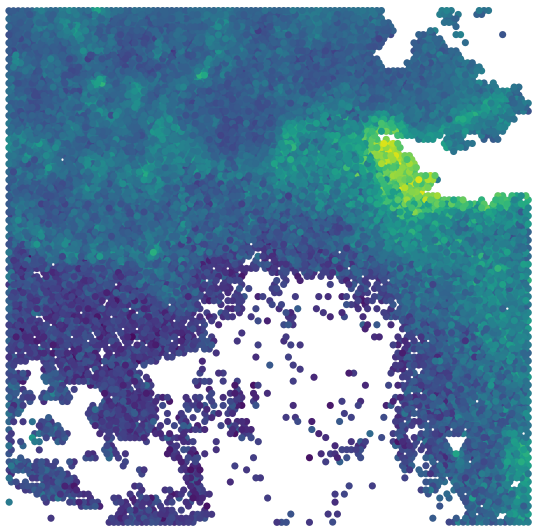
G2M Checkpoint



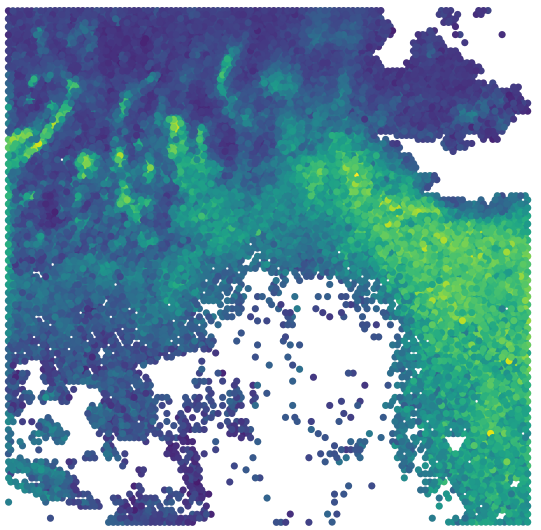
Hypoxia



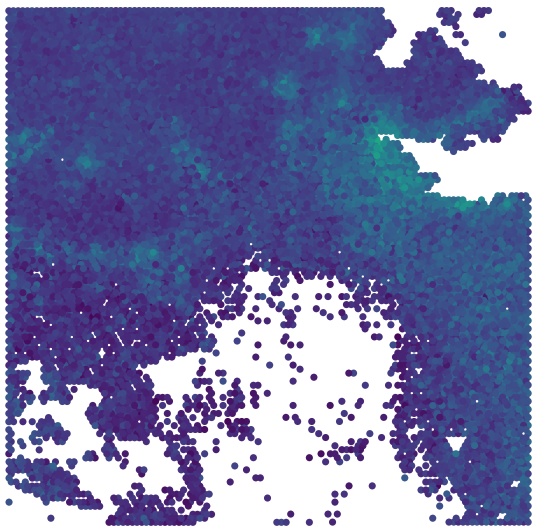
Tnfa Signaling Via Nfkb



Emt



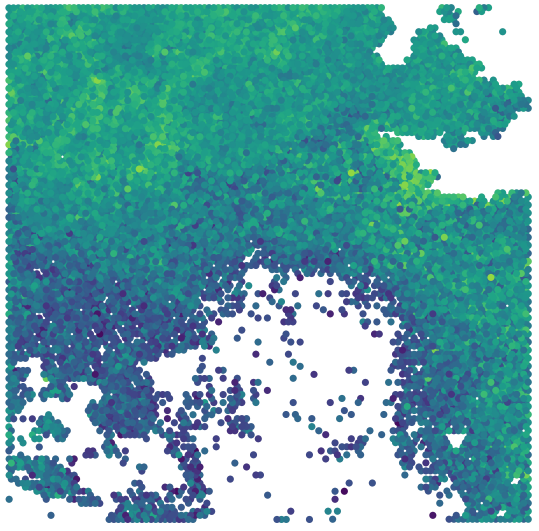
Interferon Gamma Response



CanDis_2026_Sample2 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

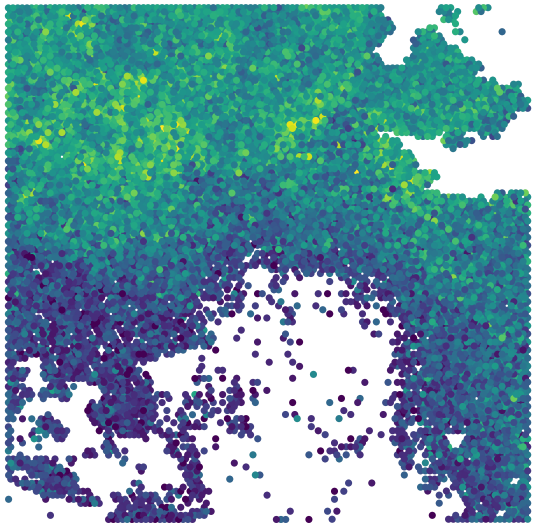
Il2 Stat5 Signaling



0 0.144

UCell score

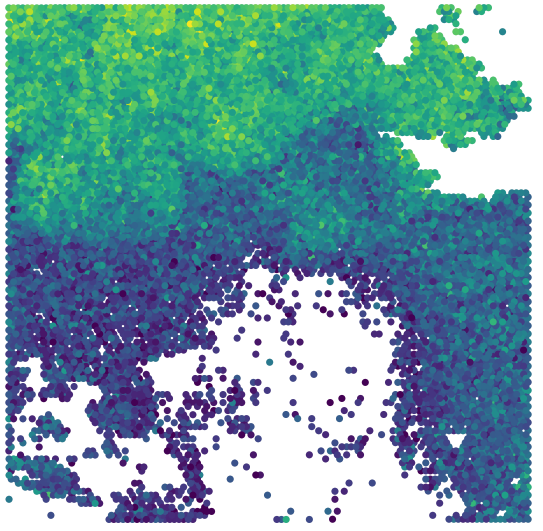
Tgf Beta Signaling



0 0.232

UCell score

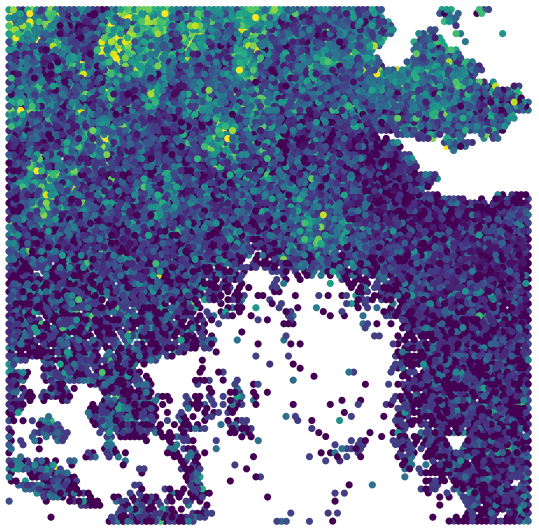
Cholesterol Homeostasis



0 0.21

UCell score

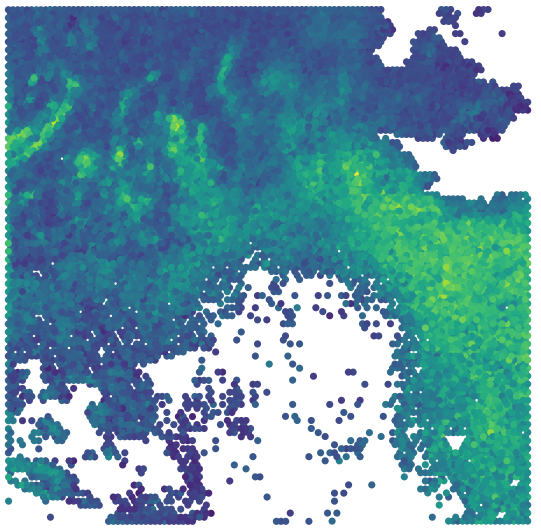
Cholesterol Biosynthesis



0 0.166

UCell score

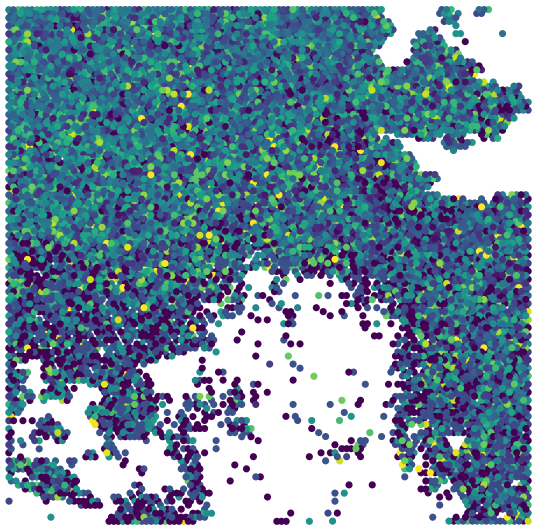
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



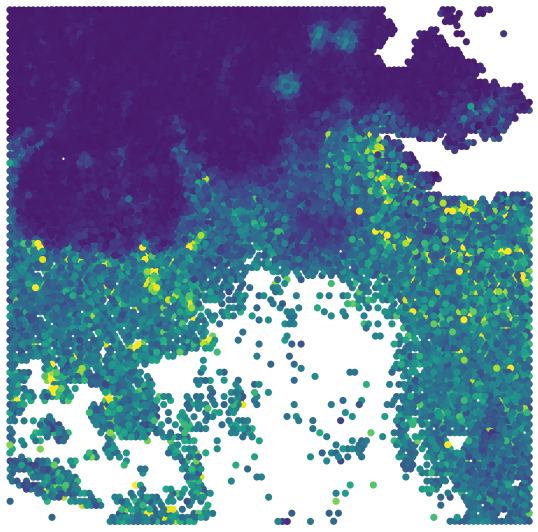
0 0.151

UCell score

CanDis_2026_Sample2 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

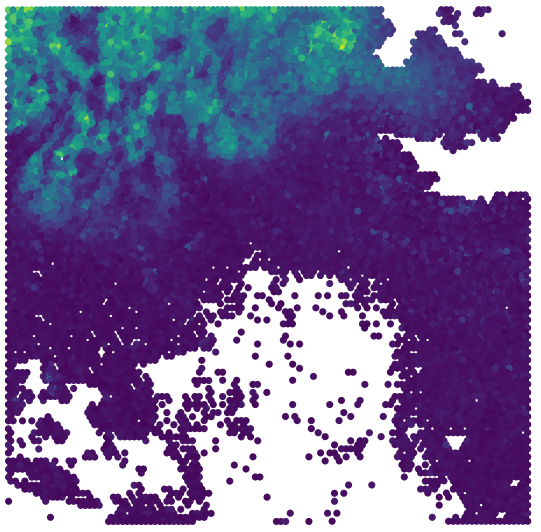
Treg composition



0 0.102

Fraction of all 30

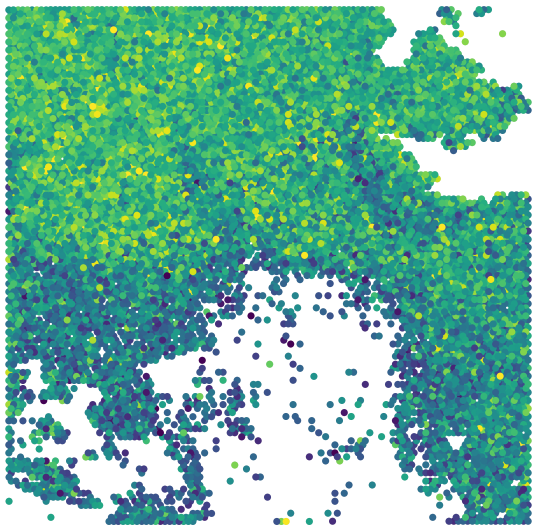
Treg estimated abundance



0 2.85

Posterior mean estimate

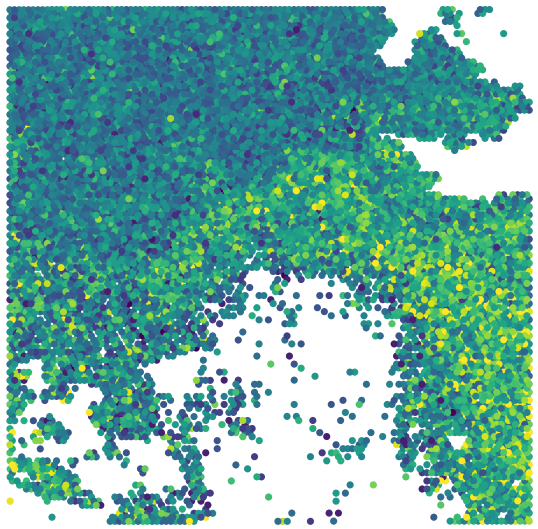
Treg Resting Vs Activated Dn



0 0.0778

UCell score

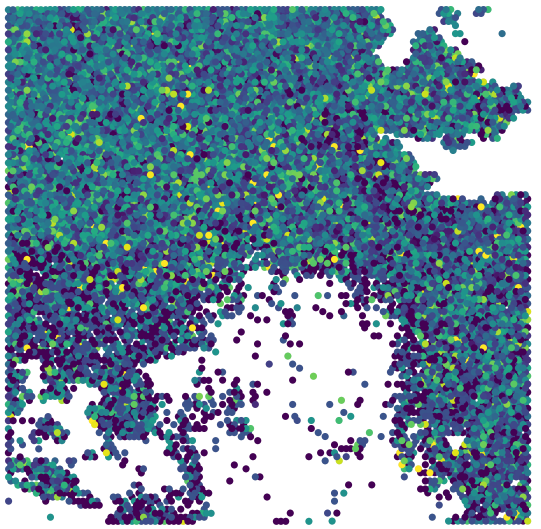
Treg Resting Vs Activated Up



0 0.0609

UCell score

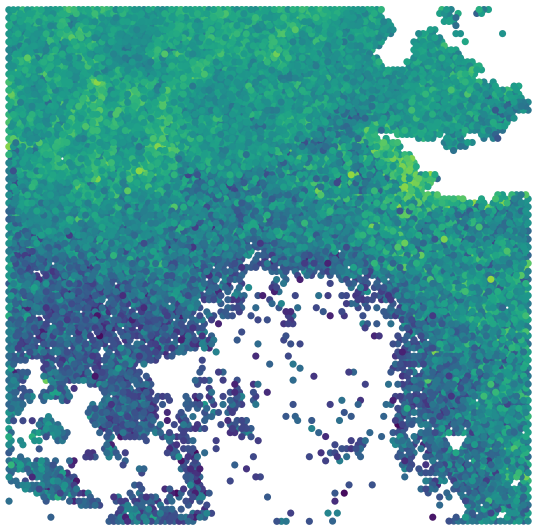
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

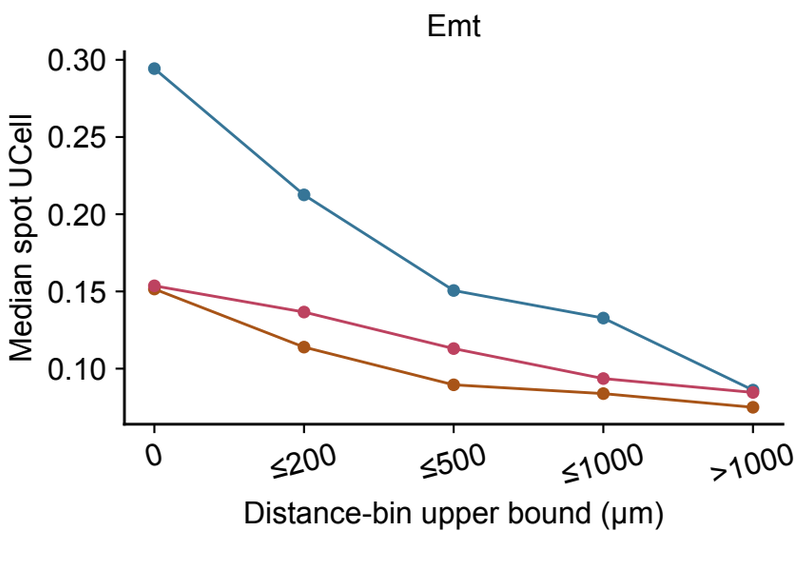


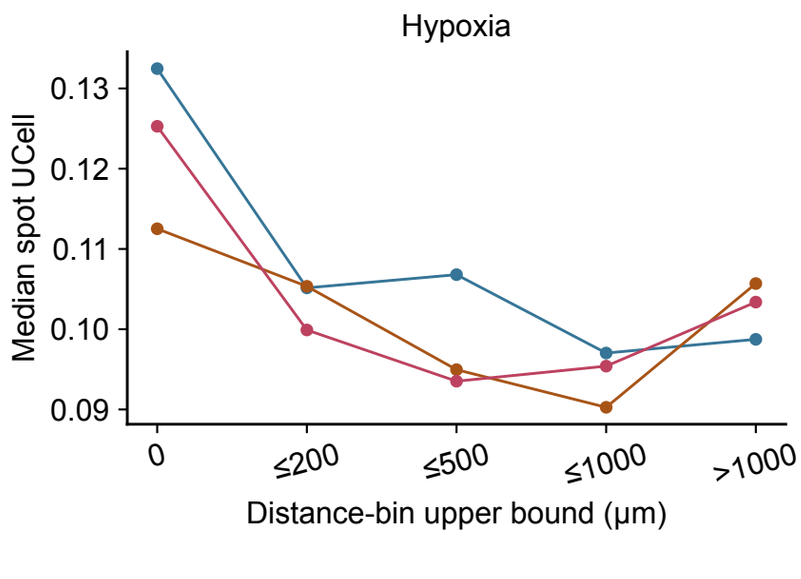
0 0.144

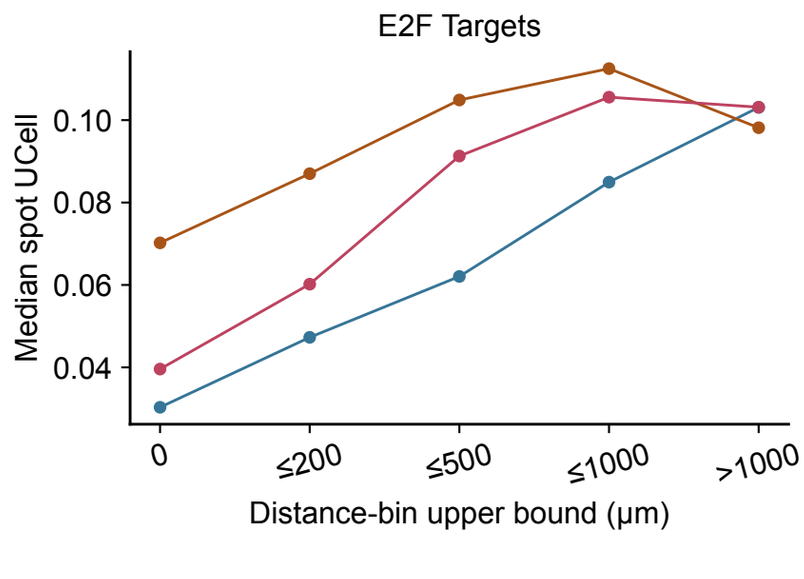
UCell score

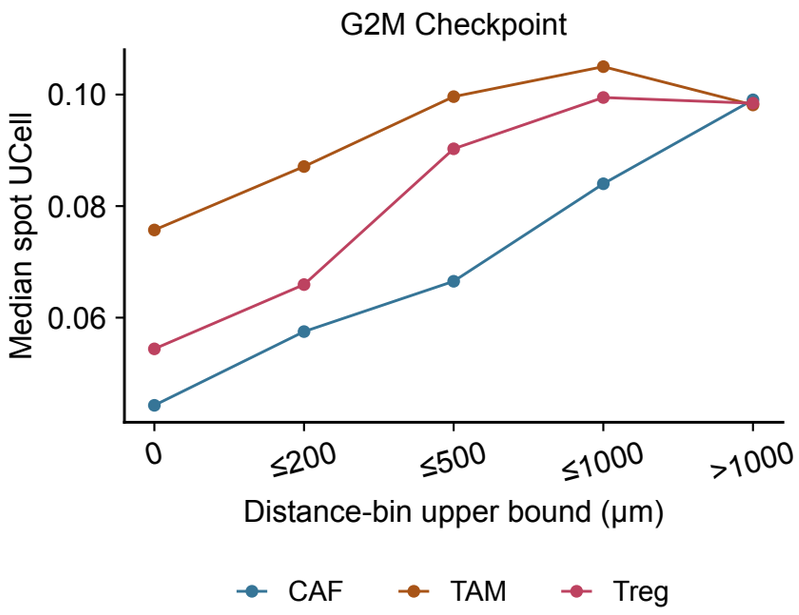
CanDis_2026_Sample2 | Programs by target distance

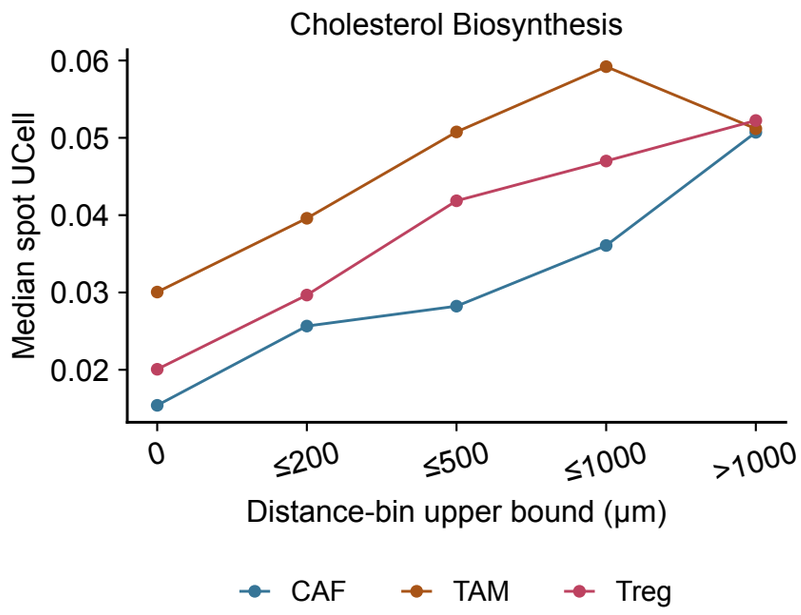
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

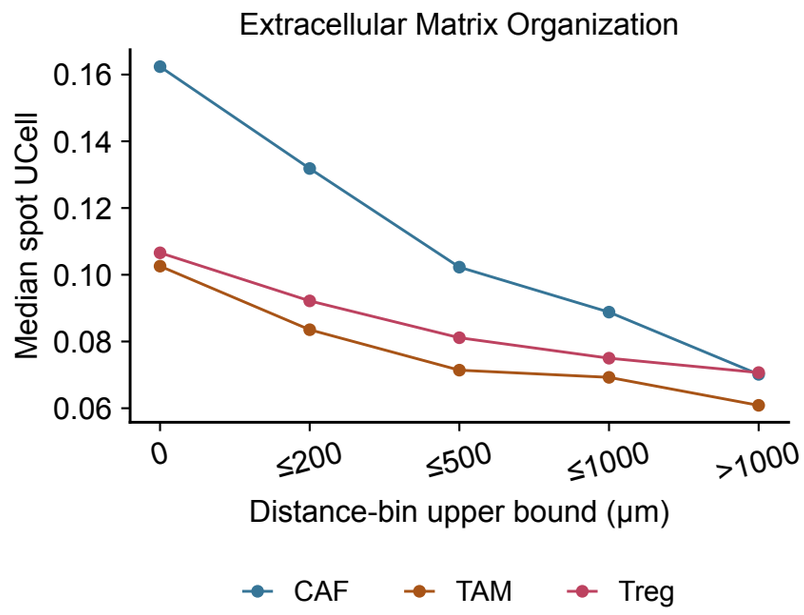






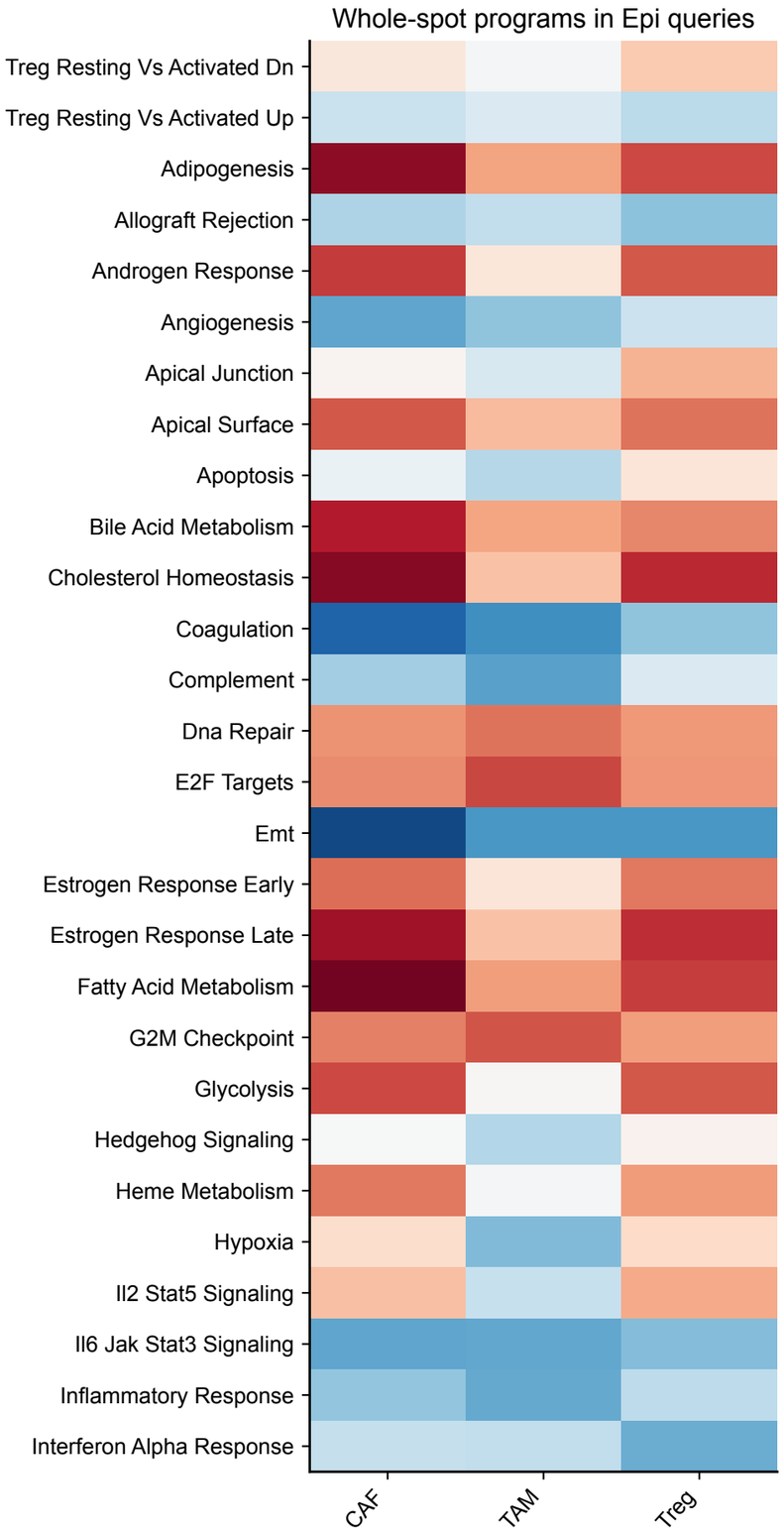


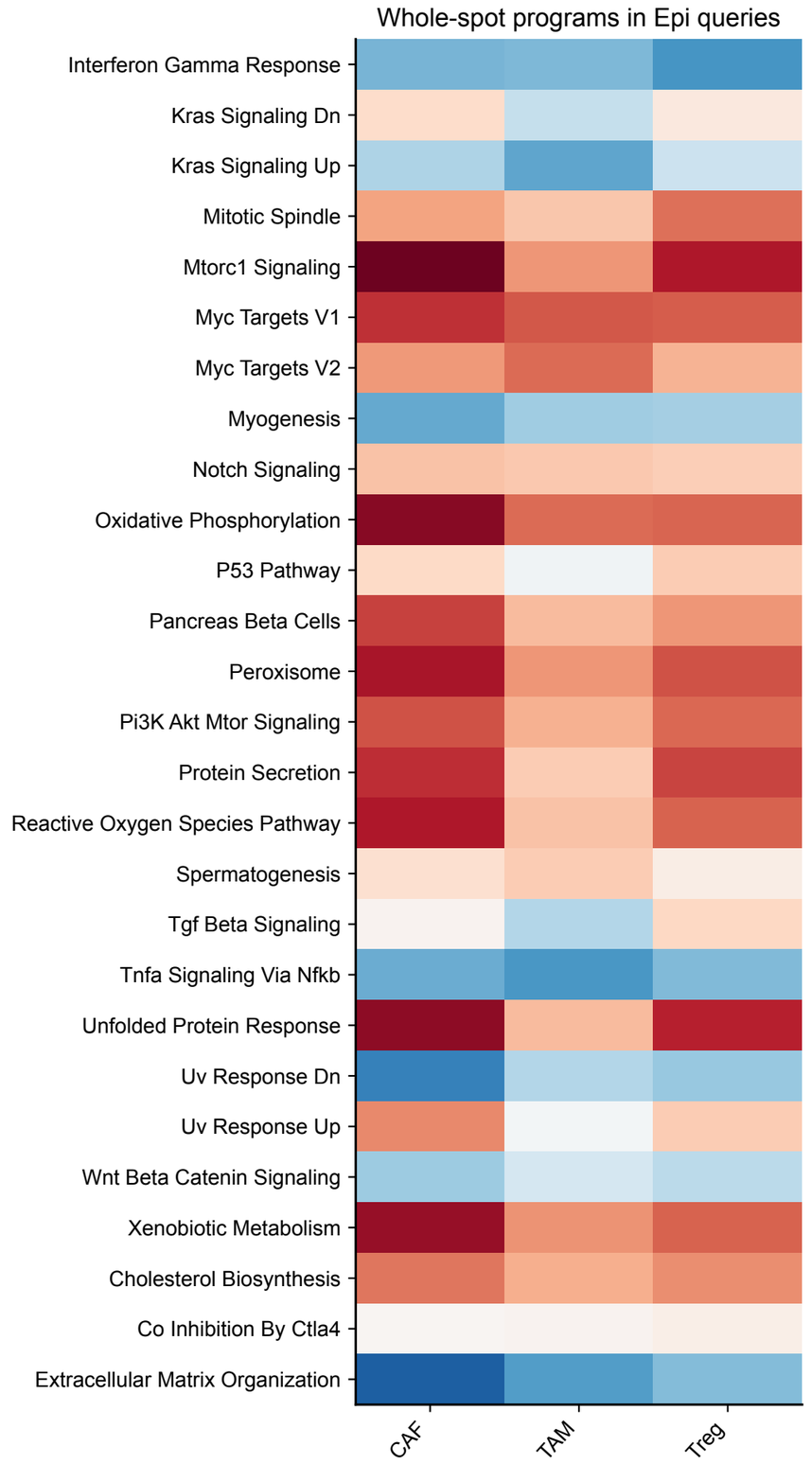




CanDis_2026_Sample2 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

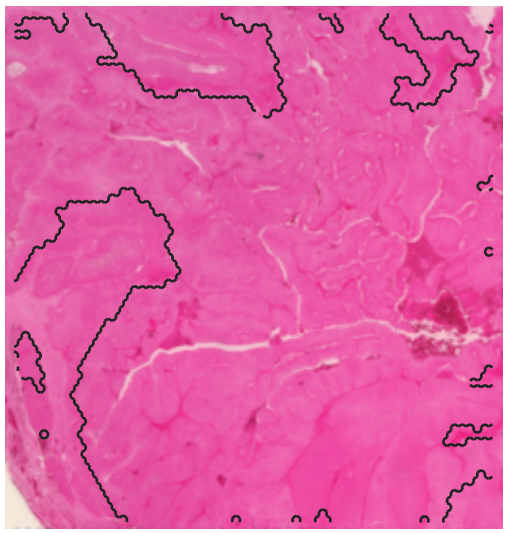




CanDis_2026_Sample3 | Tissue context

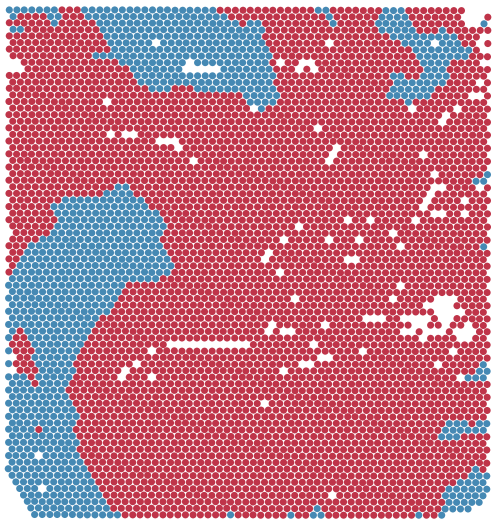
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



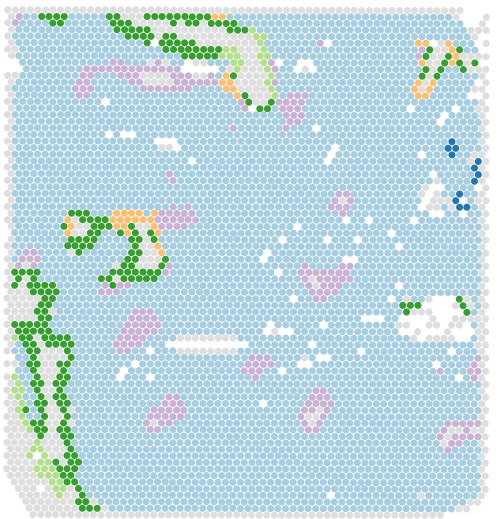
Black line: observed candidate outer edge

Final candidate location



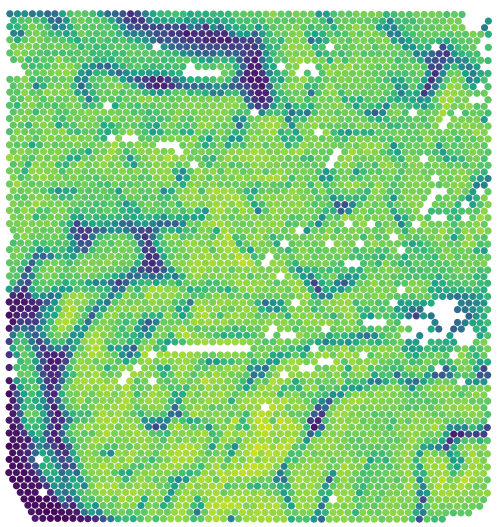
inside outside

Accepted R12 (original 0–8)



0 2 3 4 8 1

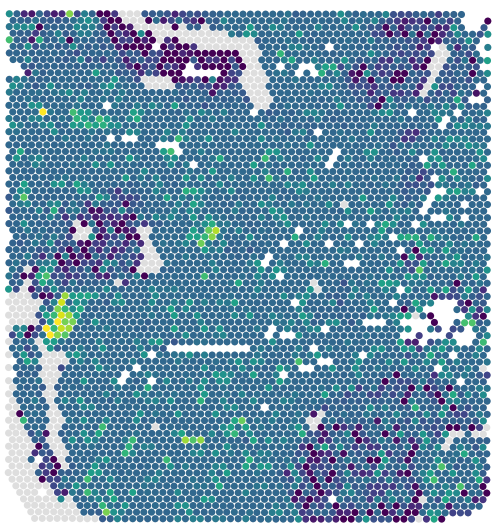
Epithelial support



0 1

Fraction of all 30

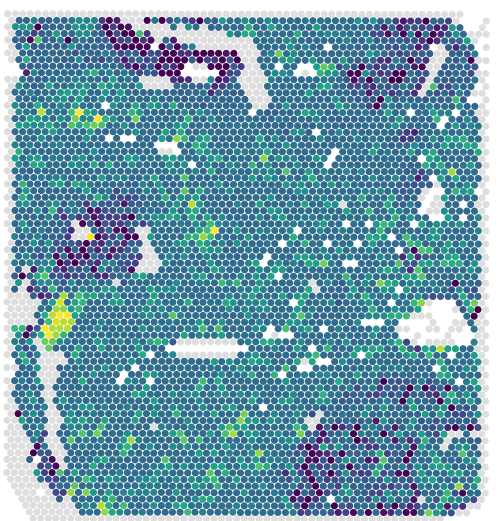
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



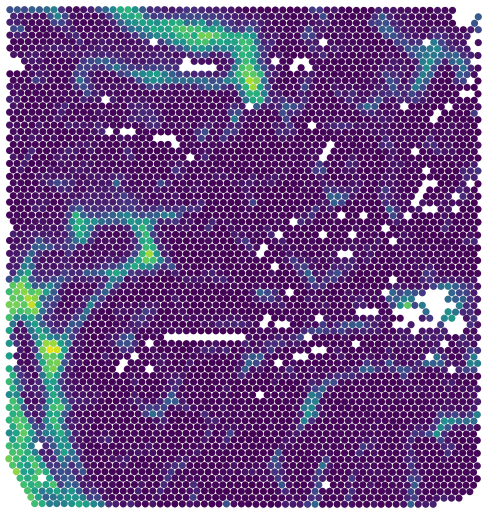
0 0.14

Matched-model AUCell

CanDis_2026_Sample3 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



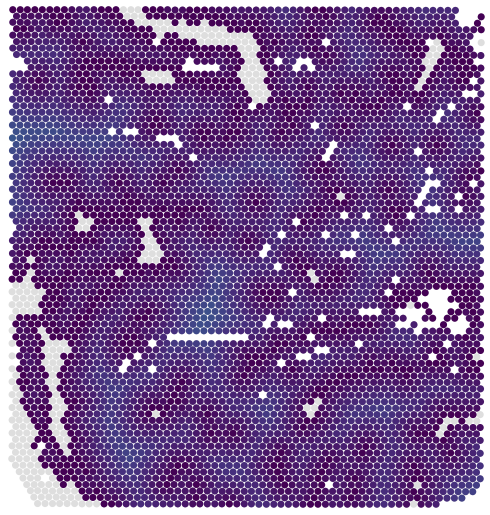
0 0.418

Fraction of all 30

CAF P80 targets

False True

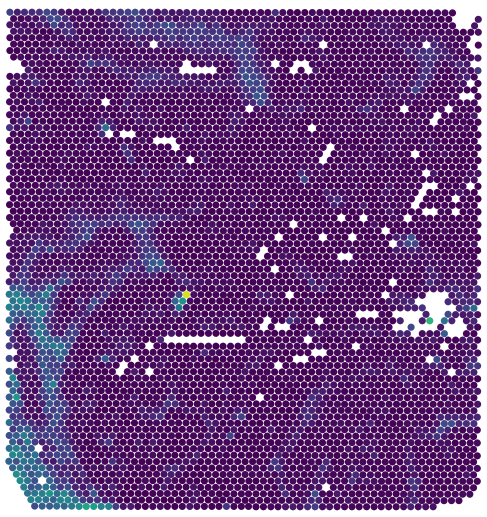
CAF distance



0 3.1e+03

Nearest spot center (µm)

TAM composition



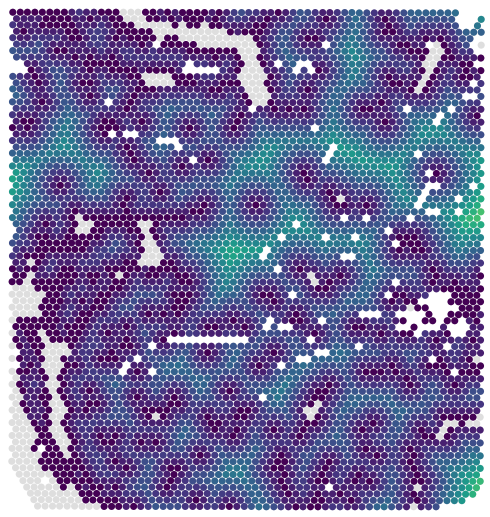
0 0.291

Fraction of all 30

TAM P80 targets

False True

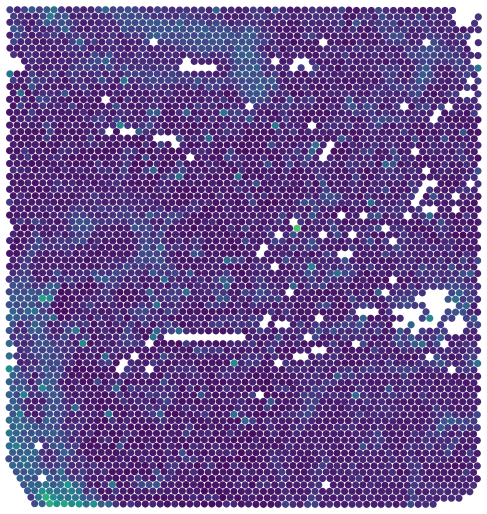
TAM distance



0 1.13e+03

Nearest spot center (µm)

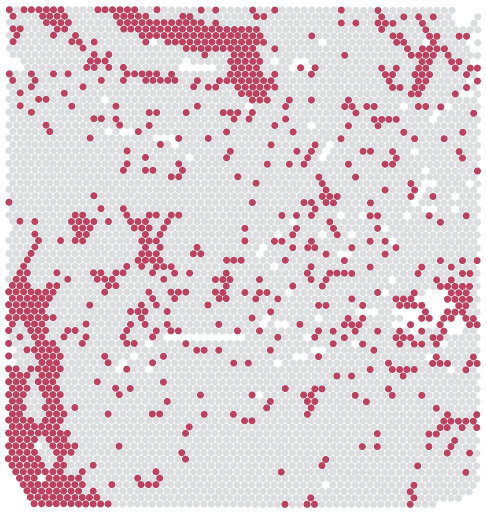
Treg composition



0 0.102

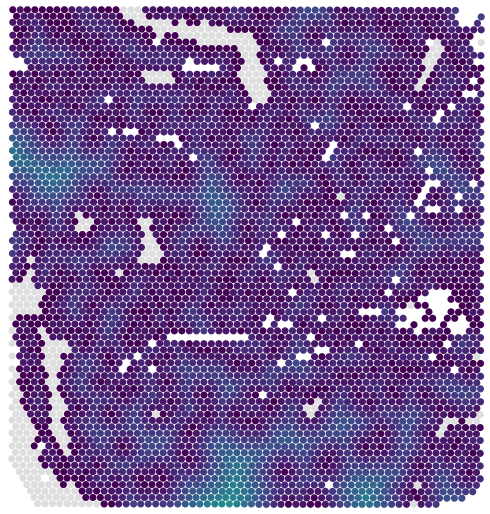
Fraction of all 30

Treg P80 targets



False True

Treg distance

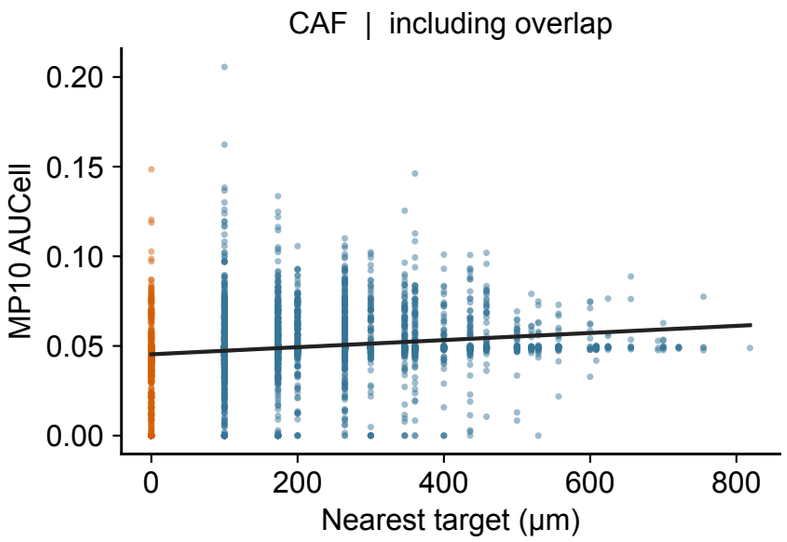


0 1.61e+03

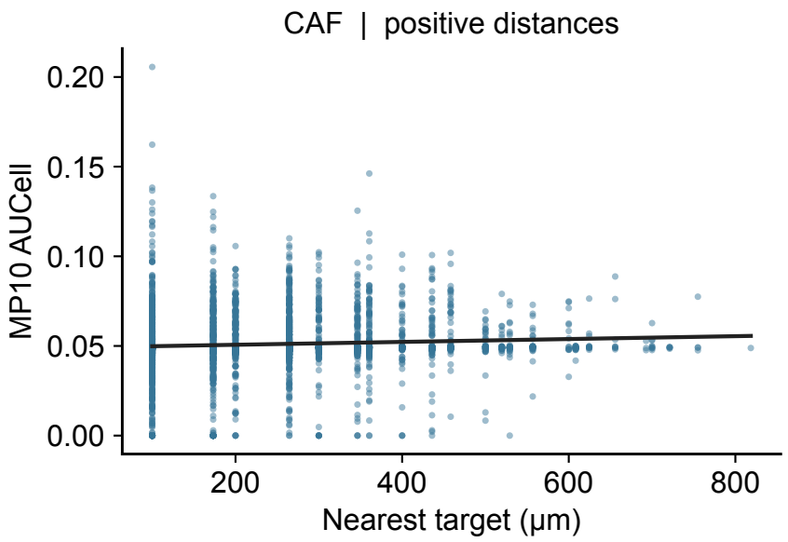
Nearest spot center (µm)

CanDis_2026_Sample3 | MP10 versus nearest target

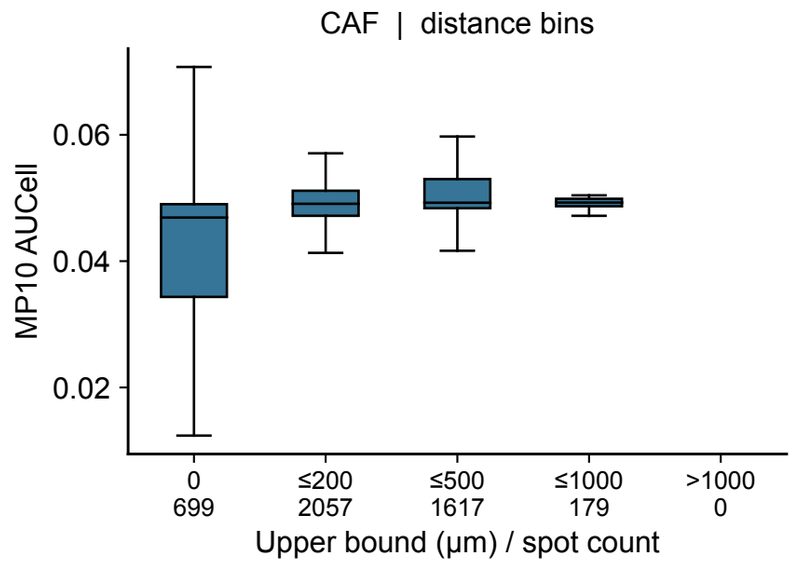
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



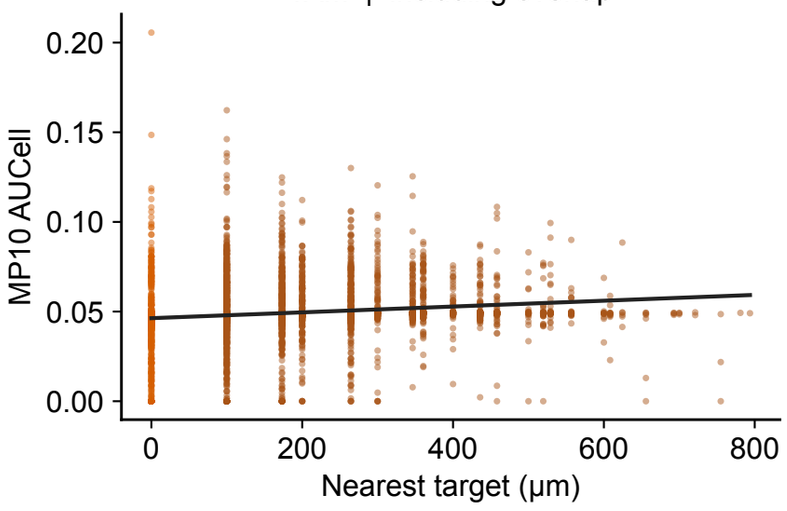
r +0.17 ρ +0.24
n 4552/4553 queries • 964 targets



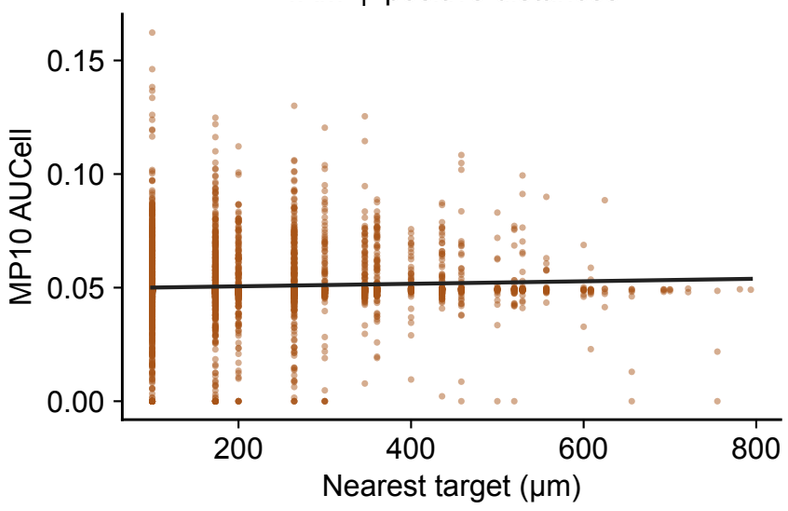
r +0.07 ρ +0.11
n 3853/4553 queries • 964 targets



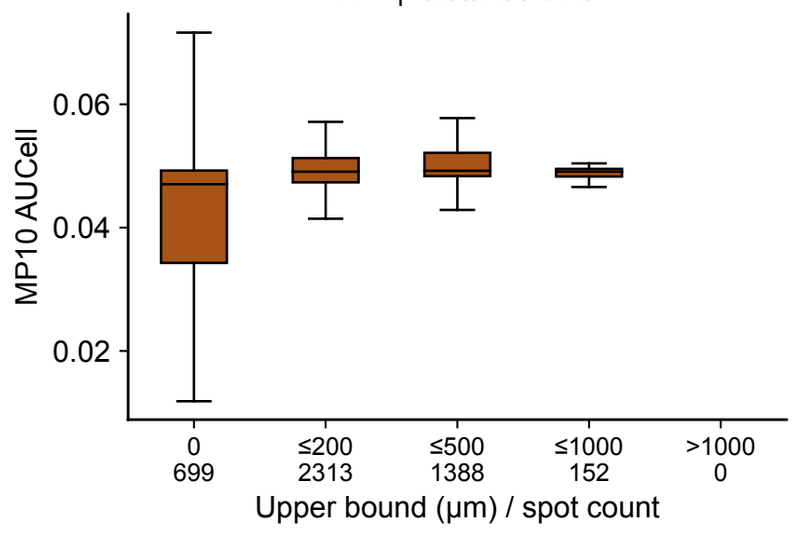
Orange = zero distance
Line = within-section linear fit



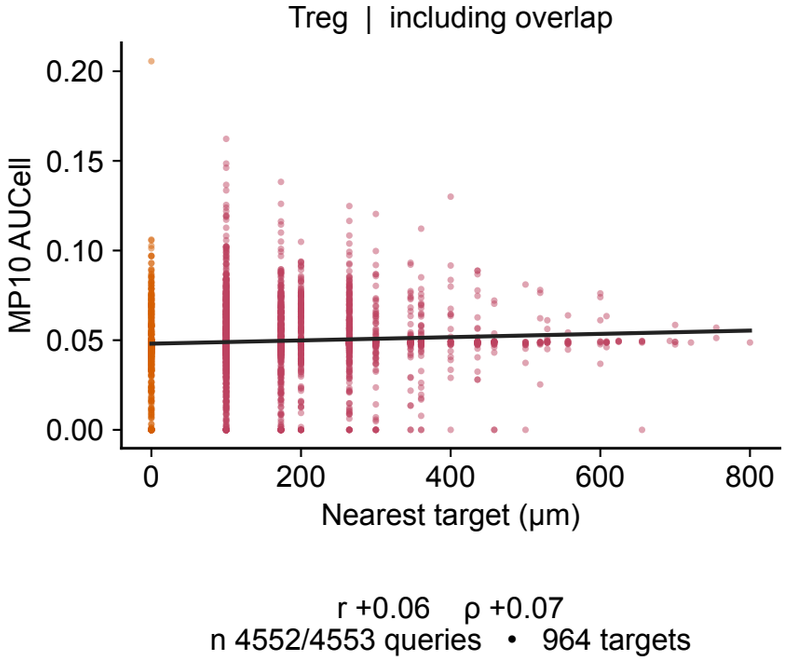
r +0.13 ρ +0.19
n 4552/4553 queries • 964 targets



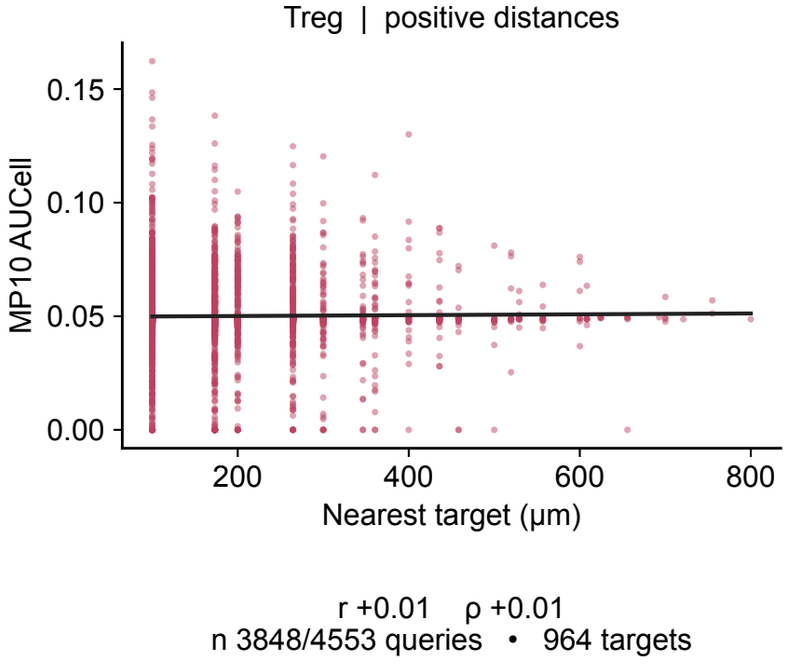
r +0.05 ρ +0.07
n 3853/4553 queries • 964 targets



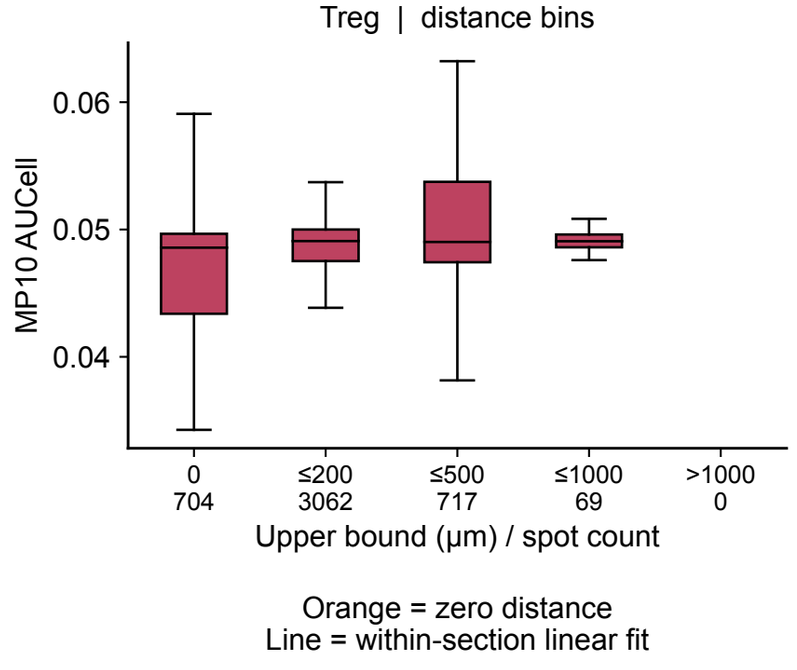
Orange = zero distance
Line = within-section linear fit



r +0.06 ρ +0.07
n 4552/4553 queries • 964 targets



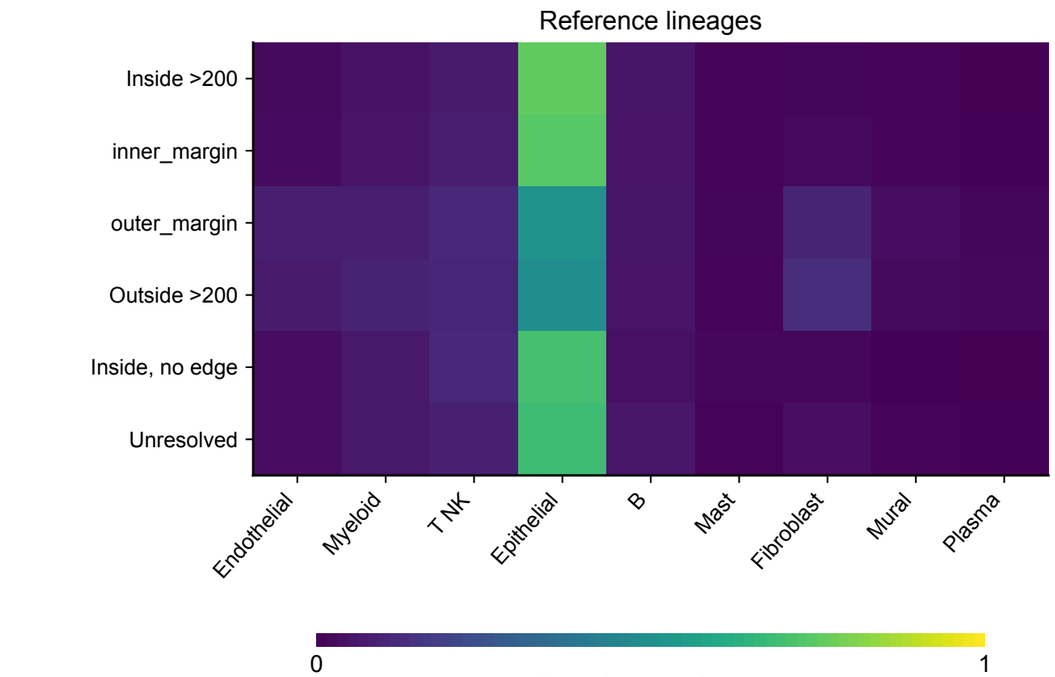
r +0.01 ρ +0.01
n 3848/4553 queries • 964 targets

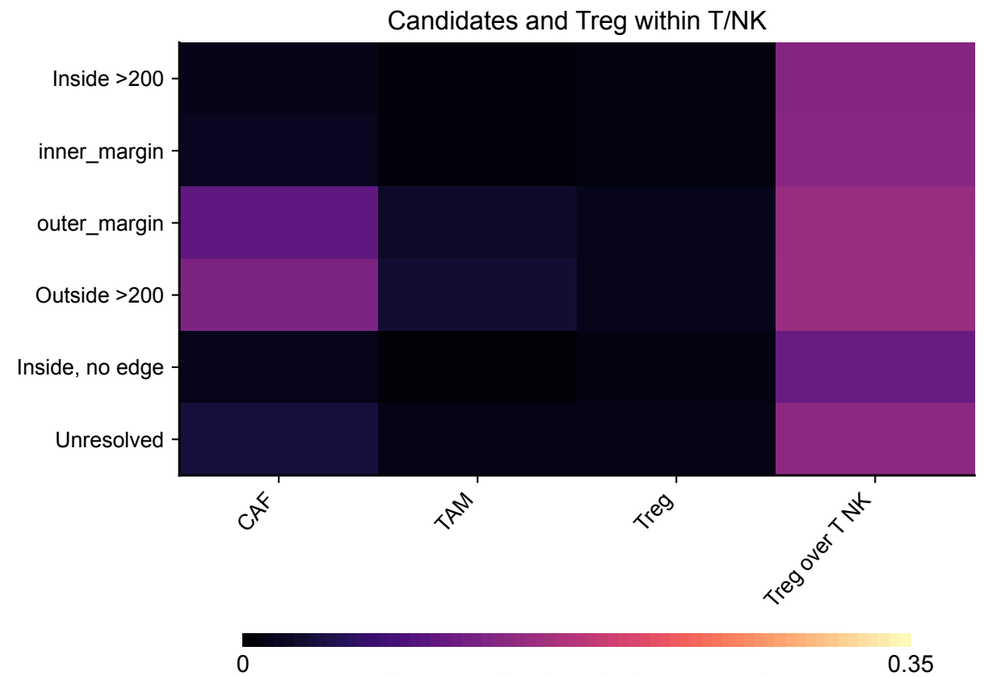


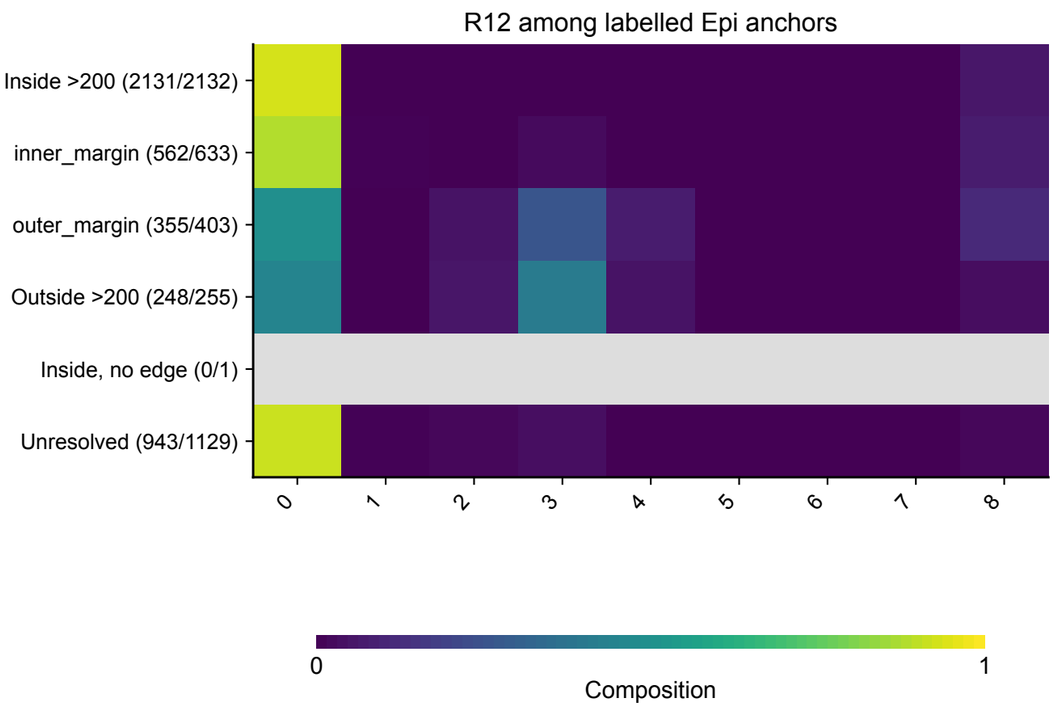
Orange = zero distance
Line = within-section linear fit

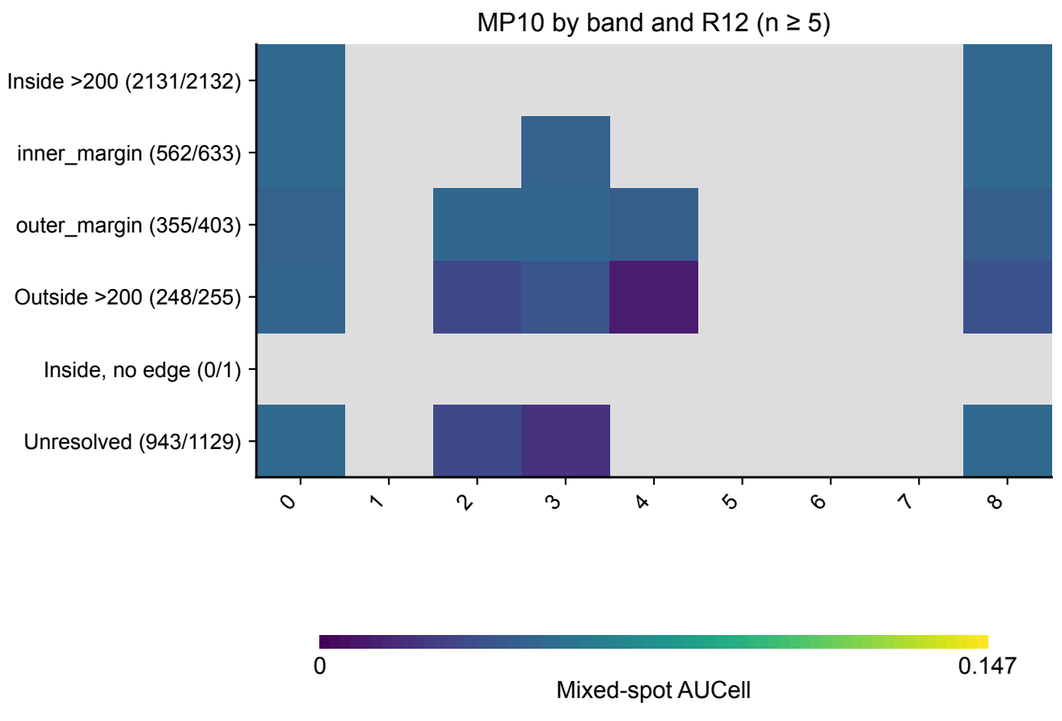
CanDis_2026_Sample3 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





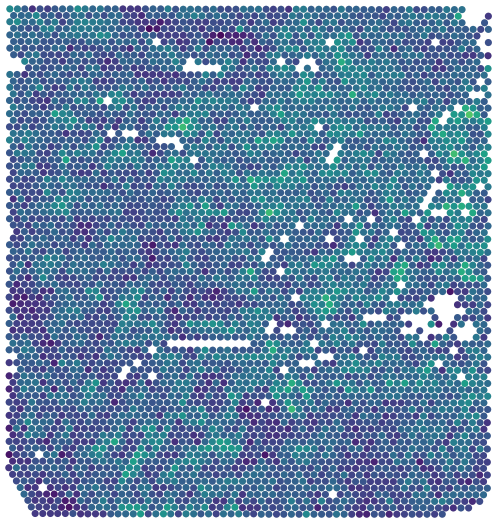




CanDis_2026_Sample3 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

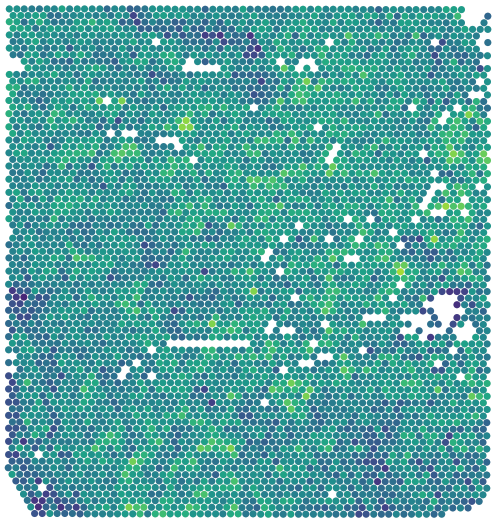
E2F Targets



0 0.152

UCell score

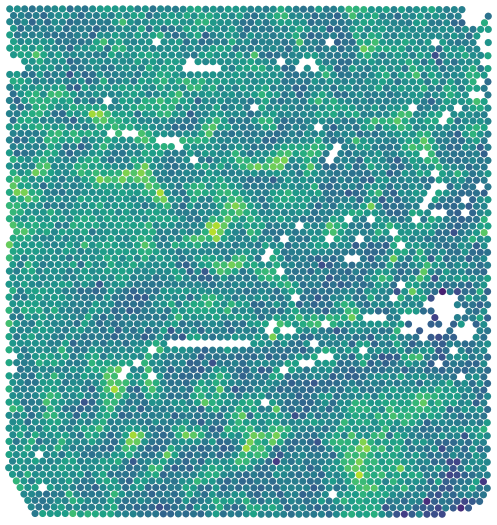
G2M Checkpoint



0 0.134

UCell score

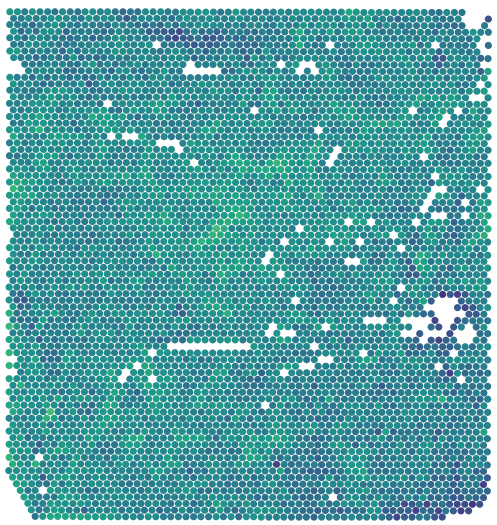
Hypoxia



0 0.211

UCell score

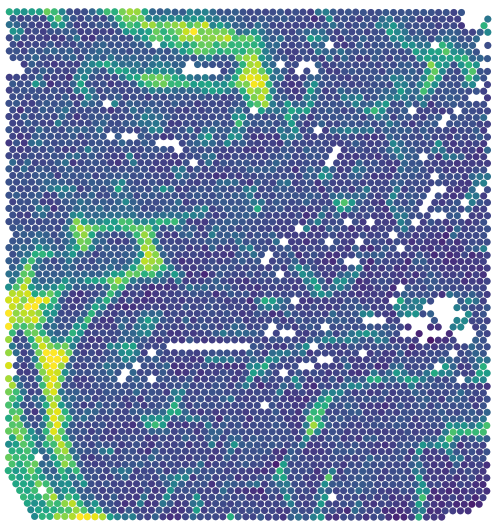
Tnfa Signaling Via Nfkb



0 0.275

UCell score

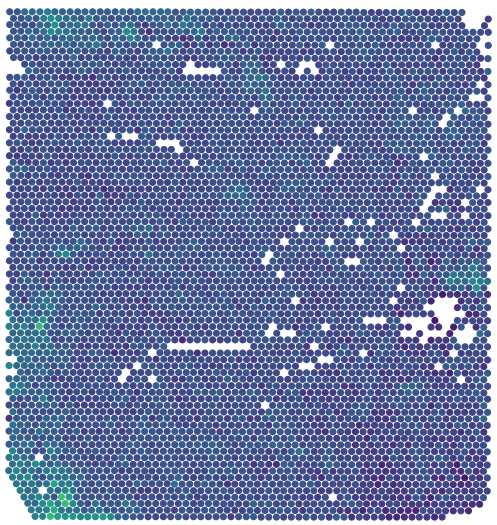
Emt



0 0.386

UCell score

Interferon Gamma Response



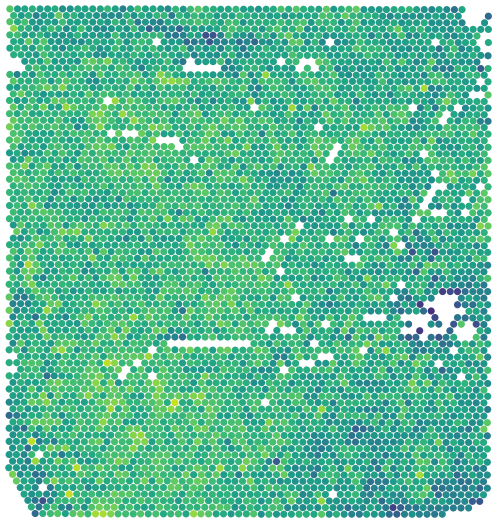
0 0.275

UCell score

CanDis_2026_Sample3 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

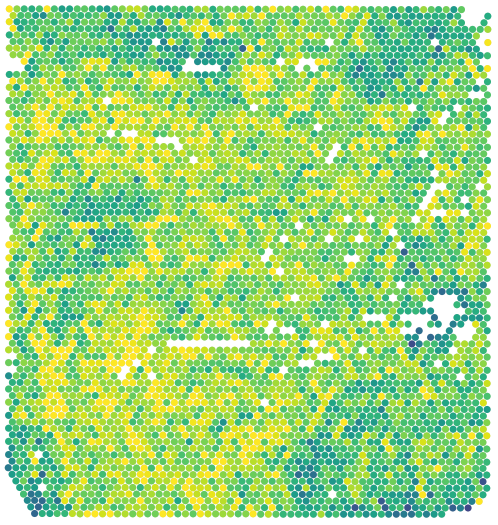
Il2 Stat5 Signaling



0 0.144

UCell score

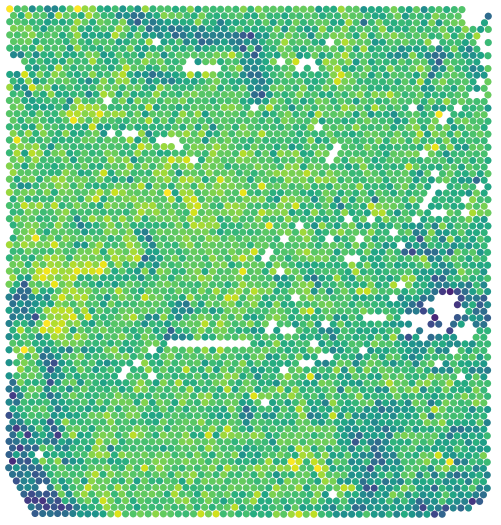
Tgf Beta Signaling



0 0.232

UCell score

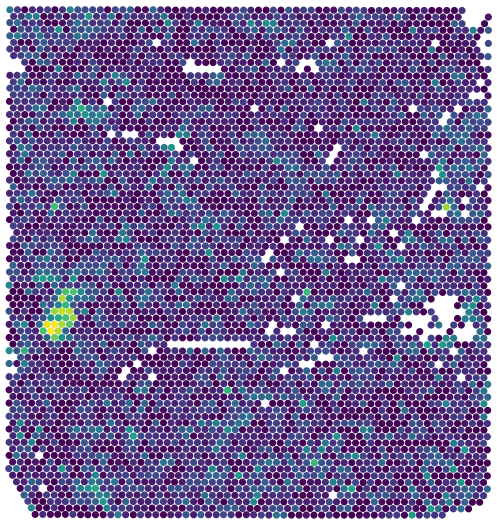
Cholesterol Homeostasis



0 0.21

UCell score

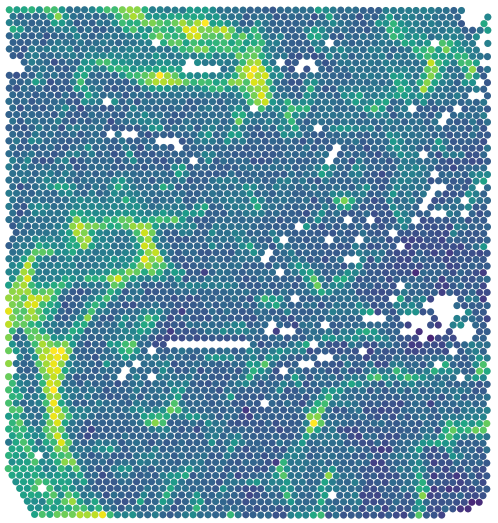
Cholesterol Biosynthesis



0 0.166

UCell score

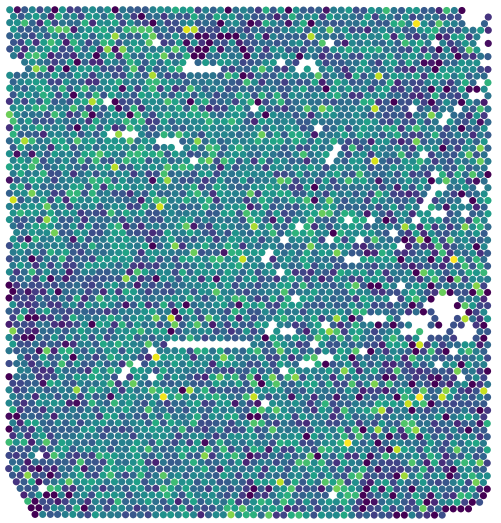
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



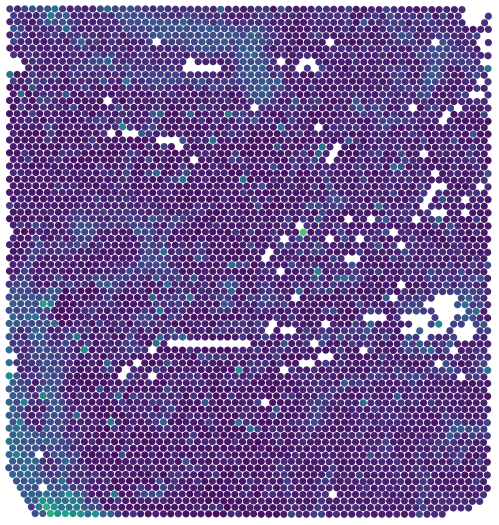
0 0.151

UCell score

CanDis_2026_Sample3 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

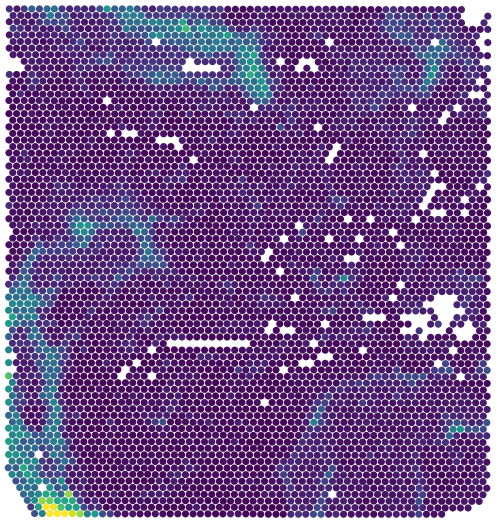
Treg composition



0 0.102

Fraction of all 30

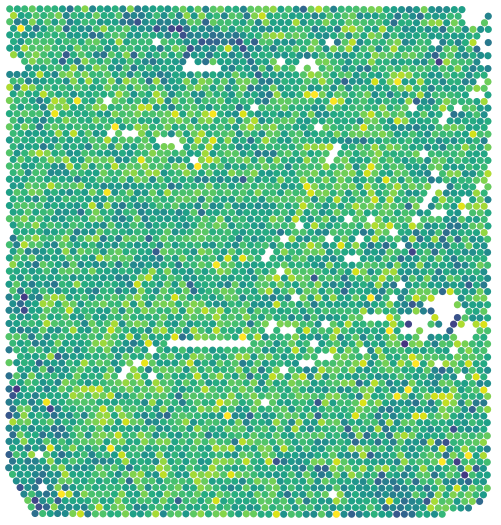
Treg estimated abundance



0 2.85

Posterior mean estimate

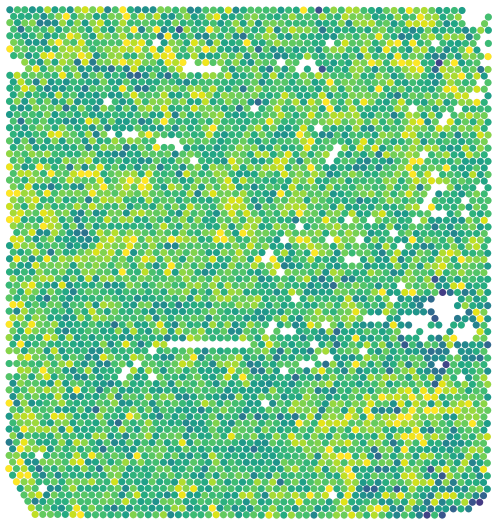
Treg Resting Vs Activated Dn



0 0.0778

UCell score

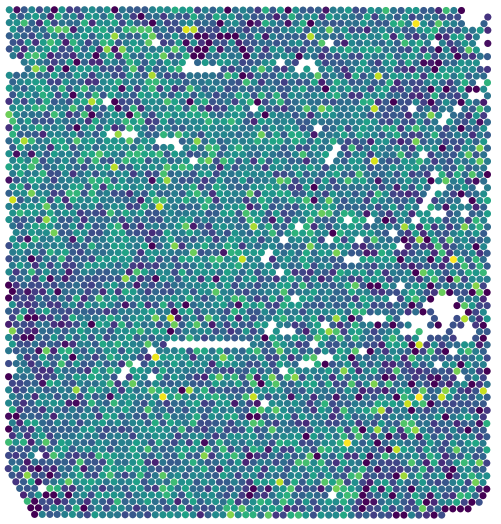
Treg Resting Vs Activated Up



0 0.0609

UCell score

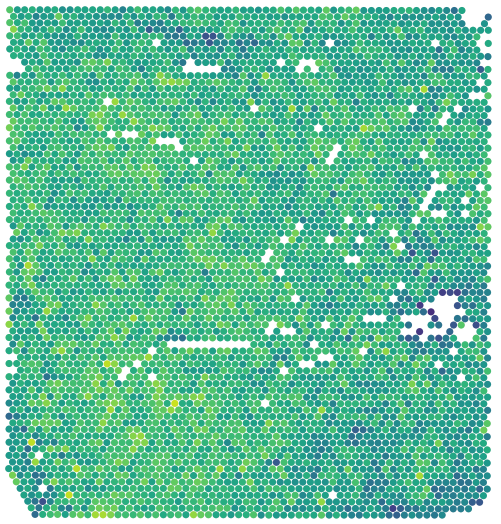
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

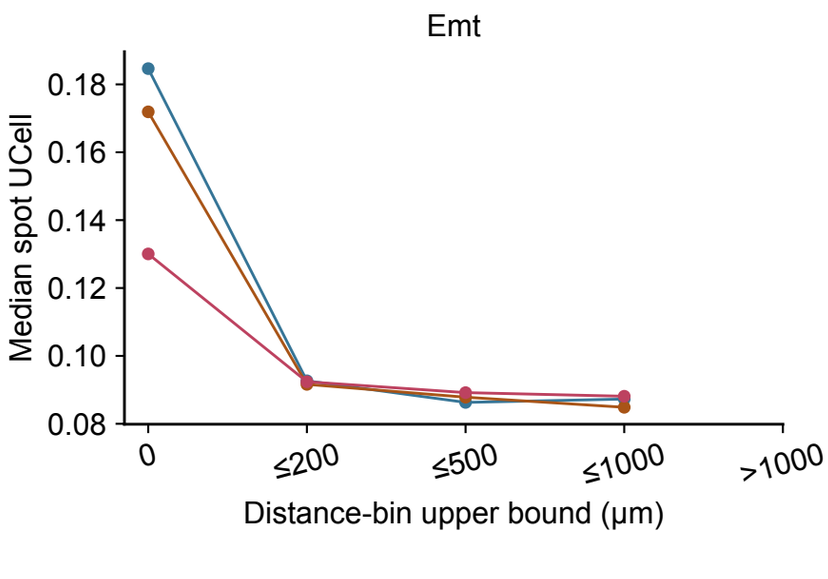


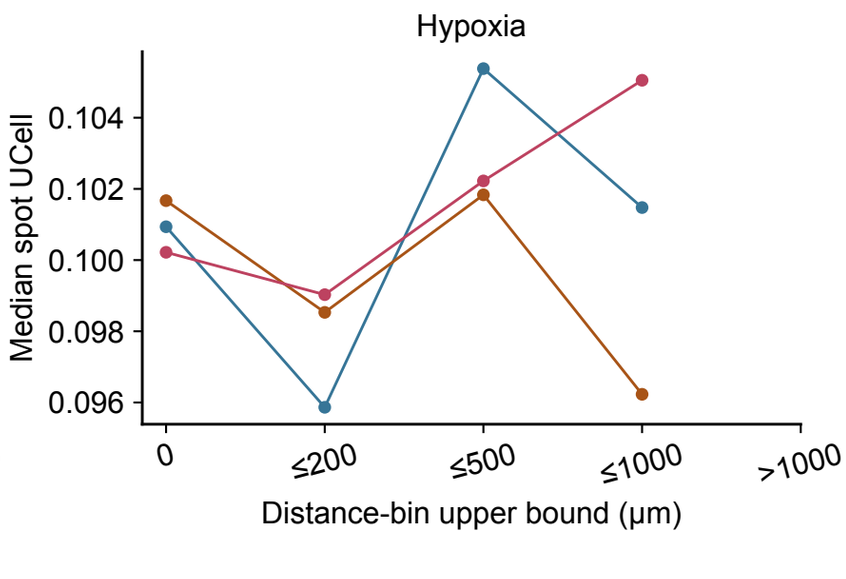
0 0.144

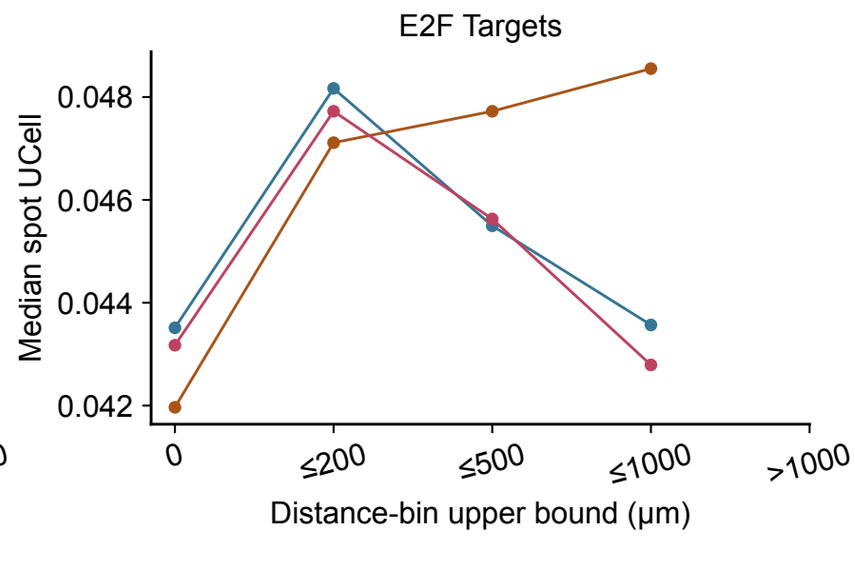
UCell score

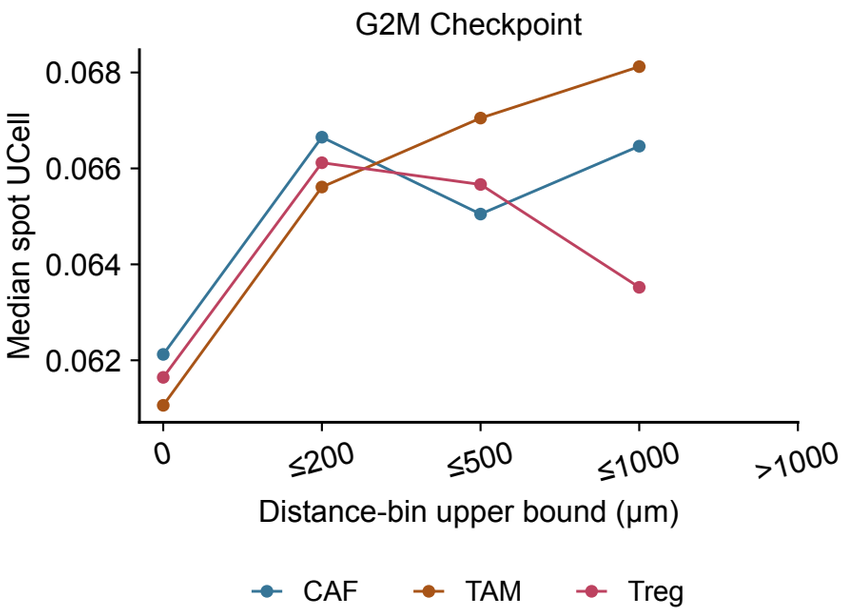
CanDis_2026_Sample3 | Programs by target distance

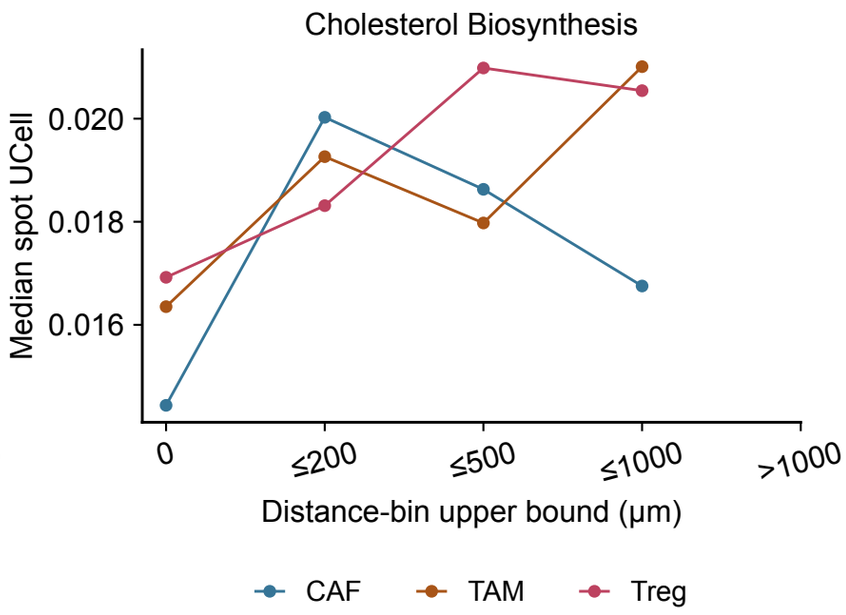
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

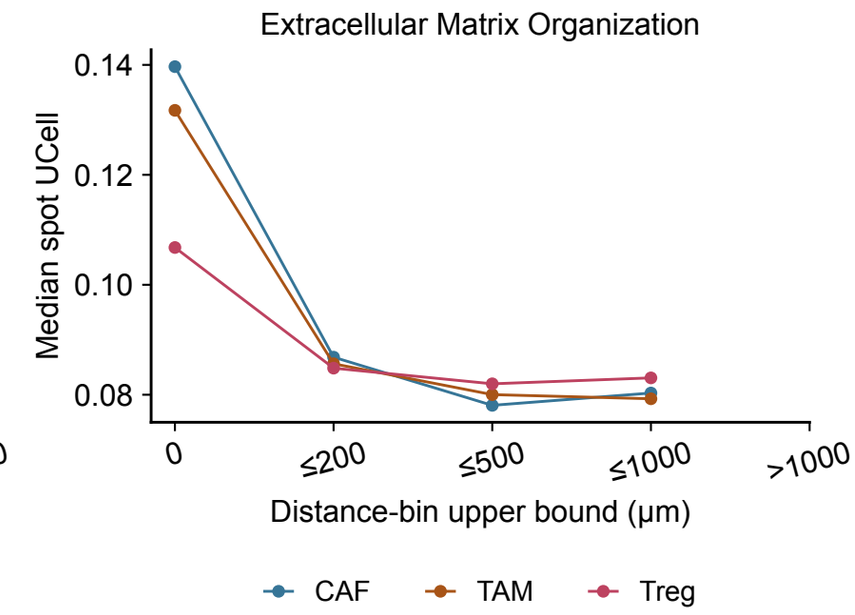






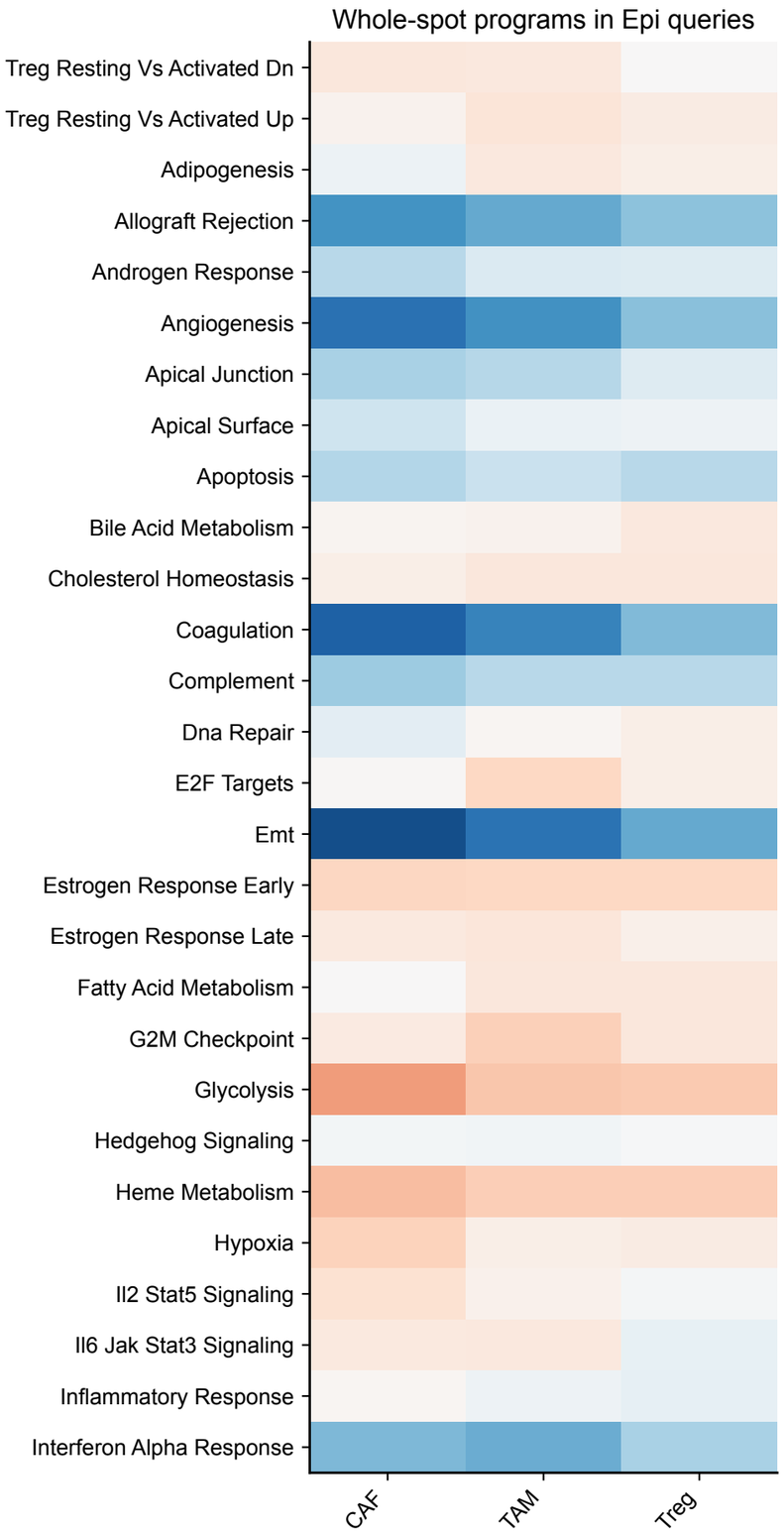


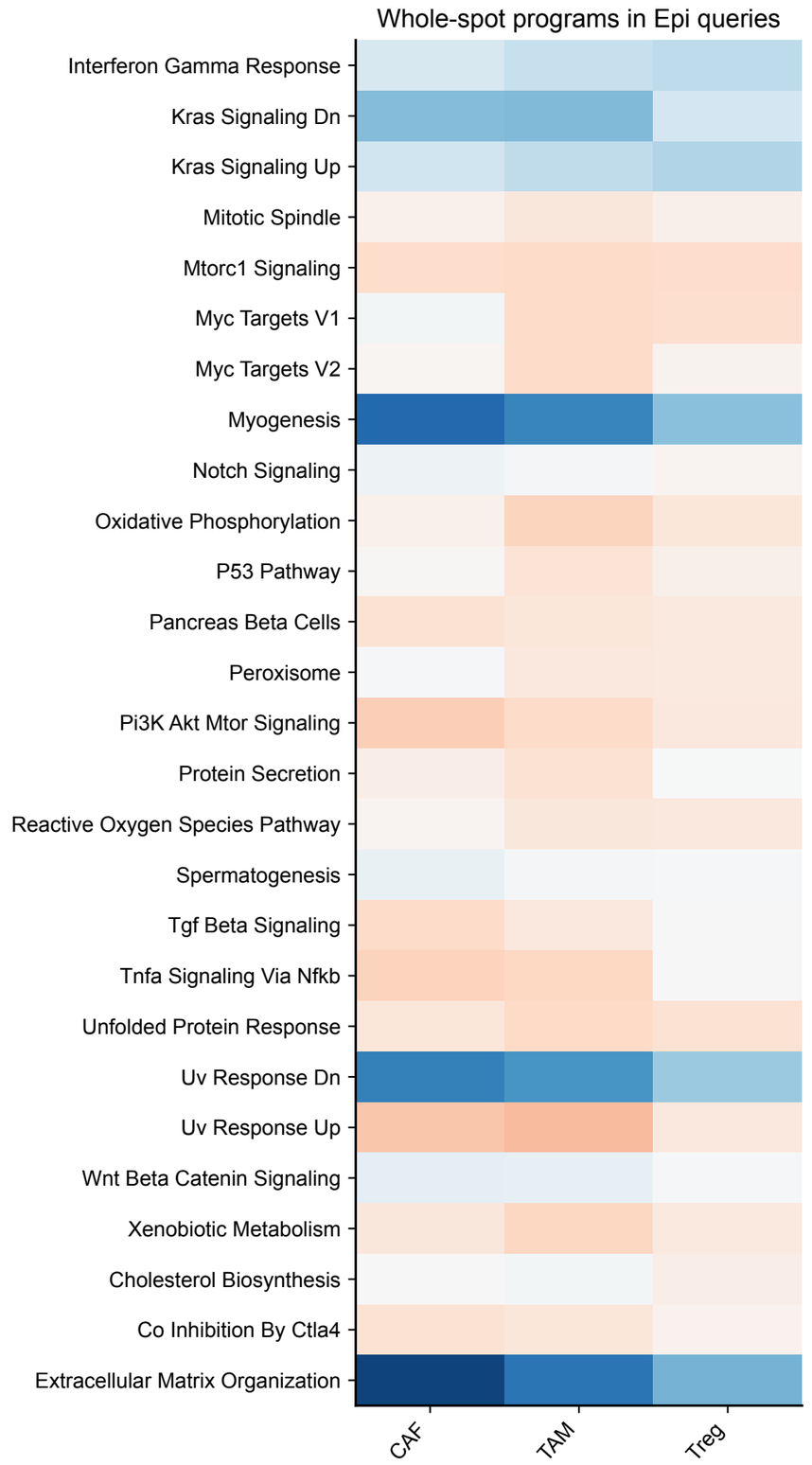




CanDis_2026_Sample3 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

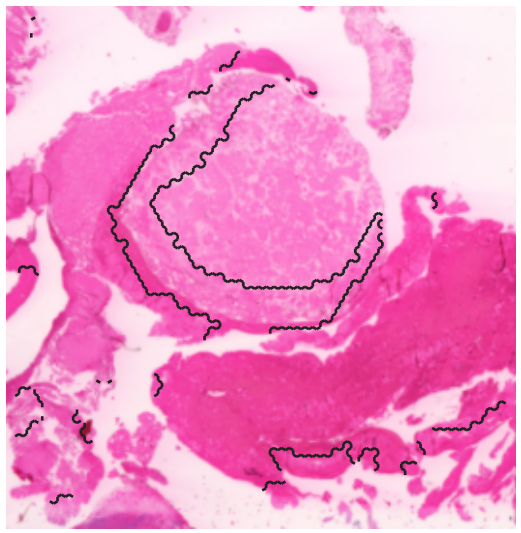




CanDis_2026_Sample4 | Tissue context

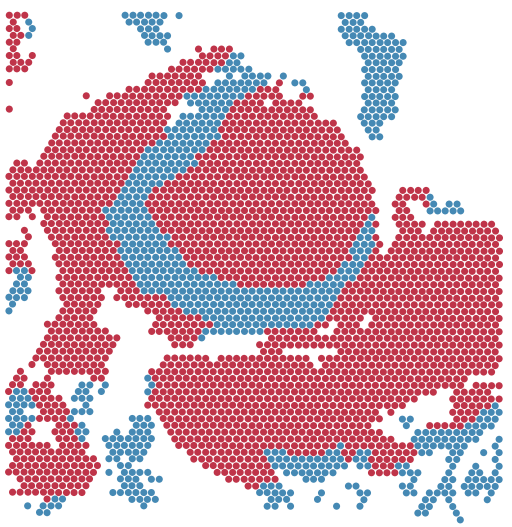
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



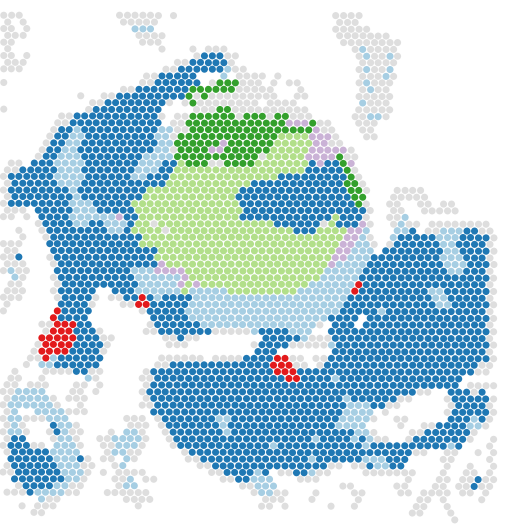
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)

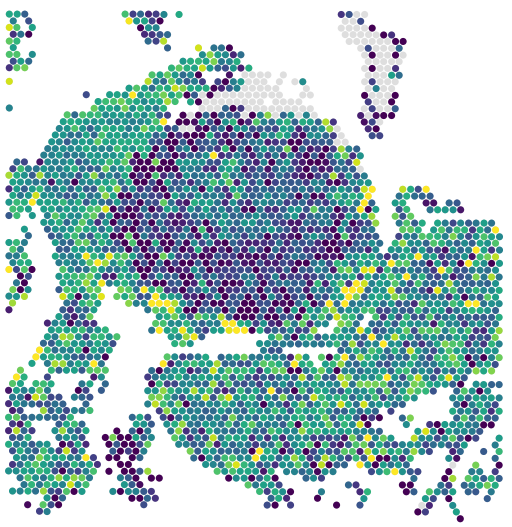


0 2 3 7 8
1

Epithelial support

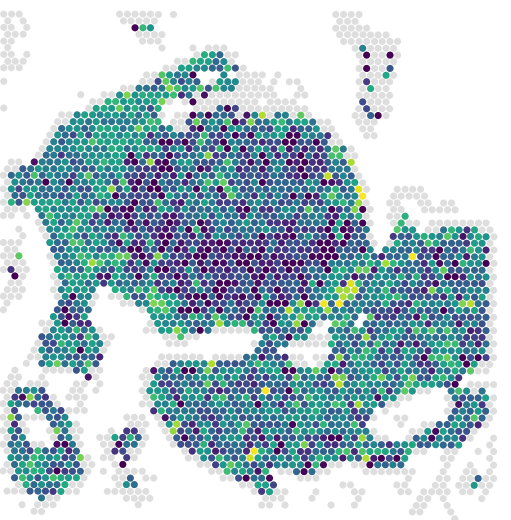
0 1
Fraction of all 30

MP10 in Epi queries



0 0.147
Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

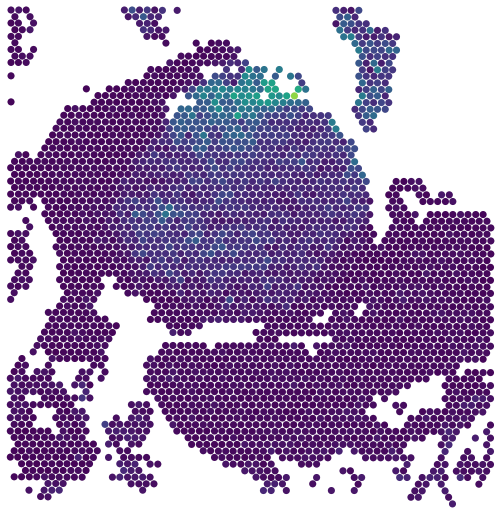


0 0.14
Matched-model AUCell

CanDis_2026_Sample4 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

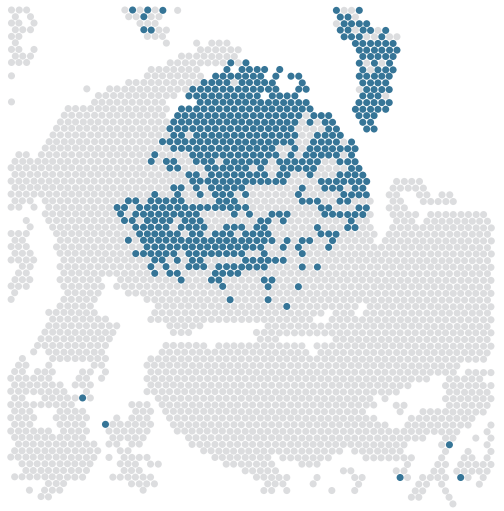
CAF composition



0 0.418

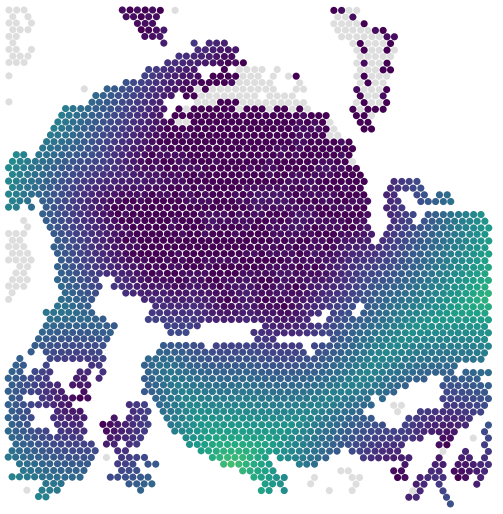
Fraction of all 30

CAF P80 targets



False True

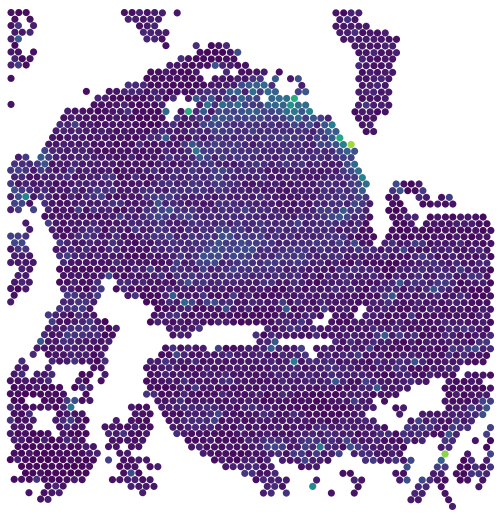
CAF distance



0 3.1e+03

Nearest spot center (μm)

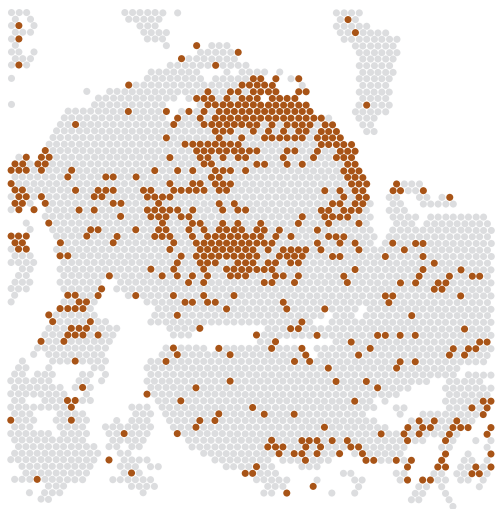
TAM composition



0 0.291

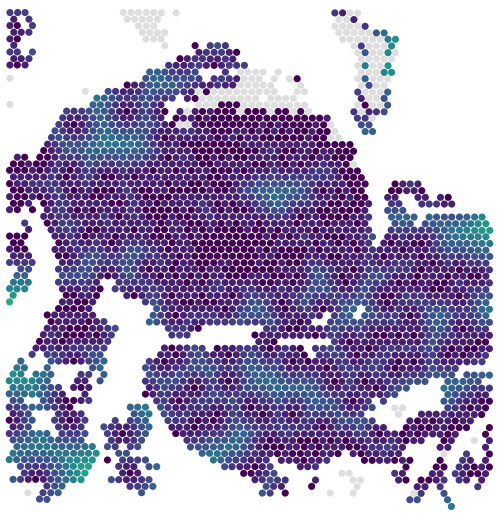
Fraction of all 30

TAM P80 targets



False True

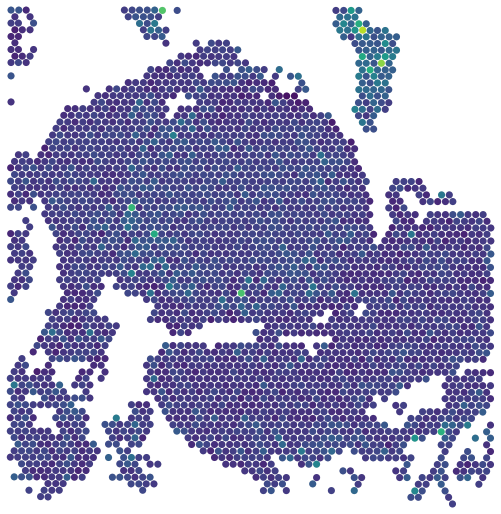
TAM distance



0 1.13e+03

Nearest spot center (μm)

Treg composition



0 0.102

Fraction of all 30

Treg P80 targets

False True

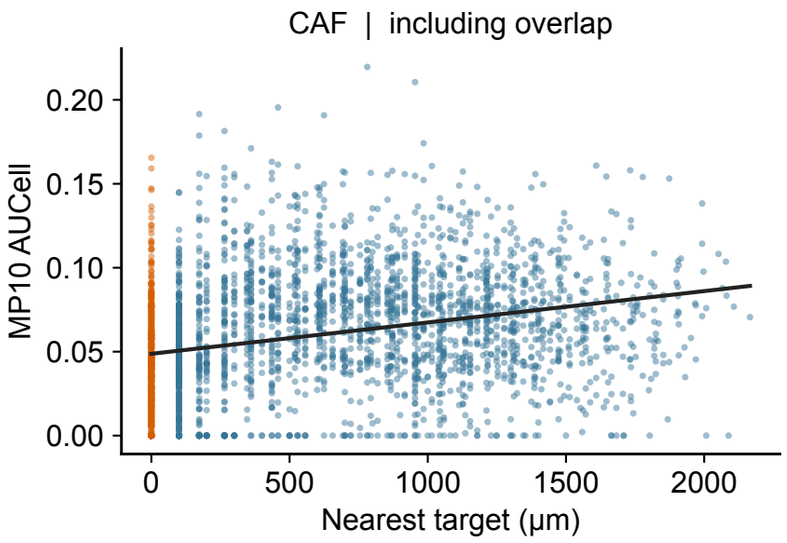
Treg distance

0 1.61e+03

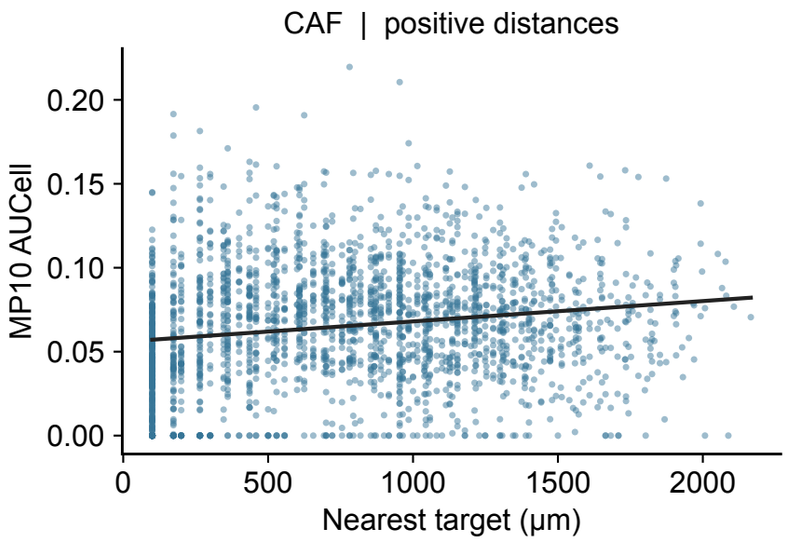
Nearest spot center (μm)

CanDis_2026_Sample4 | MP10 versus nearest target

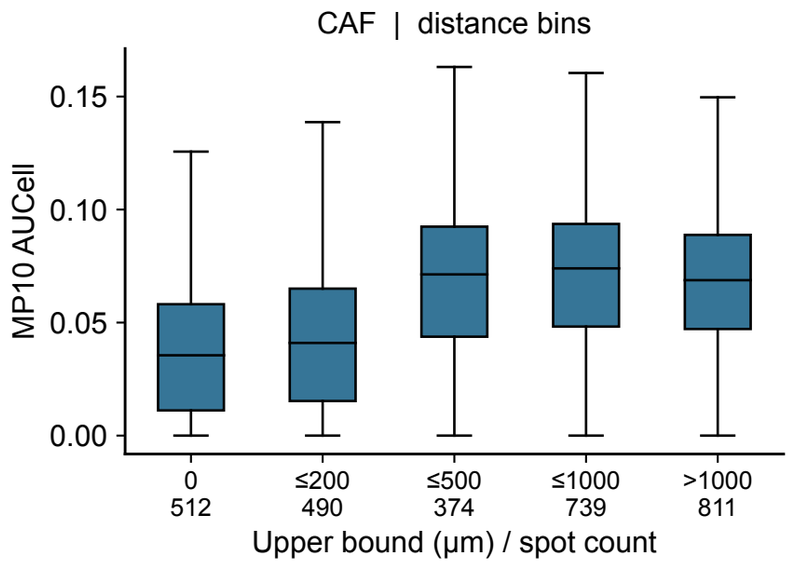
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



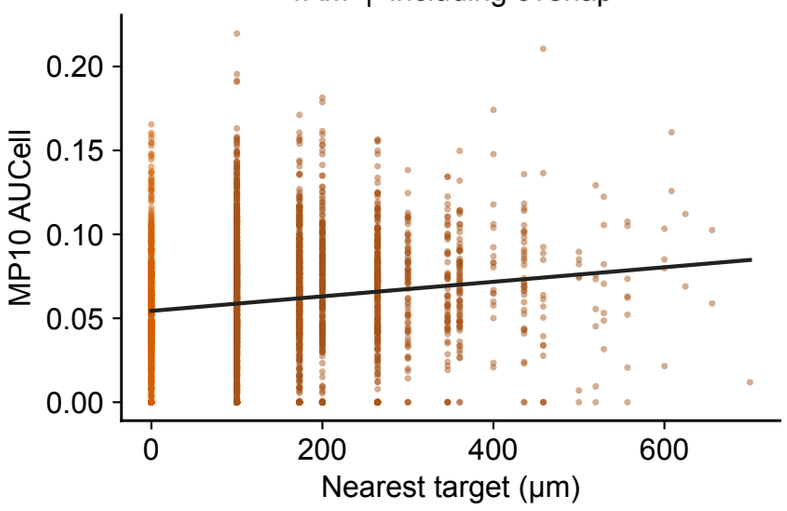
r +0.27 ρ +0.32
n 2926/2997 queries • 624 targets



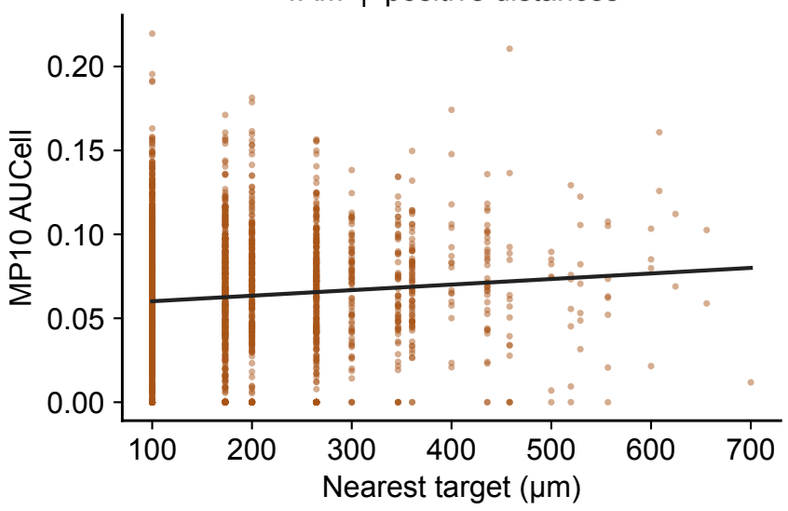
r +0.17 ρ +0.19
n 2414/2997 queries • 624 targets



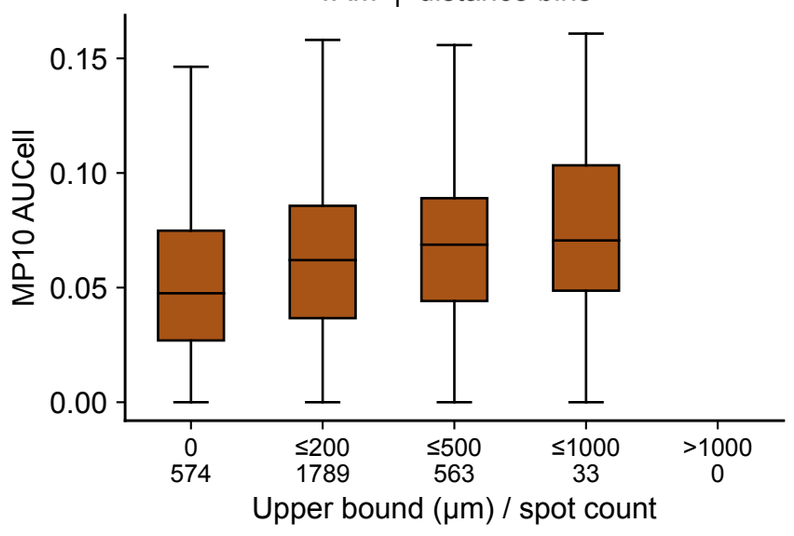
Orange = zero distance
Line = within-section linear fit



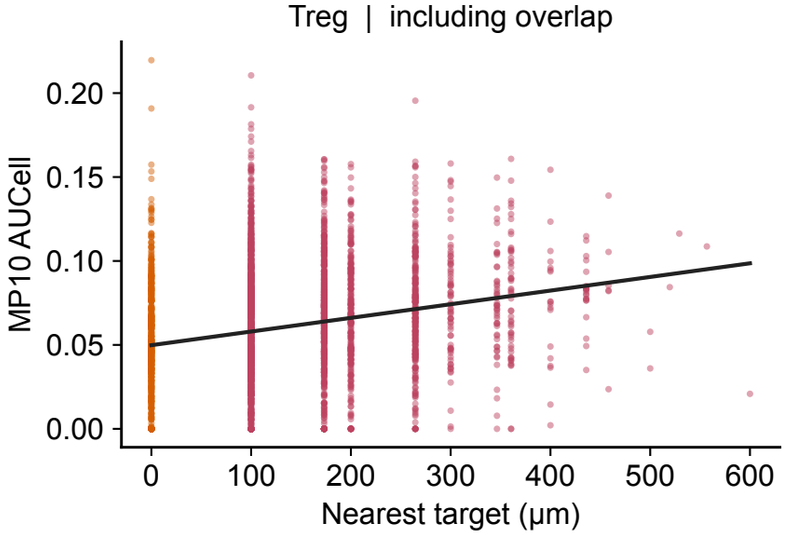
r +0.14 ρ +0.16
n 2959/2997 queries • 624 targets



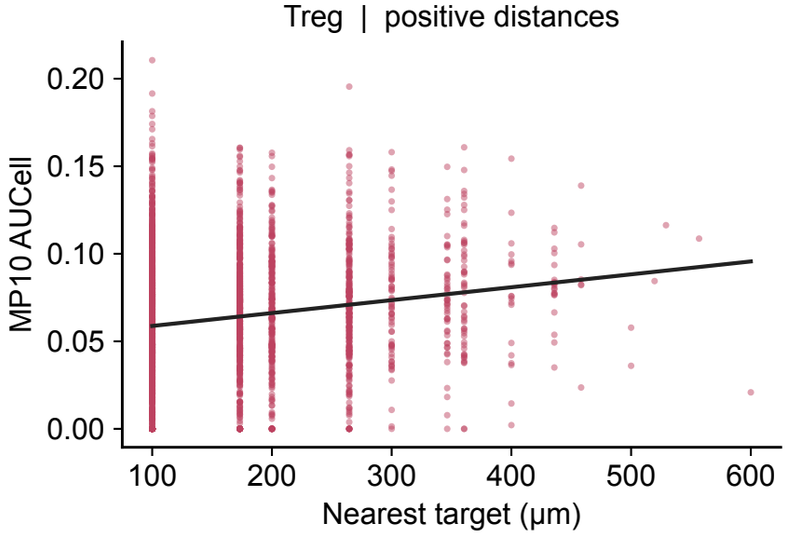
r +0.09 ρ +0.12
n 2385/2997 queries • 624 targets



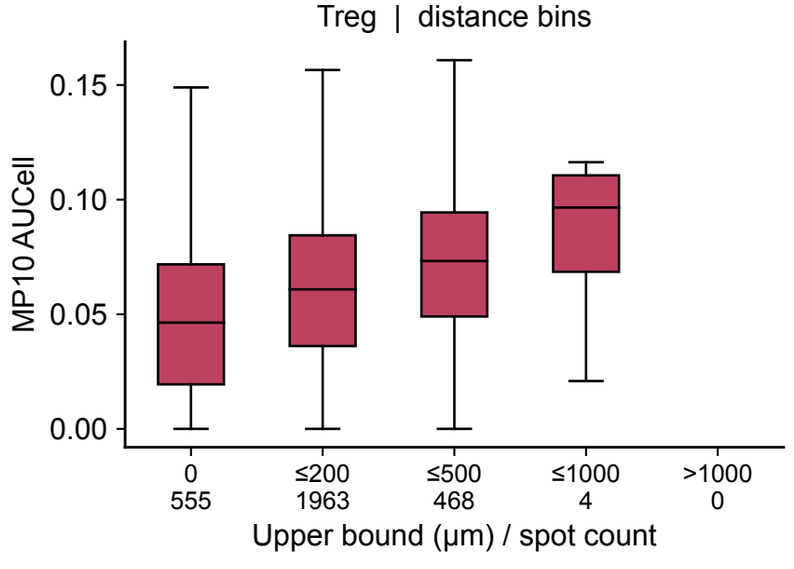
Orange = zero distance
Line = within-section linear fit



r +0.22 ρ +0.24
n 2990/2997 queries • 624 targets



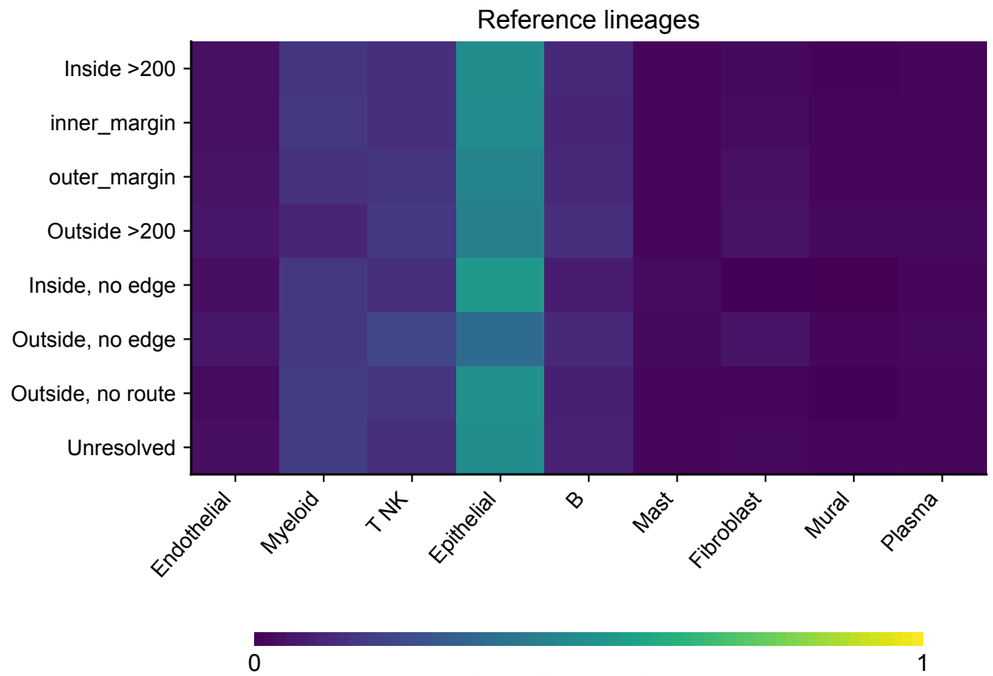
r +0.17 ρ +0.19
n 2435/2997 queries • 624 targets

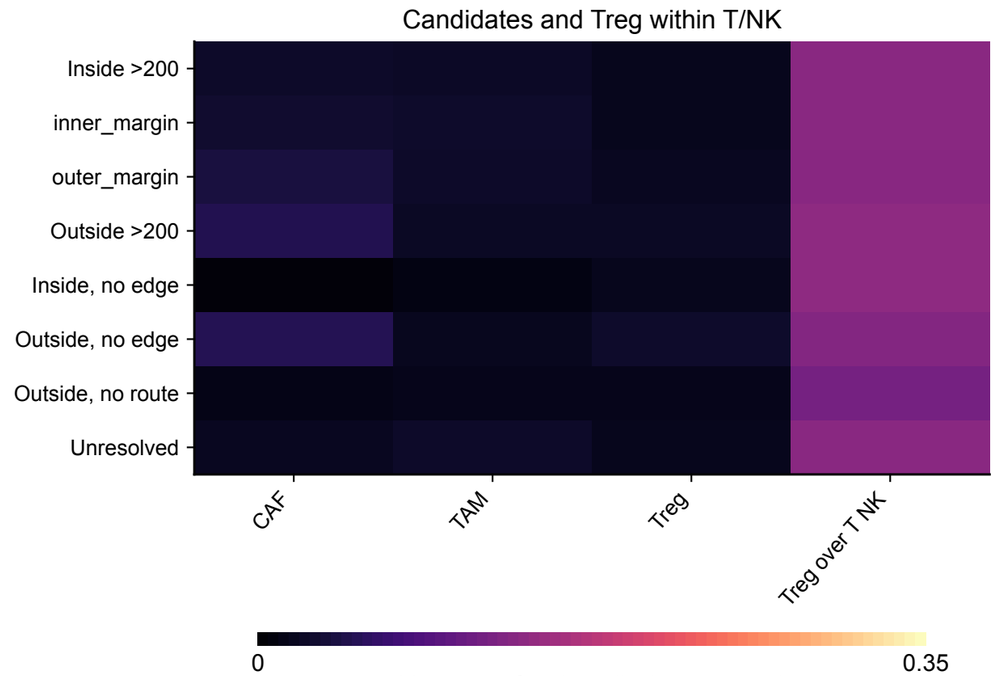


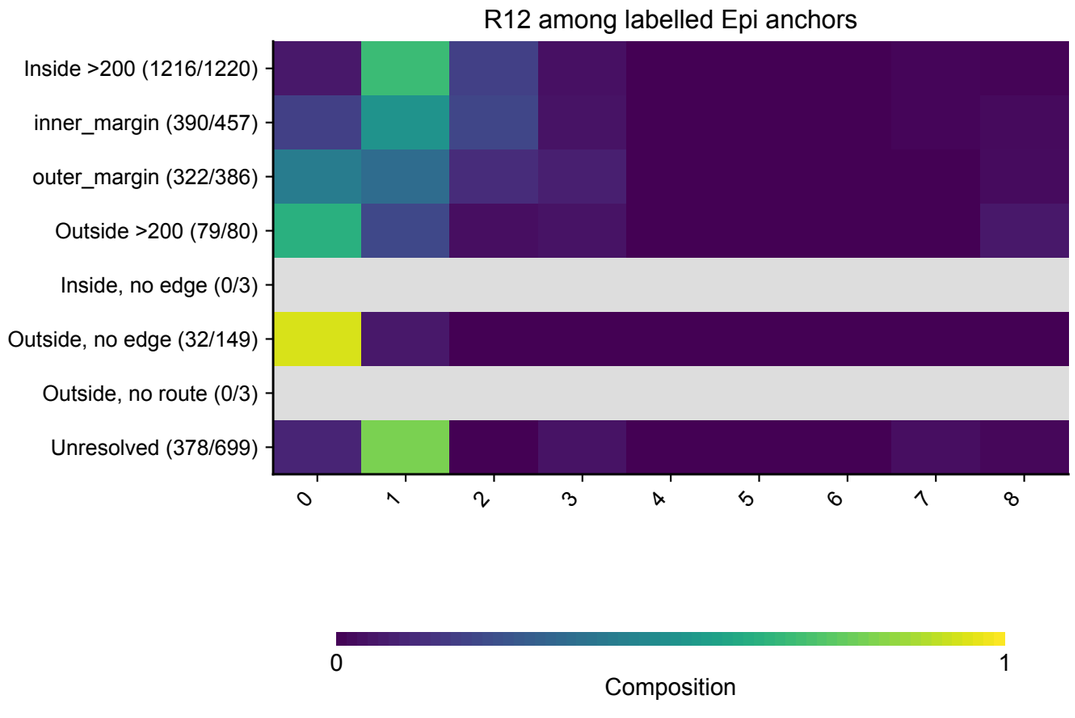
Orange = zero distance
Line = within-section linear fit

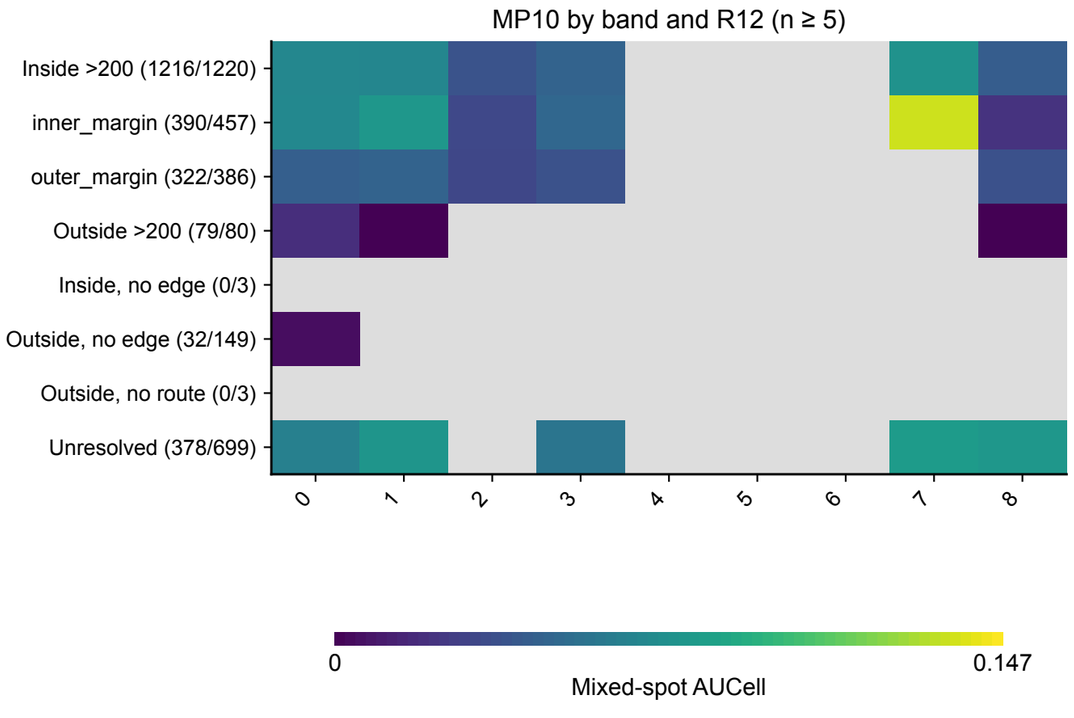
CanDis_2026_Sample4 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





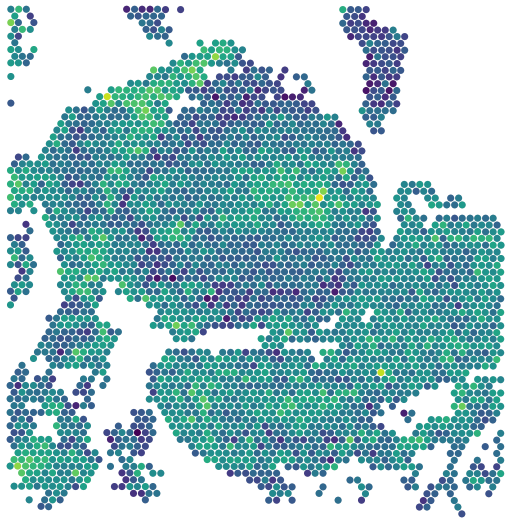




CanDis_2026_Sample4 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

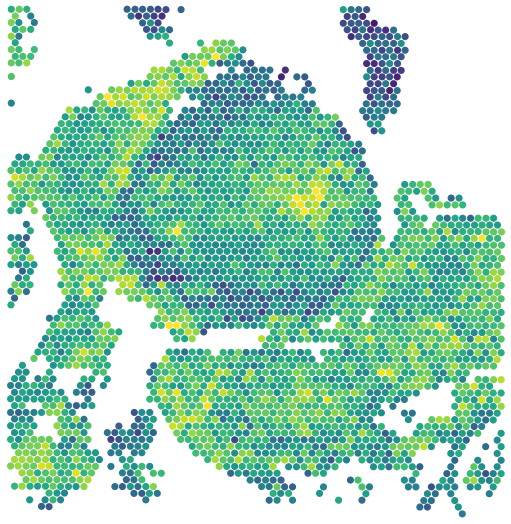
E2F Targets



0 0.152

UCell score

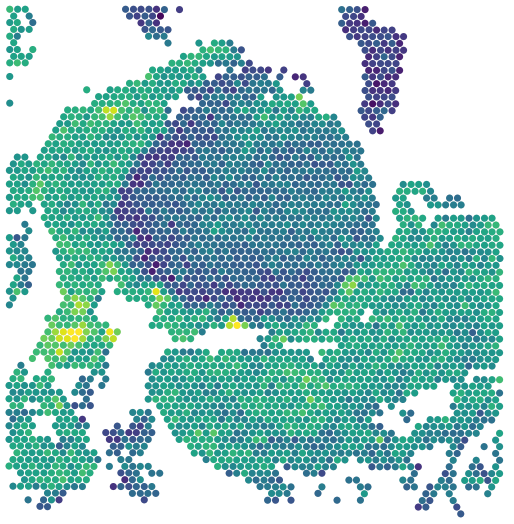
G2M Checkpoint



0 0.134

UCell score

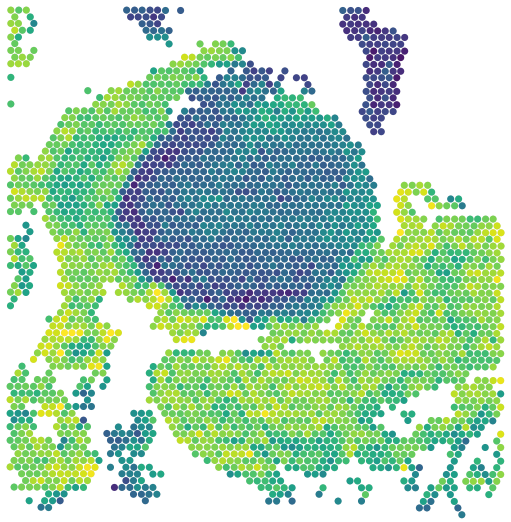
Hypoxia



0 0.211

UCell score

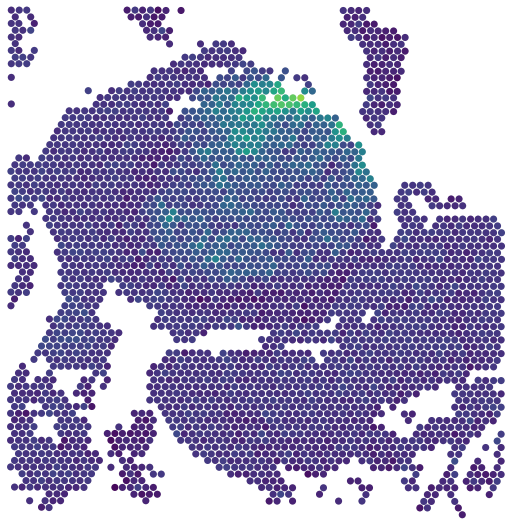
Tnfa Signaling Via Nfkb



0 0.275

UCell score

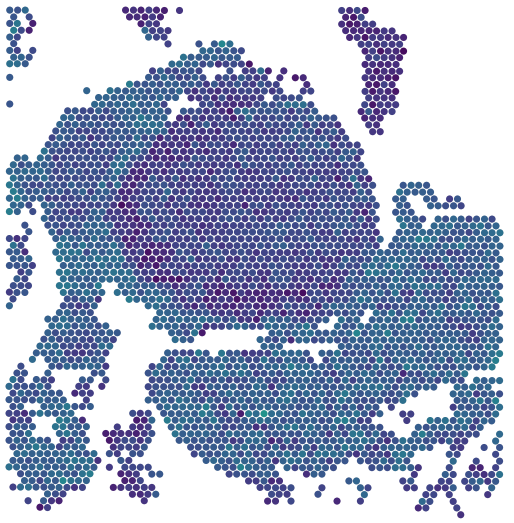
Emt



0 0.386

UCell score

Interferon Gamma Response



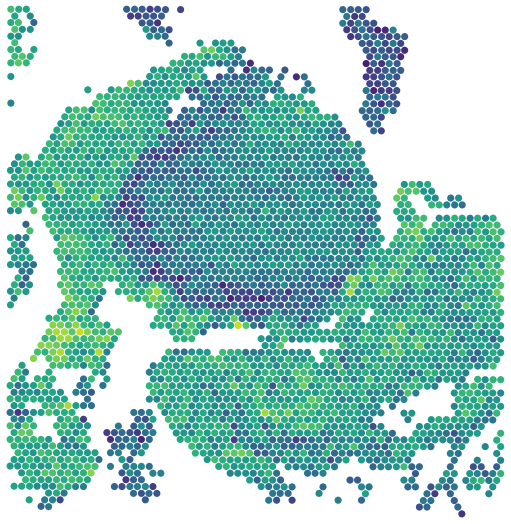
0 0.275

UCell score

CanDis_2026_Sample4 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

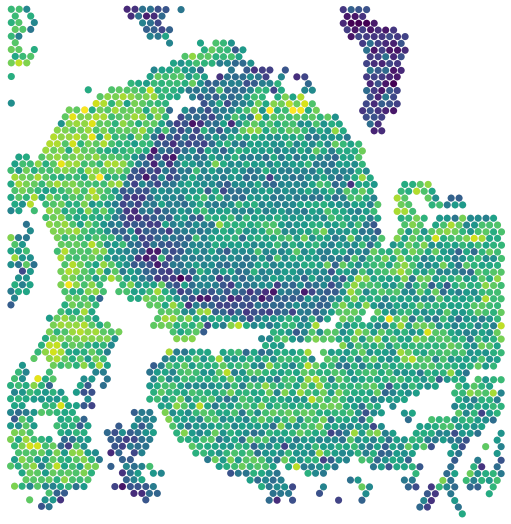
Il2 Stat5 Signaling



0 0.144

UCell score

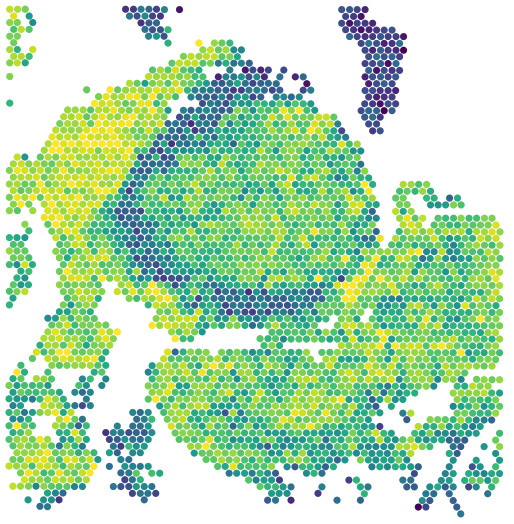
Tgf Beta Signaling



0 0.232

UCell score

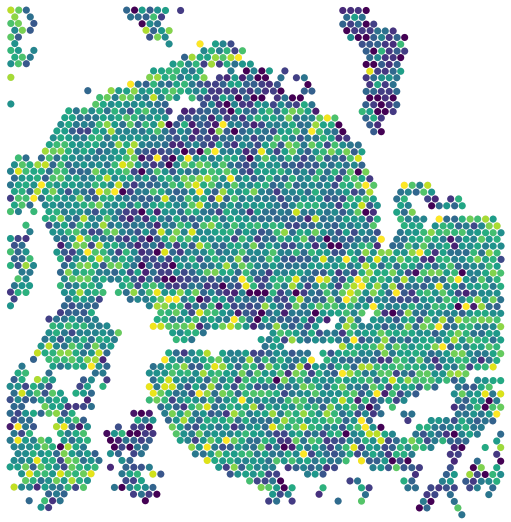
Cholesterol Homeostasis



0 0.21

UCell score

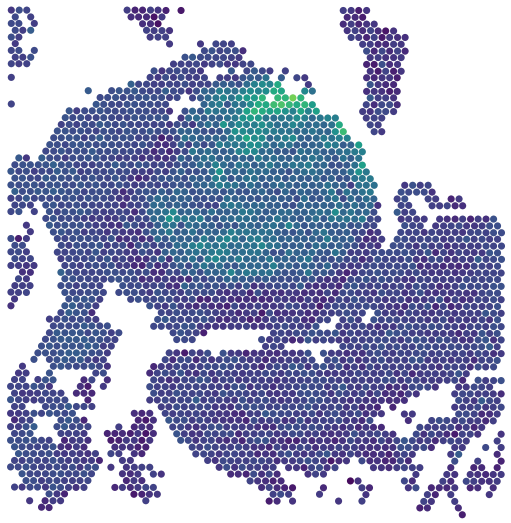
Cholesterol Biosynthesis



0 0.166

UCell score

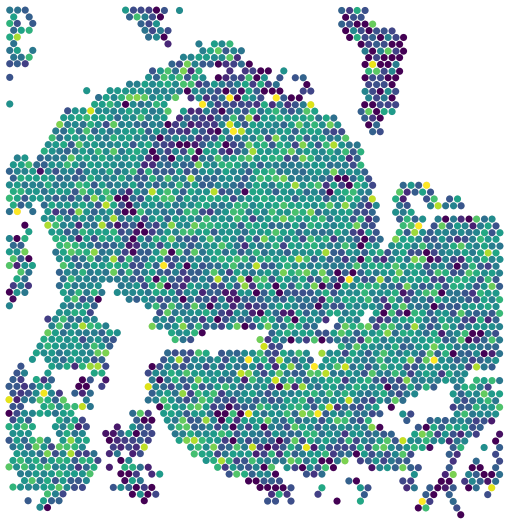
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



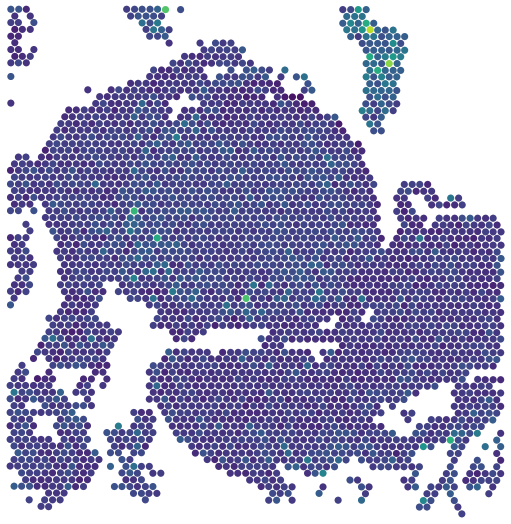
0 0.151

UCell score

CanDis_2026_Sample4 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

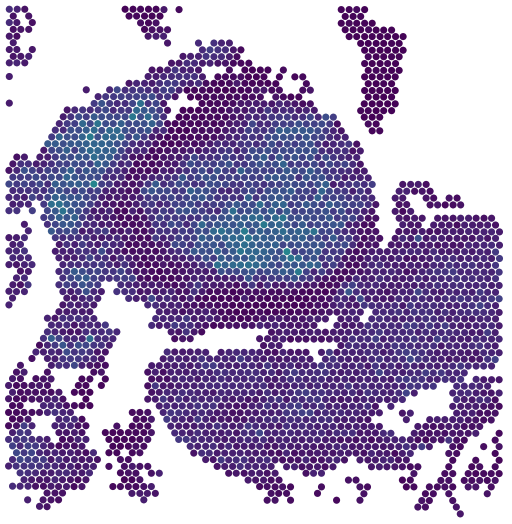
Treg composition



0 0.102

Fraction of all 30

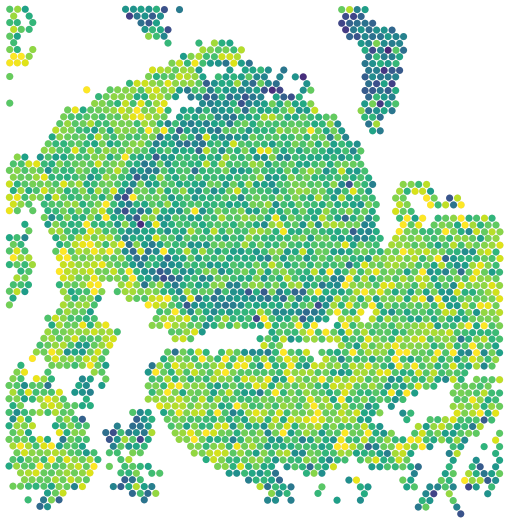
Treg estimated abundance



0 2.85

Posterior mean estimate

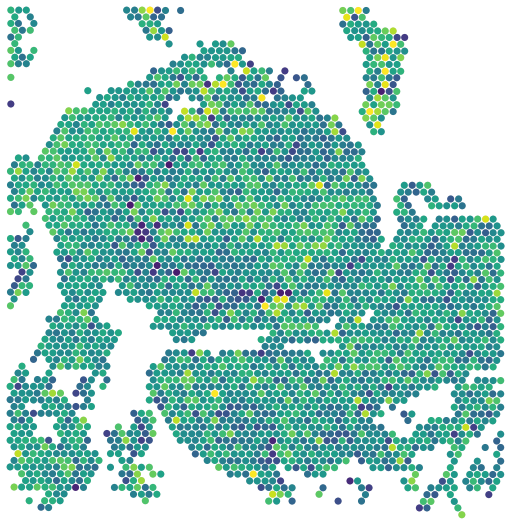
Treg Resting Vs Activated Dn



0 0.0778

UCell score

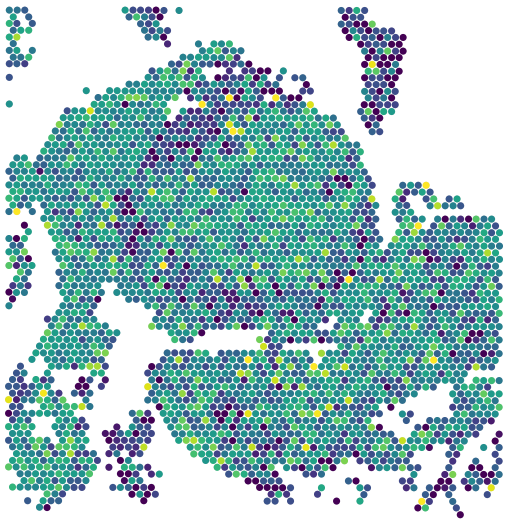
Treg Resting Vs Activated Up



0 0.0609

UCell score

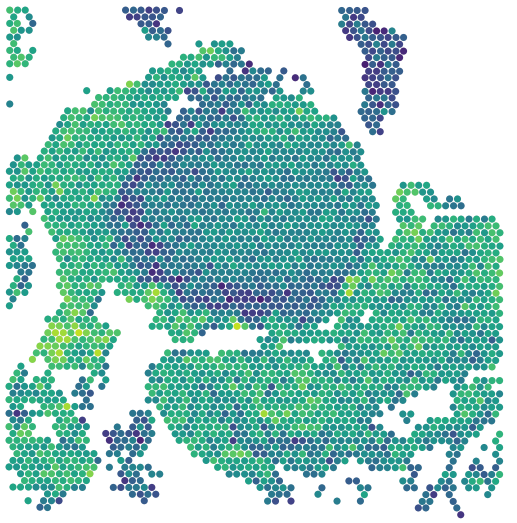
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

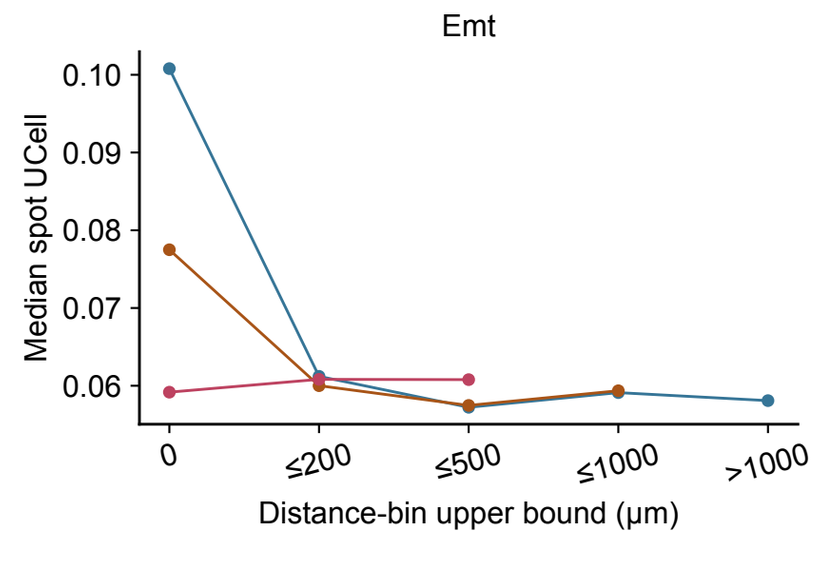


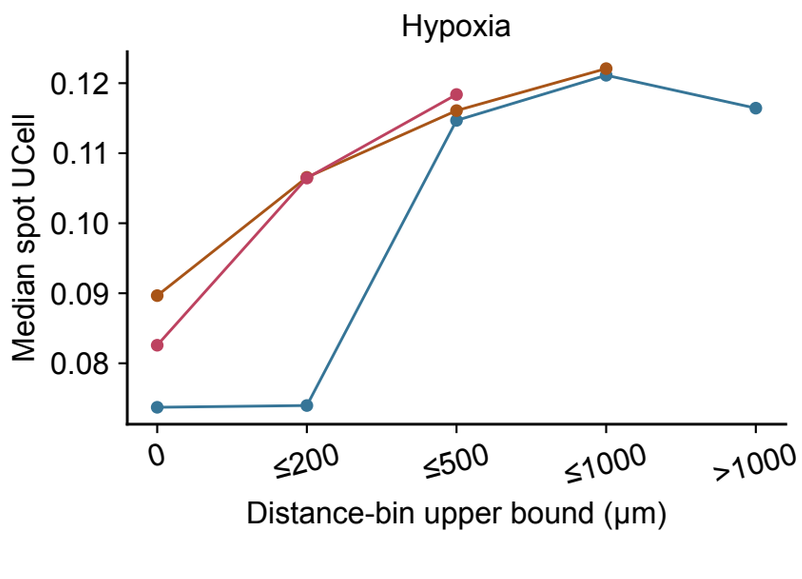
0 0.144

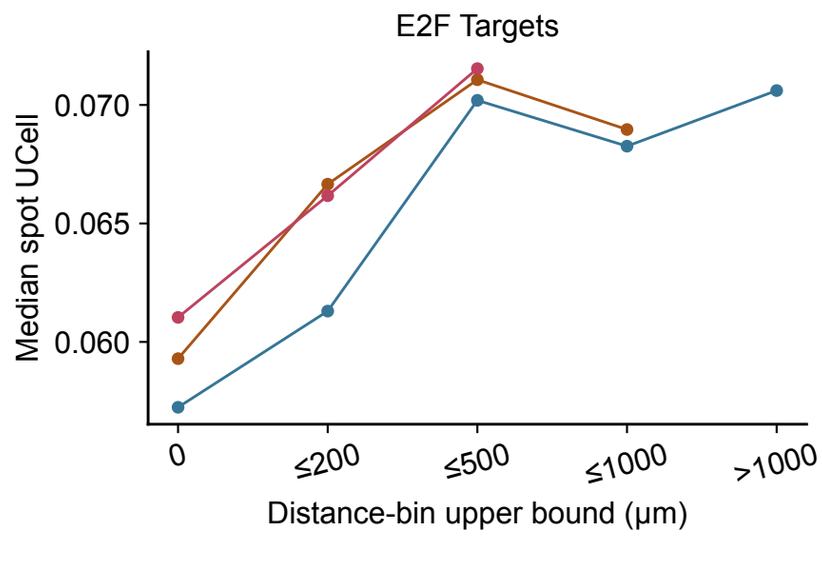
UCell score

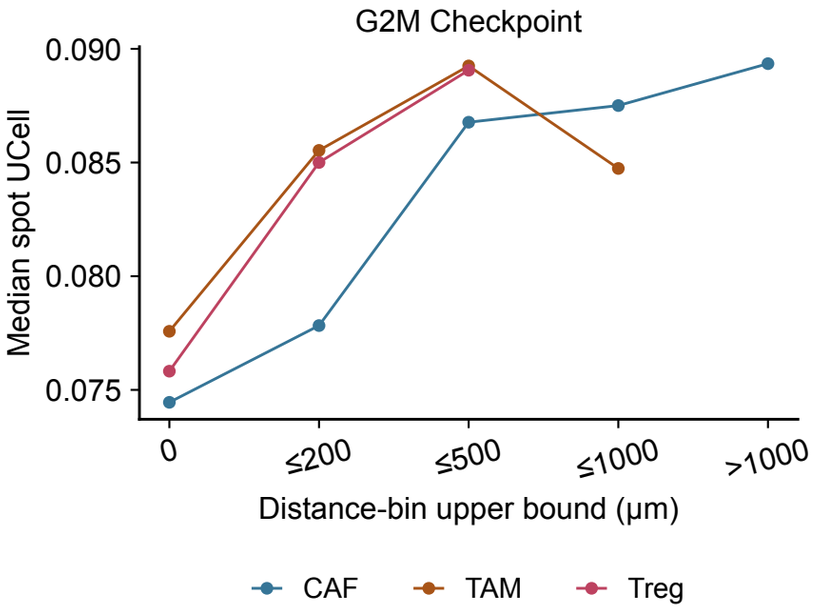
CanDis_2026_Sample4 | Programs by target distance

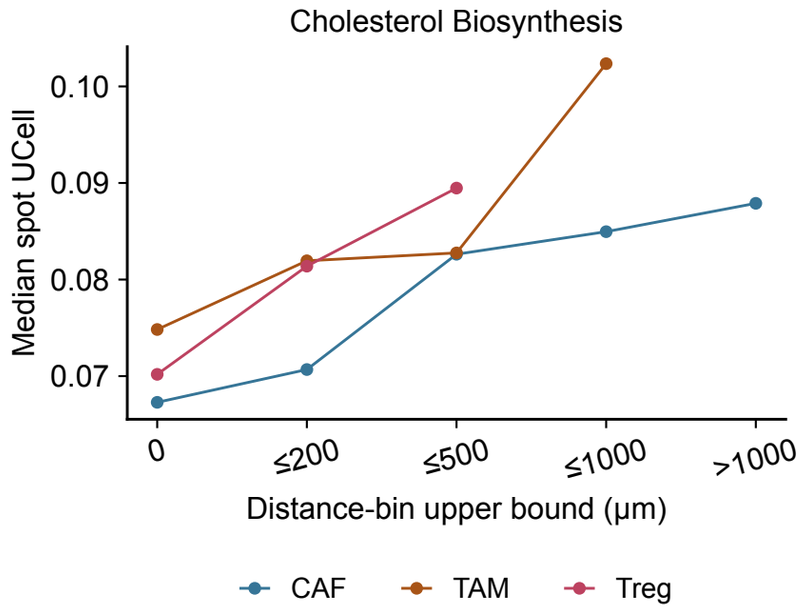
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

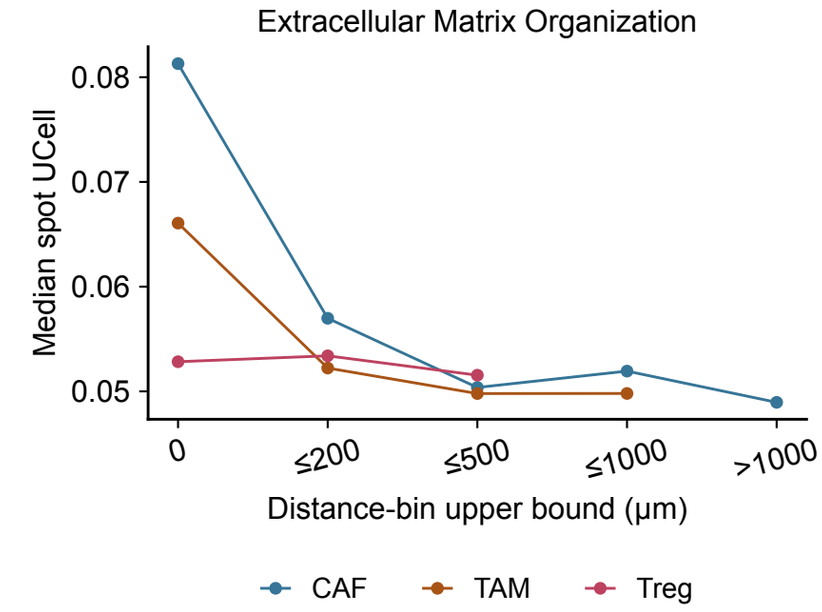






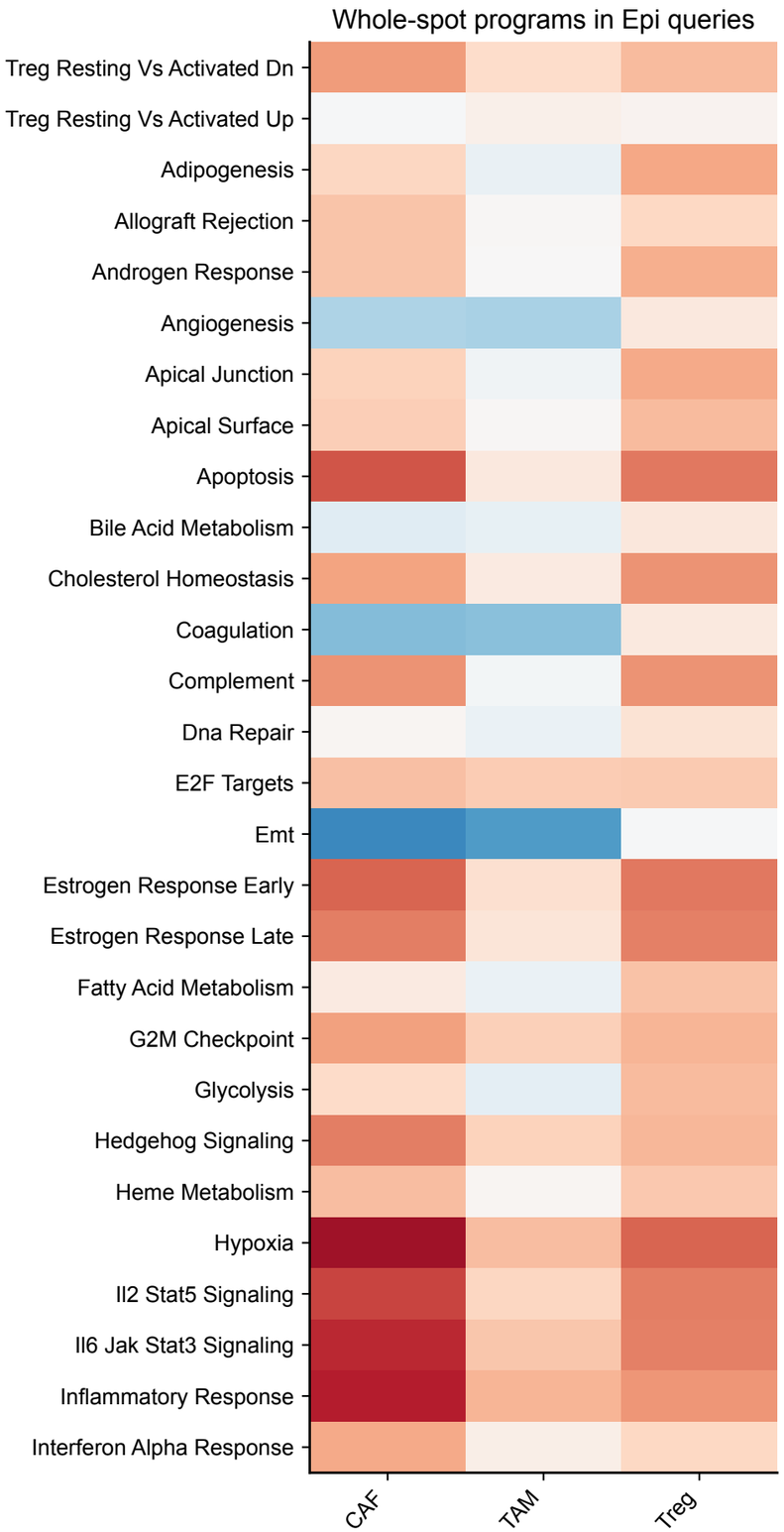


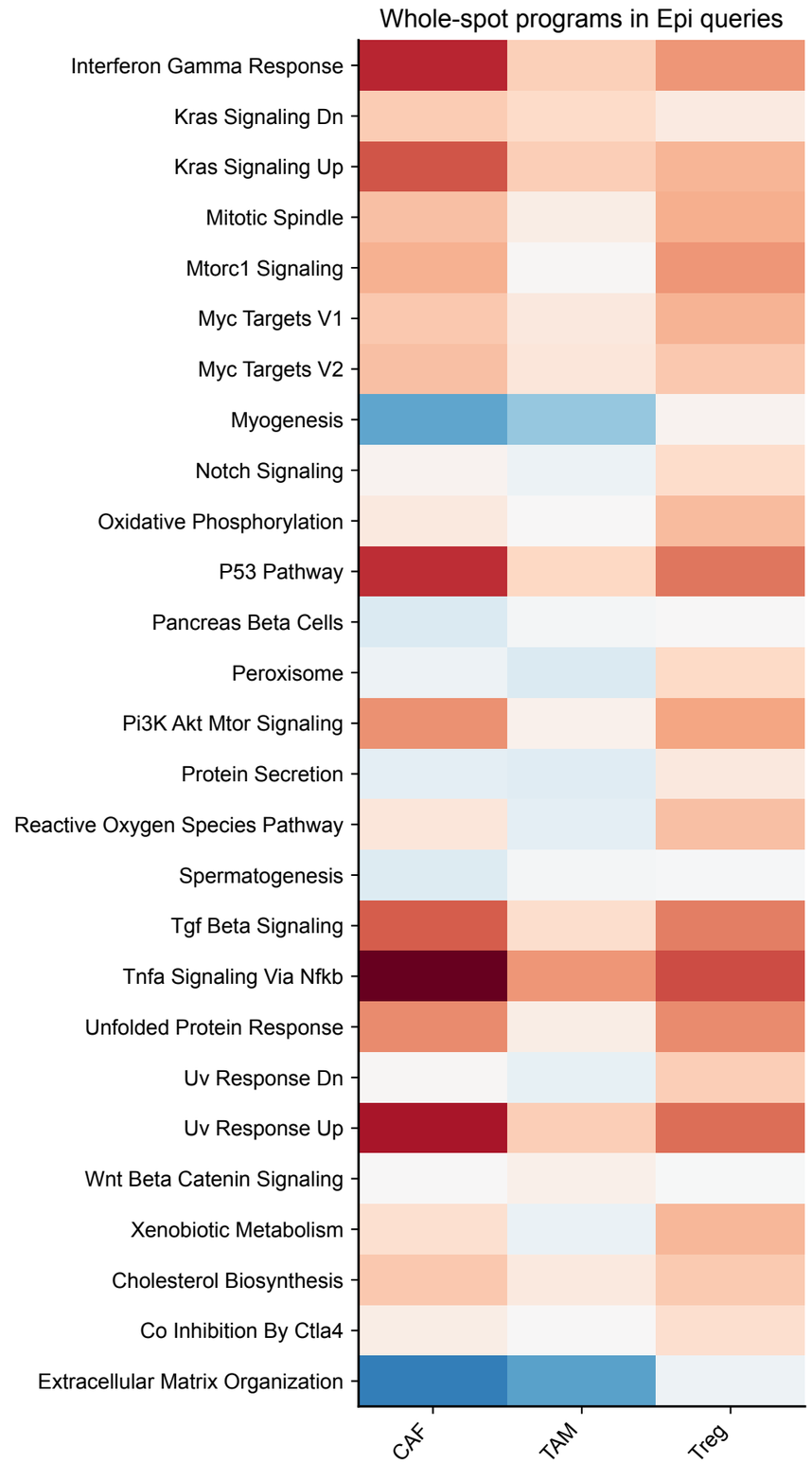




CanDis_2026_Sample4 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

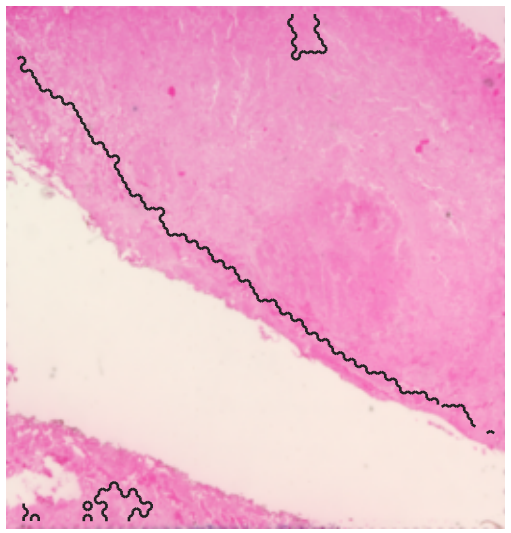




CanDis_2026_Sample5 | Tissue context

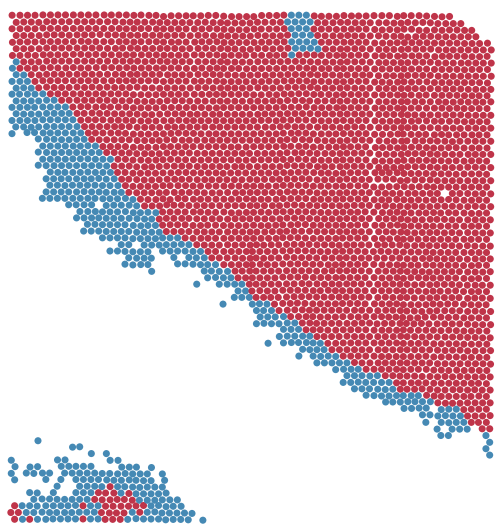
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



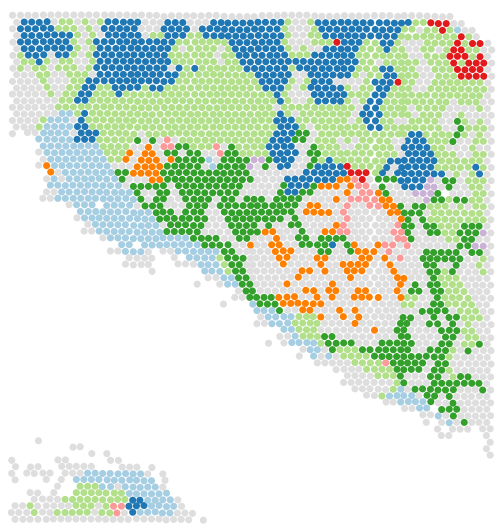
Black line: observed candidate outer edge

Final candidate location



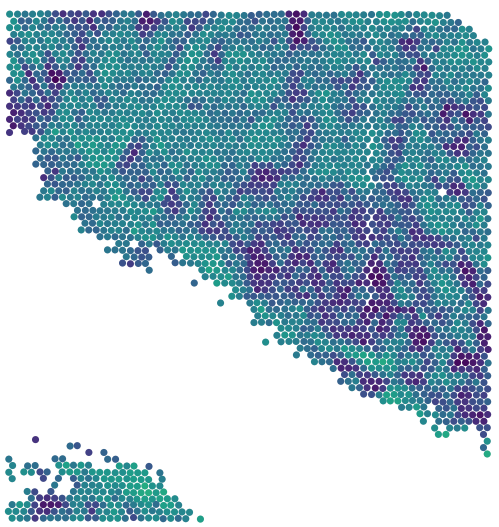
inside outside

Accepted R12 (original 0–8)



0 2 5 7 8
1 3 6

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries

0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

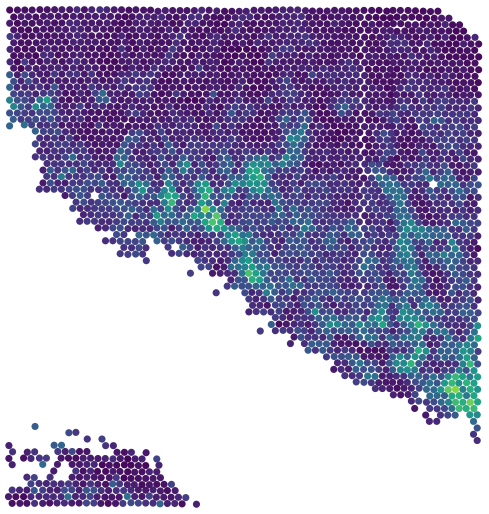
0 0.14

Matched-model AUCell

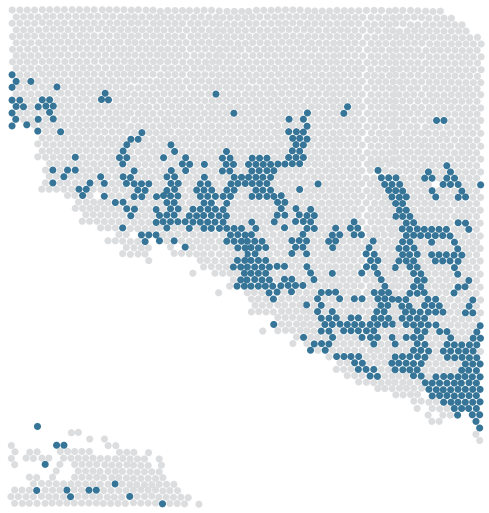
CanDis_2026_Sample5 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

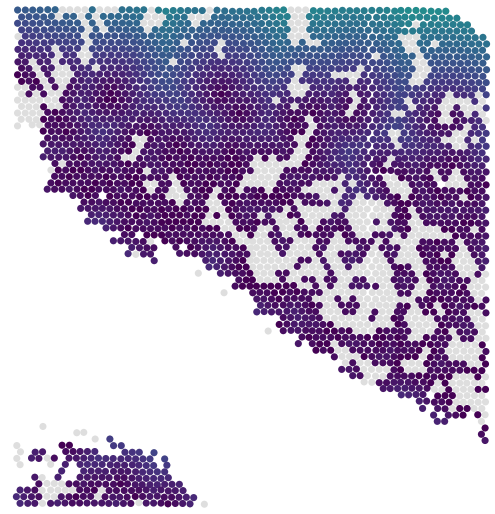
CAF composition



CAF P80 targets



CAF distance

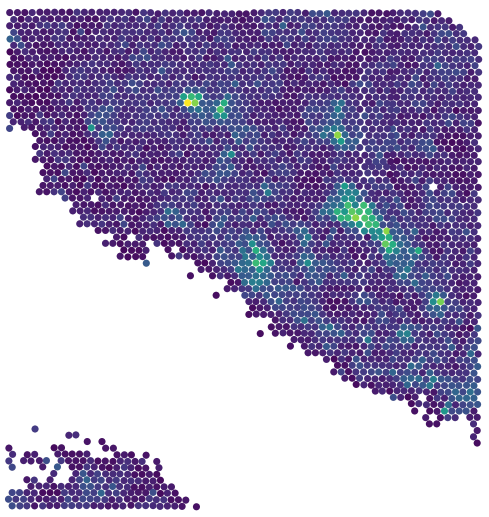


0 0.418
Fraction of all 30

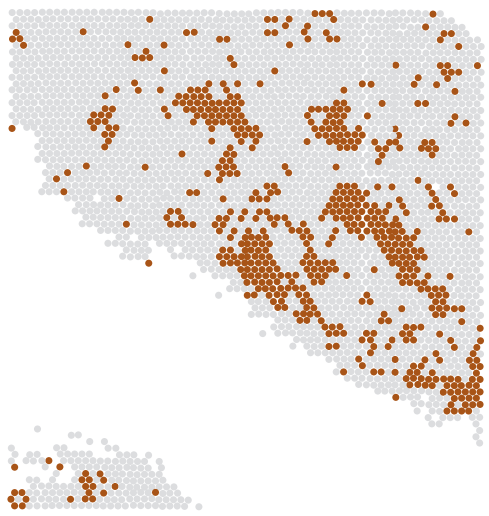
False True

0 3.1e+03
Nearest spot center (μm)

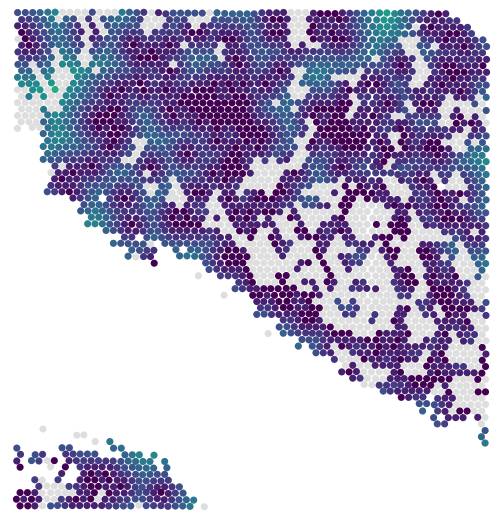
TAM composition



TAM P80 targets



TAM distance

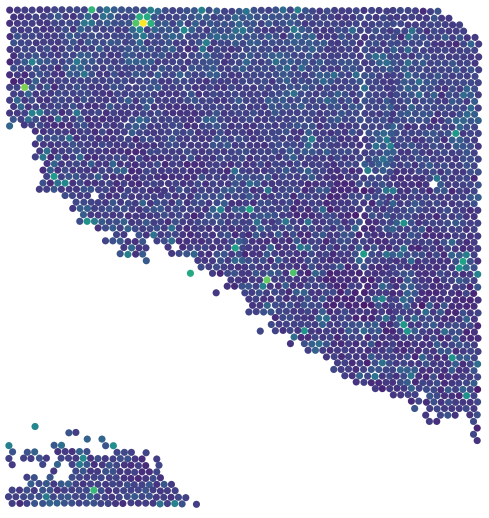


0 0.291
Fraction of all 30

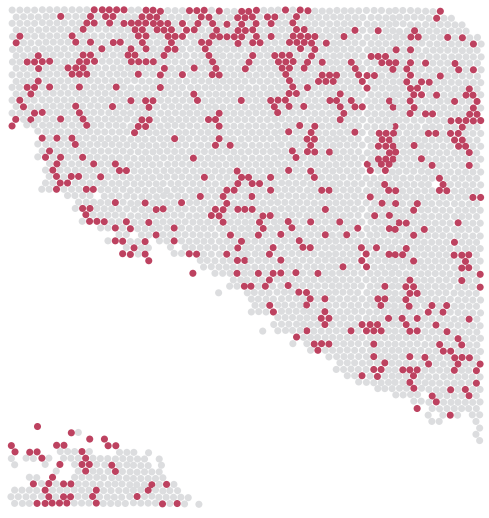
False True

0 1.13e+03
Nearest spot center (μm)

Treg composition



Treg P80 targets



Treg distance

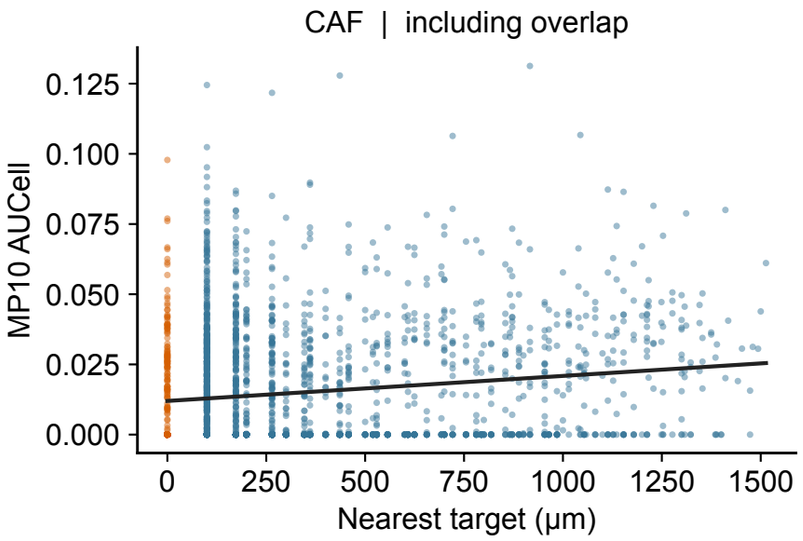
0 0.102
Fraction of all 30

False True

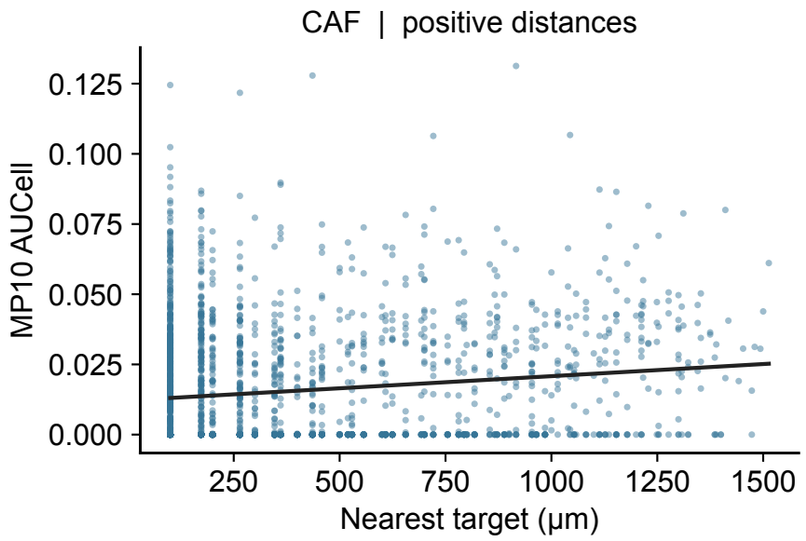
0 1.61e+03
Nearest spot center (μm)

CanDis_2026_Sample5 | MP10 versus nearest target

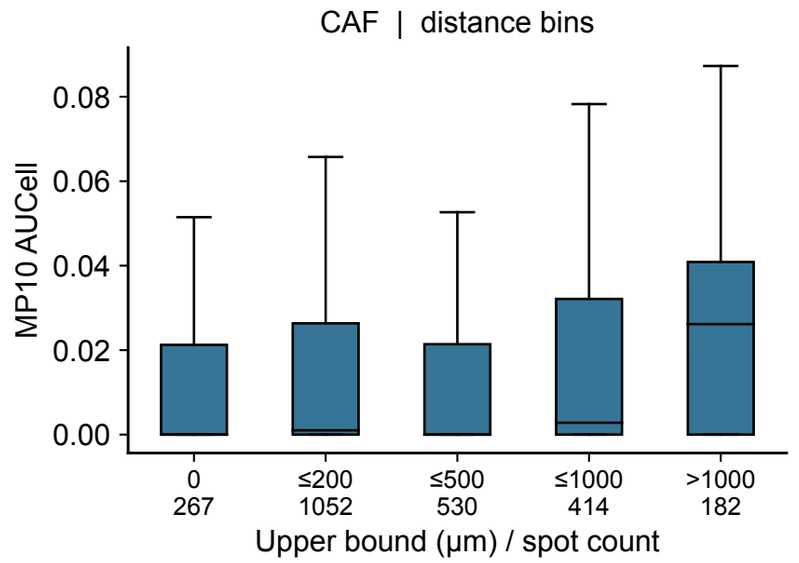
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



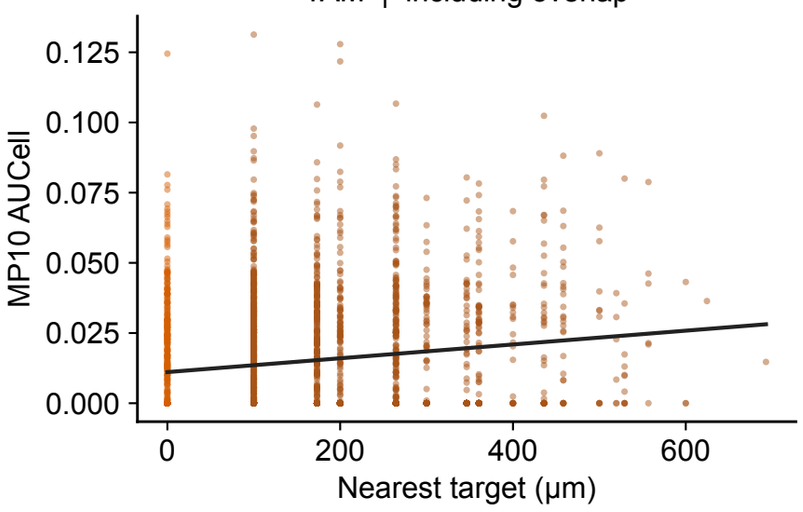
r +0.15 ρ +0.11
n 2445/2455 queries • 625 targets



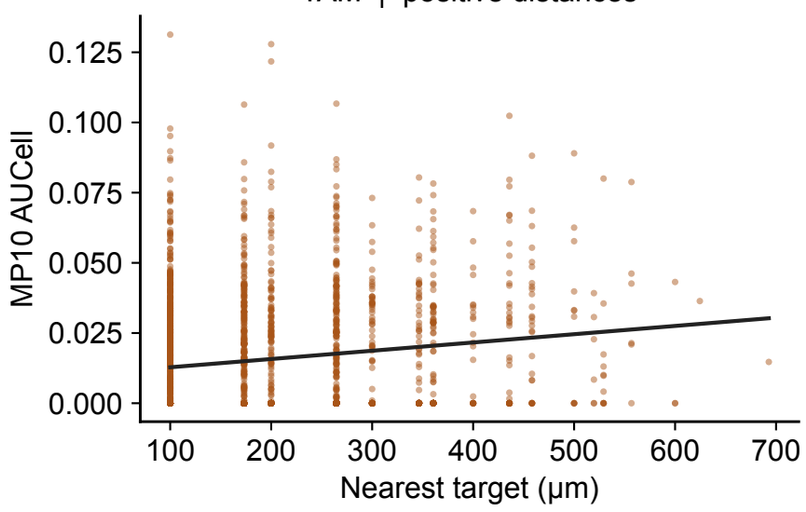
r +0.14 ρ +0.08
n 2178/2455 queries • 625 targets



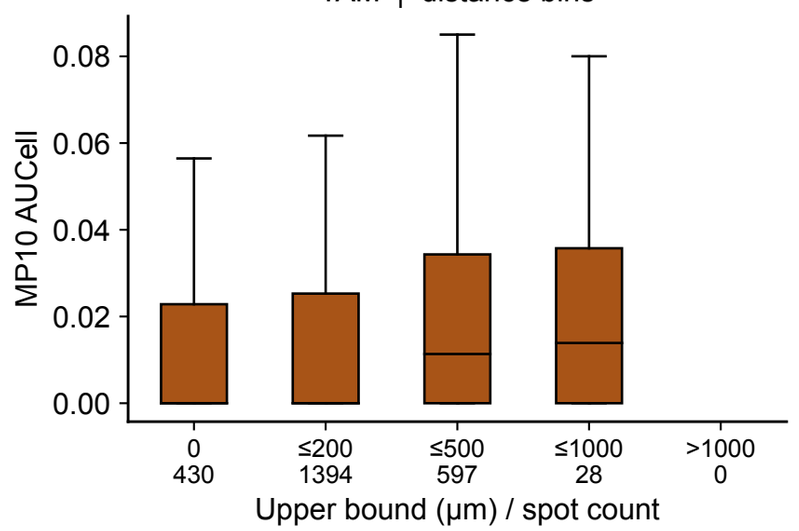
Orange = zero distance
Line = within-section linear fit



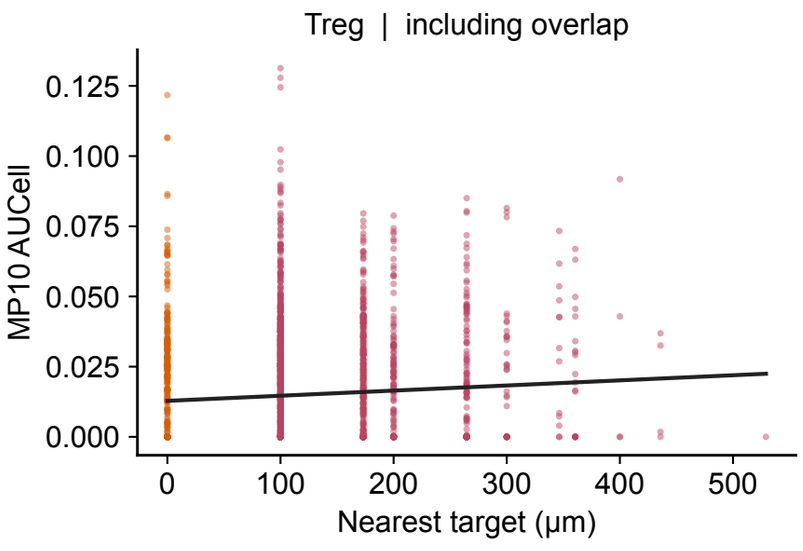
r +0.15 ρ +0.13
n 2449/2455 queries • 625 targets



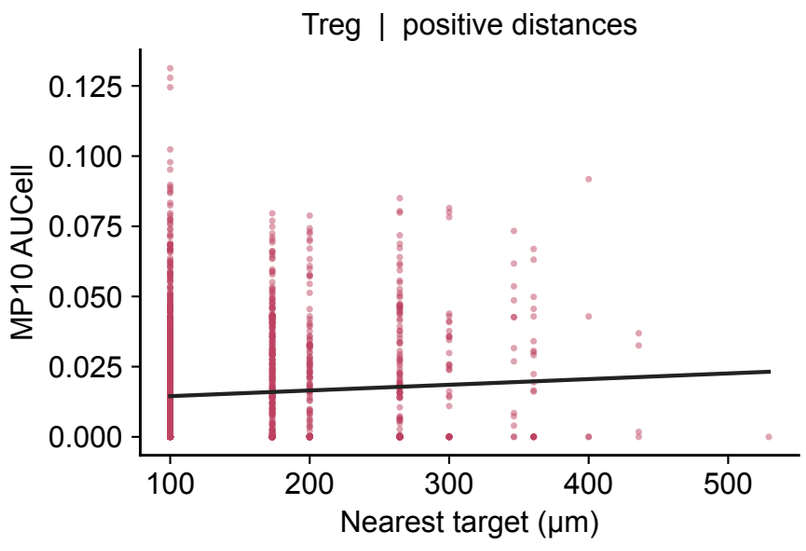
r +0.15 ρ +0.13
n 2019/2455 queries • 625 targets



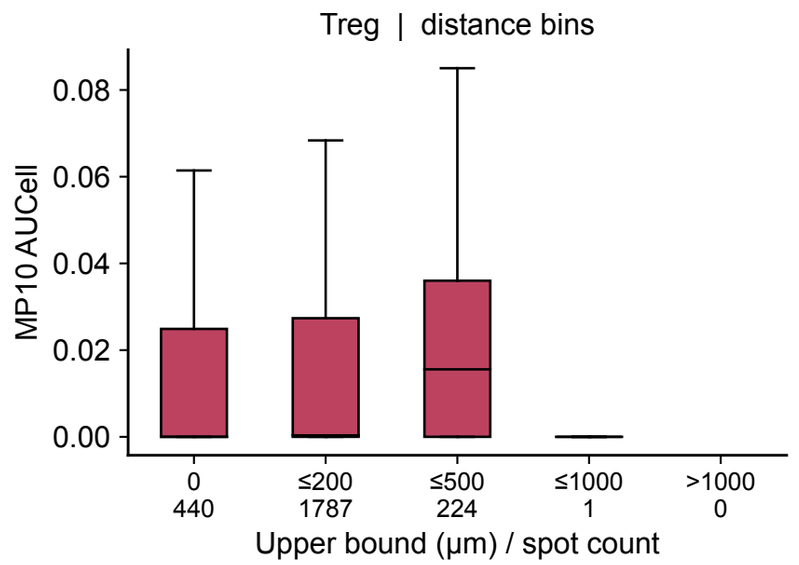
Orange = zero distance
Line = within-section linear fit



r +0.07 ρ +0.07
n 2452/2455 queries • 625 targets



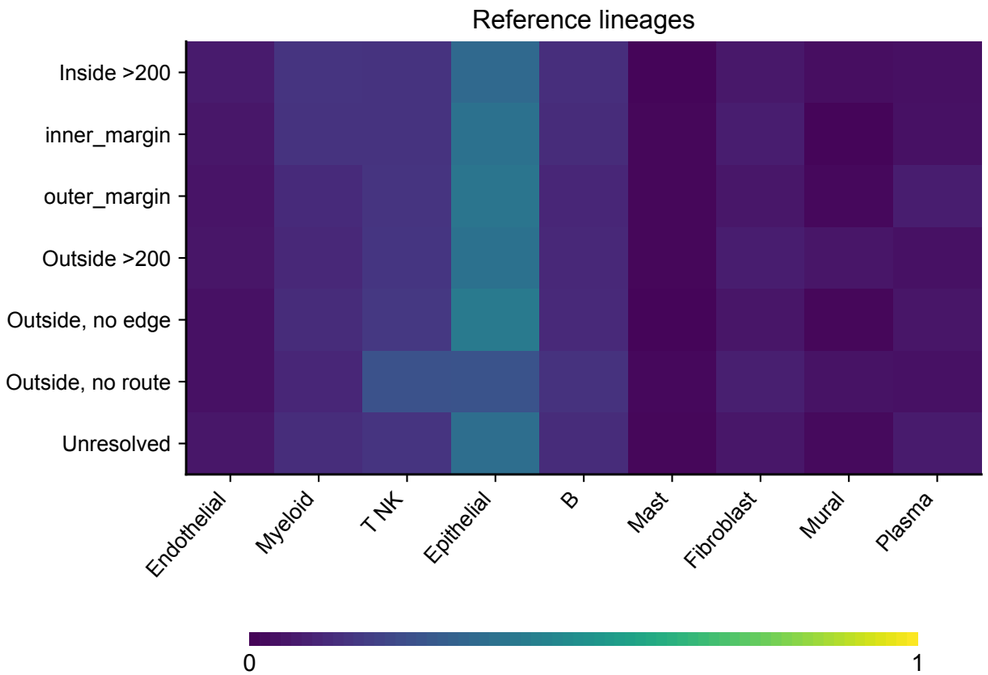
r +0.06 ρ +0.05
n 2012/2455 queries • 625 targets

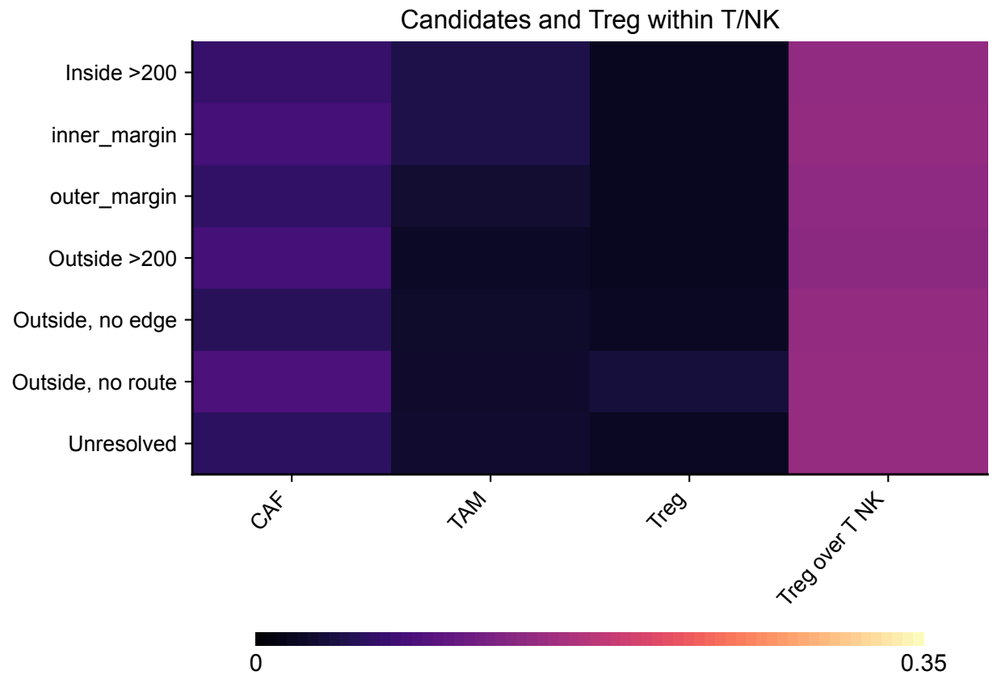


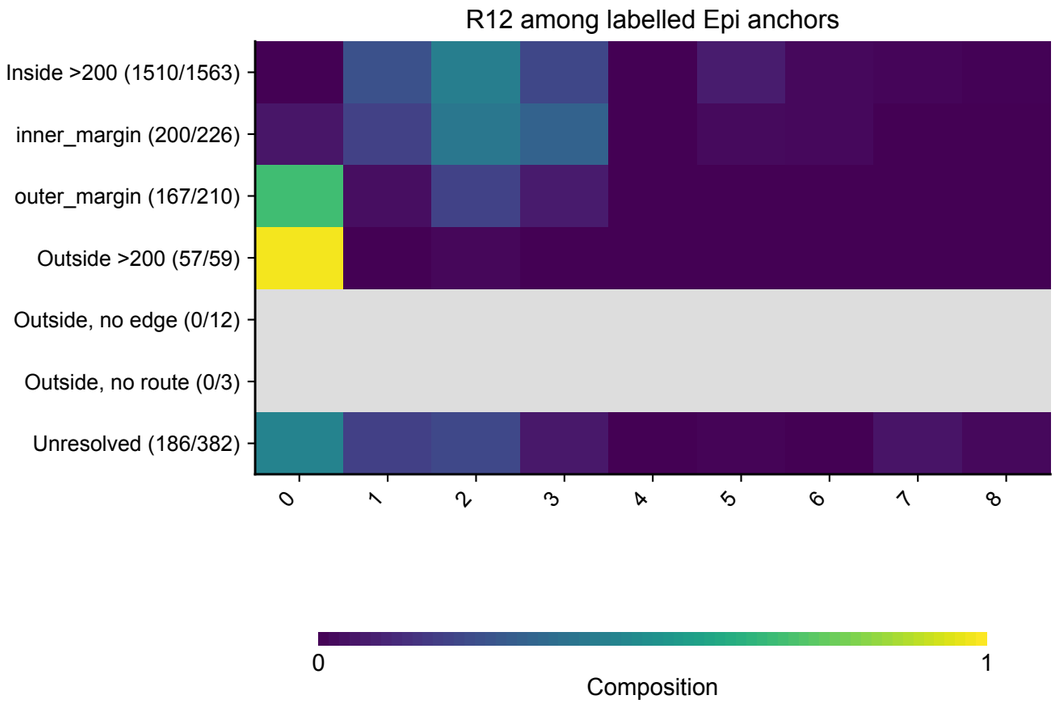
Orange = zero distance
Line = within-section linear fit

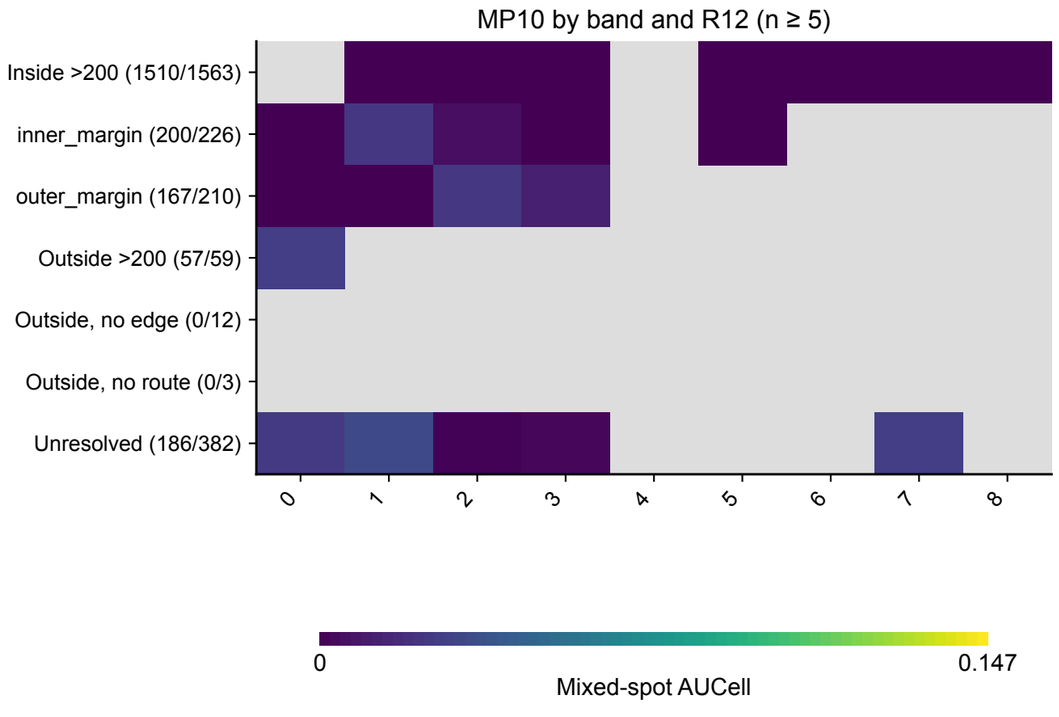
CanDis_2026_Sample5 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





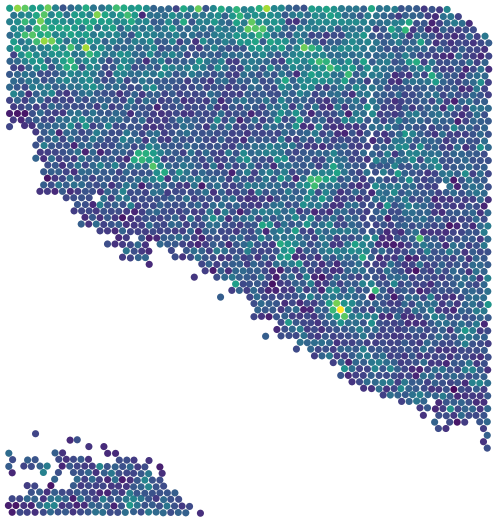




CanDis_2026_Sample5 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

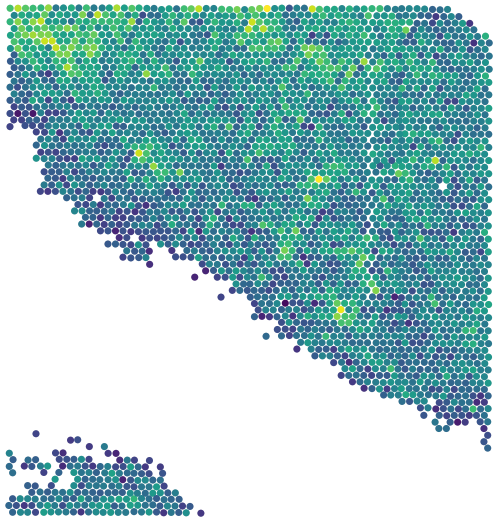
E2F Targets



0 0.152

UCell score

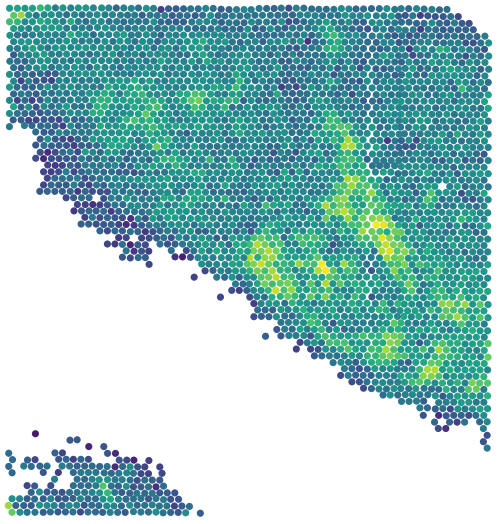
G2M Checkpoint



0 0.134

UCell score

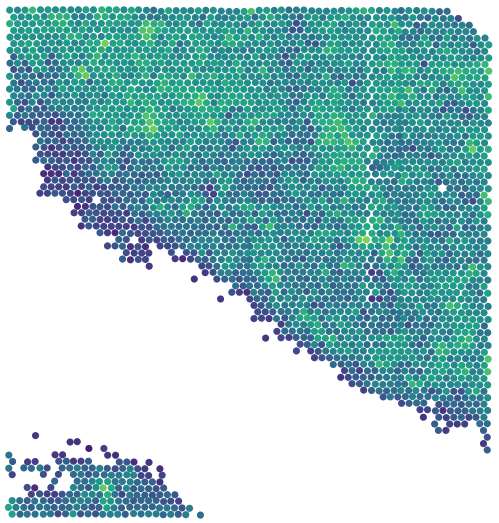
Hypoxia



0 0.211

UCell score

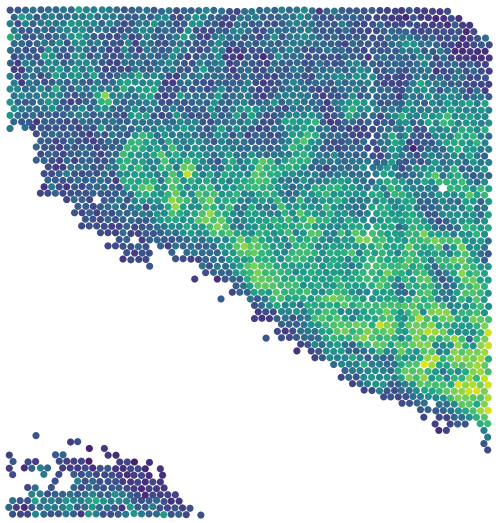
Tnfa Signaling Via Nfkb



0 0.275

UCell score

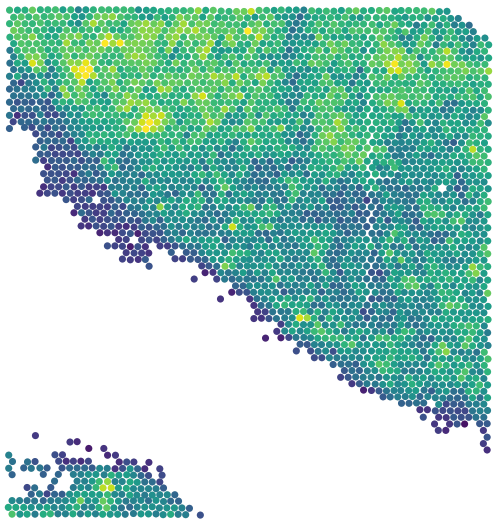
Emt



0 0.386

UCell score

Interferon Gamma Response



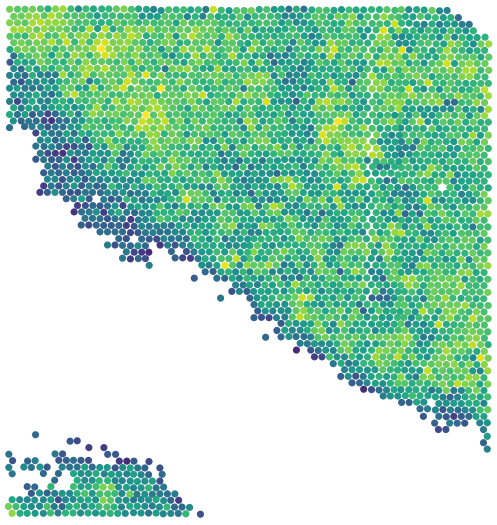
0 0.275

UCell score

CanDis_2026_Sample5 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

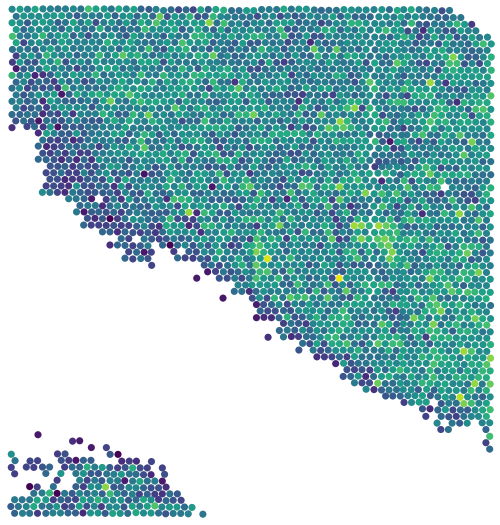
Il2 Stat5 Signaling



0 0.144

UCell score

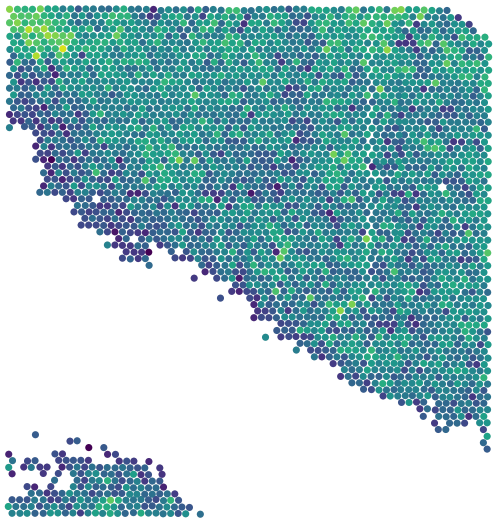
Tgf Beta Signaling



0 0.232

UCell score

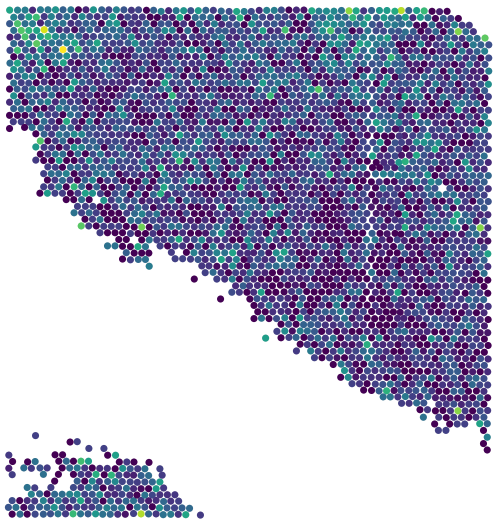
Cholesterol Homeostasis



0 0.21

UCell score

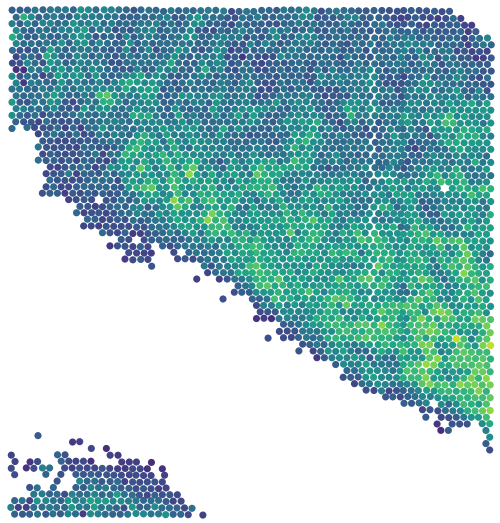
Cholesterol Biosynthesis



0 0.166

UCell score

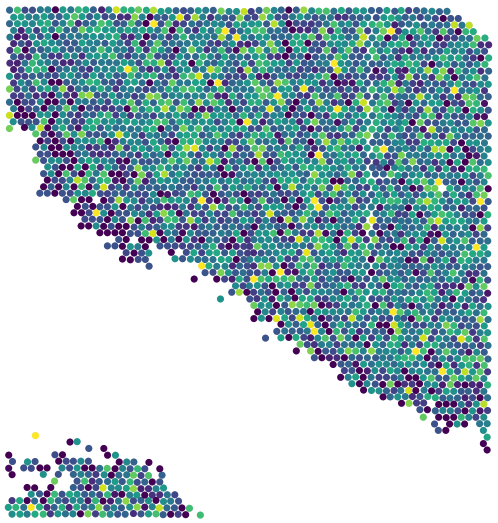
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



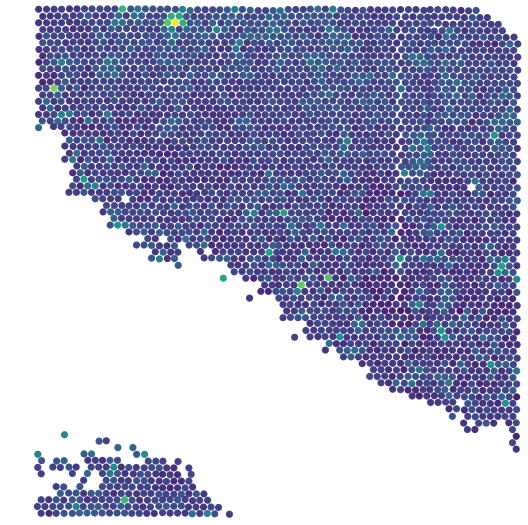
0 0.151

UCell score

CanDis_2026_Sample5 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

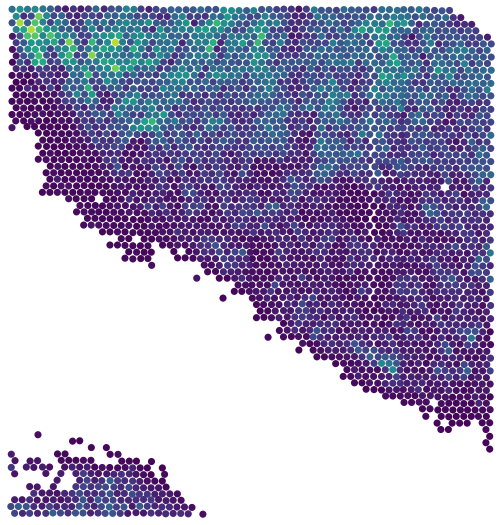
Treg composition



0 0.102

Fraction of all 30

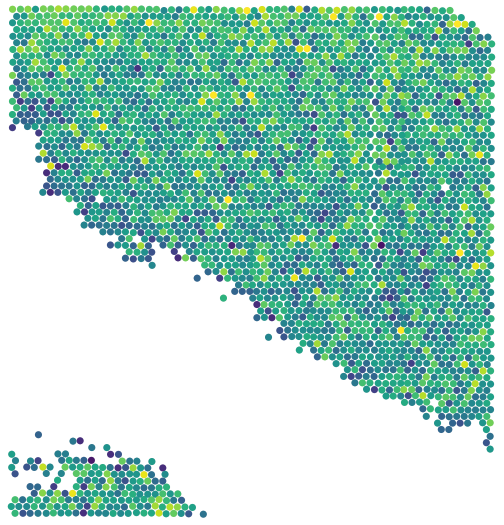
Treg estimated abundance



0 2.85

Posterior mean estimate

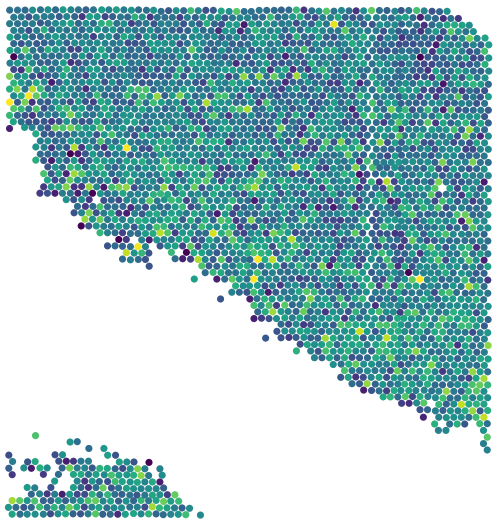
Treg Resting Vs Activated Dn



0 0.0778

UCell score

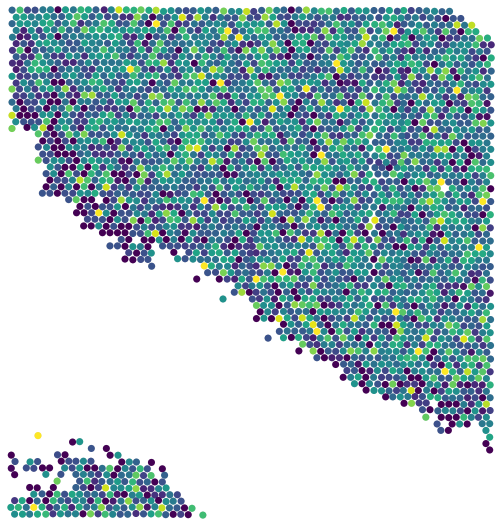
Treg Resting Vs Activated Up



0 0.0609

UCell score

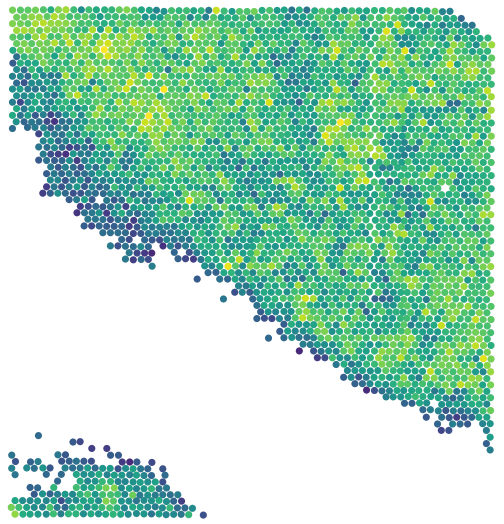
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

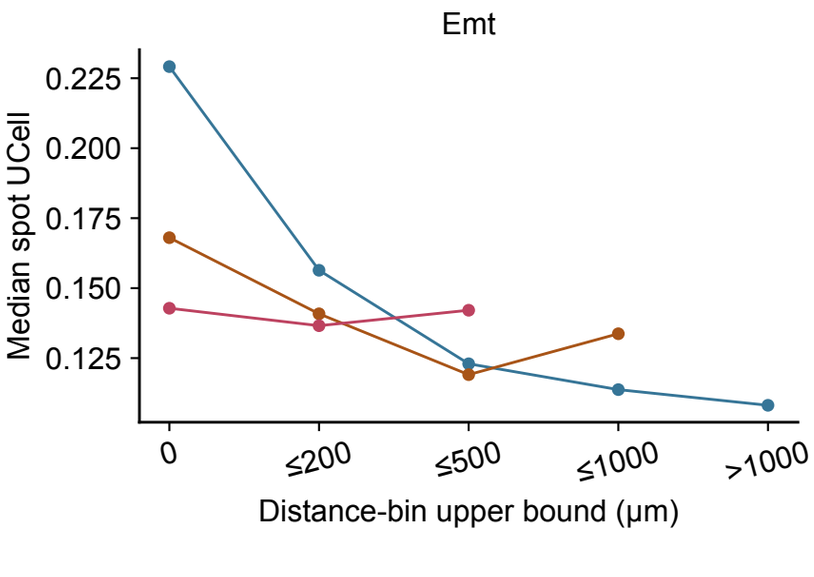


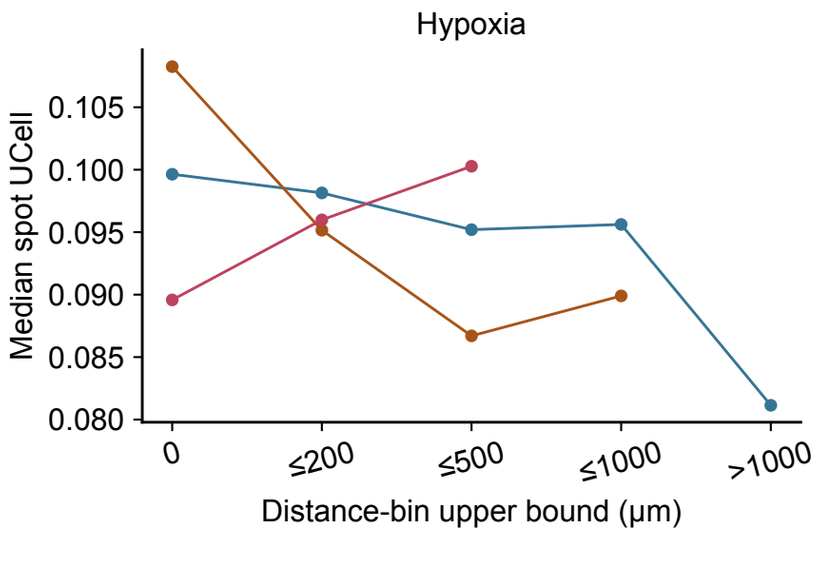
0 0.144

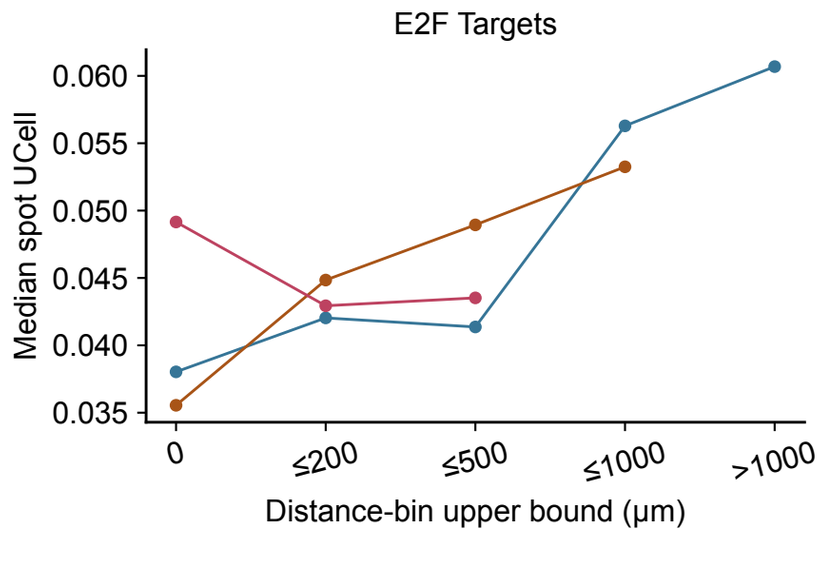
UCell score

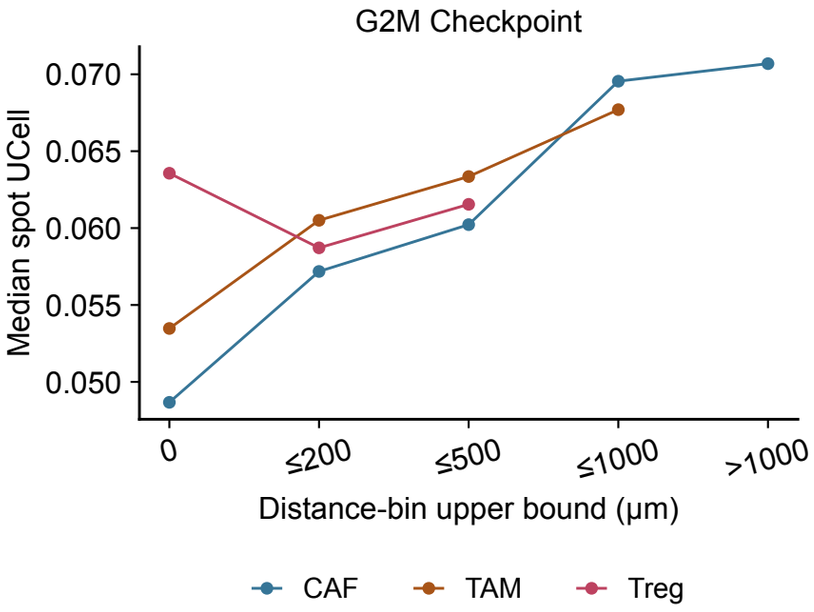
CanDis_2026_Sample5 | Programs by target distance

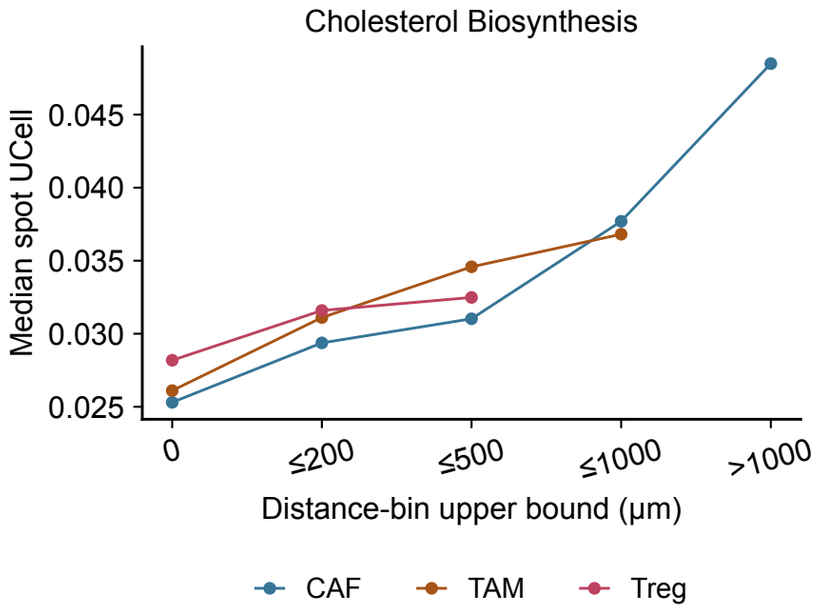
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

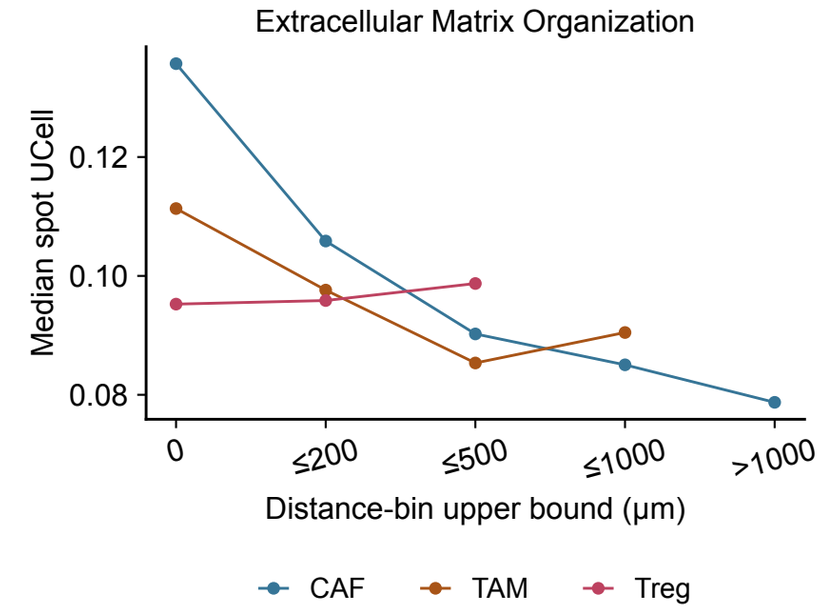






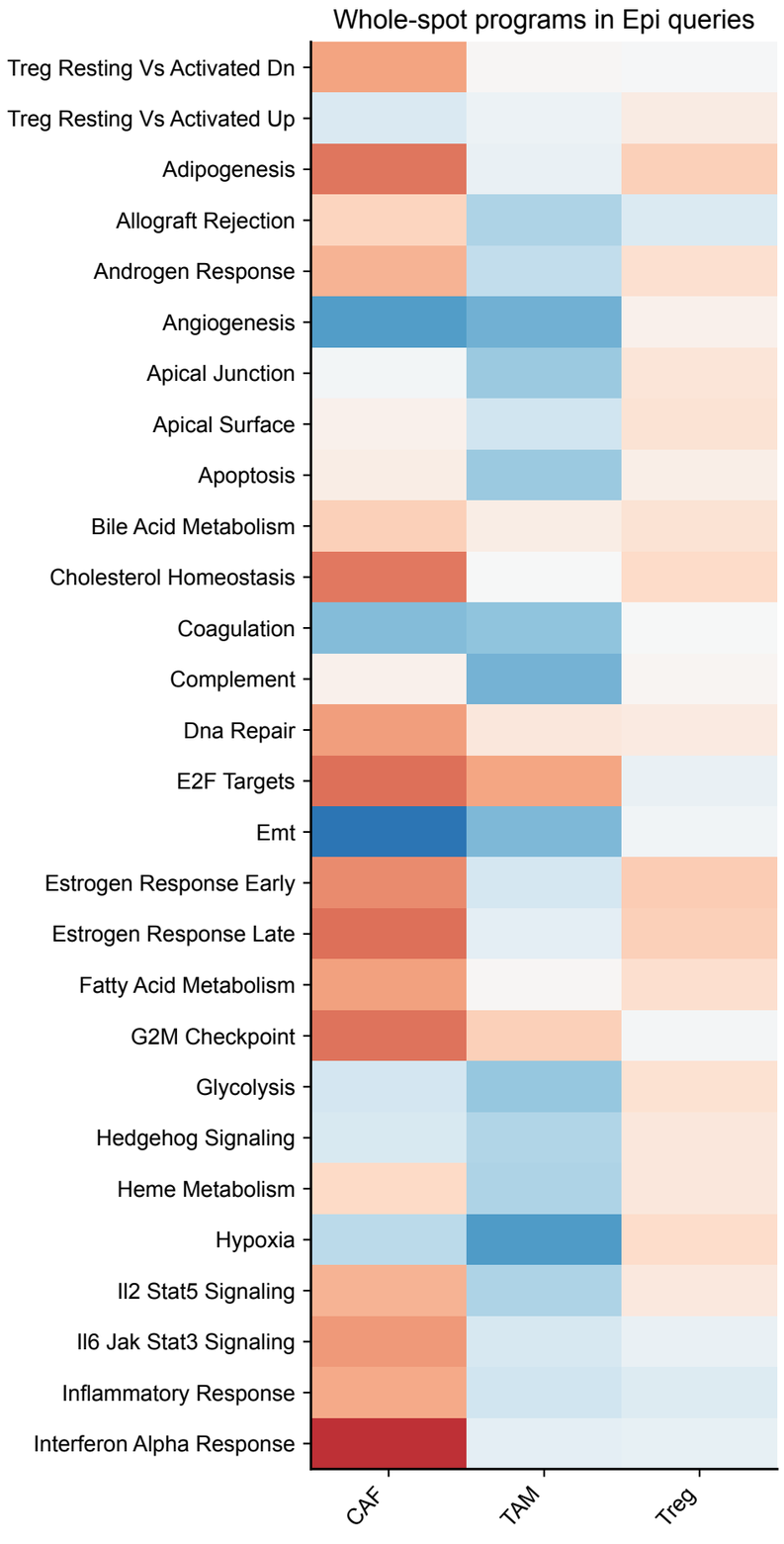


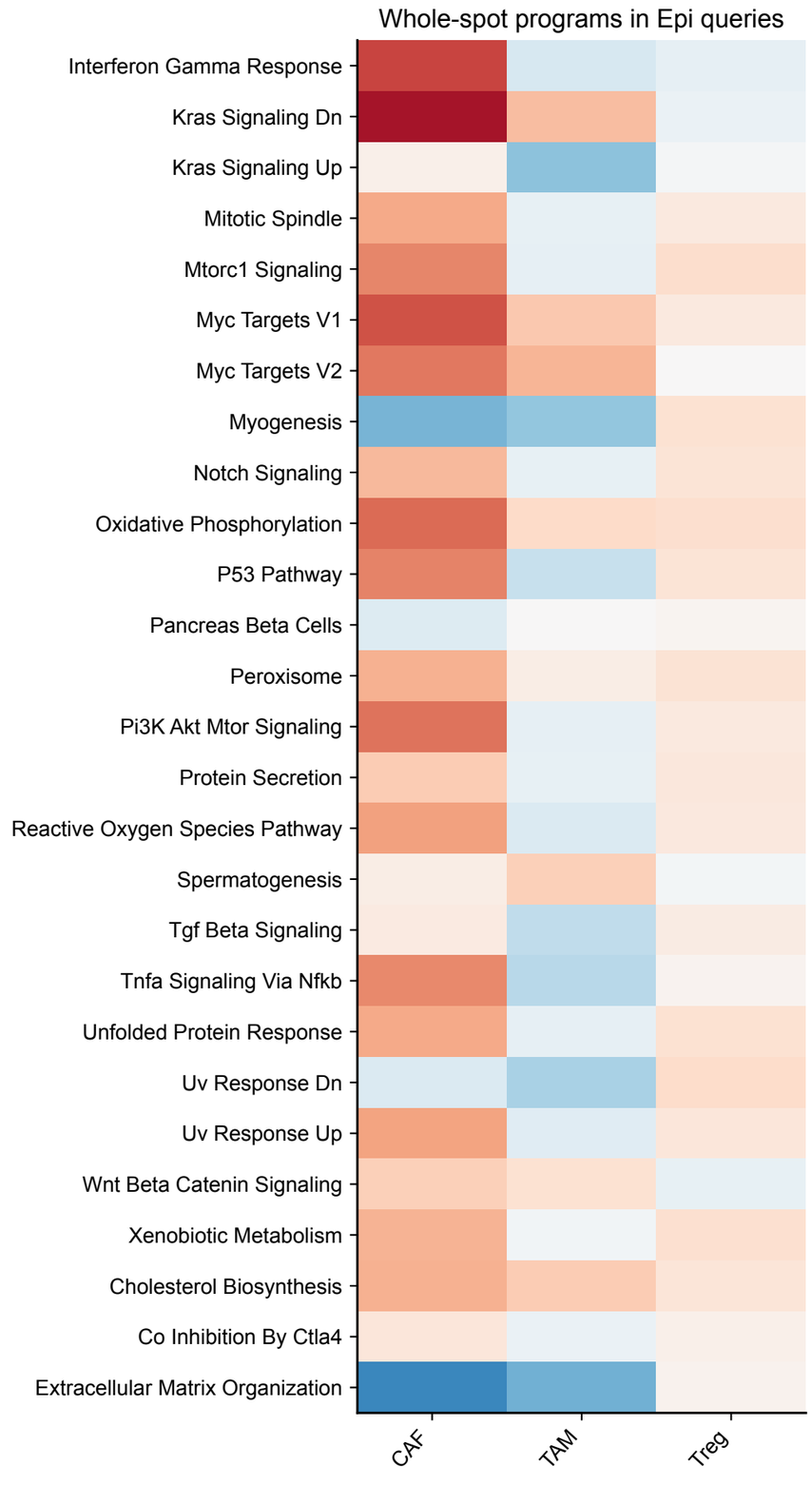




CanDis_2026_Sample5 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

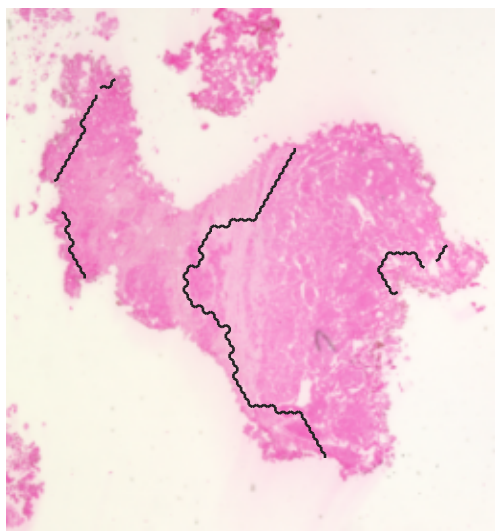




CanDis_2026_Sample6 | Tissue context

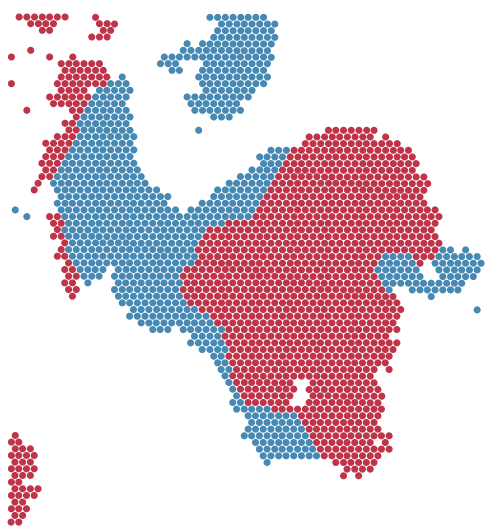
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



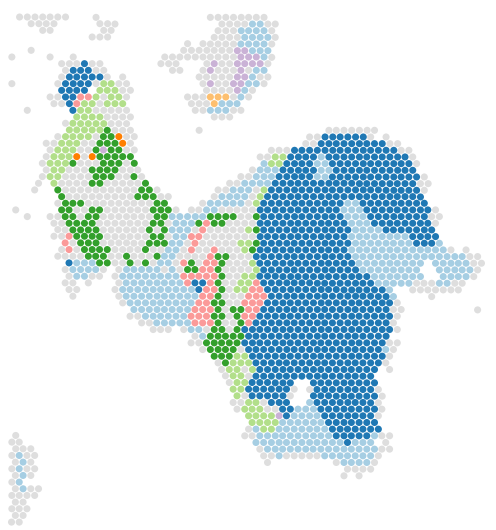
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



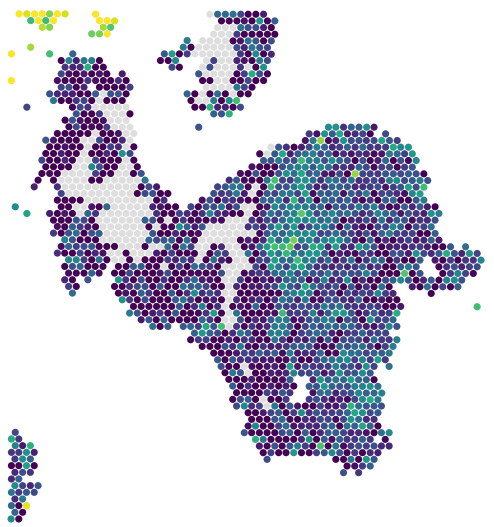
0 2 4 6 8
1 3 5

Epithelial support

0 1

Fraction of all 30

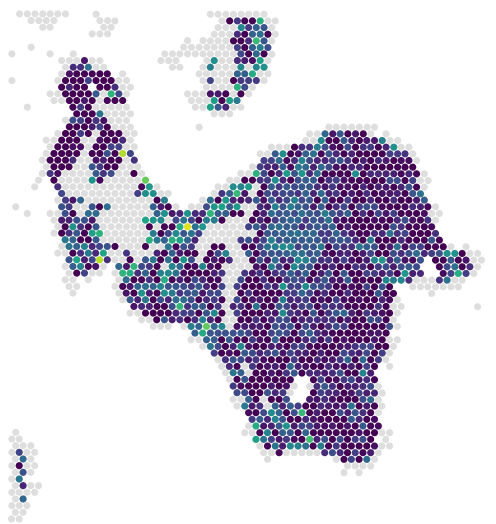
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



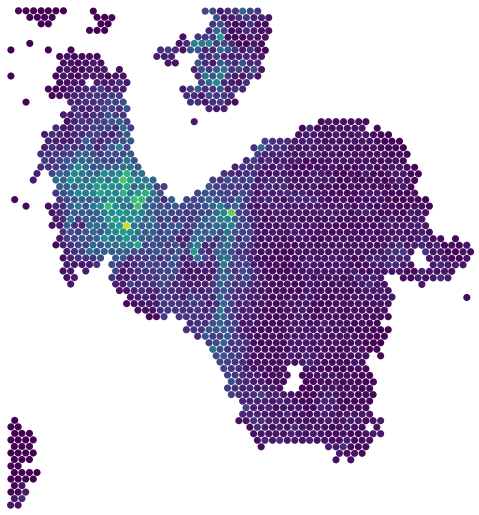
0 0.14

Matched-model AUCell

CanDis_2026_Sample6 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

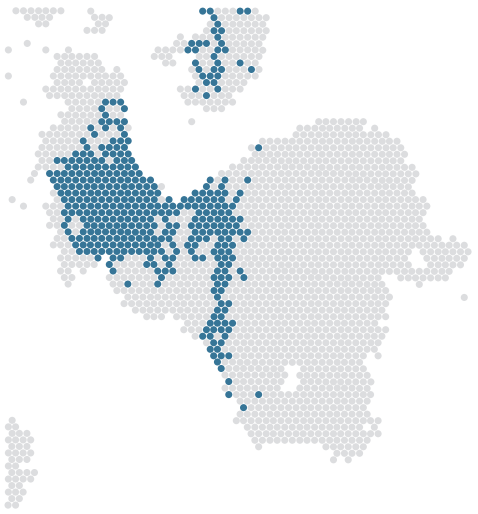
CAF composition



0 0.418

Fraction of all 30

CAF P80 targets



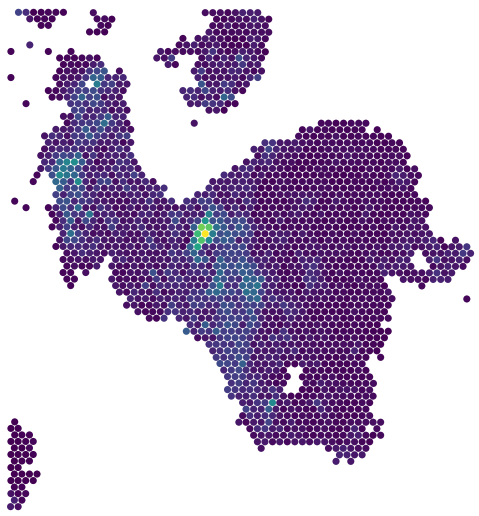
False True

CAF distance

0 3.1e+03

Nearest spot center (μm)

TAM composition



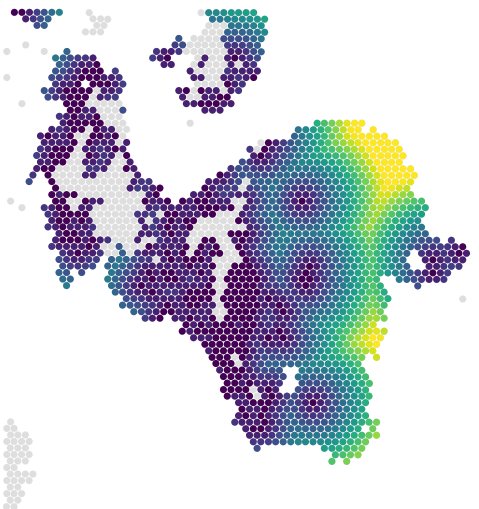
0 0.291

Fraction of all 30

TAM P80 targets

False True

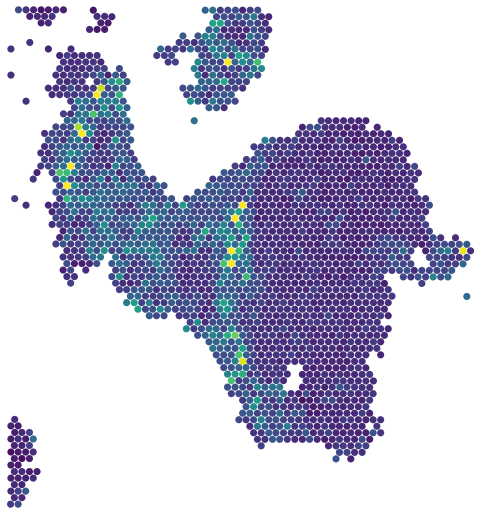
TAM distance



0 1.13e+03

Nearest spot center (μm)

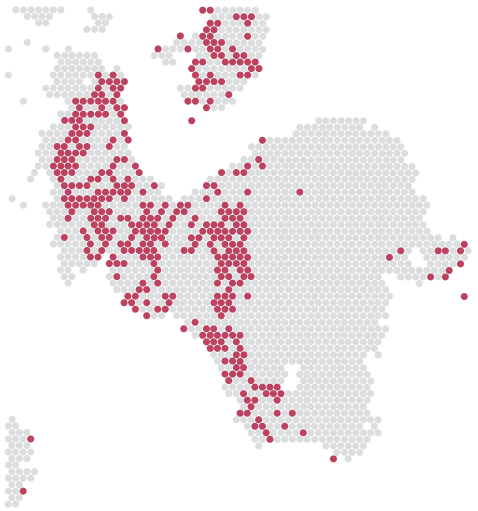
Treg composition



0 0.102

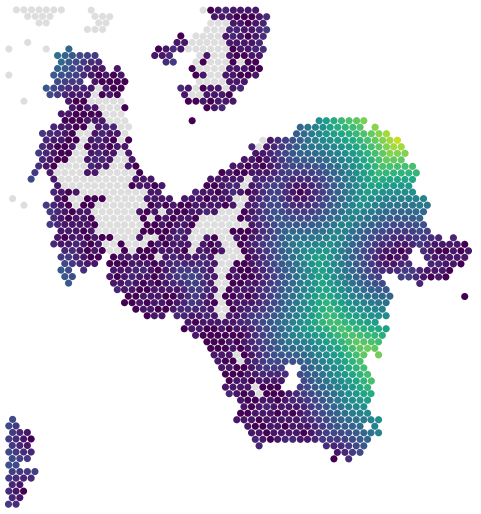
Fraction of all 30

Treg P80 targets



False True

Treg distance

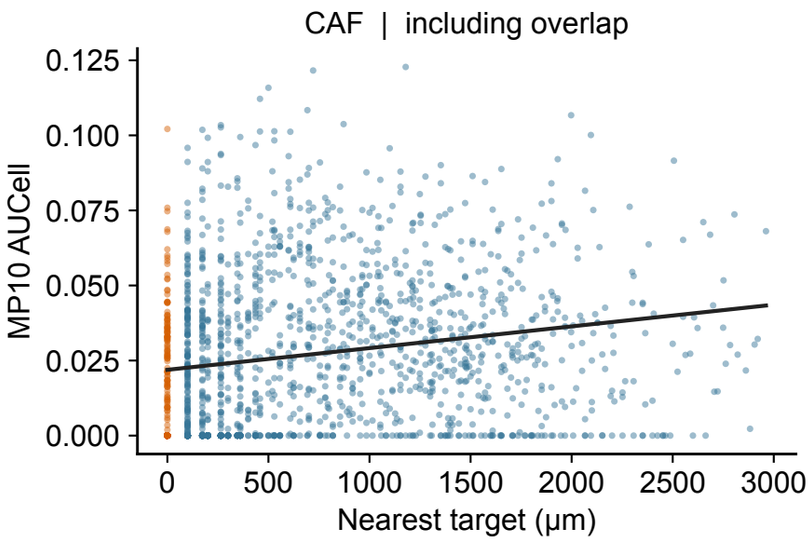


0 1.61e+03

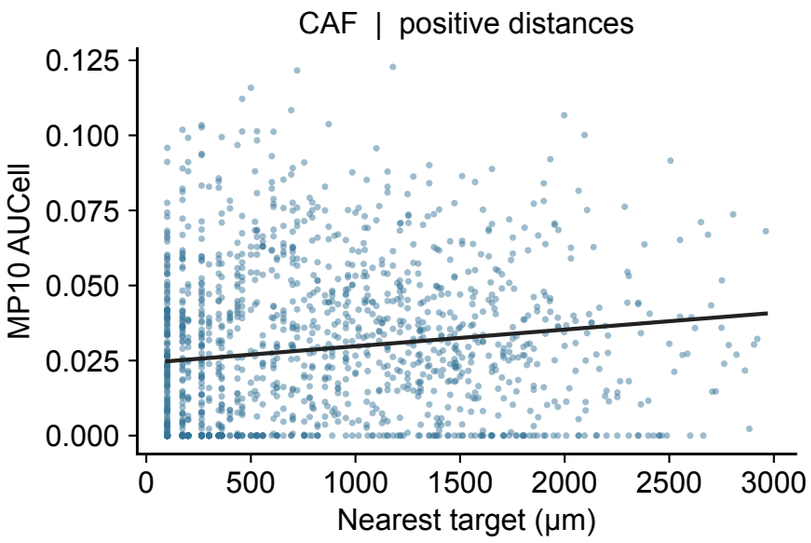
Nearest spot center (μm)

CanDis_2026_Sample6 | MP10 versus nearest target

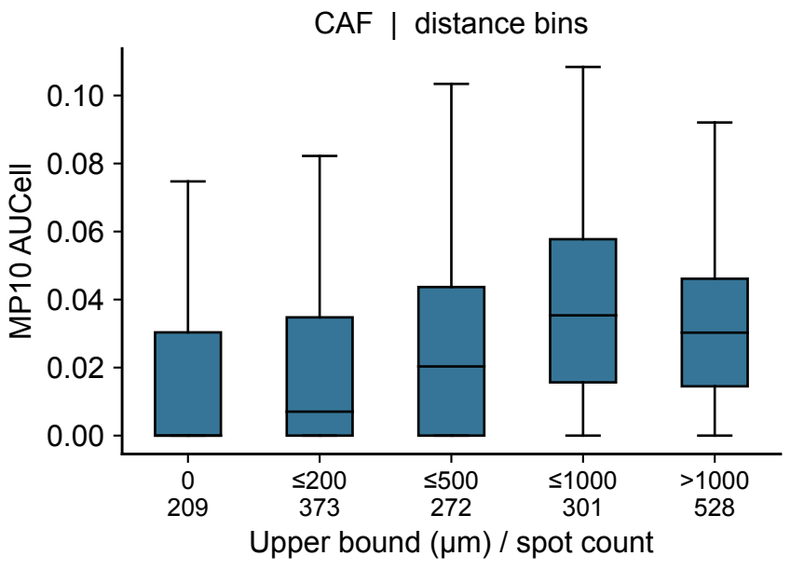
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



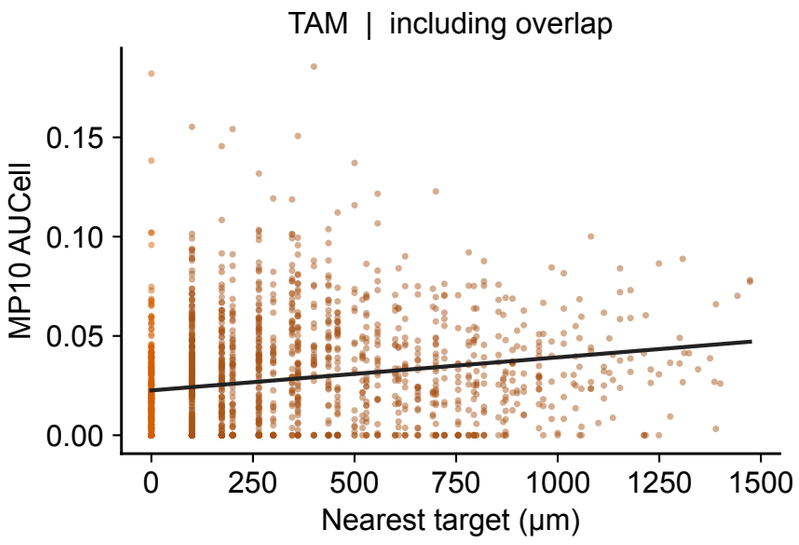
r +0.19 ρ +0.28
n 1683/1753 queries • 397 targets



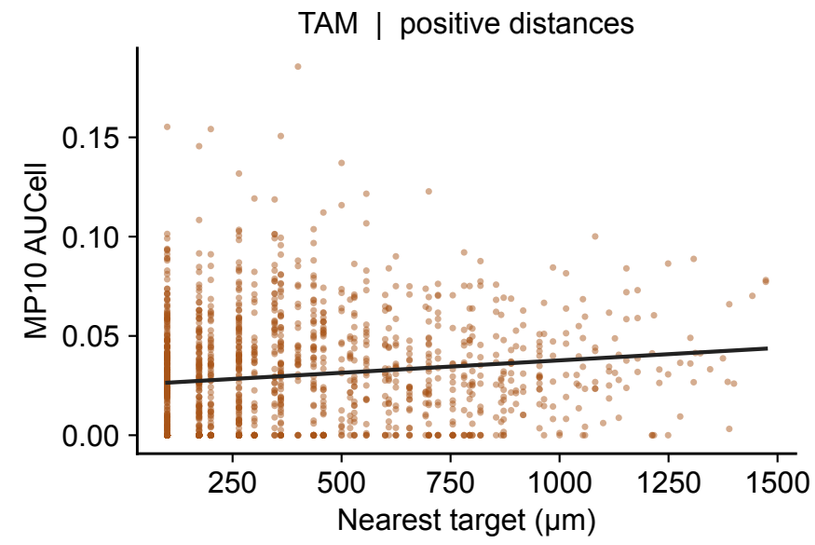
r +0.14 ρ +0.22
n 1474/1753 queries • 397 targets



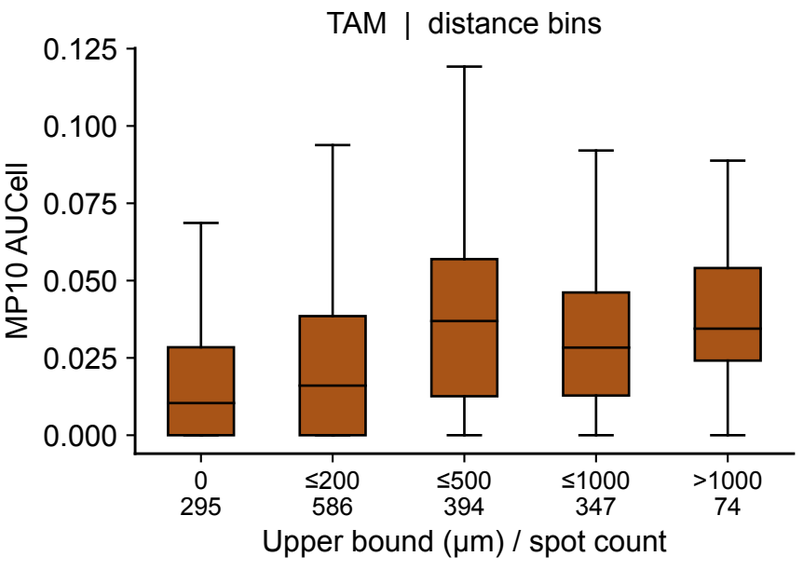
Orange = zero distance
Line = within-section linear fit



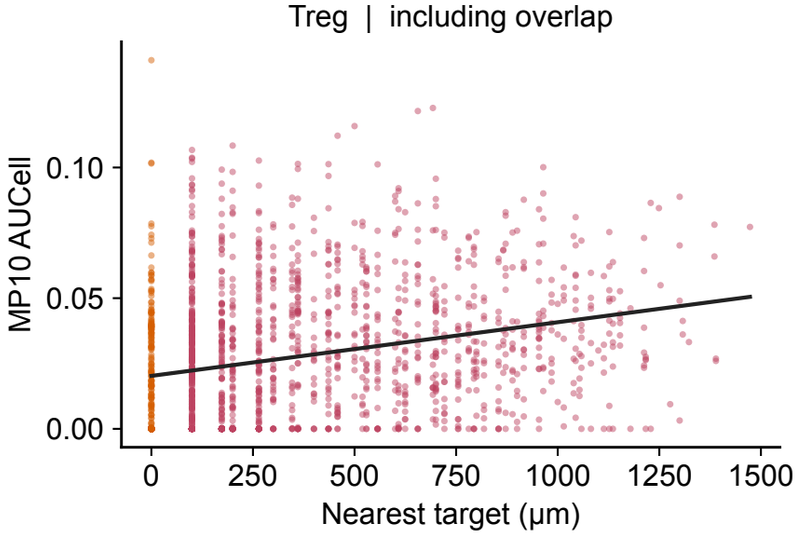
r +0.19 ρ +0.28
n 1696/1753 queries • 397 targets



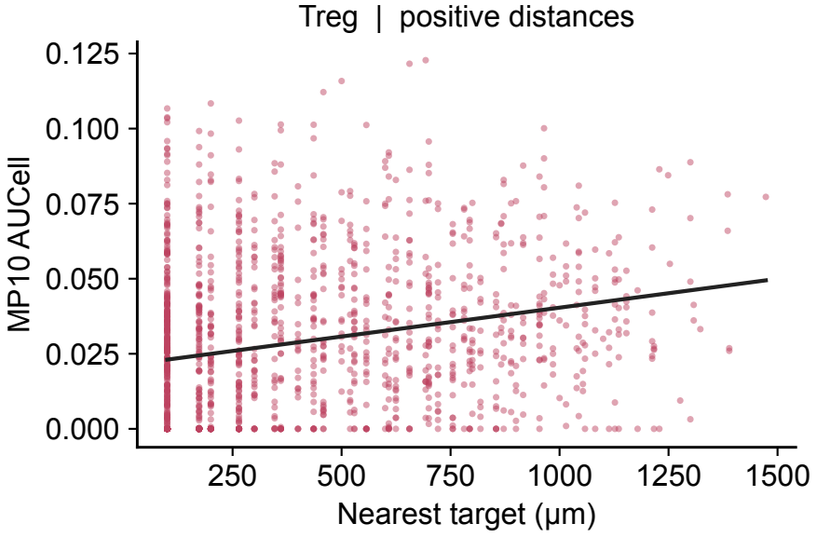
r +0.14 ρ +0.22
n 1401/1753 queries • 397 targets



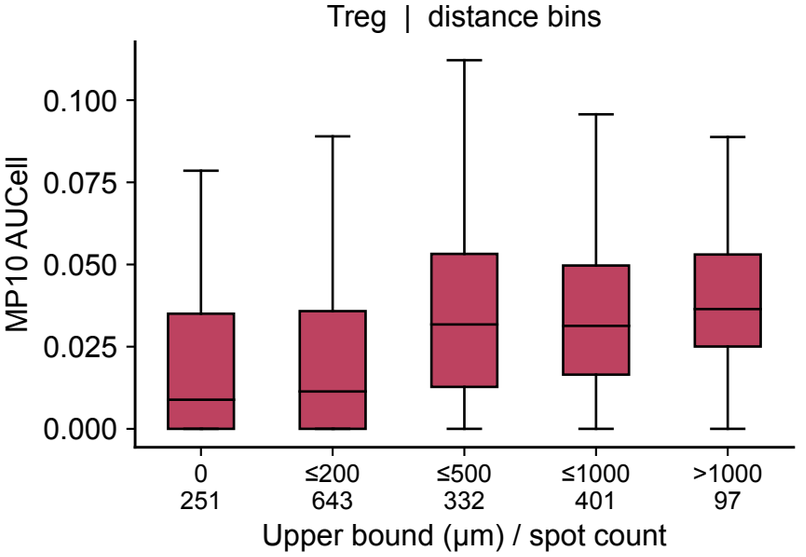
Orange = zero distance
Line = within-section linear fit



r +0.27 ρ +0.33
n 1724/1753 queries • 397 targets



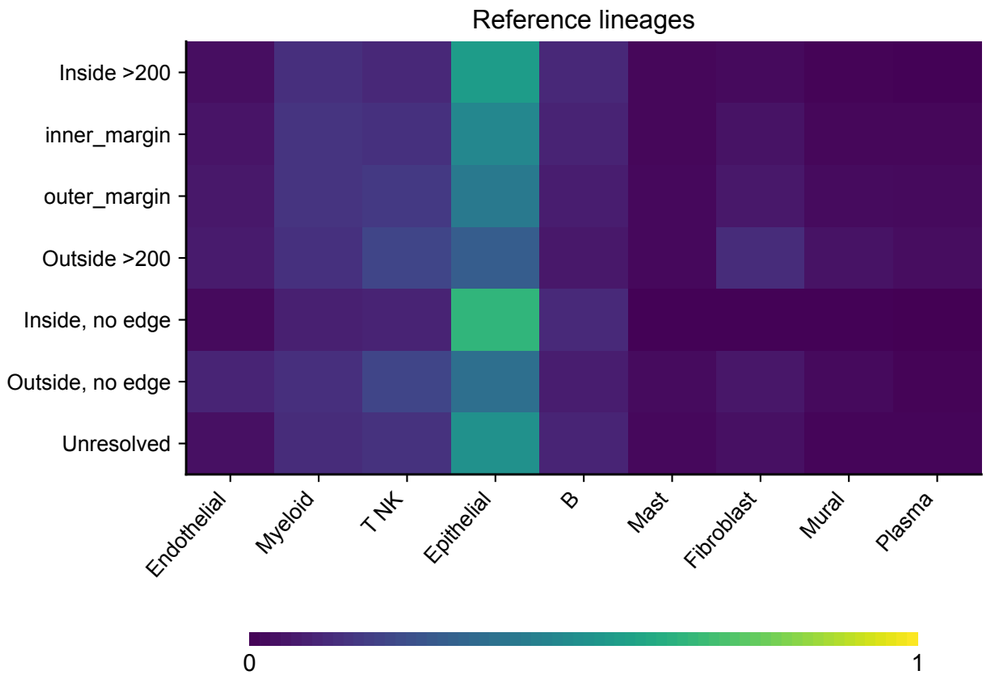
r +0.24 ρ +0.31
n 1473/1753 queries • 397 targets

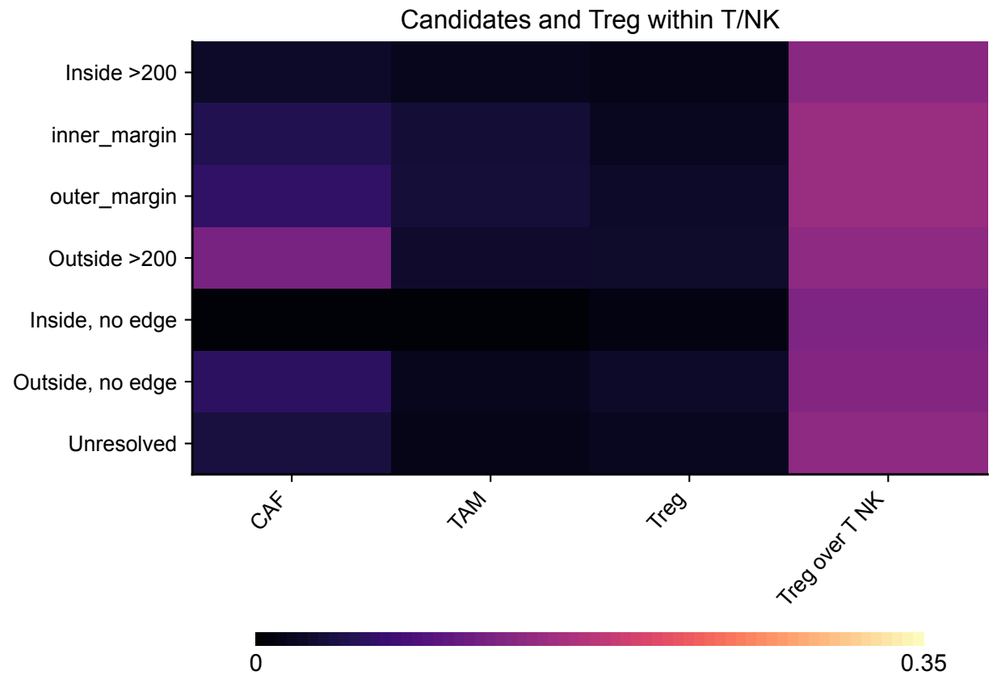


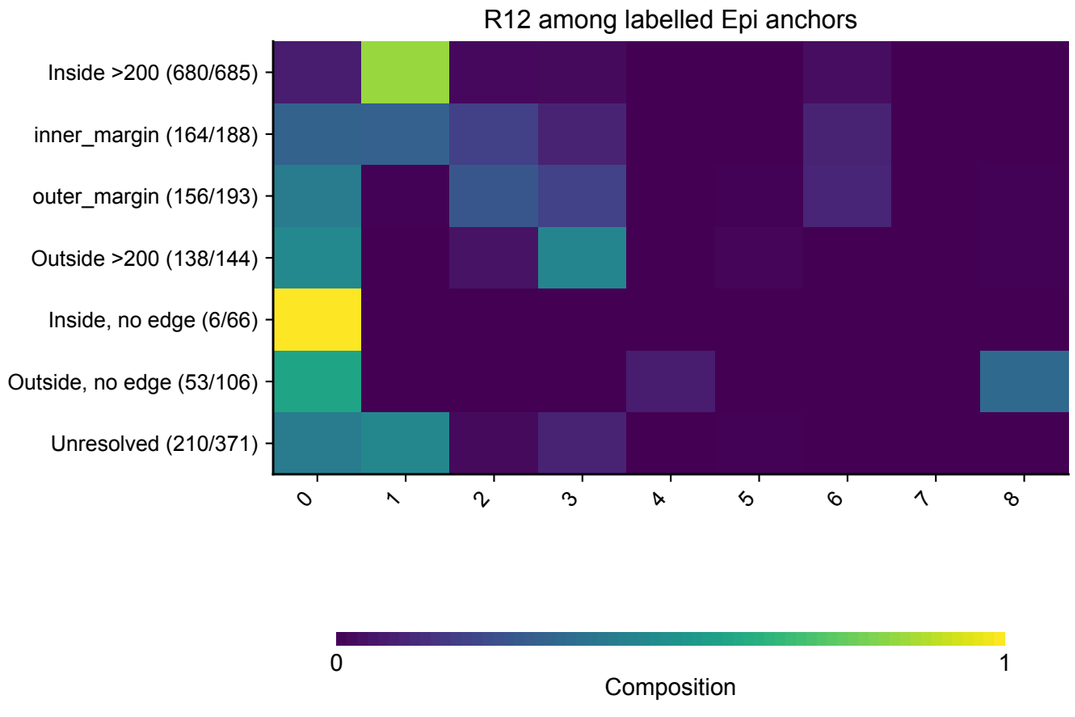
Orange = zero distance
Line = within-section linear fit

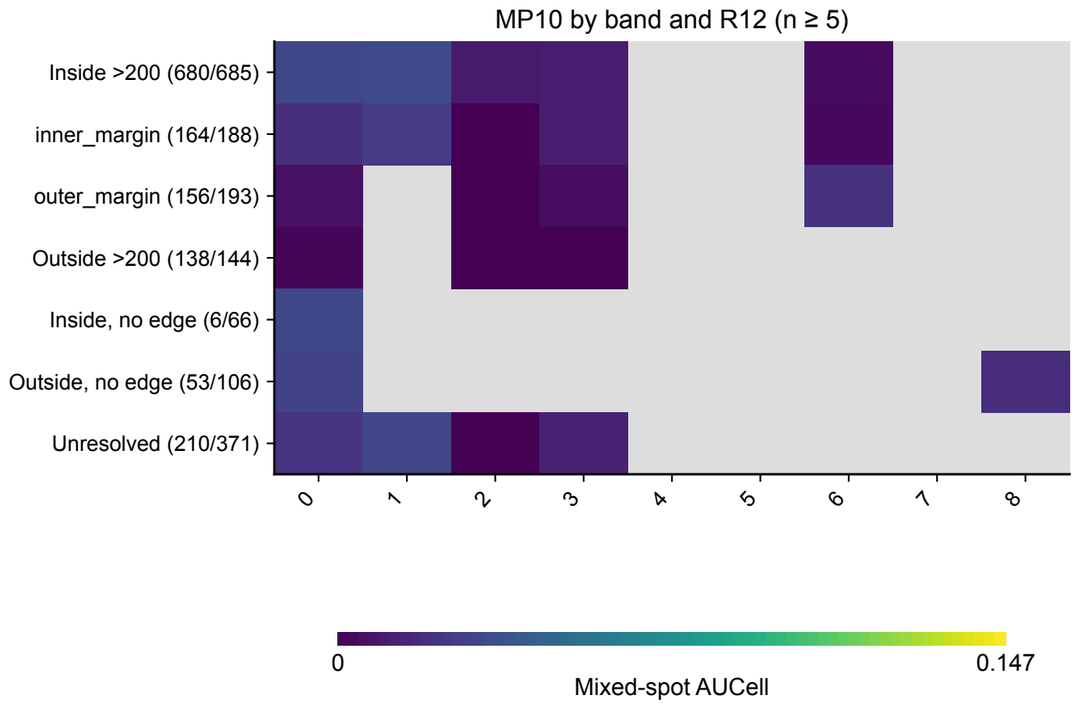
CanDis_2026_Sample6 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





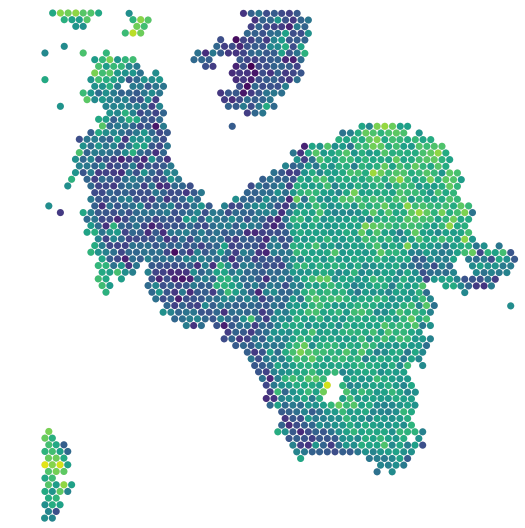




CanDis_2026_Sample6 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets



0 0.152

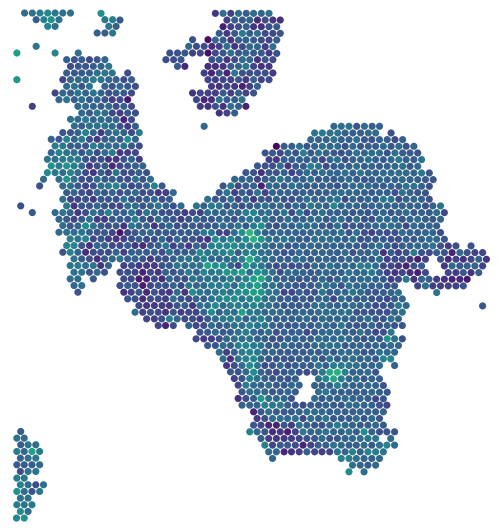
UCell score

G2M Checkpoint

0 0.134

UCell score

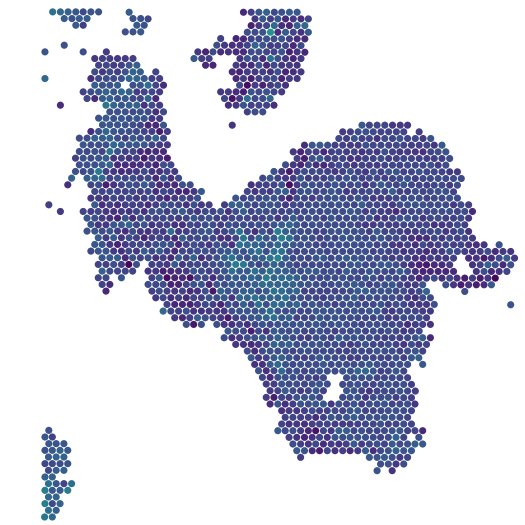
Hypoxia



0 0.211

UCell score

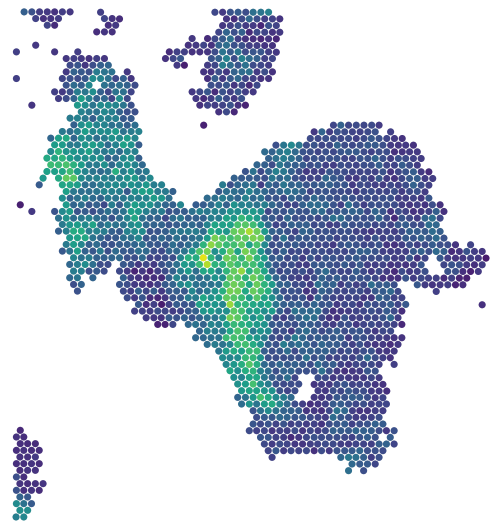
Tnfa Signaling Via Nfkb



0 0.275

UCell score

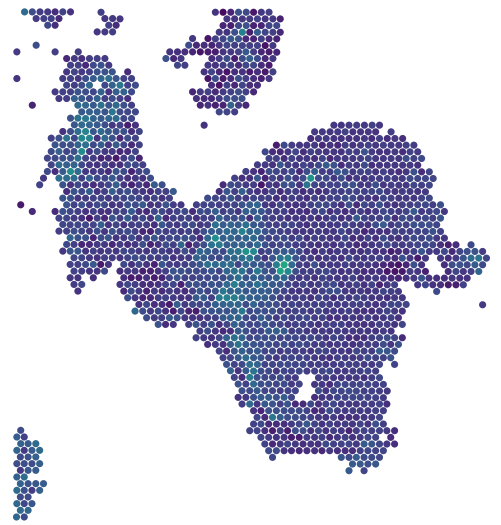
Emt



0 0.386

UCell score

Interferon Gamma Response



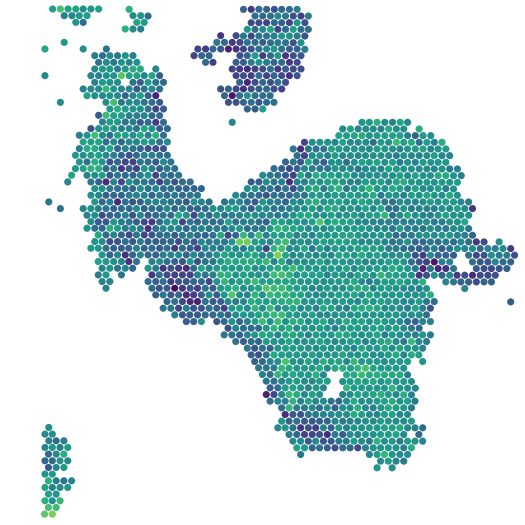
0 0.275

UCell score

CanDis_2026_Sample6 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

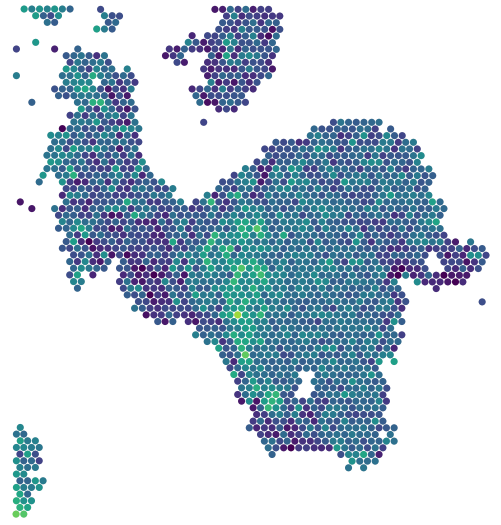
Il2 Stat5 Signaling



0 0.144

UCell score

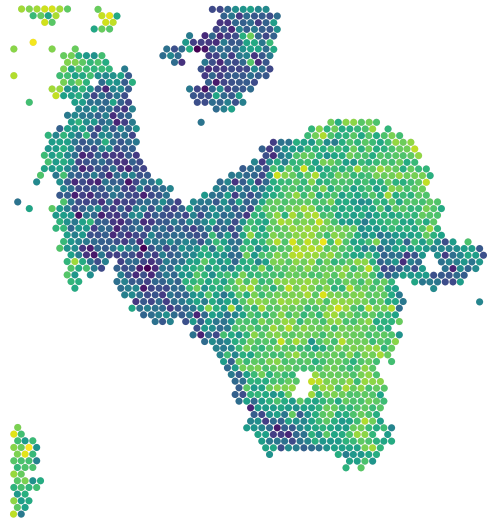
Tgf Beta Signaling



0 0.232

UCell score

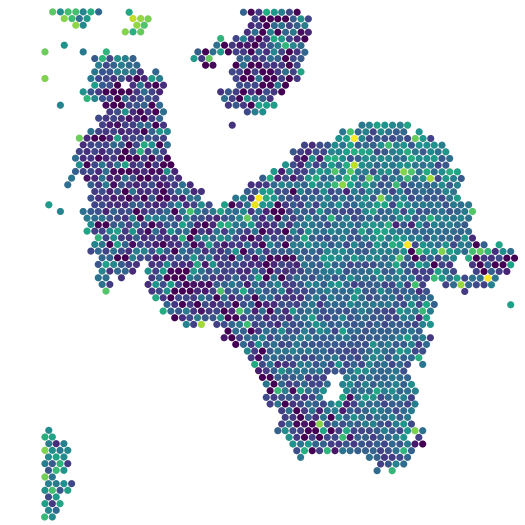
Cholesterol Homeostasis



0 0.21

UCell score

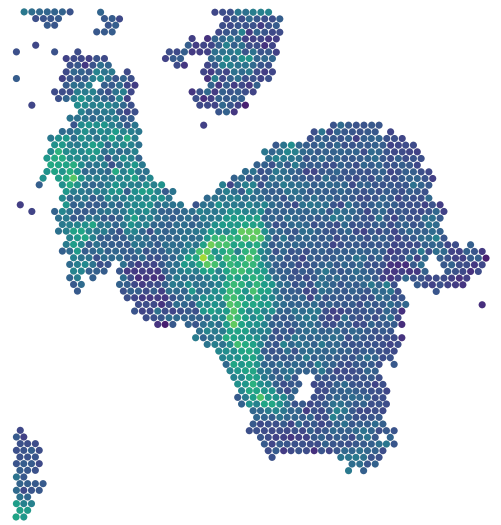
Cholesterol Biosynthesis



0 0.166

UCell score

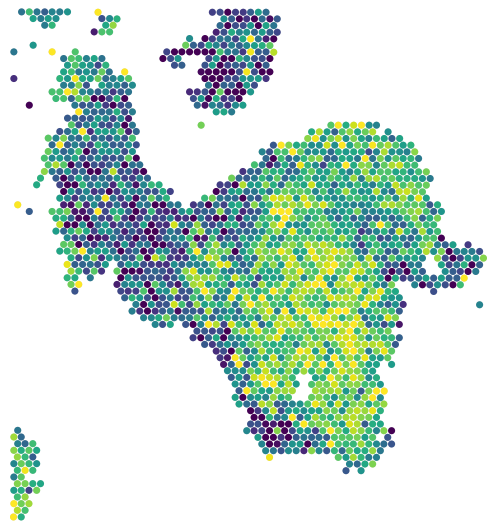
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



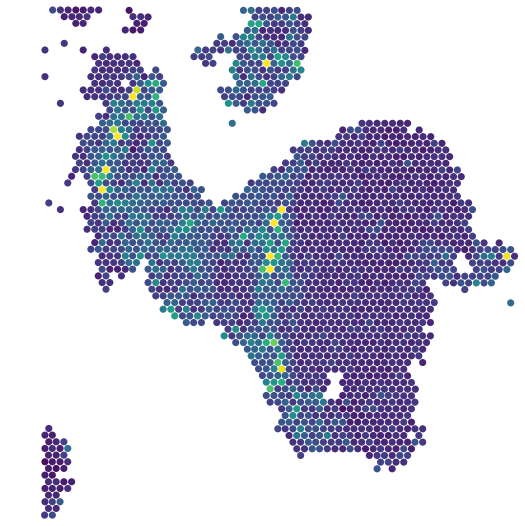
0 0.151

UCell score

CanDis_2026_Sample6 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

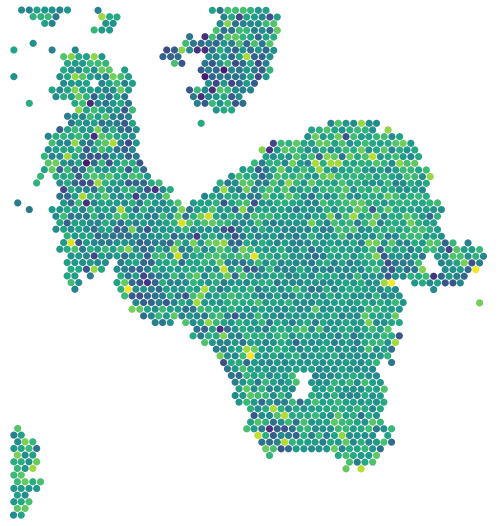
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

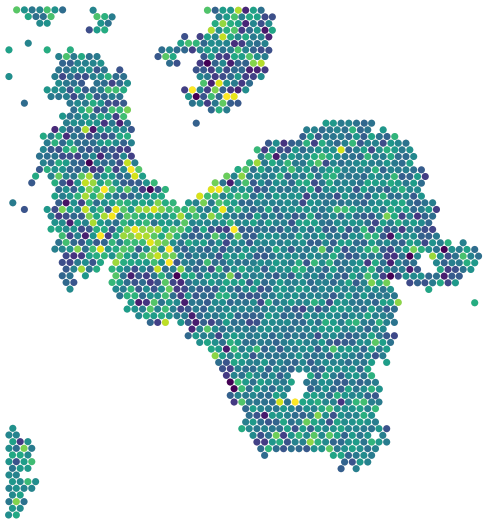
Treg Resting Vs Activated Dn



0 0.0778

UCell score

Treg Resting Vs Activated Up



0 0.0609

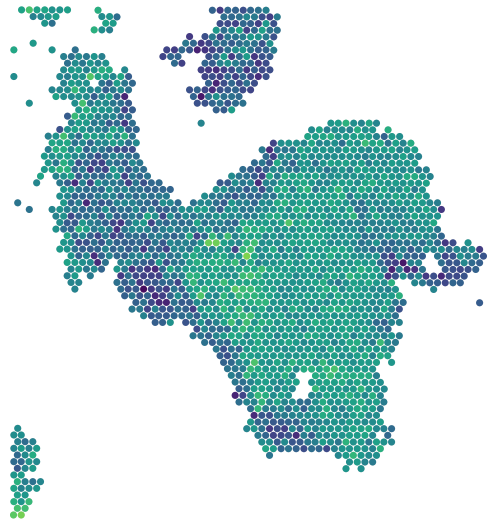
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

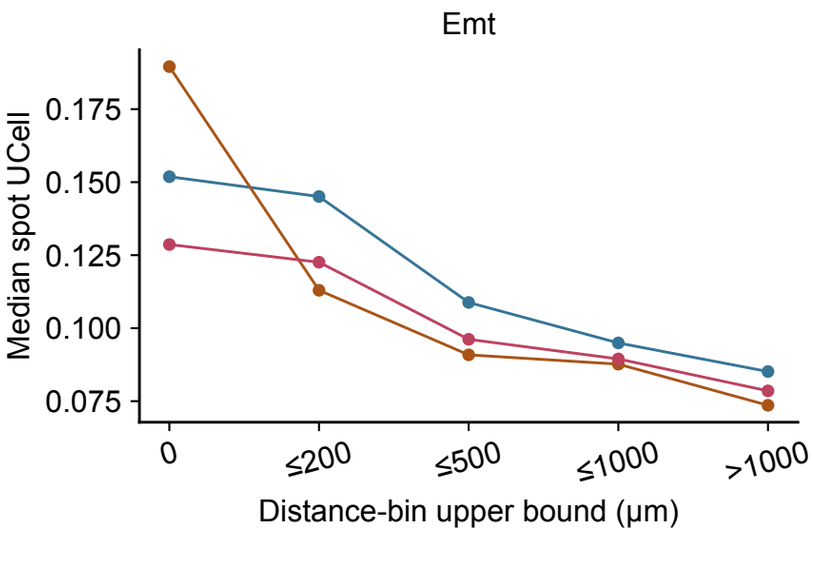


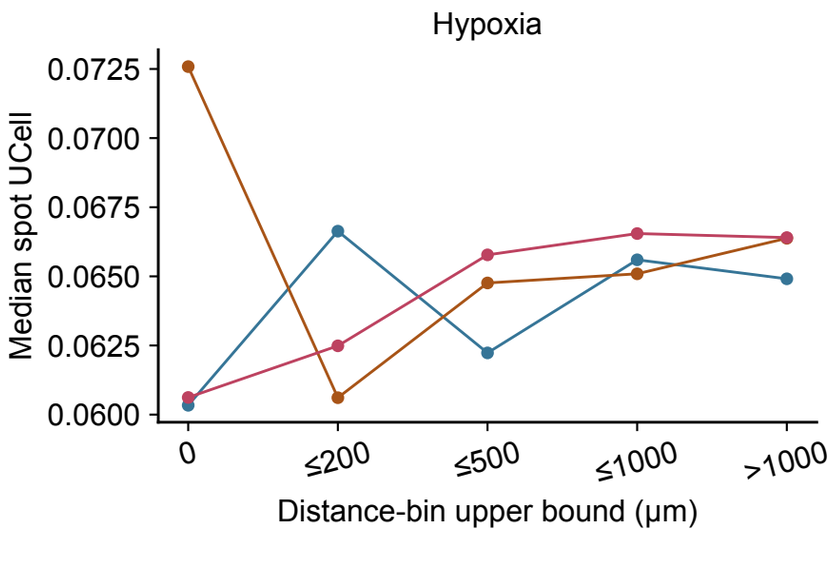
0 0.144

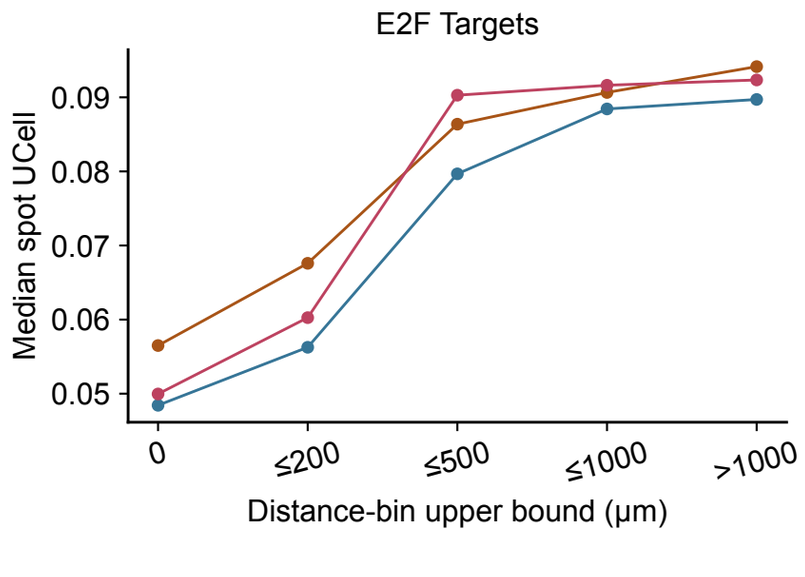
UCell score

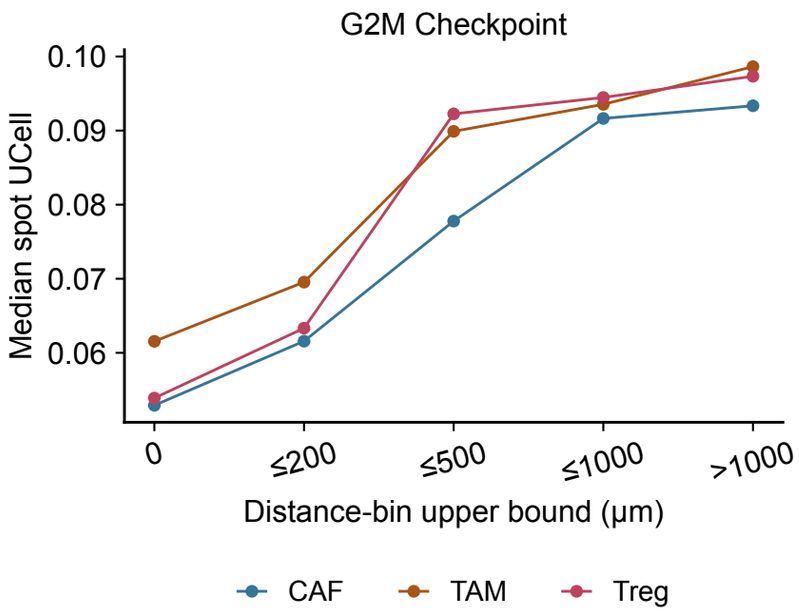
CanDis_2026_Sample6 | Programs by target distance

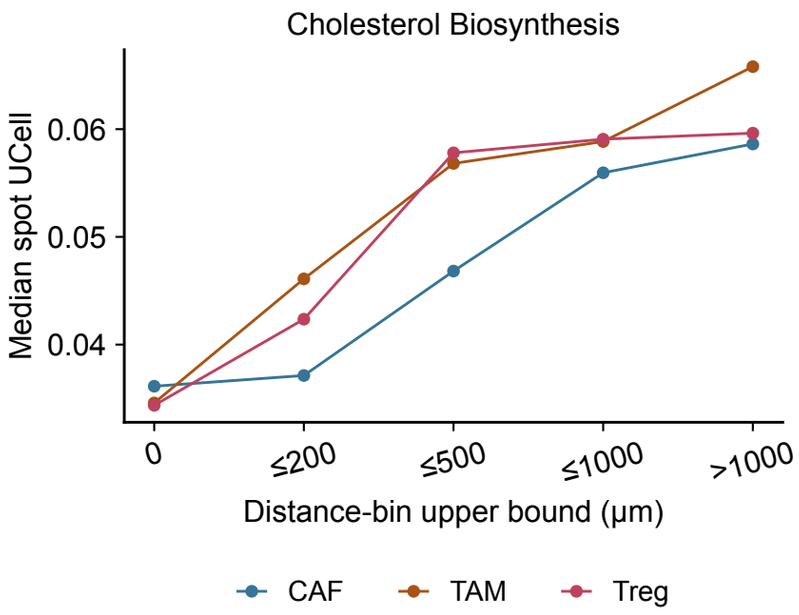
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

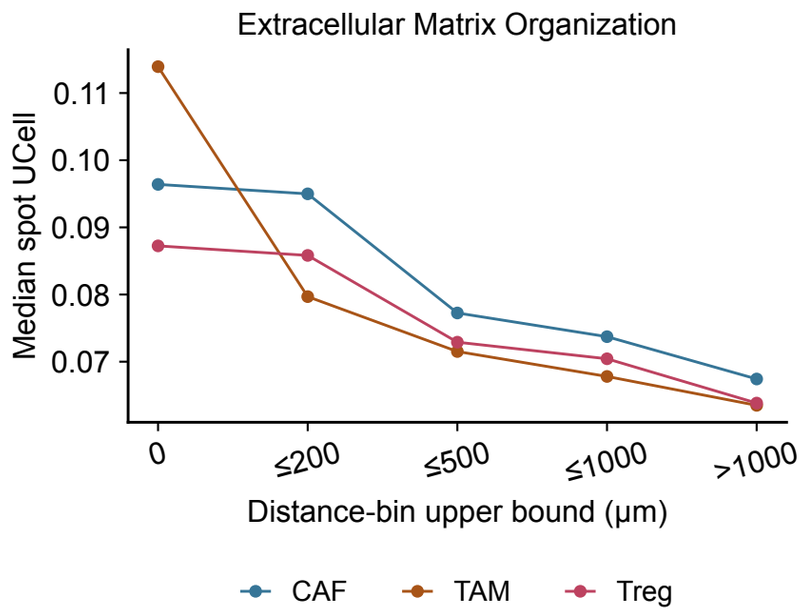






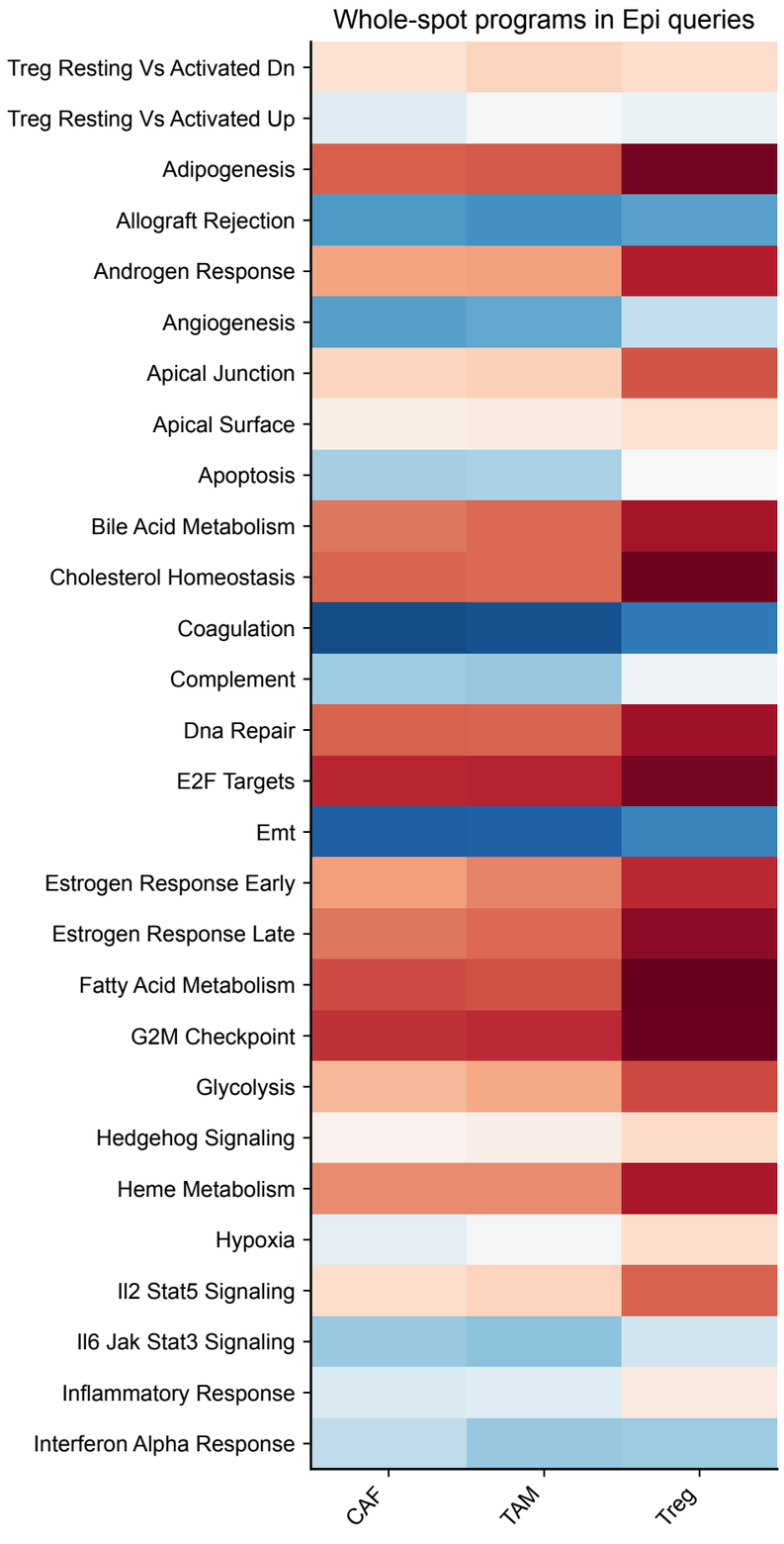


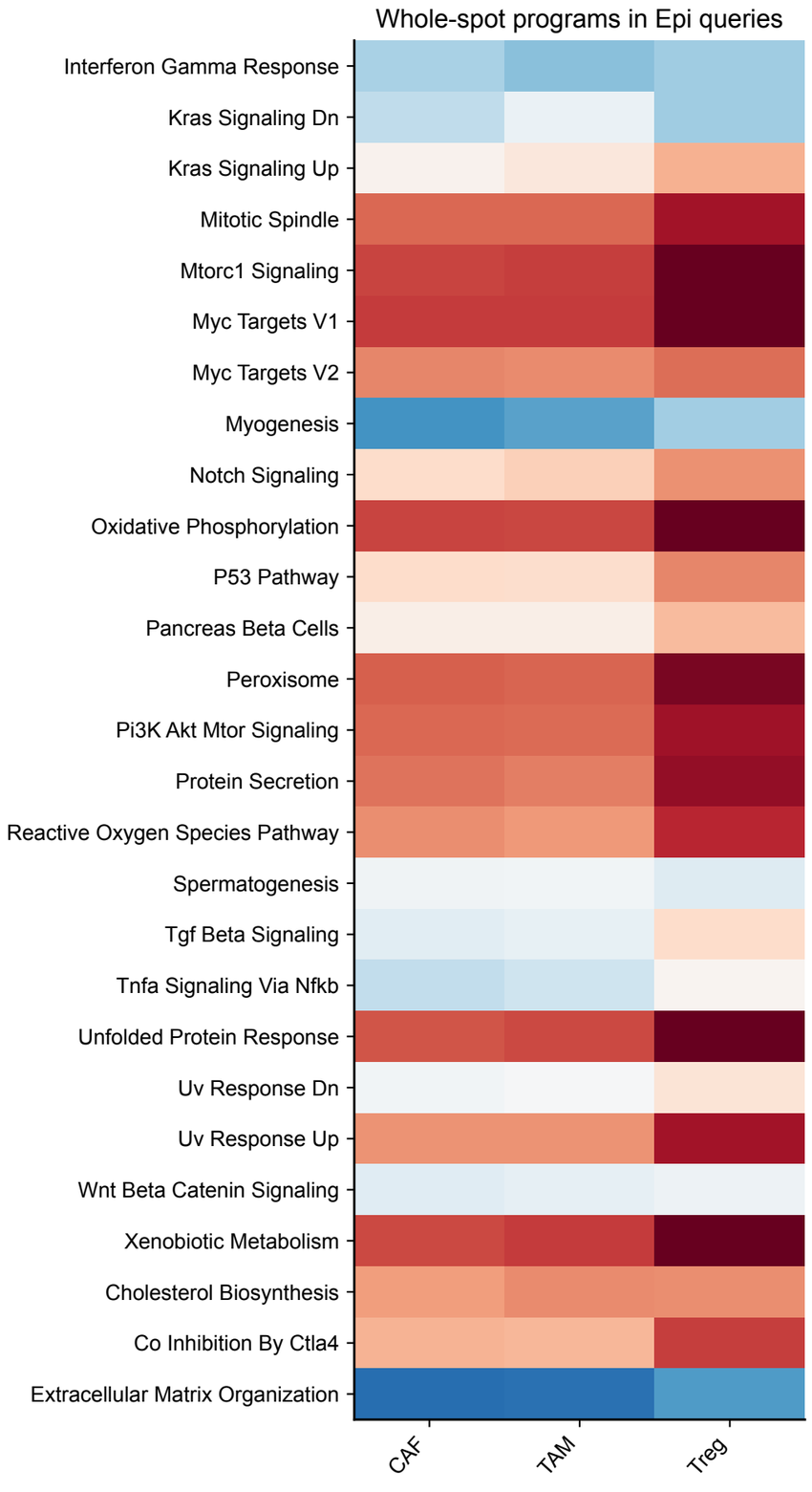




CanDis_2026_Sample6 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

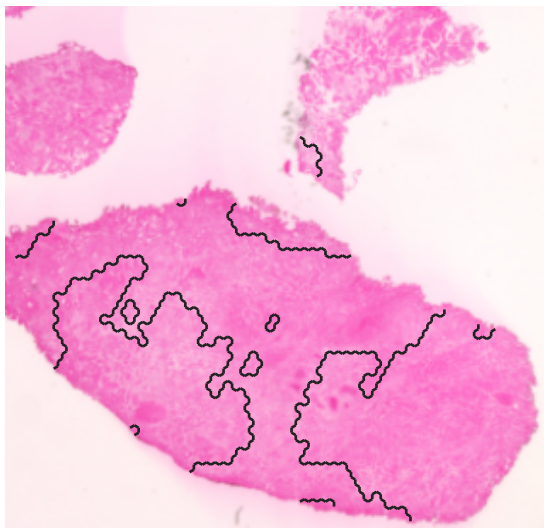




CanDis_2026_Sample7 | Tissue context

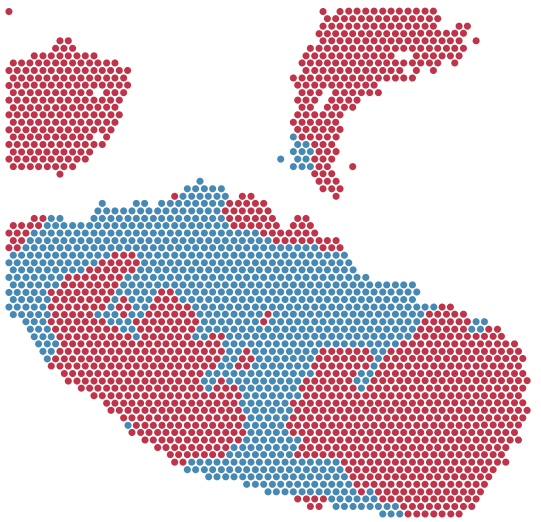
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



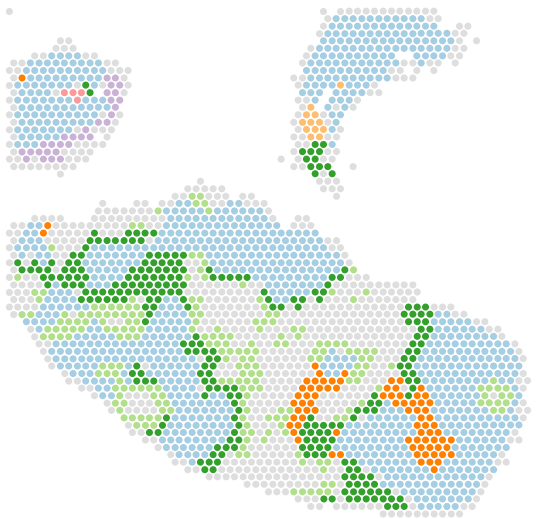
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



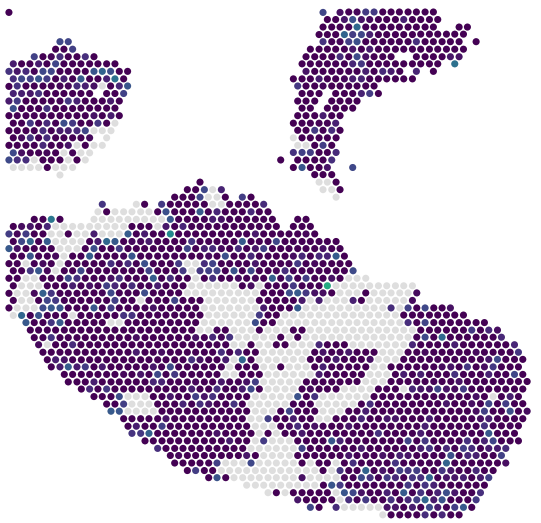
0 3 5 6 8
2 4

Epithelial support

0 1

Fraction of all 30

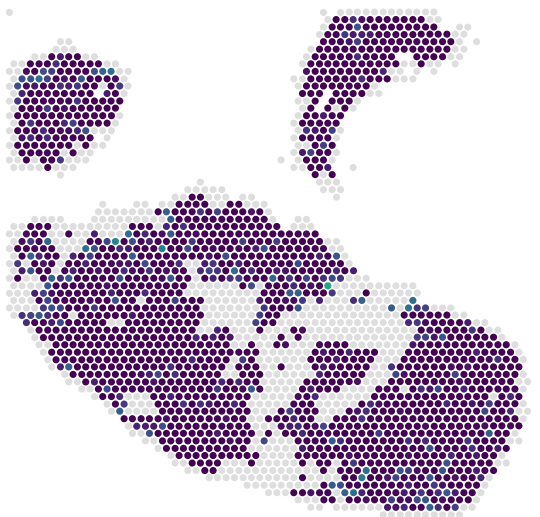
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



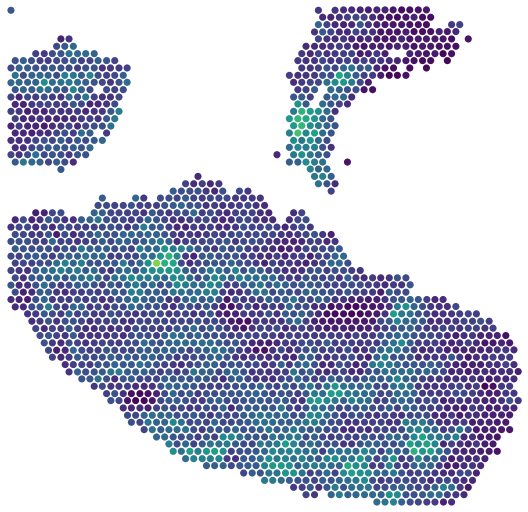
0 0.14

Matched-model AUCell

CanDis_2026_Sample7 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

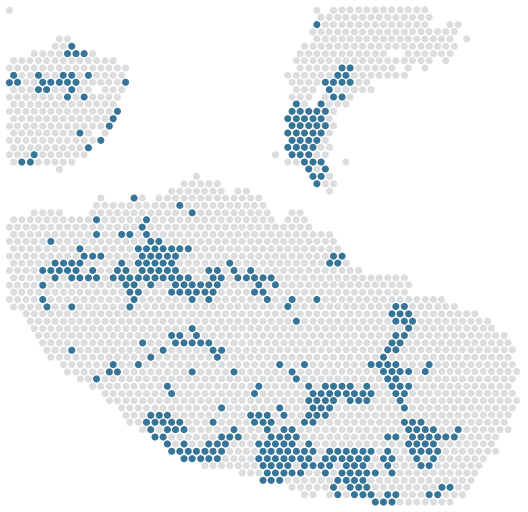
CAF composition



0 0.418

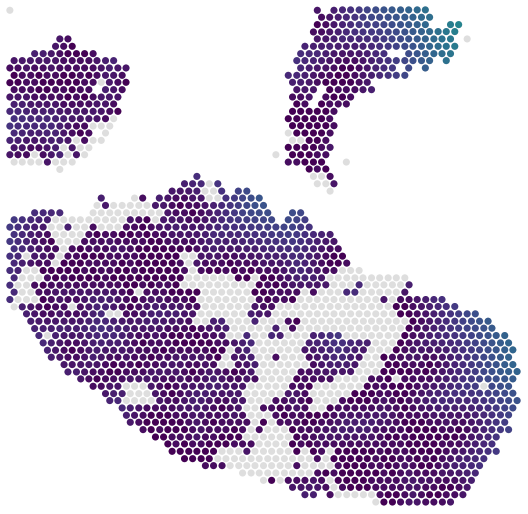
Fraction of all 30

CAF P80 targets



False True

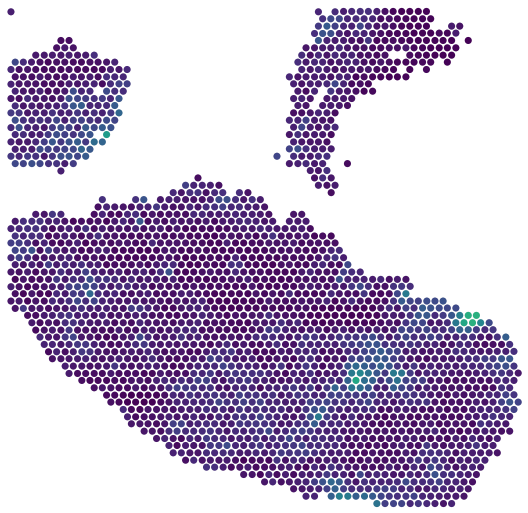
CAF distance



0 3.1e+03

Nearest spot center (µm)

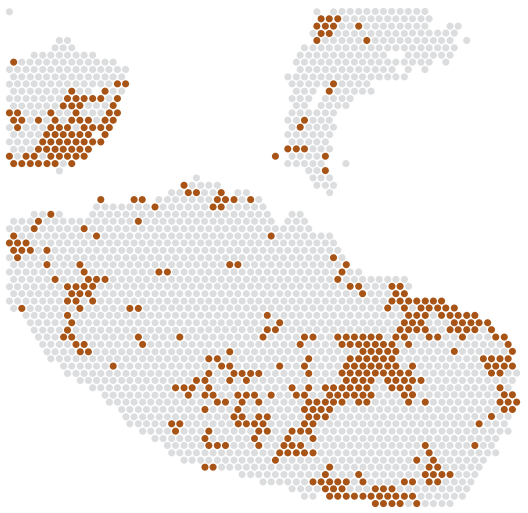
TAM composition



0 0.291

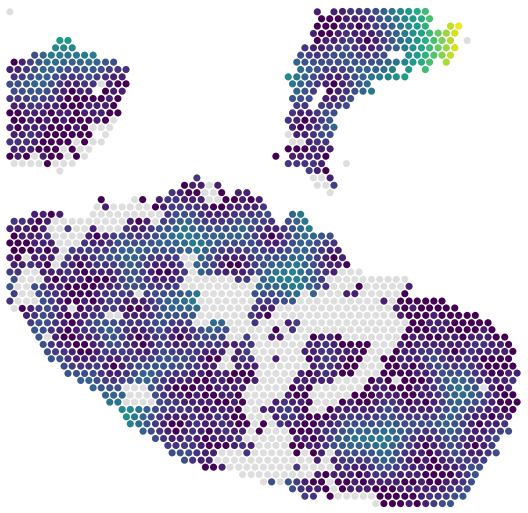
Fraction of all 30

TAM P80 targets



False True

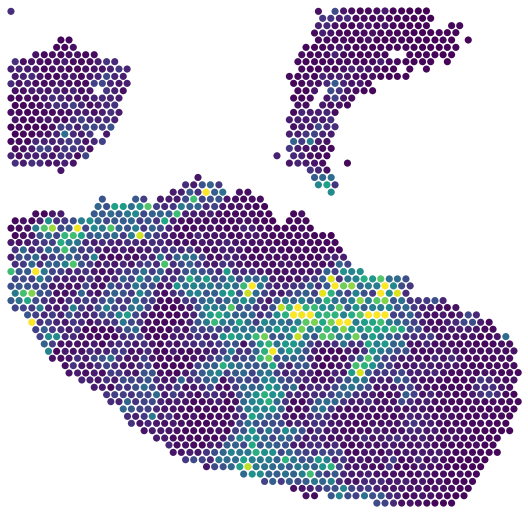
TAM distance



0 1.13e+03

Nearest spot center (µm)

Treg composition



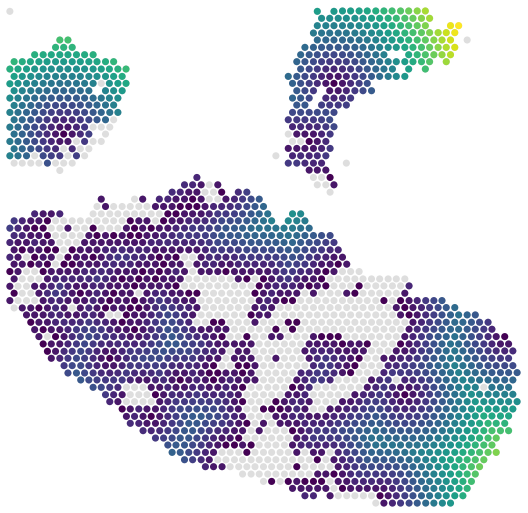
0 0.102

Fraction of all 30

Treg P80 targets

False True

Treg distance

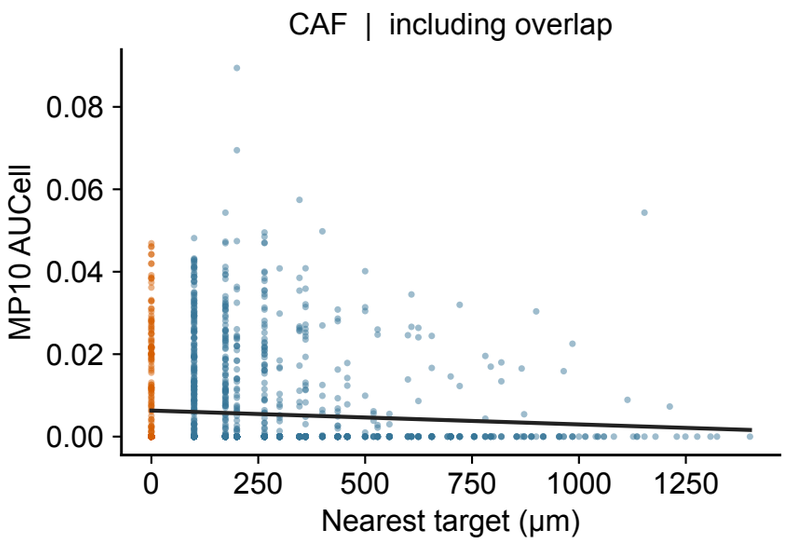


0 1.61e+03

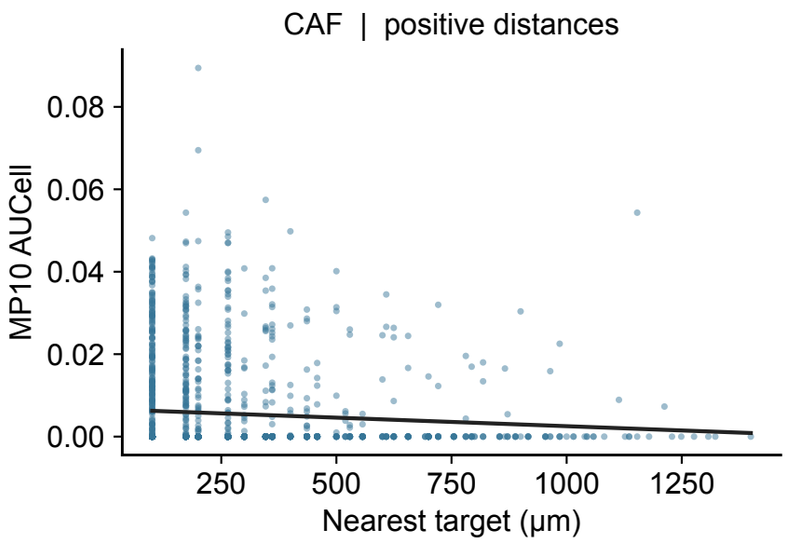
Nearest spot center (µm)

CanDis_2026_Sample7 | MP10 versus nearest target

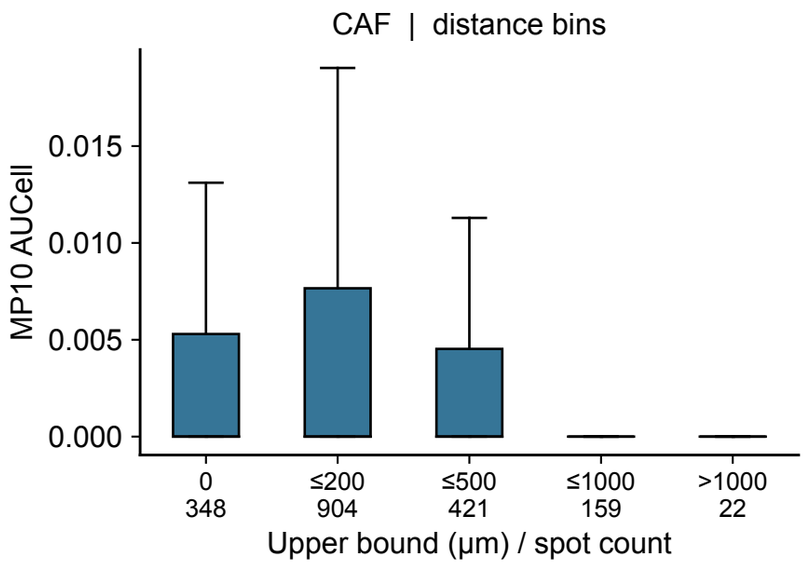
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



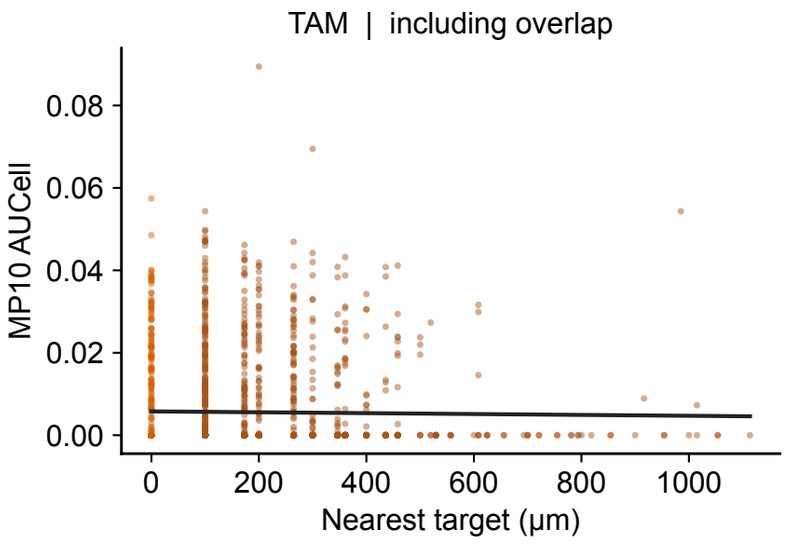
r -0.07 ρ -0.04
n 1854/1858 queries • 459 targets



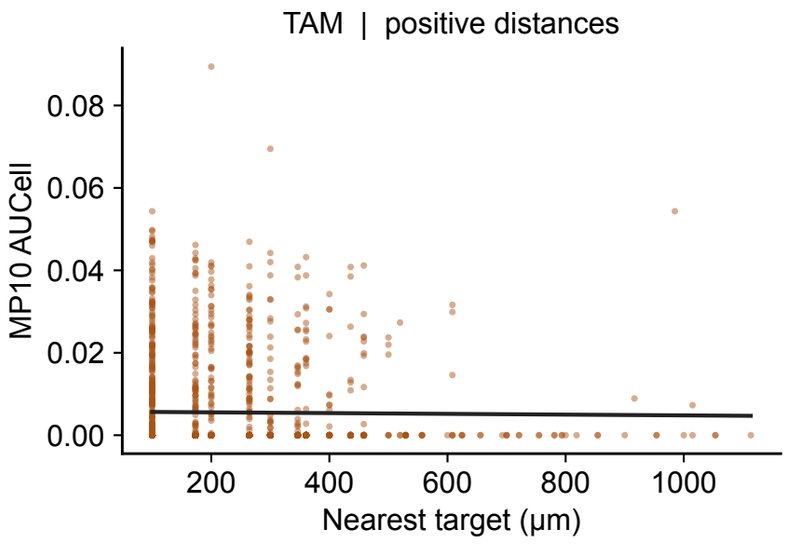
r -0.08 ρ -0.07
n 1506/1858 queries • 459 targets



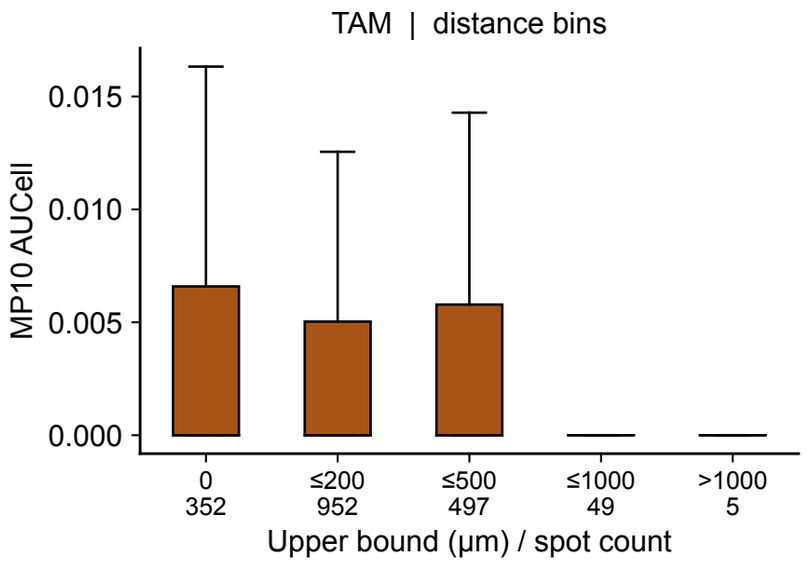
Orange = zero distance
Line = within-section linear fit



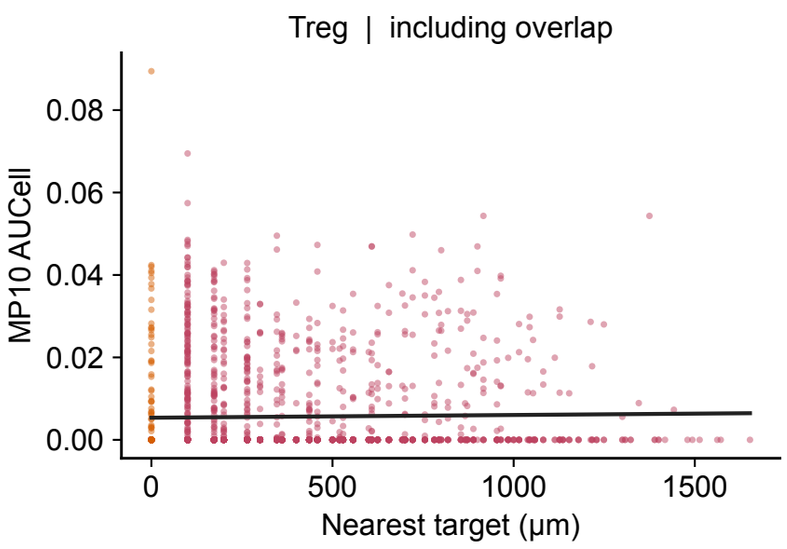
r -0.01 ρ -0.00
n 1855/1858 queries • 459 targets



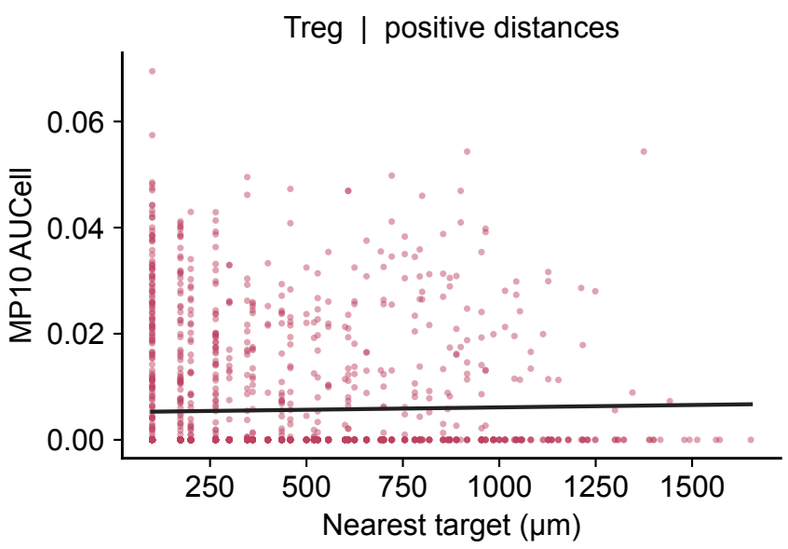
r -0.01 ρ -0.00
n 1503/1858 queries • 459 targets



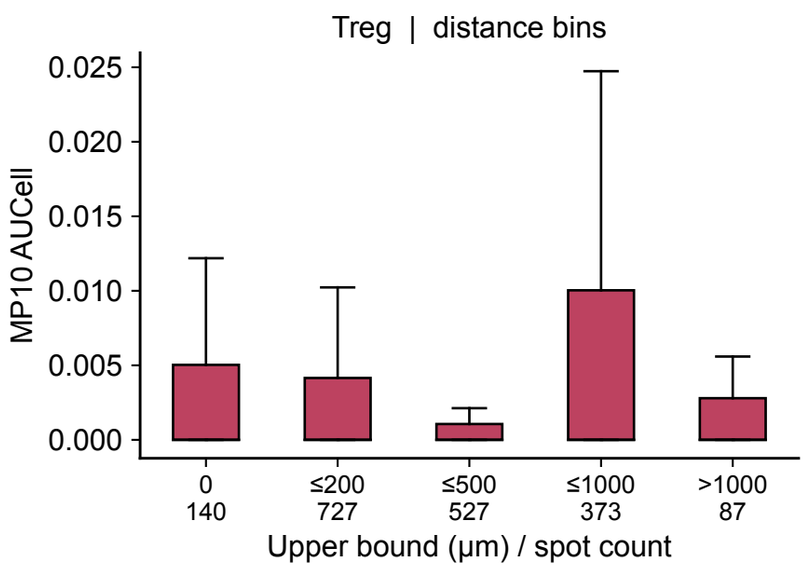
Orange = zero distance
Line = within-section linear fit



r +0.02 ρ +0.02
n 1854/1858 queries • 459 targets



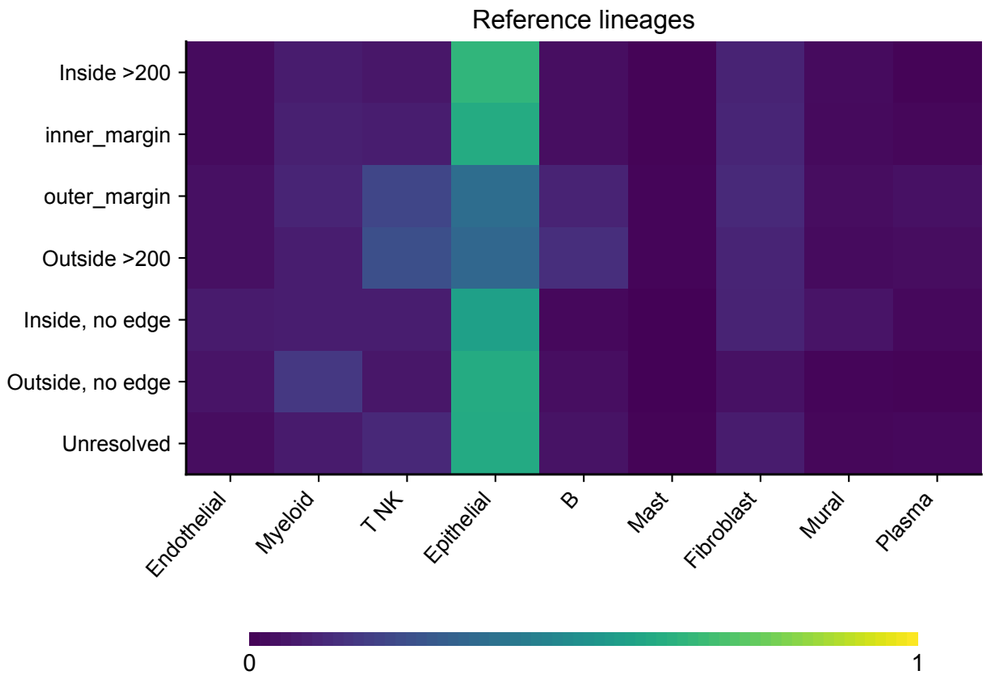
r +0.02 ρ +0.03
n 1714/1858 queries • 459 targets

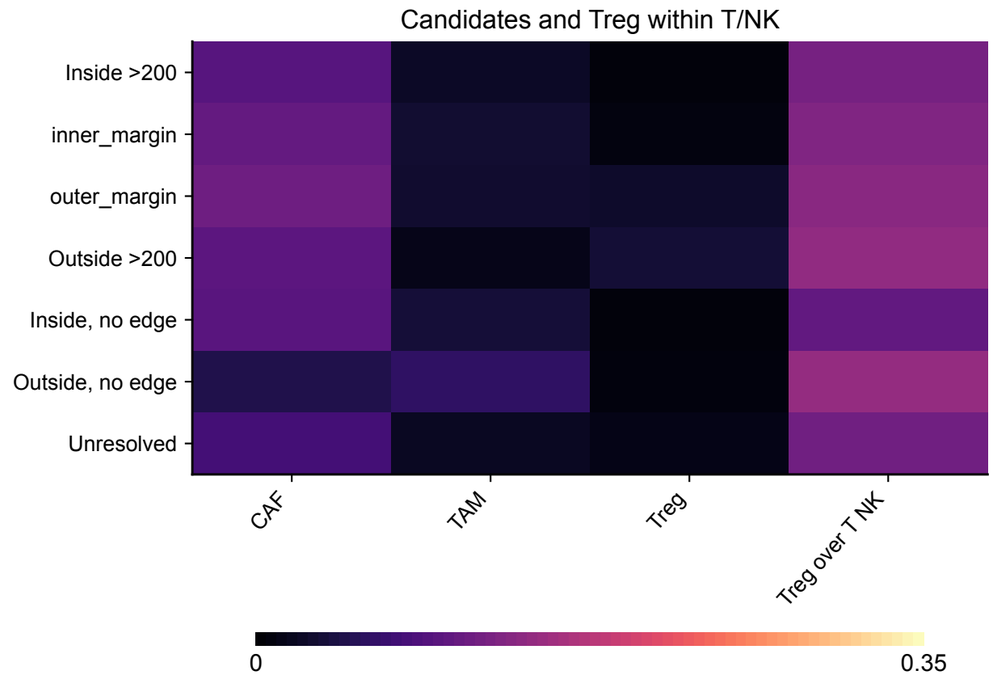


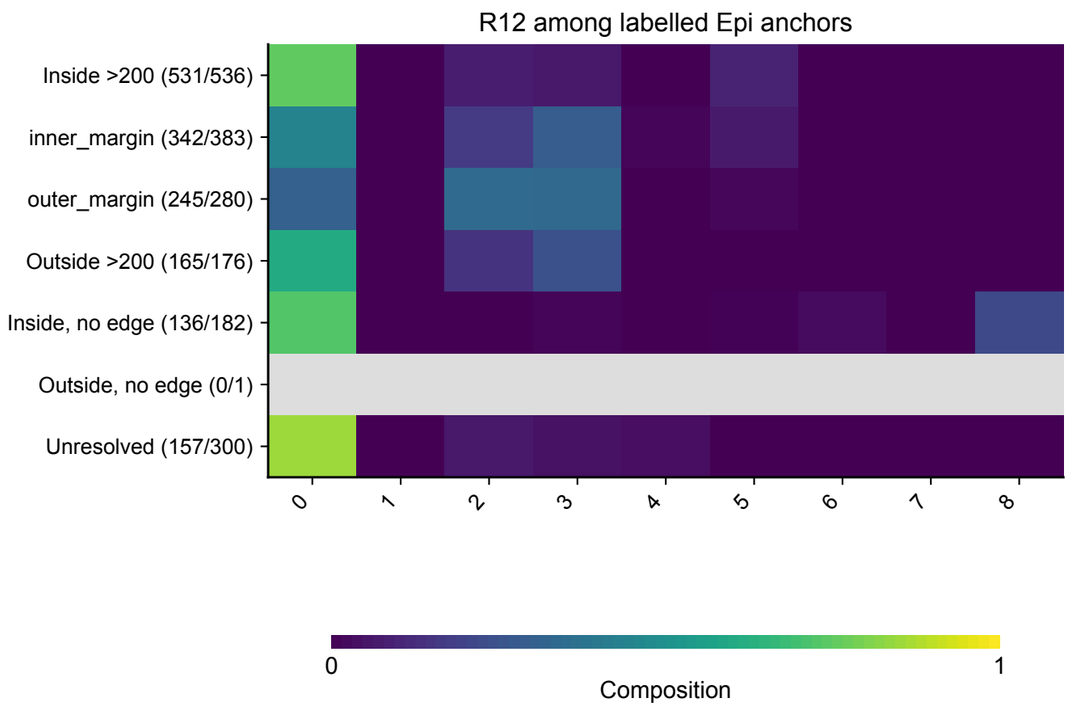
Orange = zero distance
Line = within-section linear fit

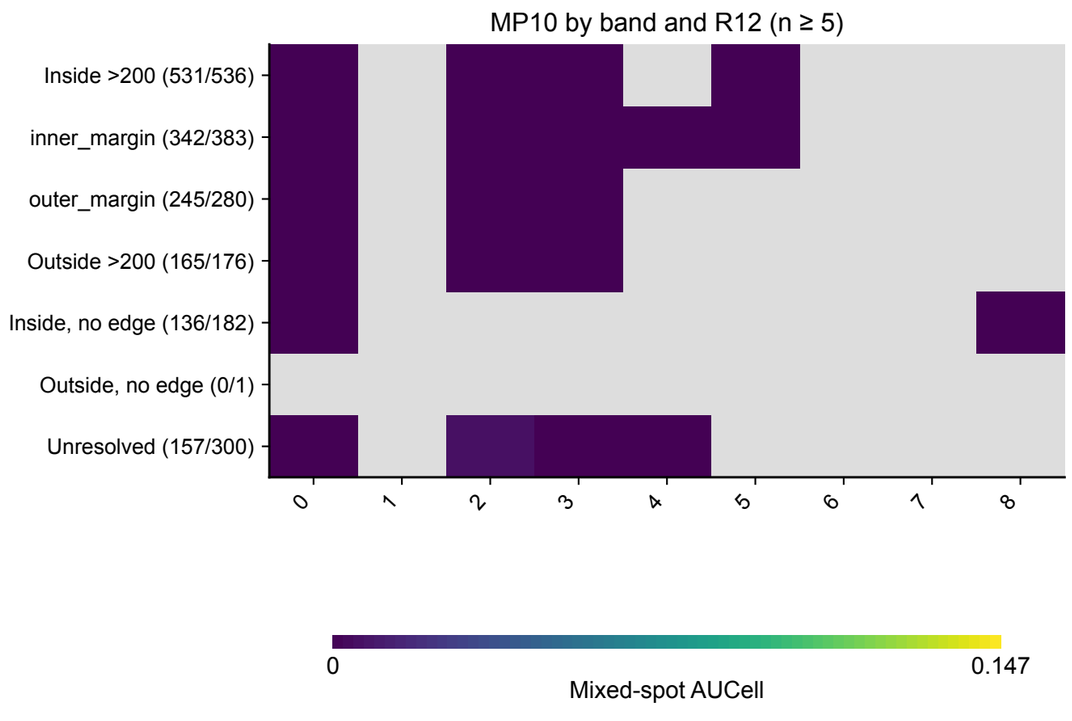
CanDis_2026_Sample7 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





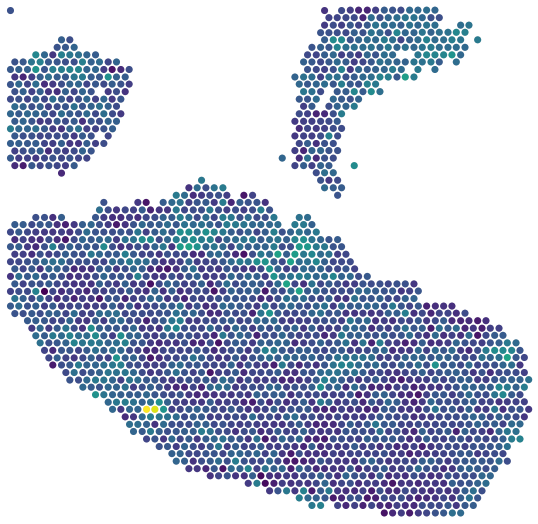




CanDis_2026_Sample7 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

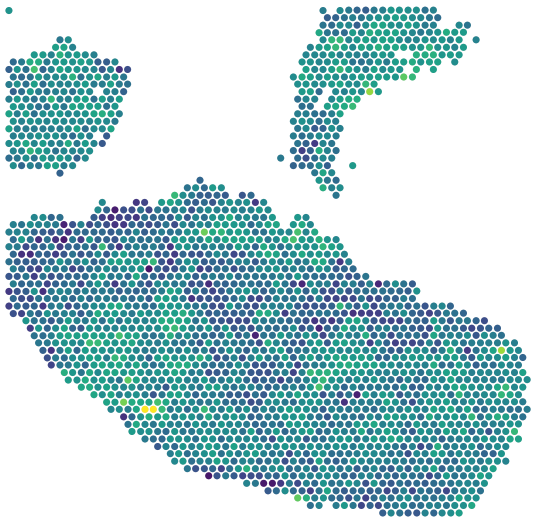
E2F Targets



0 0.152

UCell score

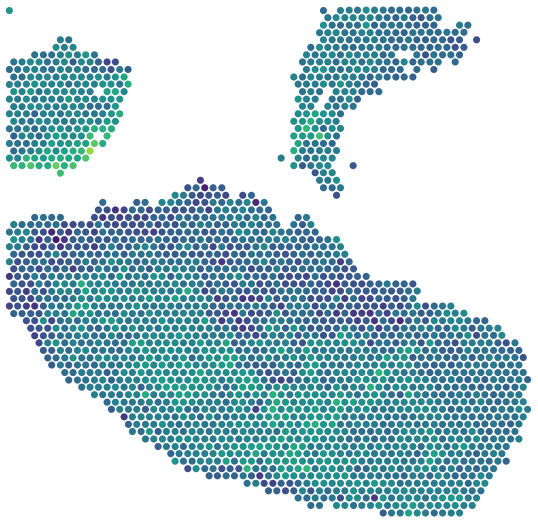
G2M Checkpoint



0 0.134

UCell score

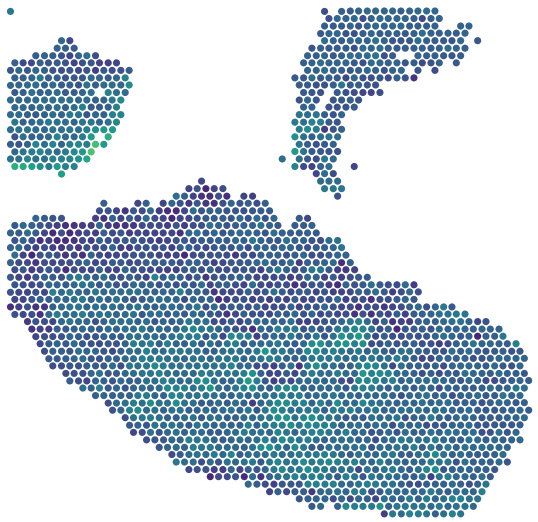
Hypoxia



0 0.211

UCell score

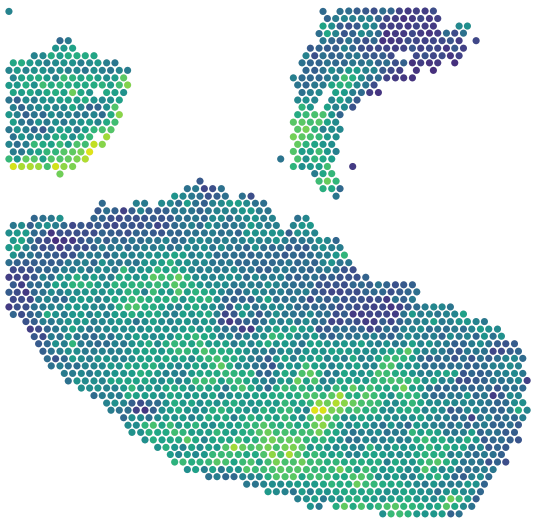
Tnfa Signaling Via Nfkb



0 0.275

UCell score

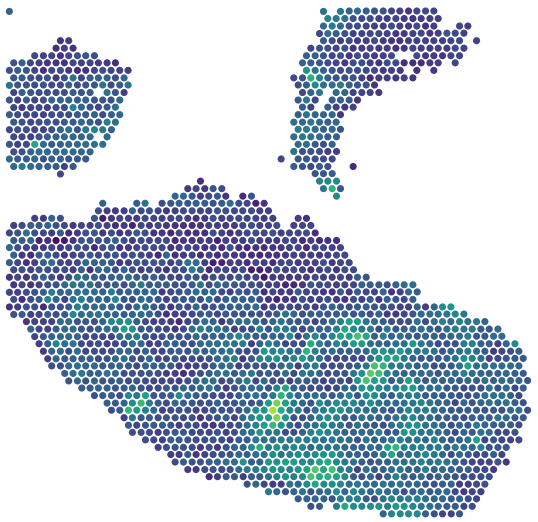
Emt



0 0.386

UCell score

Interferon Gamma Response



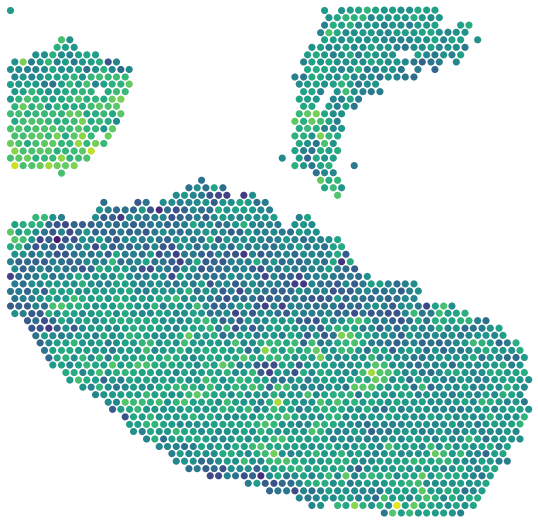
0 0.275

UCell score

CanDis_2026_Sample7 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

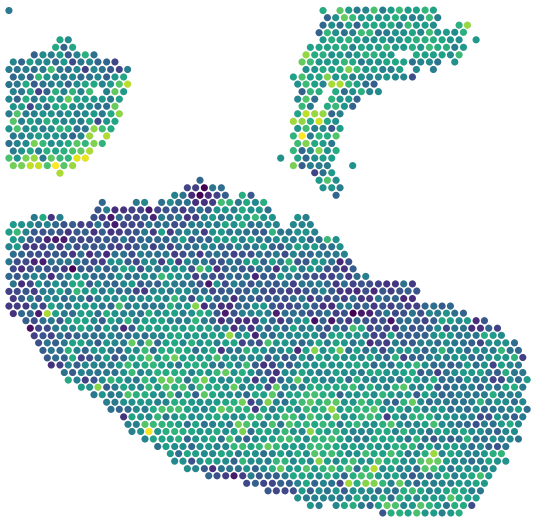
Il2 Stat5 Signaling



0 0.144

UCell score

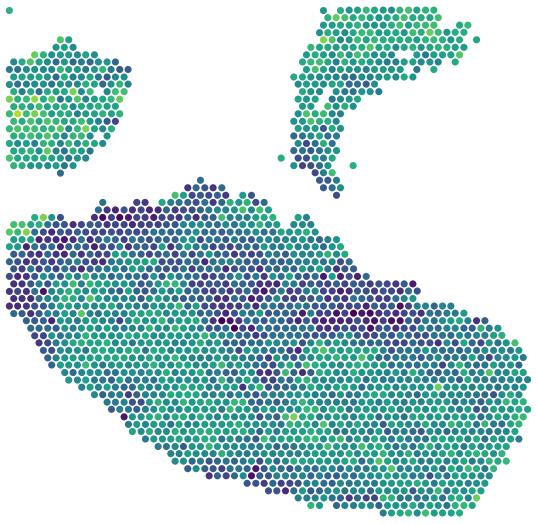
Tgf Beta Signaling



0 0.232

UCell score

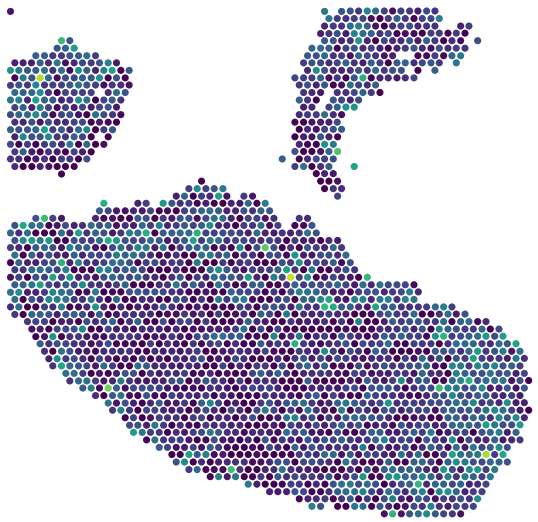
Cholesterol Homeostasis



0 0.21

UCell score

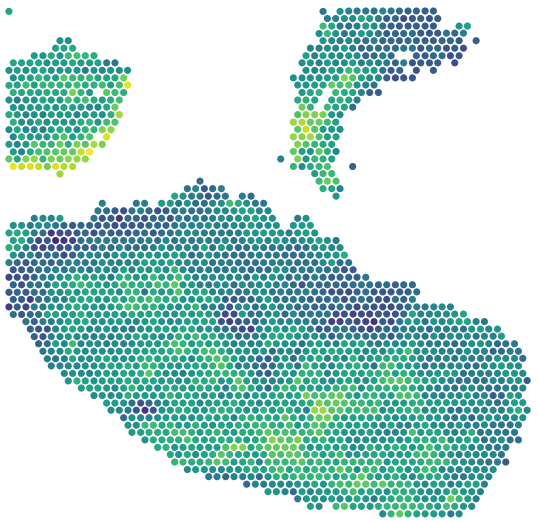
Cholesterol Biosynthesis



0 0.166

UCell score

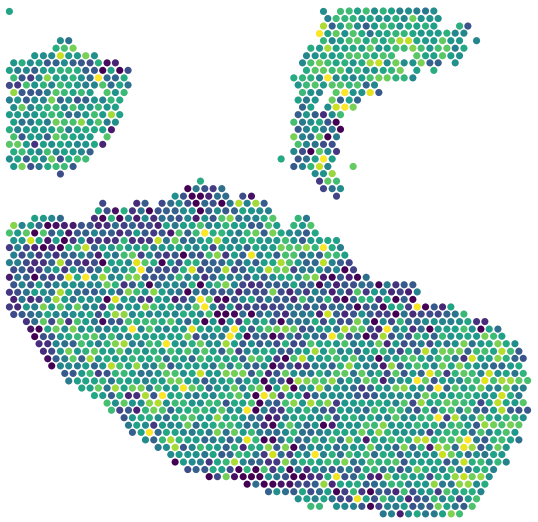
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



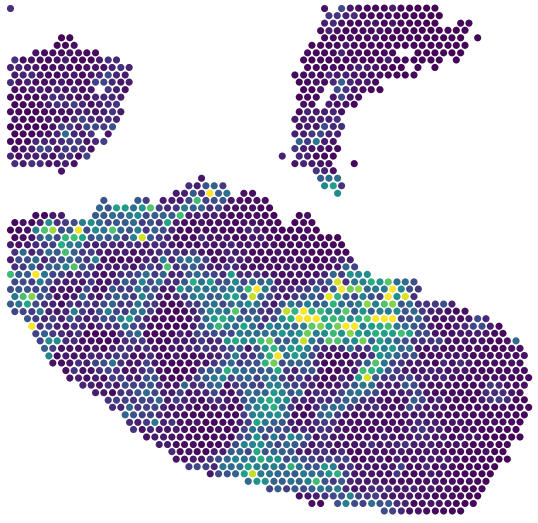
0 0.151

UCell score

CanDis_2026_Sample7 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

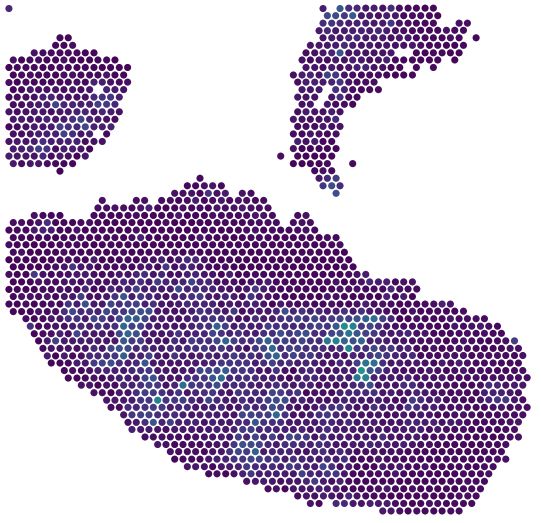
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

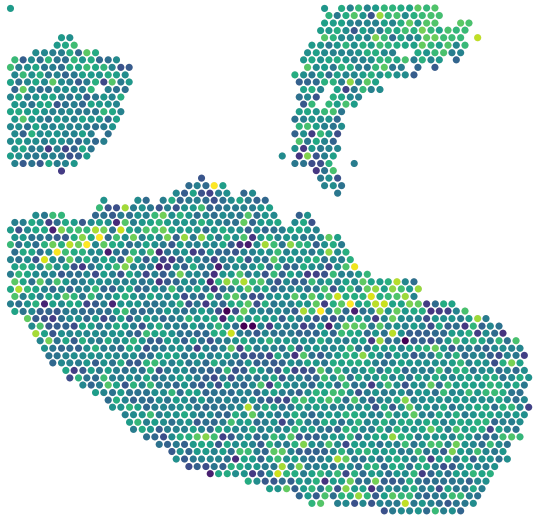
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

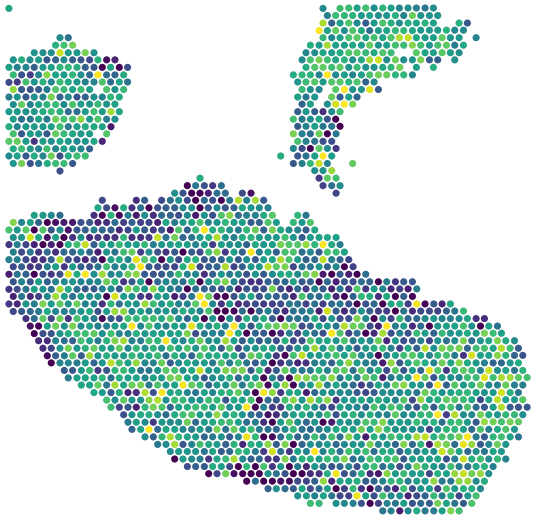
Treg Resting Vs Activated Up



0 0.0609

UCell score

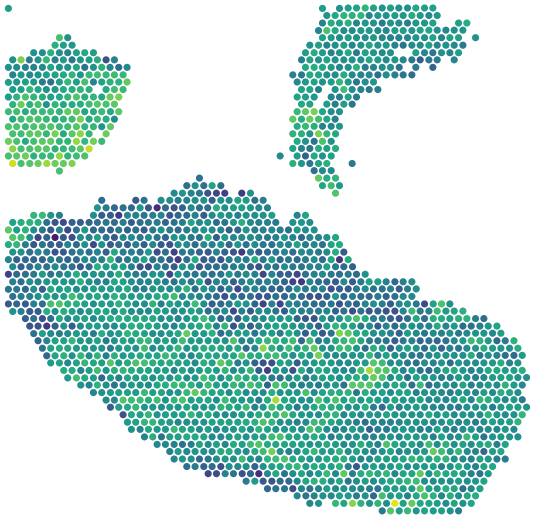
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

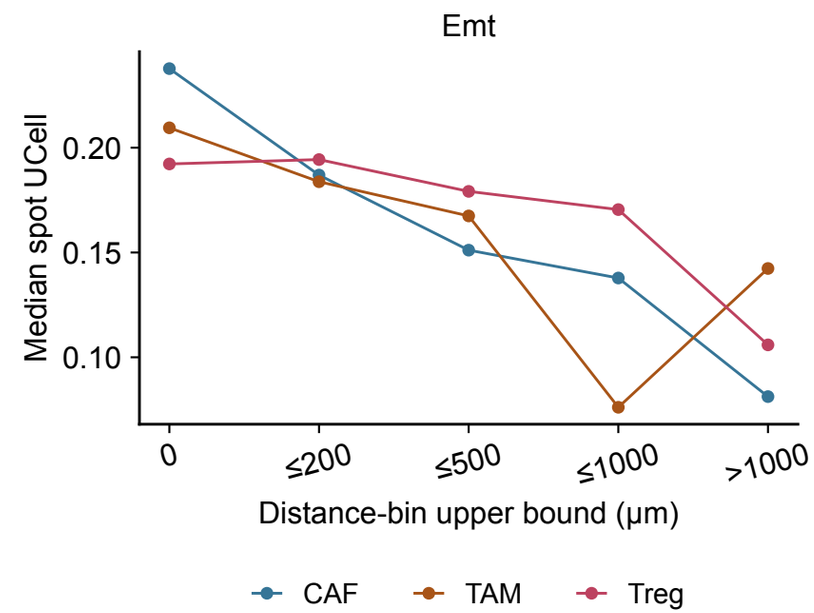


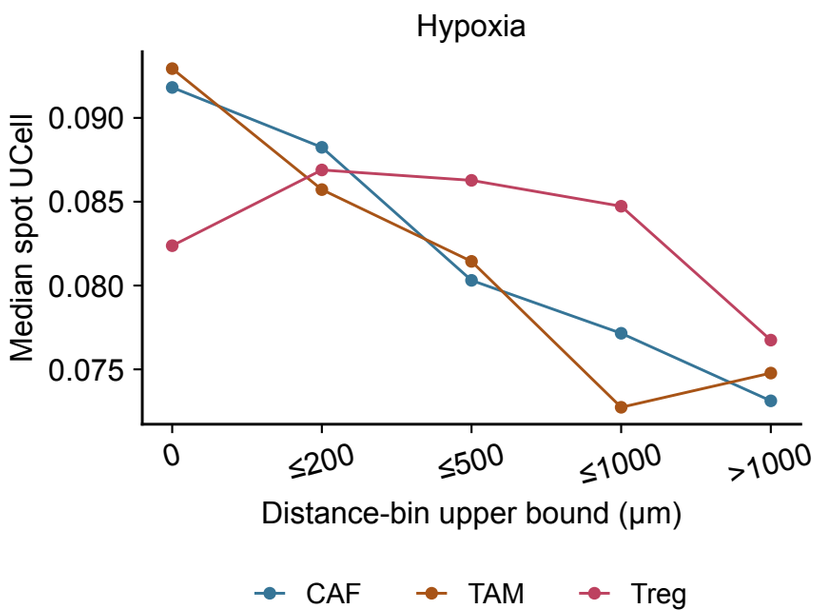
0 0.144

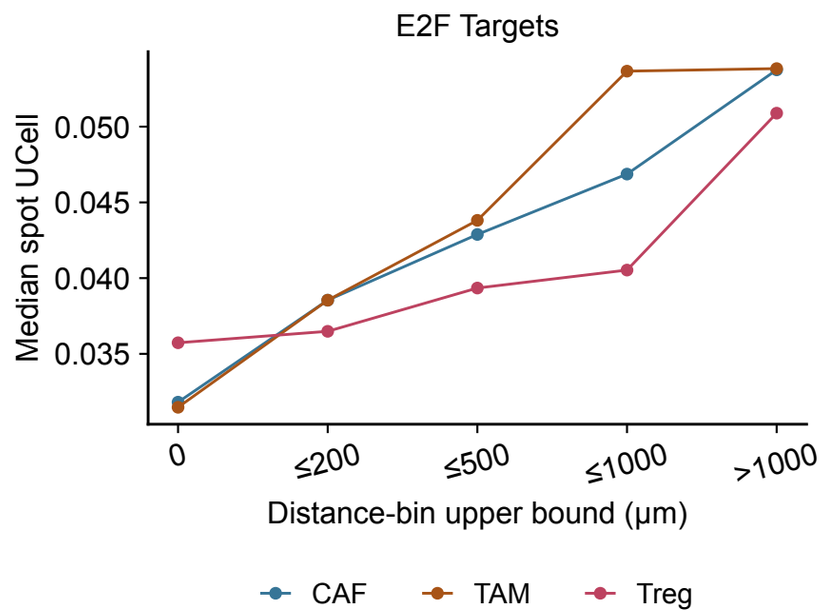
UCell score

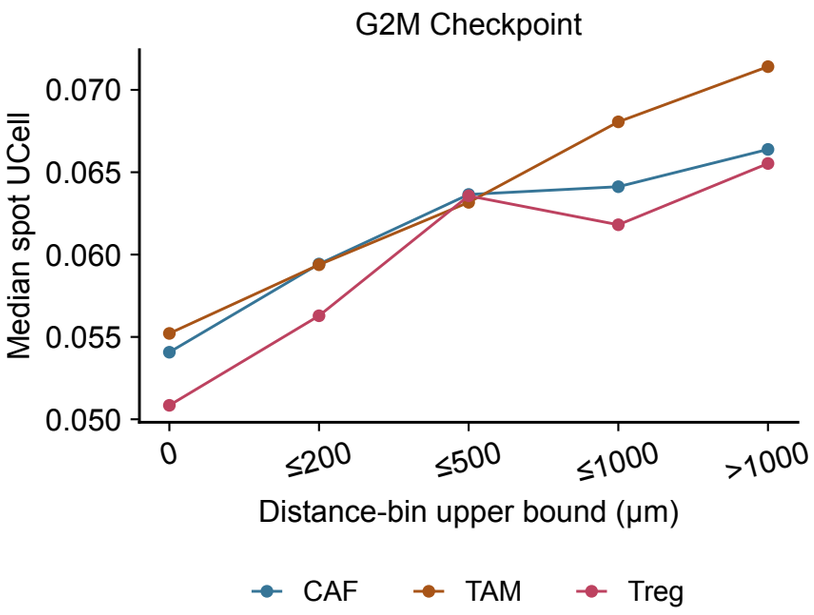
CanDis_2026_Sample7 | Programs by target distance

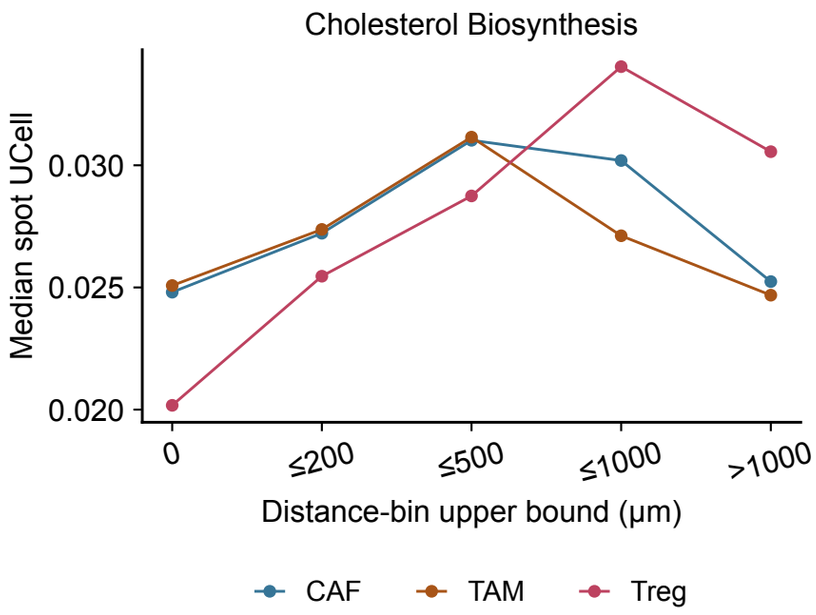
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

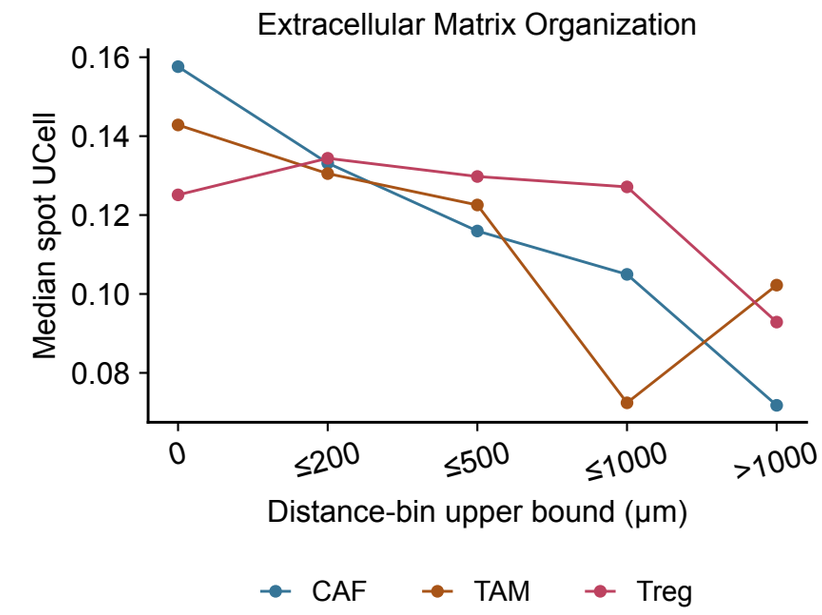






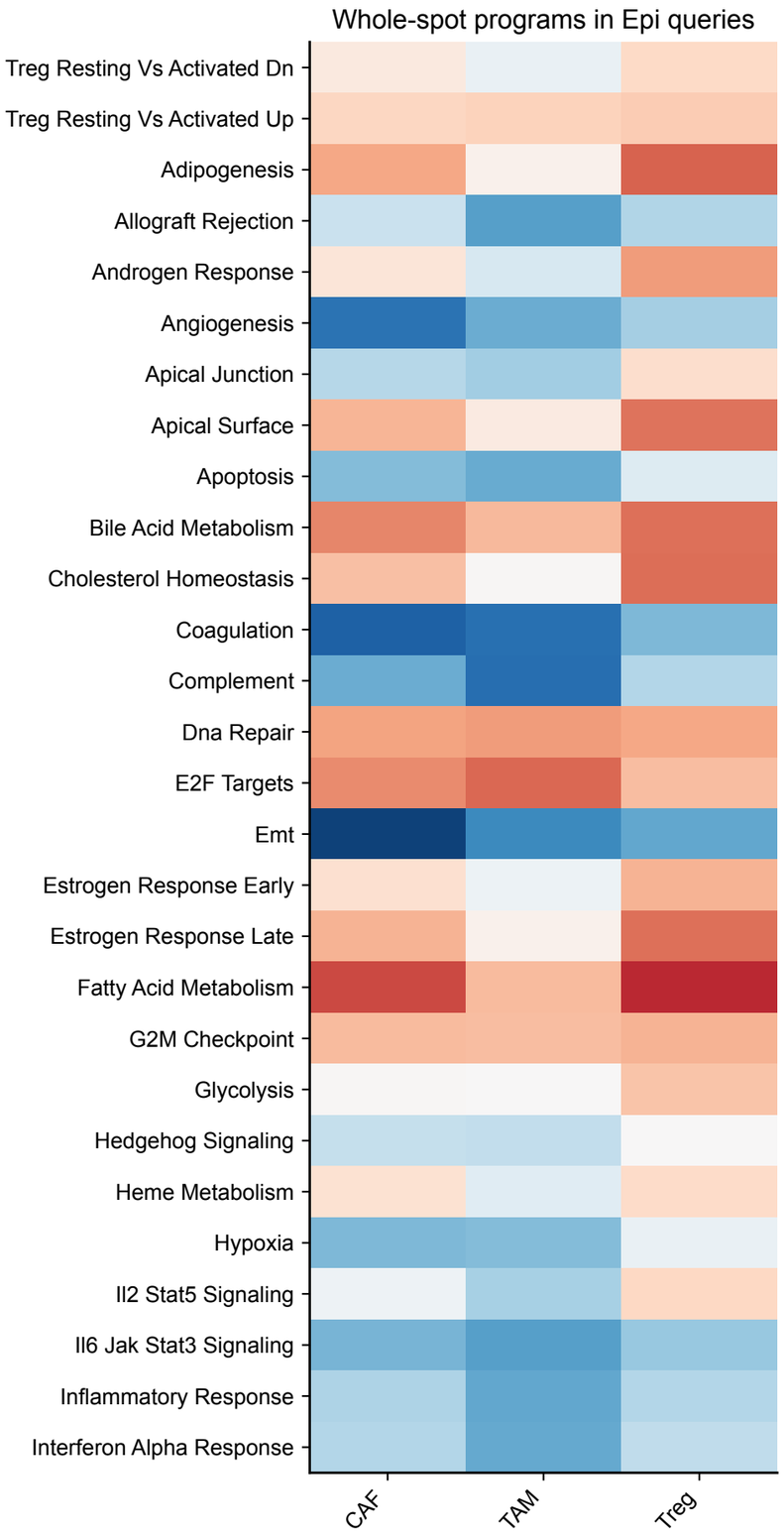


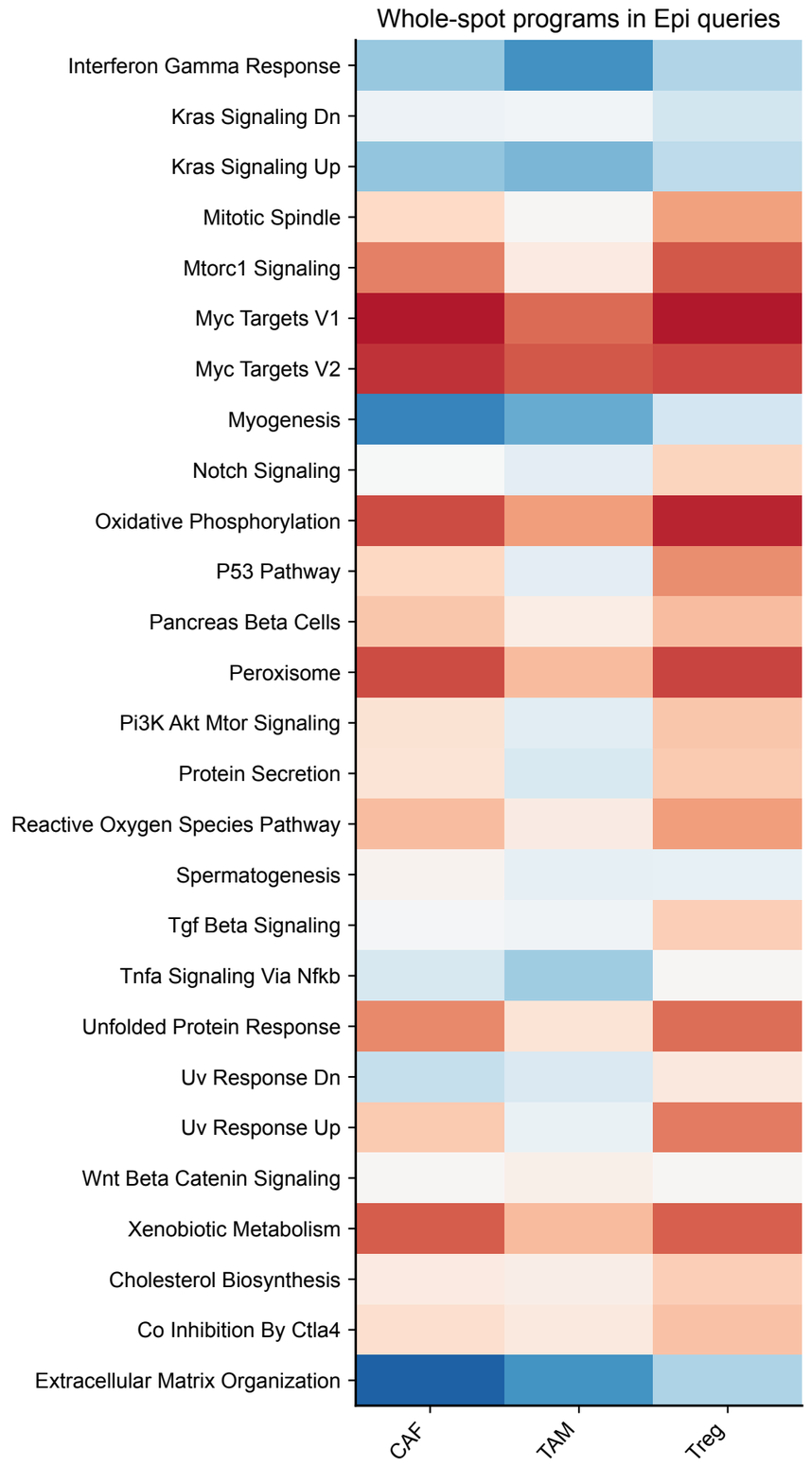




CanDis_2026_Sample7 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

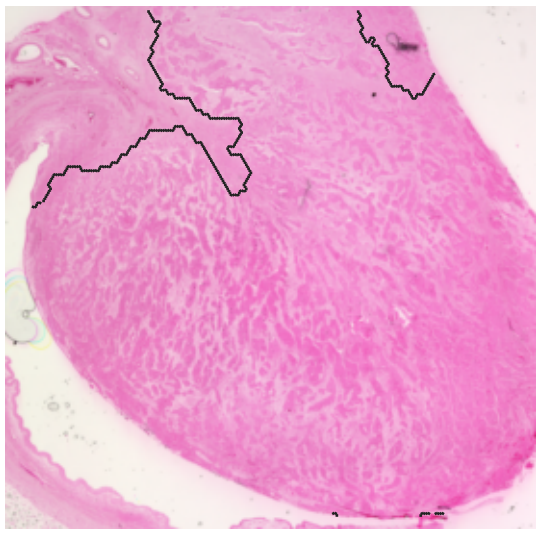




CanDis_2026_Sample8 | Tissue context

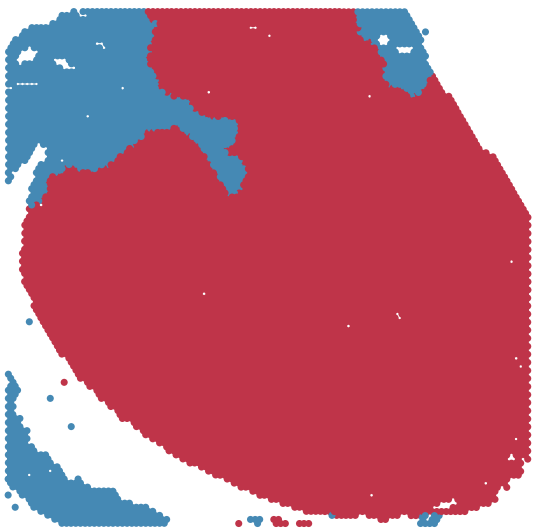
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



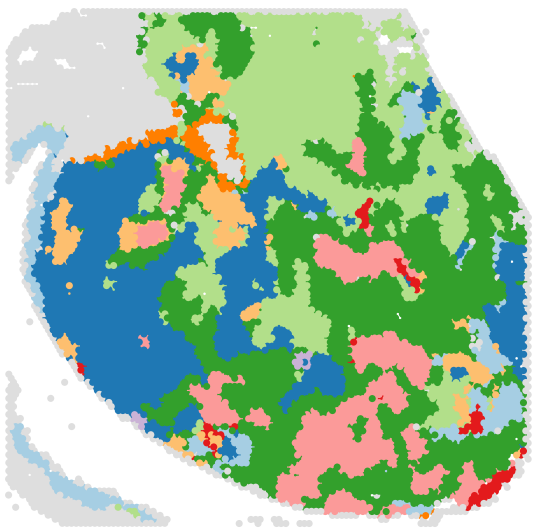
Black line: observed candidate outer edge

Final candidate location



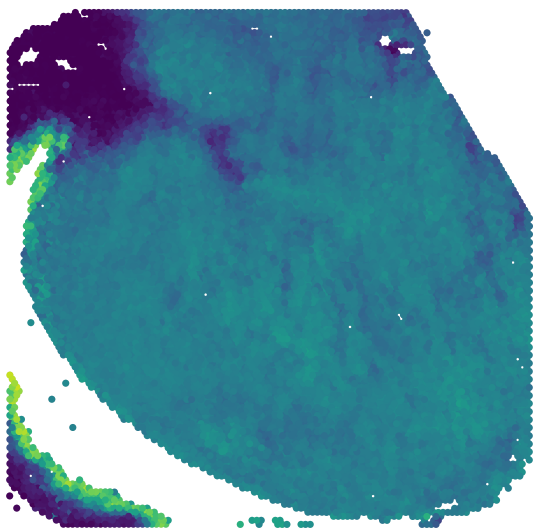
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

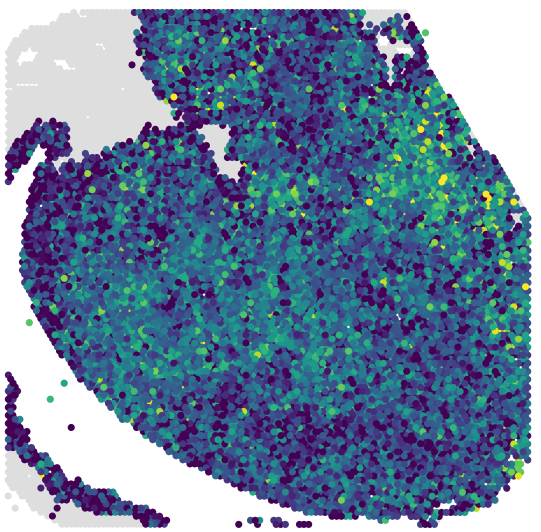
Epithelial support



0 1

Fraction of all 30

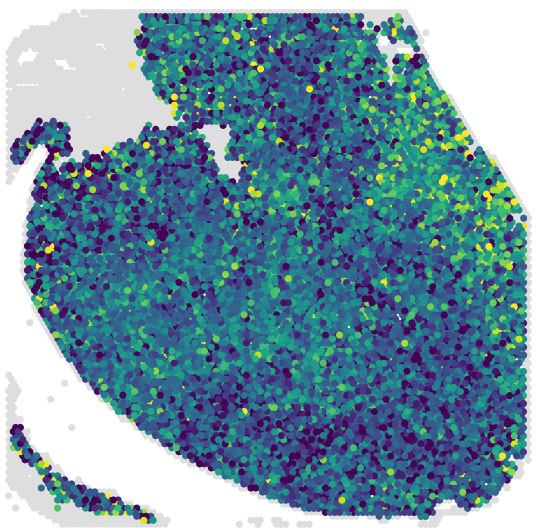
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



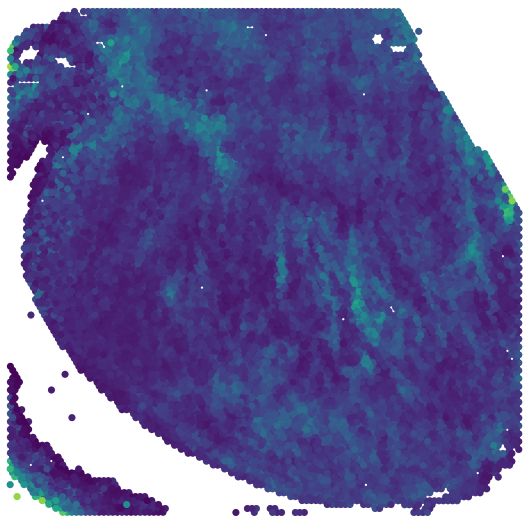
0 0.14

Matched-model AUCell

CanDis_2026_Sample8 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

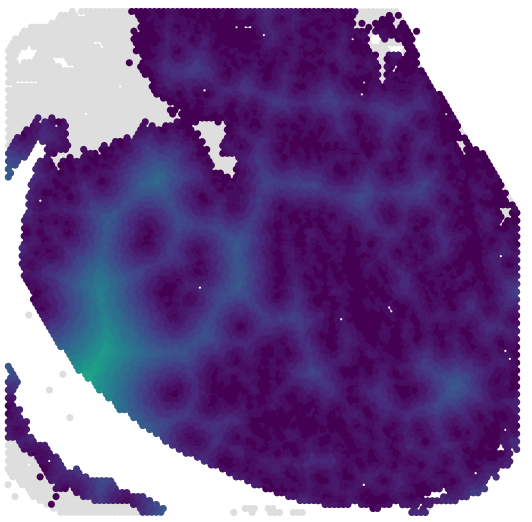


0 0.418
Fraction of all 30

CAF P80 targets

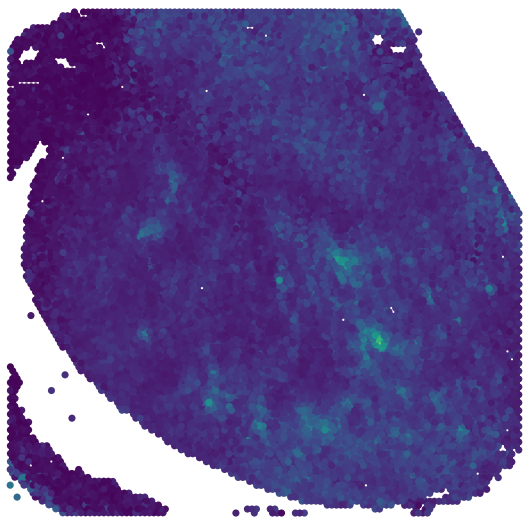
False True

CAF distance



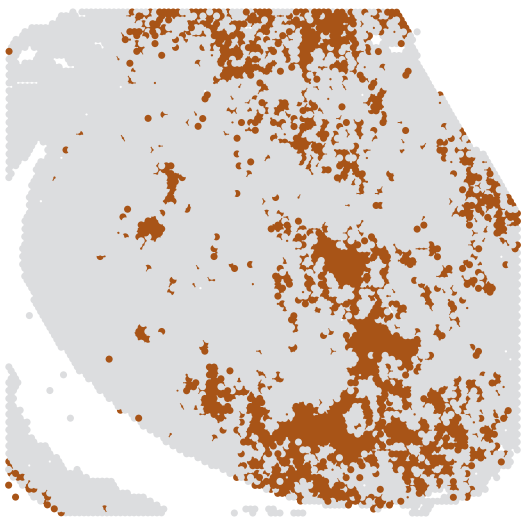
0 3.1e+03
Nearest spot center (µm)

TAM composition



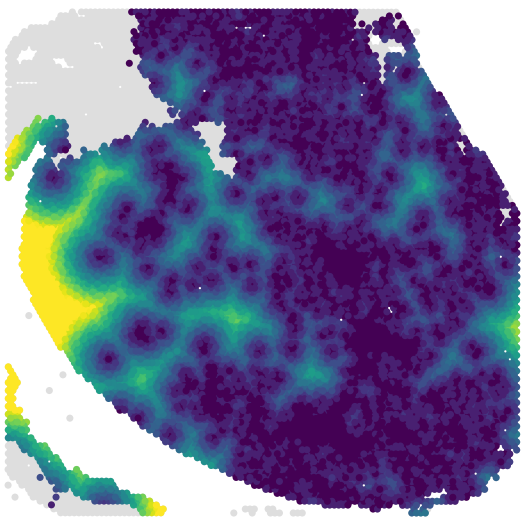
0 0.291
Fraction of all 30

TAM P80 targets



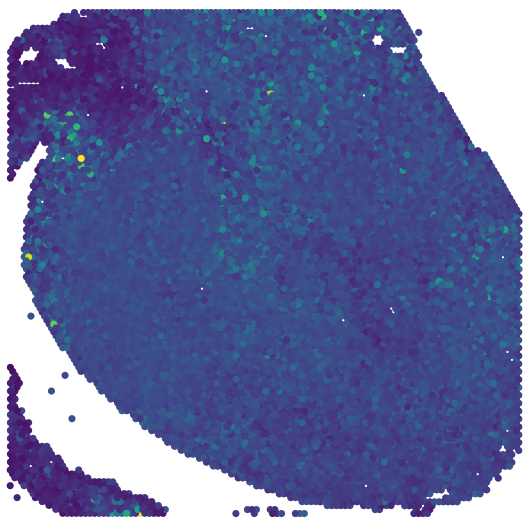
False True

TAM distance



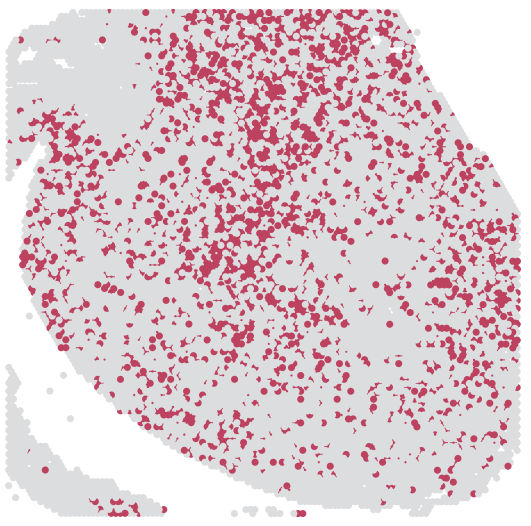
0 1.13e+03
Nearest spot center (µm)

Treg composition



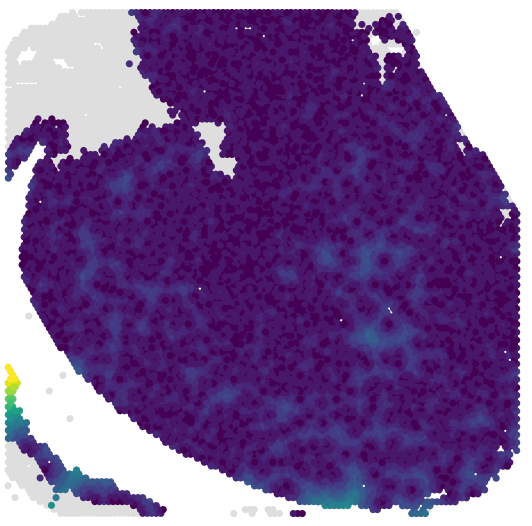
0 0.102
Fraction of all 30

Treg P80 targets



False True

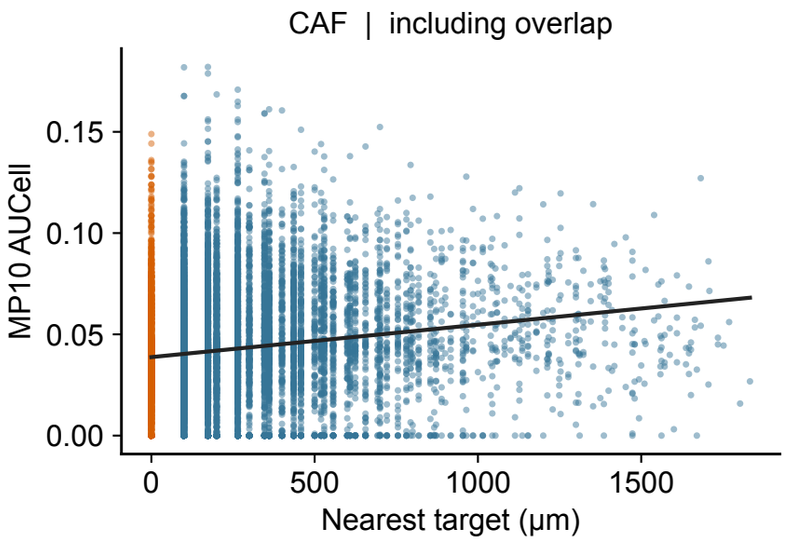
Treg distance



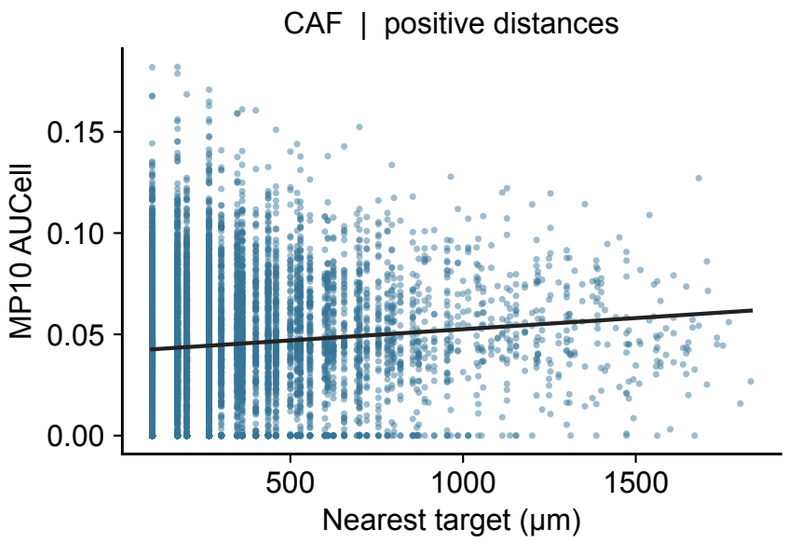
0 1.61e+03
Nearest spot center (µm)

CanDis_2026_Sample8 | MP10 versus nearest target

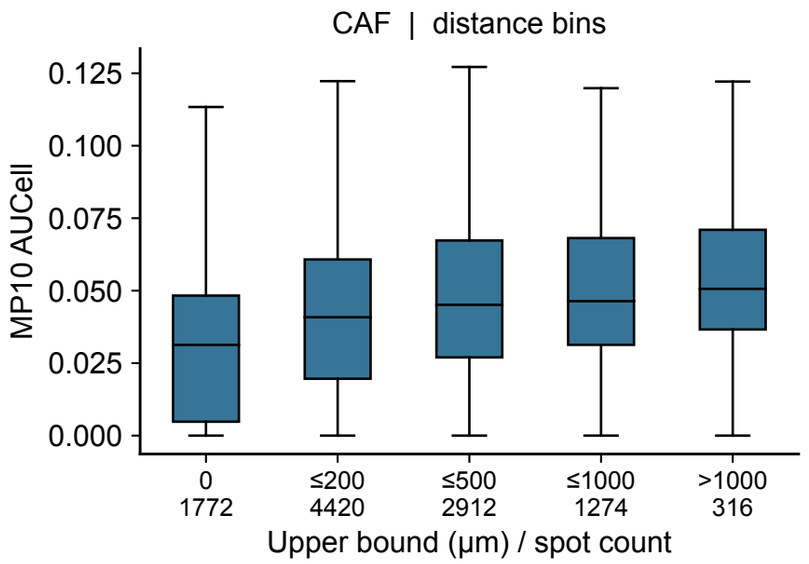
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



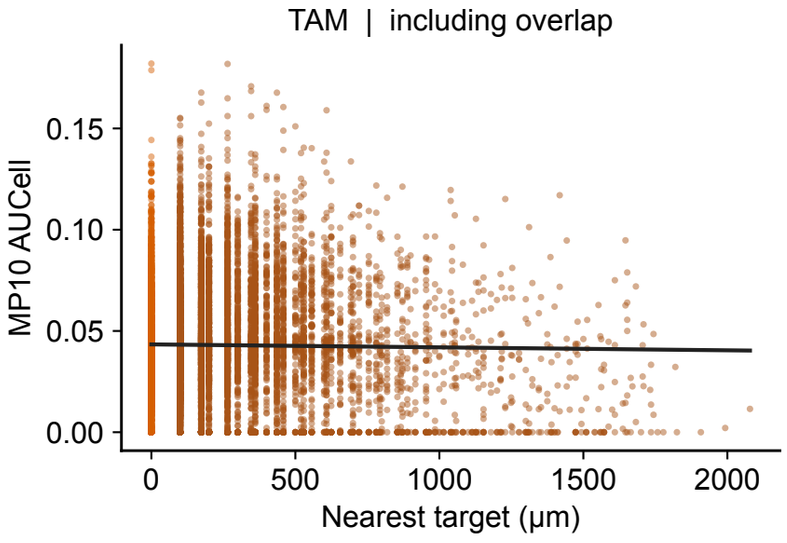
r +0.15 ρ +0.19
n 10694/10711 queries • 2392 targets



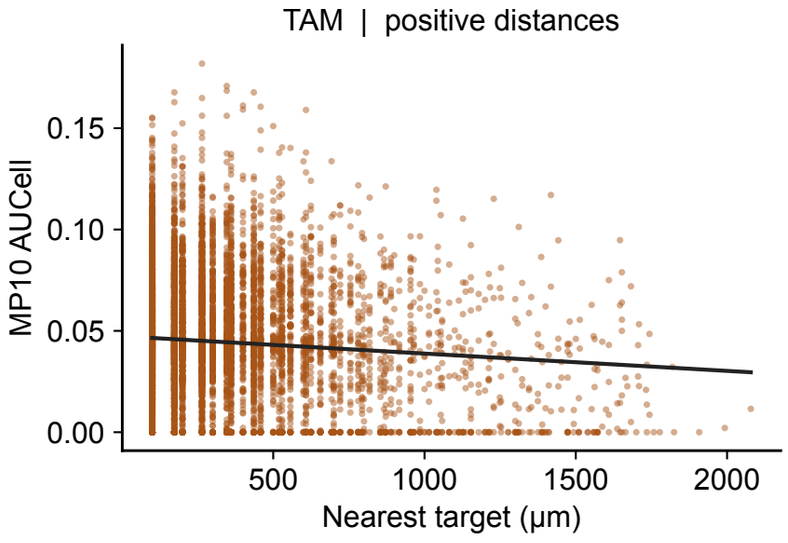
r +0.10 ρ +0.13
n 8922/10711 queries • 2392 targets



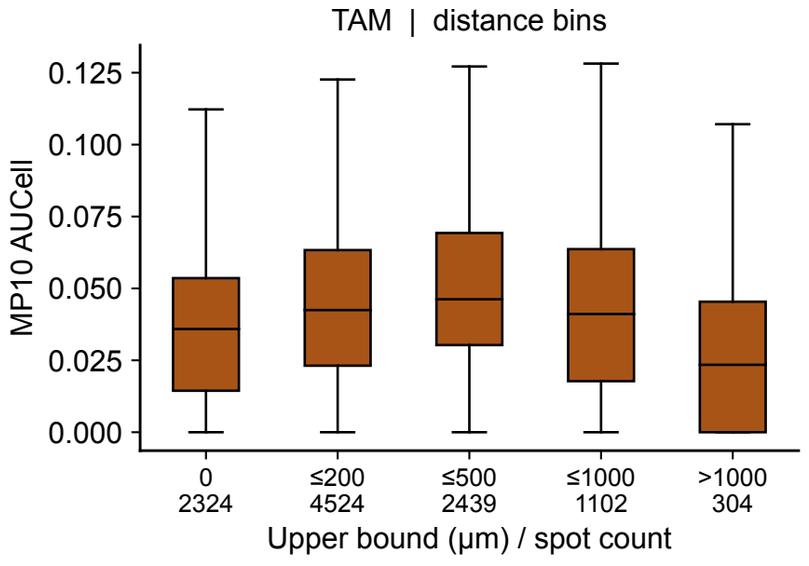
Orange = zero distance
Line = within-section linear fit



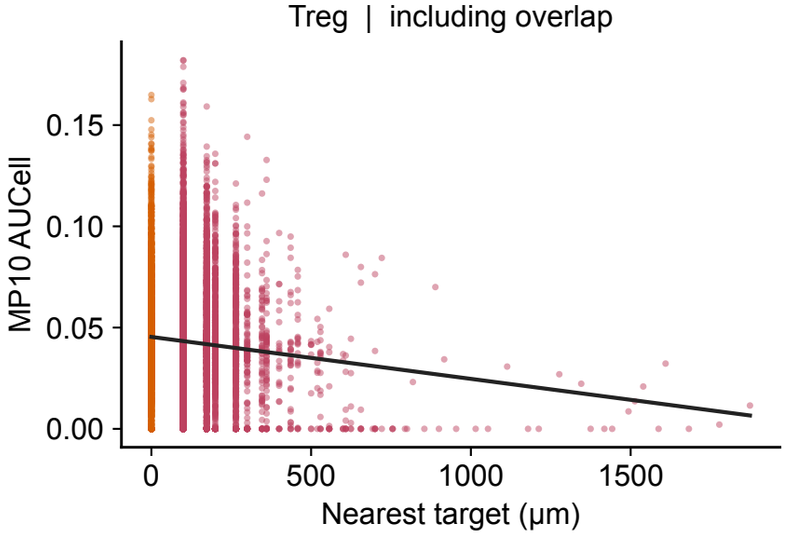
r -0.01 ρ +0.09
n 10693/10711 queries • 2392 targets



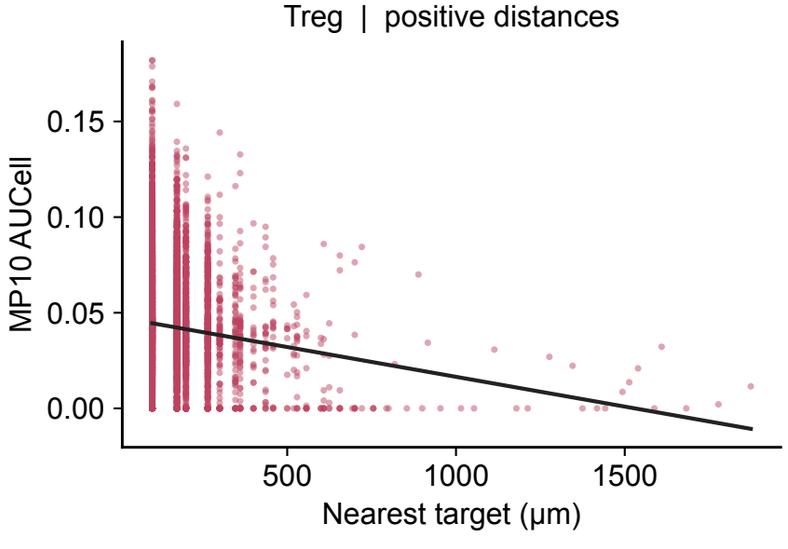
r -0.08 ρ +0.01
n 8369/10711 queries • 2392 targets



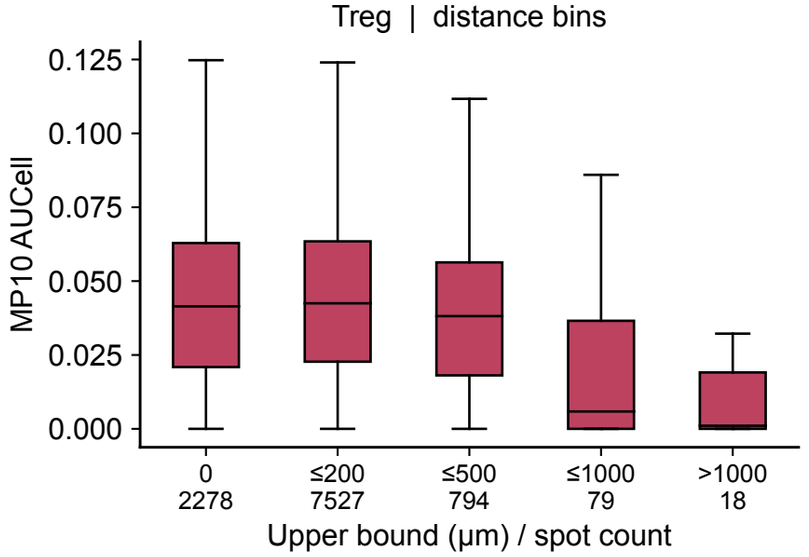
Orange = zero distance
Line = within-section linear fit



r -0.07 ρ -0.04
n 10696/10711 queries • 2392 targets



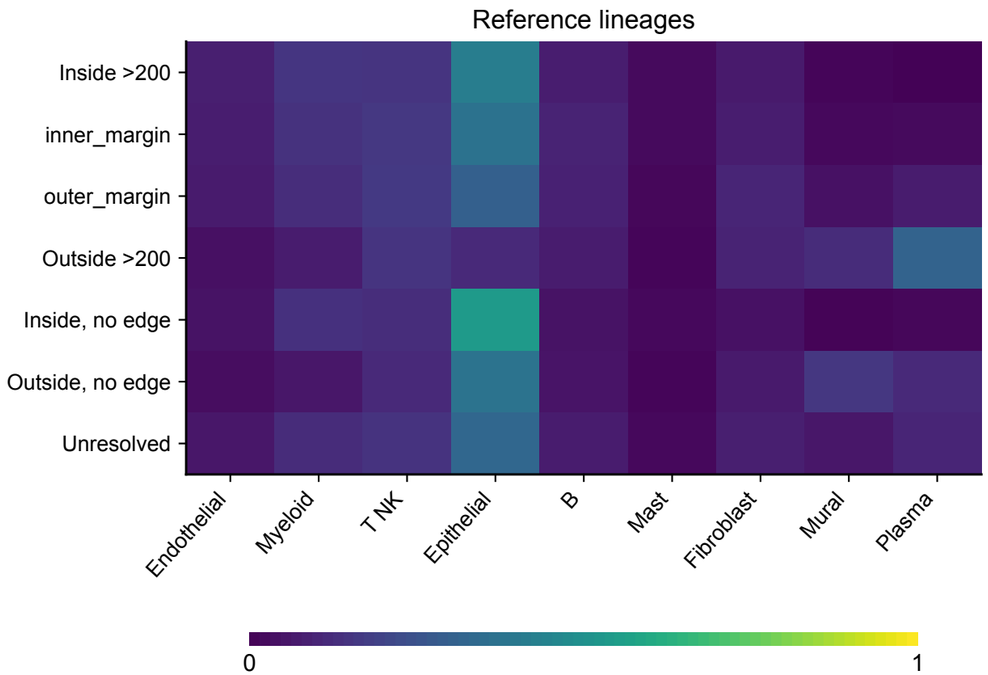
r -0.11 ρ -0.07
n 8418/10711 queries • 2392 targets

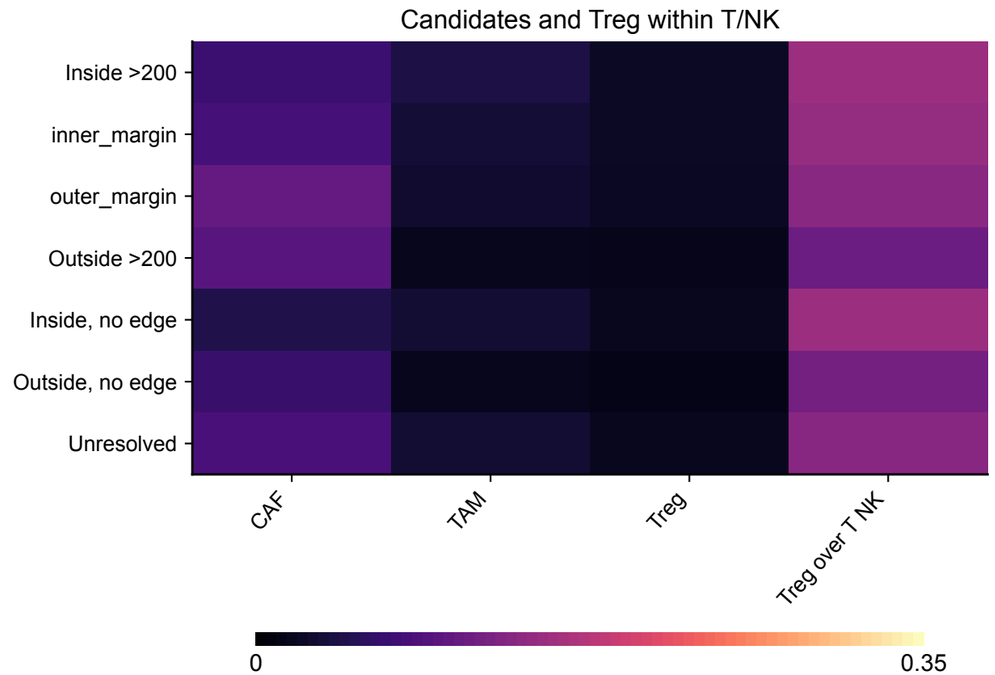


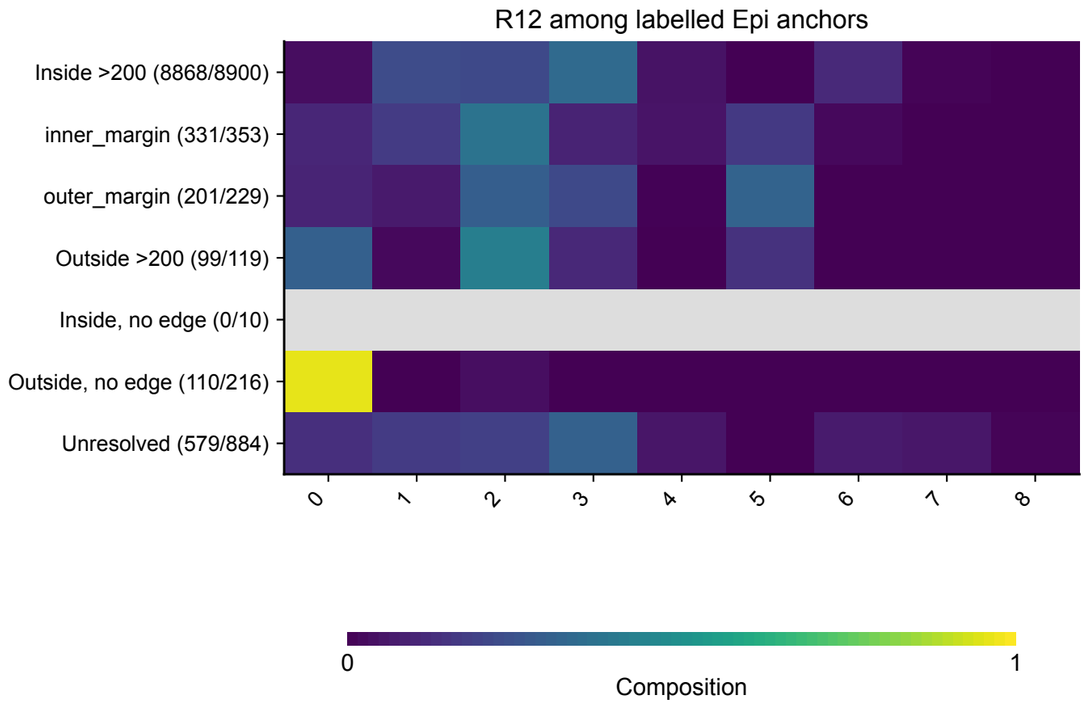
Orange = zero distance
Line = within-section linear fit

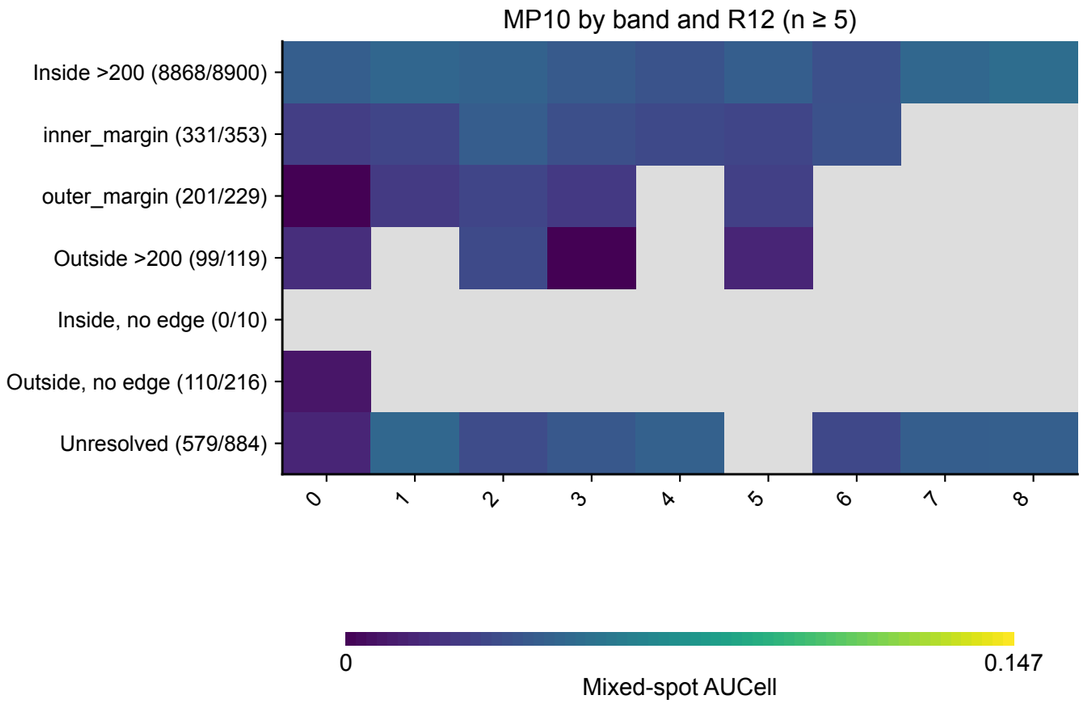
CanDis_2026_Sample8 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





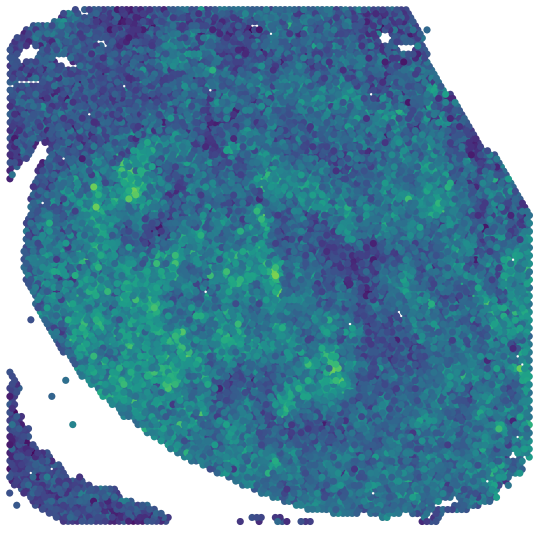




CanDis_2026_Sample8 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

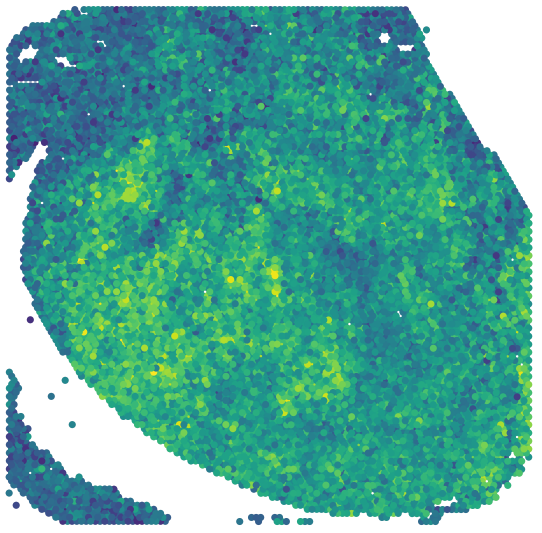
E2F Targets



0 0.152

UCell score

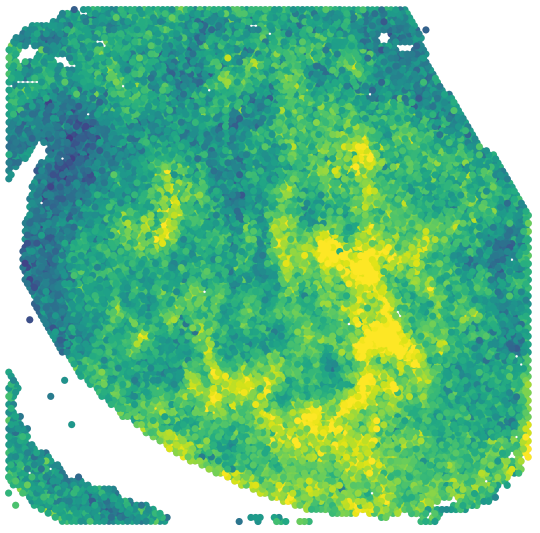
G2M Checkpoint



0 0.134

UCell score

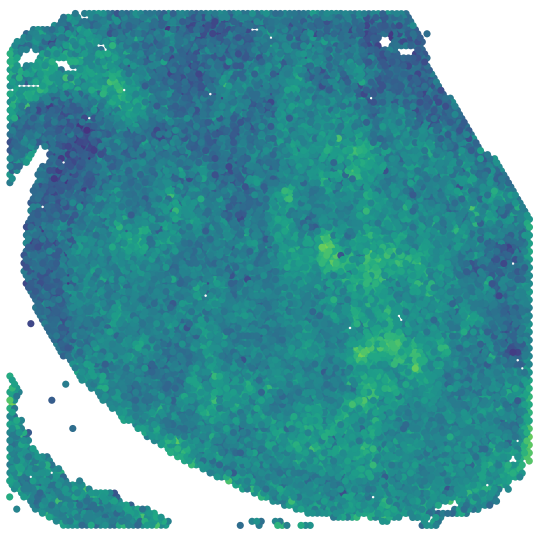
Hypoxia



0 0.211

UCell score

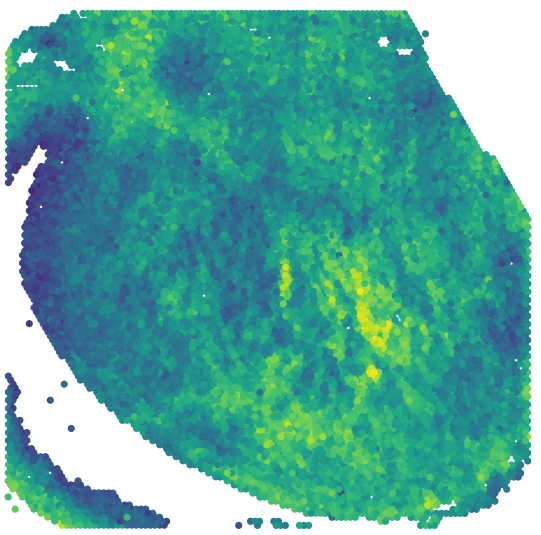
Tnfa Signaling Via Nfkb



0 0.275

UCell score

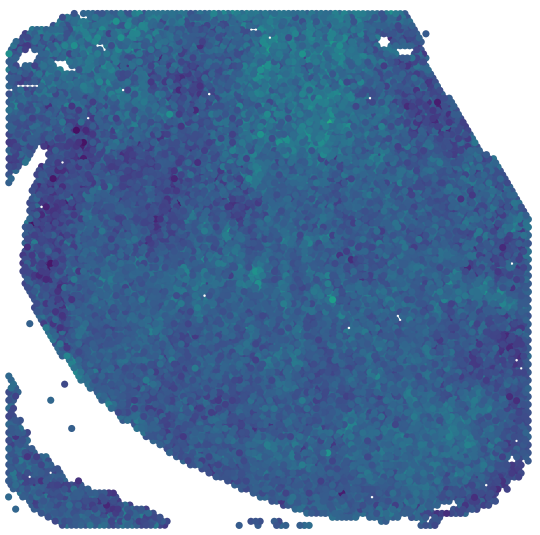
Emt



0 0.386

UCell score

Interferon Gamma Response



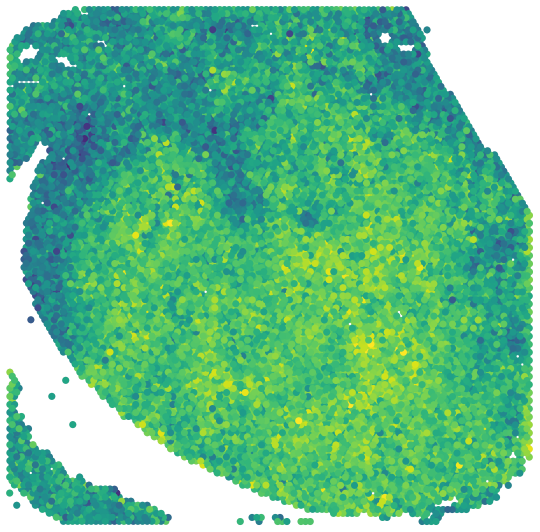
0 0.275

UCell score

CanDis_2026_Sample8 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

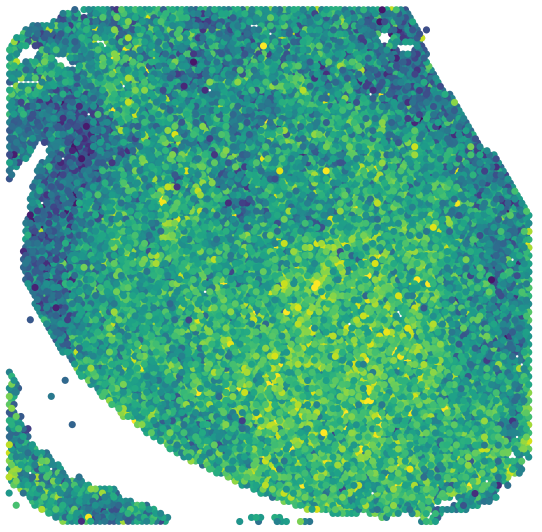
Il2 Stat5 Signaling



0 0.144

UCell score

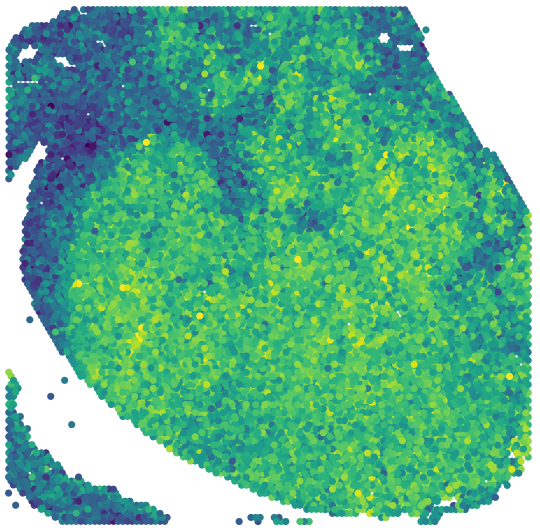
Tgf Beta Signaling



0 0.232

UCell score

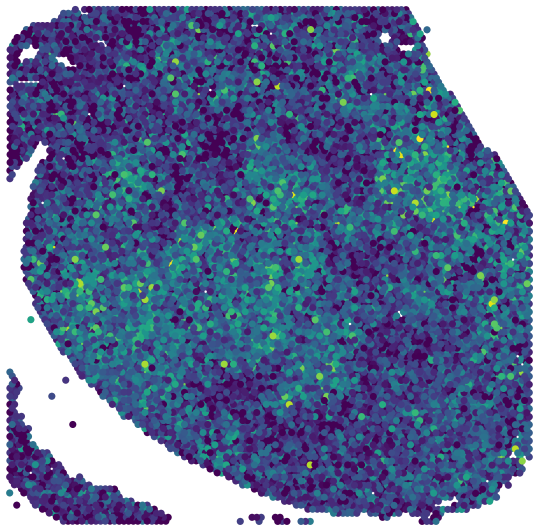
Cholesterol Homeostasis



0 0.21

UCell score

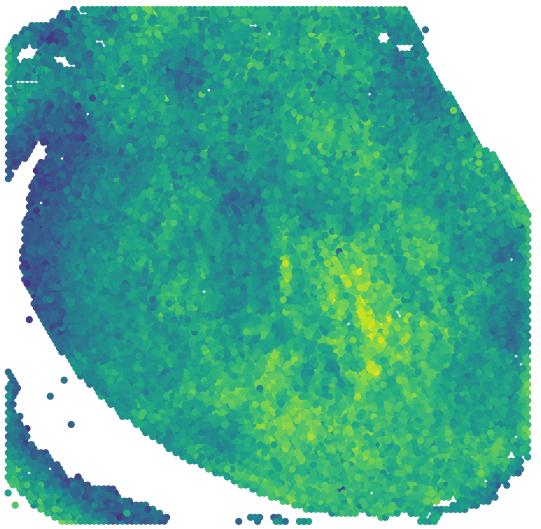
Cholesterol Biosynthesis



0 0.166

UCell score

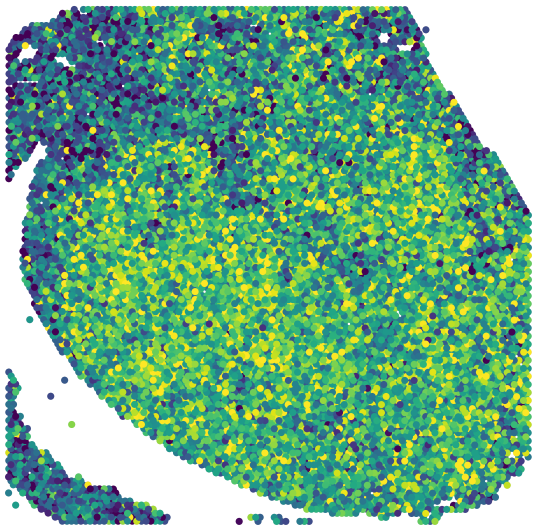
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



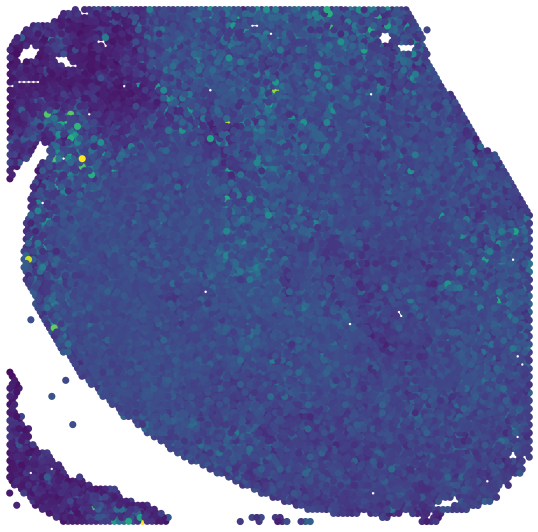
0 0.151

UCell score

CanDis_2026_Sample8 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

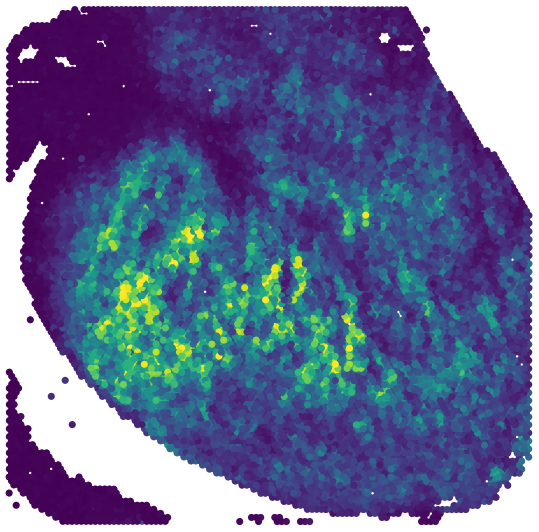
Treg composition



0 0.102

Fraction of all 30

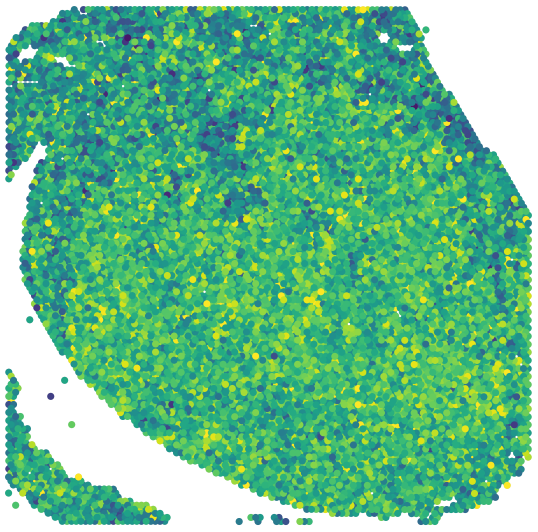
Treg estimated abundance



0 2.85

Posterior mean estimate

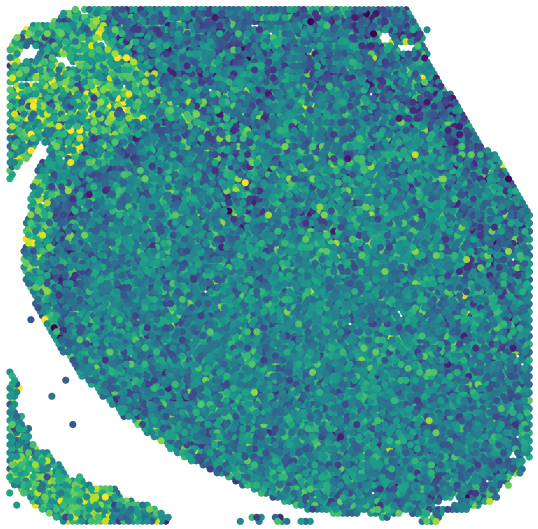
Treg Resting Vs Activated Dn



0 0.0778

UCell score

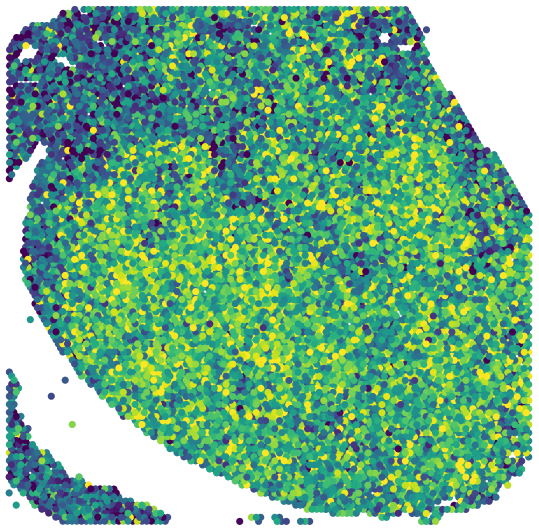
Treg Resting Vs Activated Up



0 0.0609

UCell score

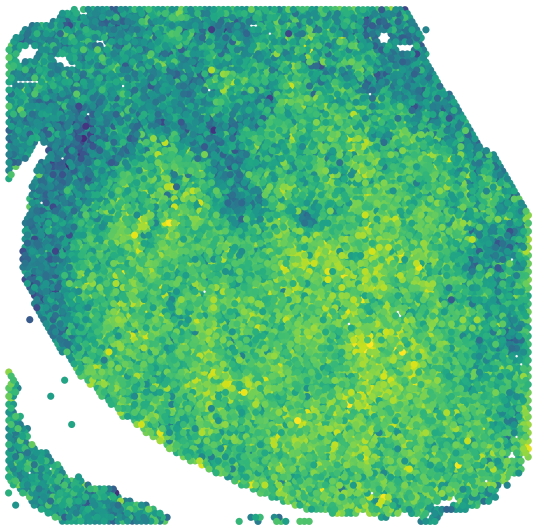
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

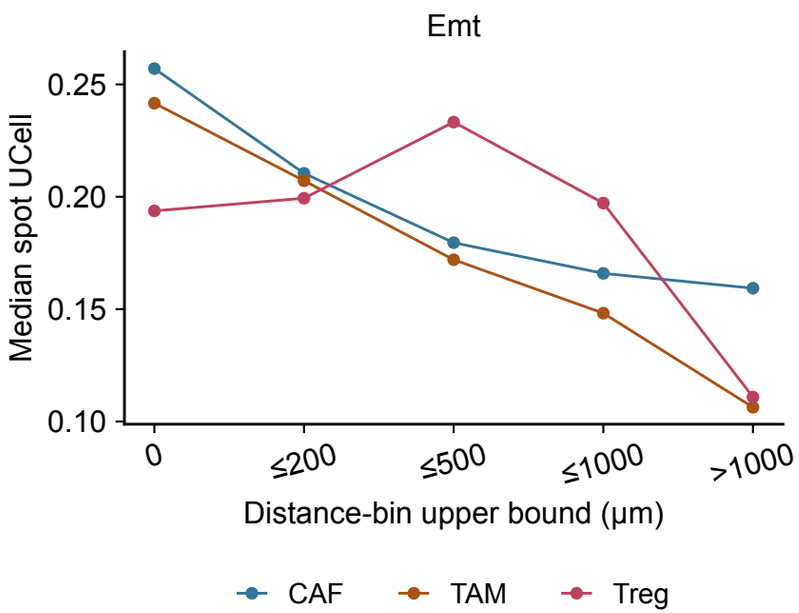


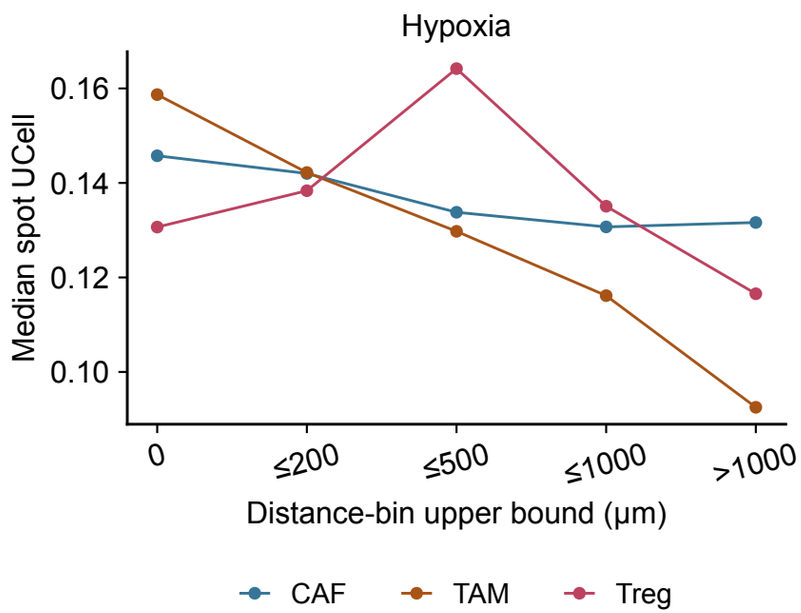
0 0.144

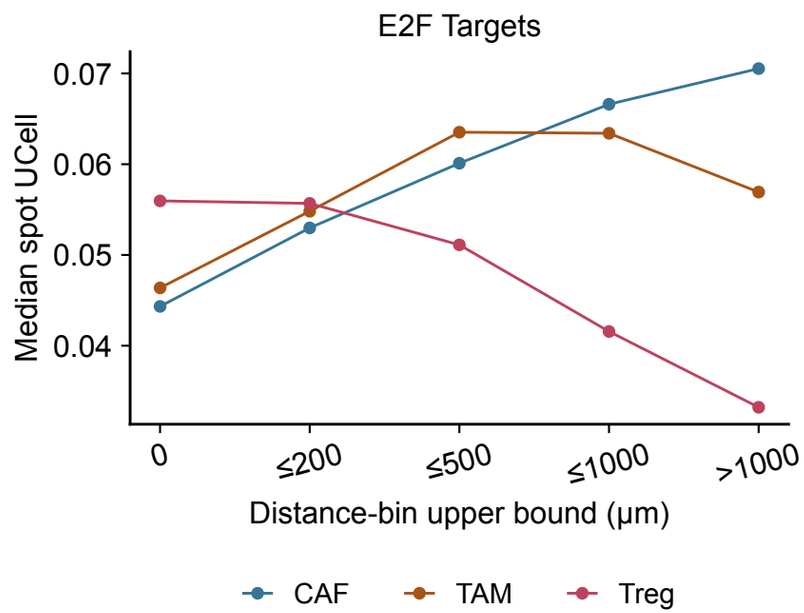
UCell score

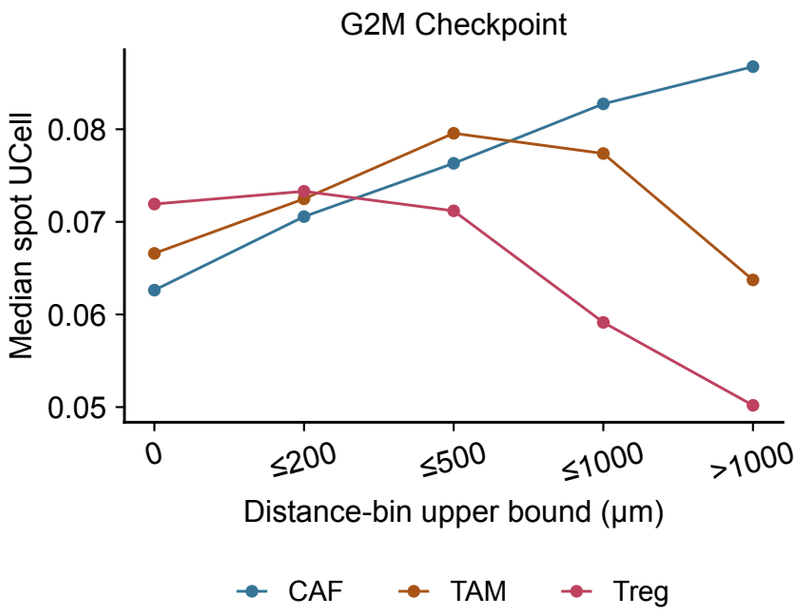
CanDis_2026_Sample8 | Programs by target distance

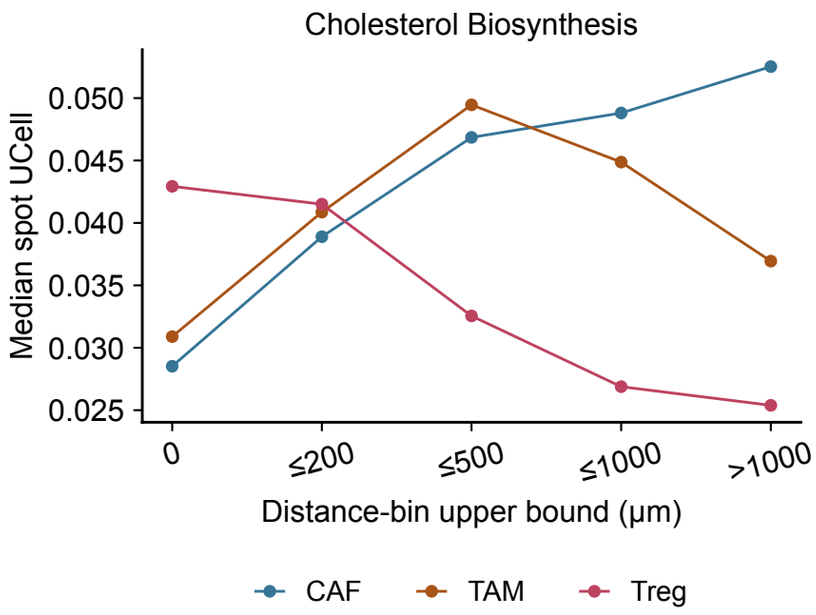
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

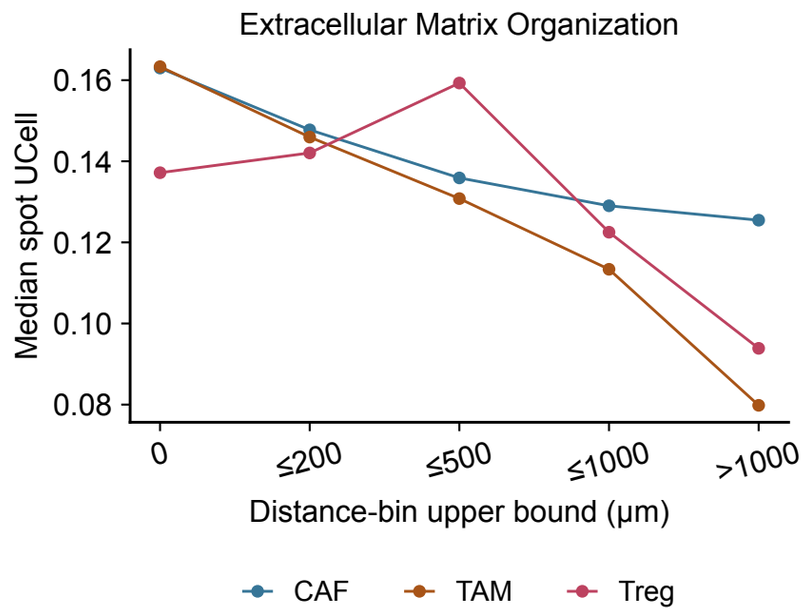






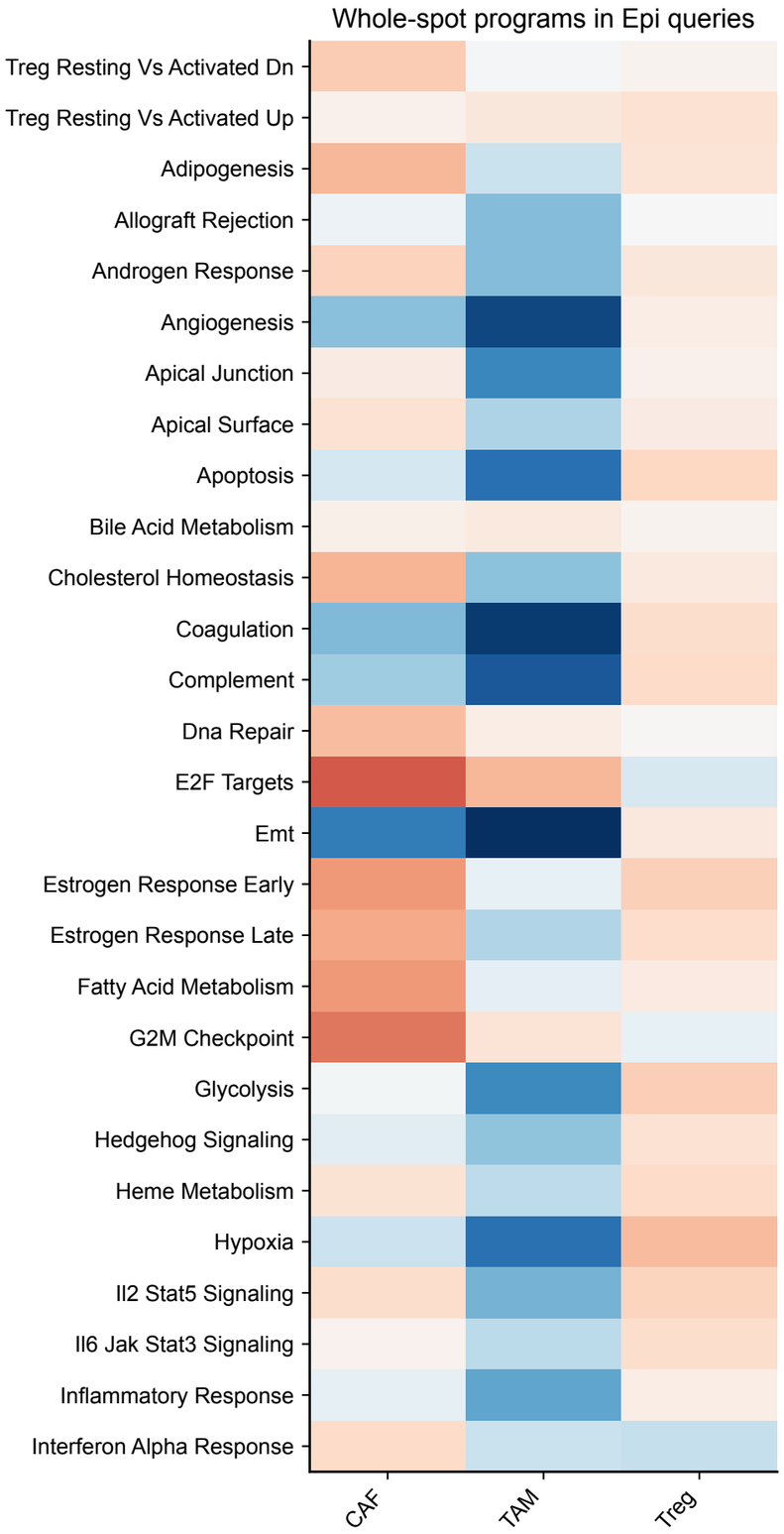


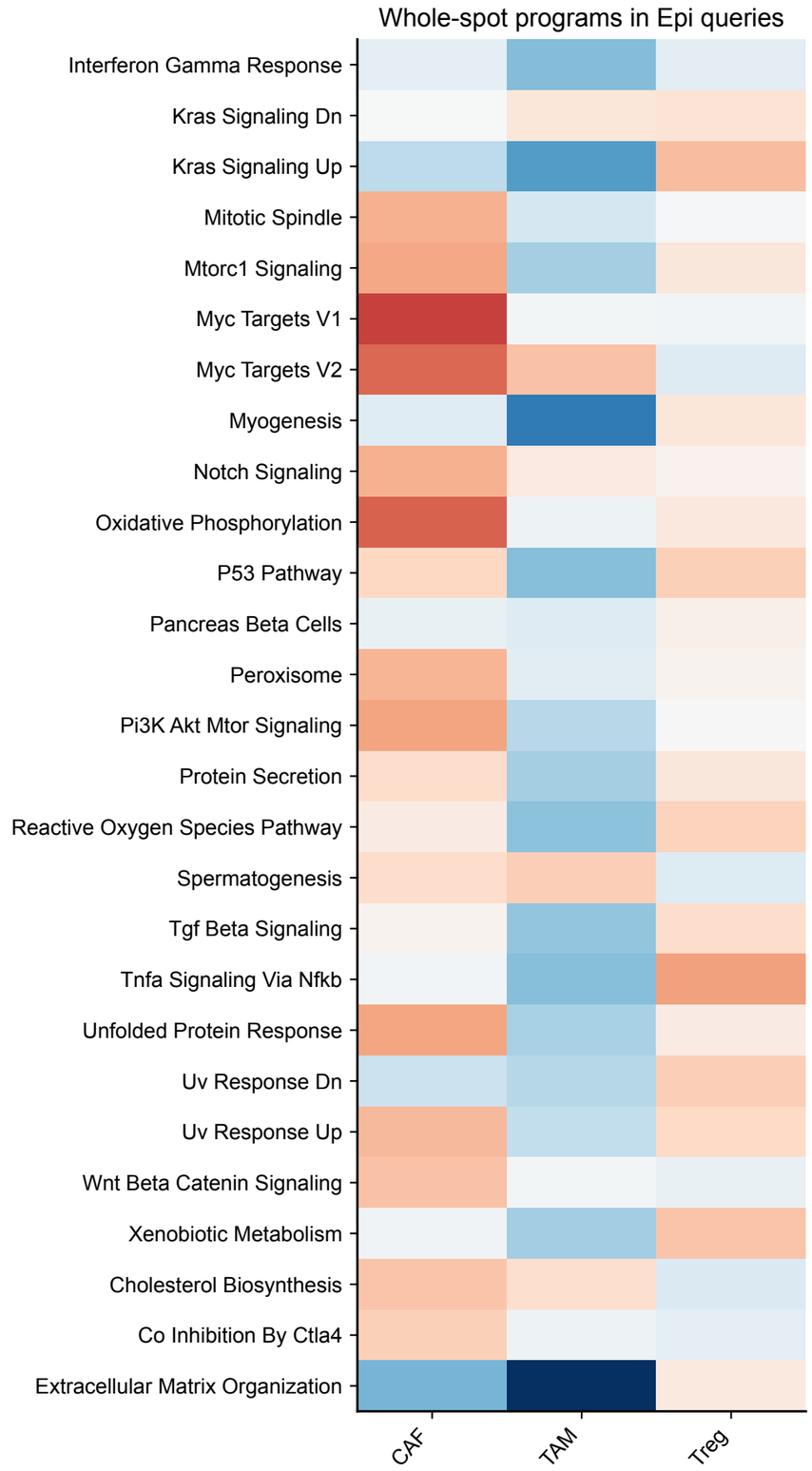




CanDis_2026_Sample8 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6





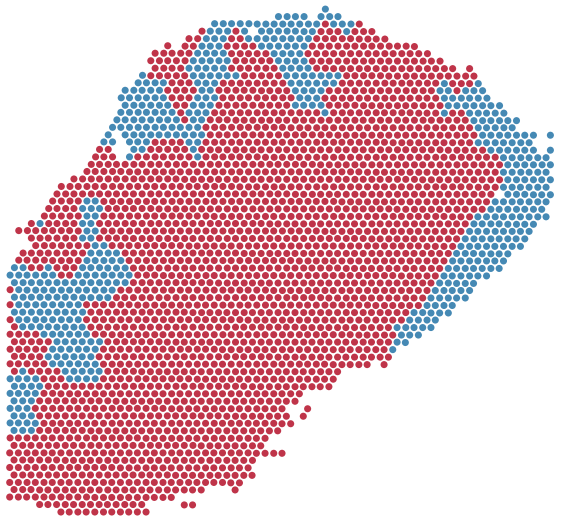
CanDis_2026_Sample9 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge

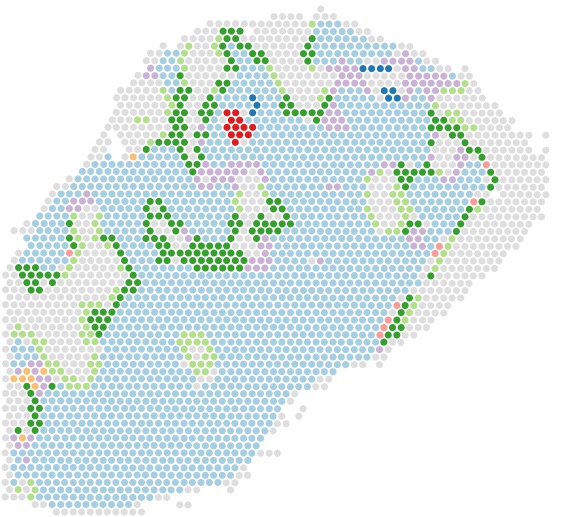
Black line: observed candidate outer edge

Final candidate location



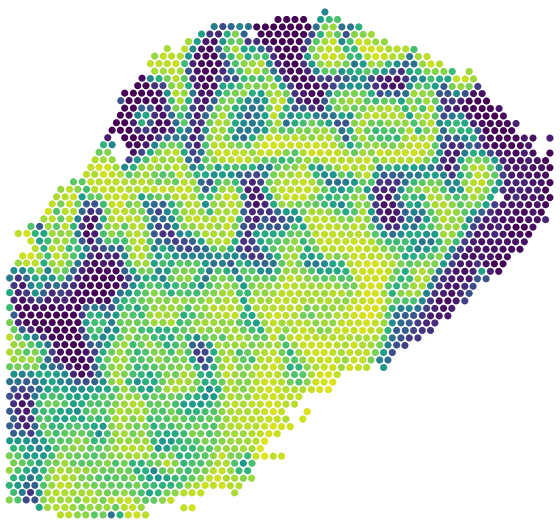
inside outside

Accepted R12 (original 0–8)



0 2 4 7 8
1 3 6

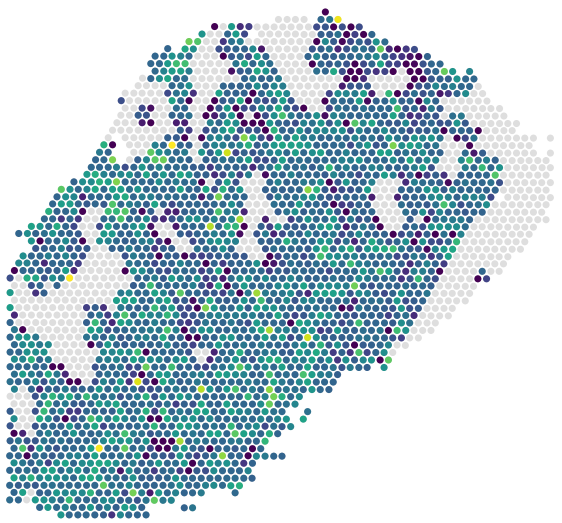
Epithelial support



0 1

Fraction of all 30

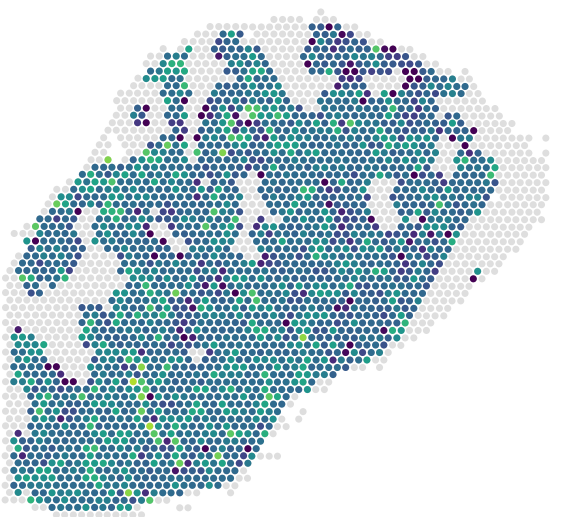
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



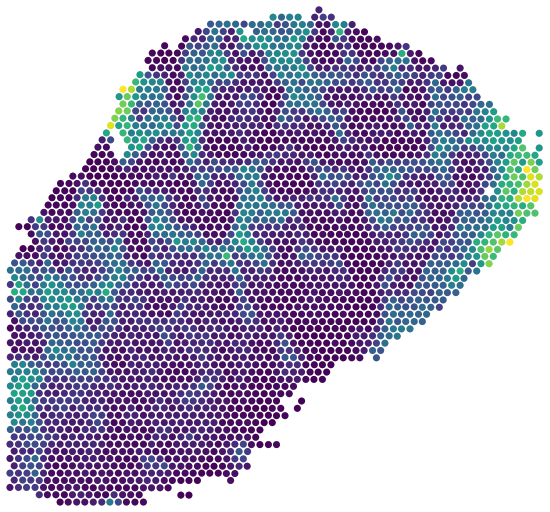
0 0.14

Matched-model AUCell

CanDis_2026_Sample9 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

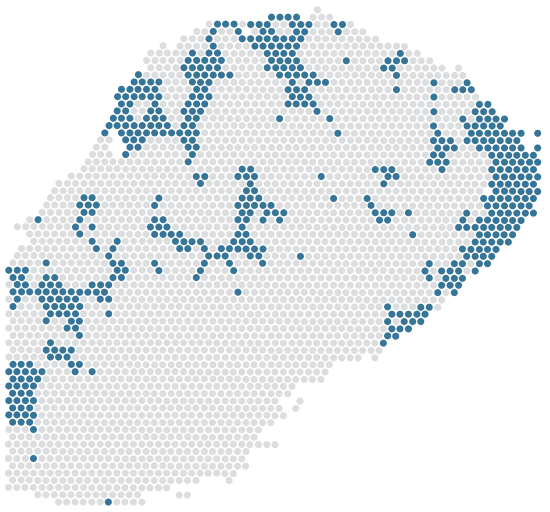
CAF composition



0 0.418

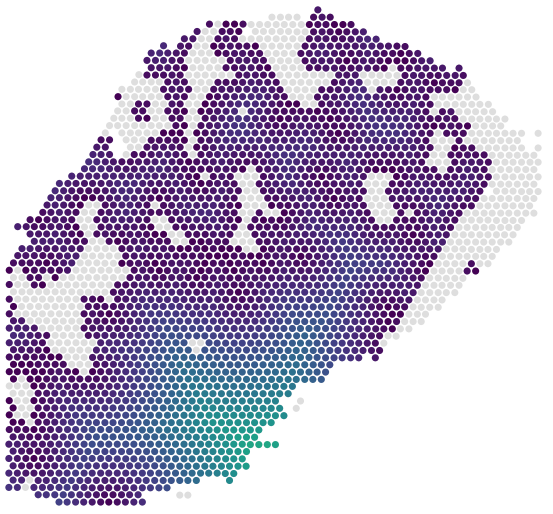
Fraction of all 30

CAF P80 targets



False True

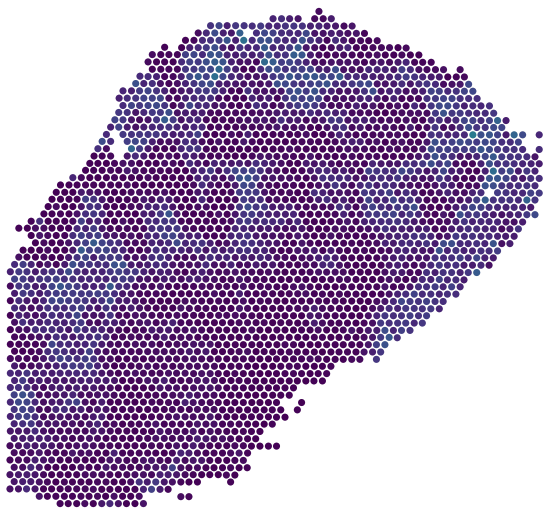
CAF distance



0 3.1e+03

Nearest spot center (µm)

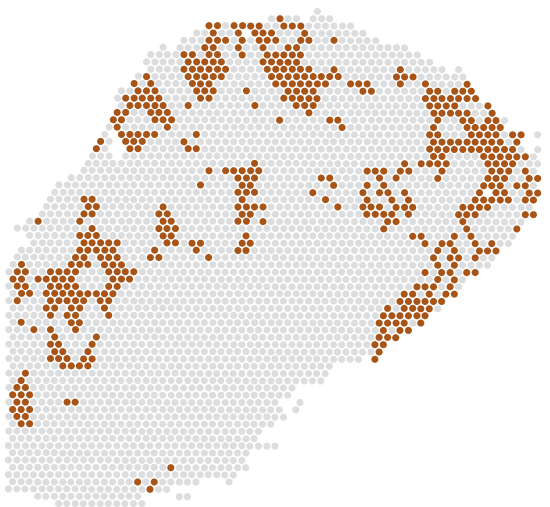
TAM composition



0 0.291

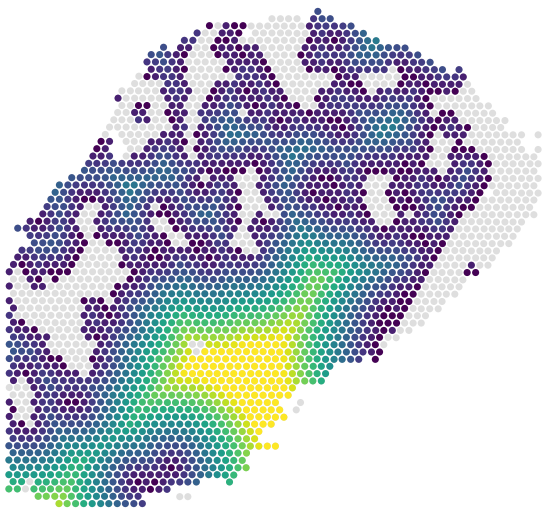
Fraction of all 30

TAM P80 targets



False True

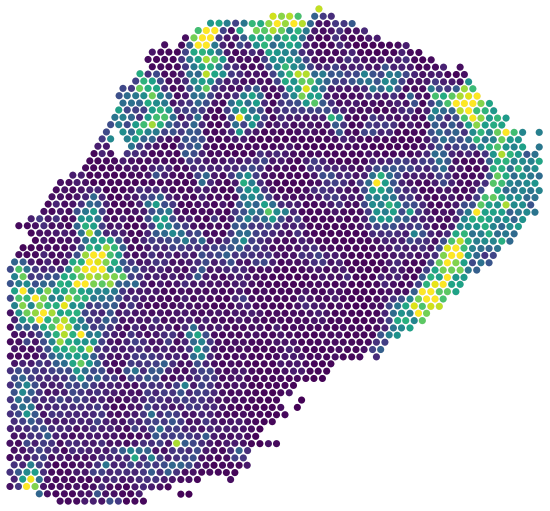
TAM distance



0 1.13e+03

Nearest spot center (µm)

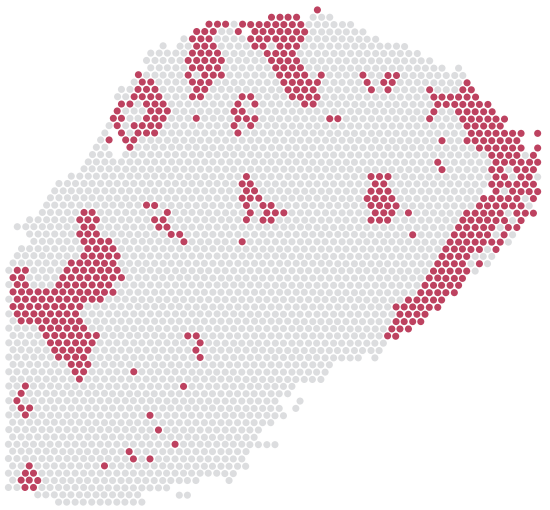
Treg composition



0 0.102

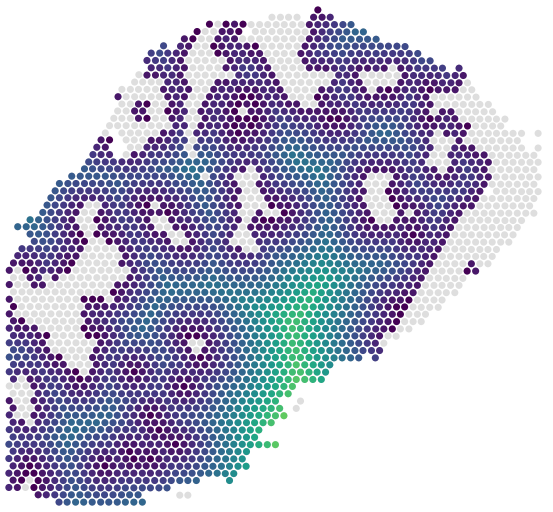
Fraction of all 30

Treg P80 targets



False True

Treg distance

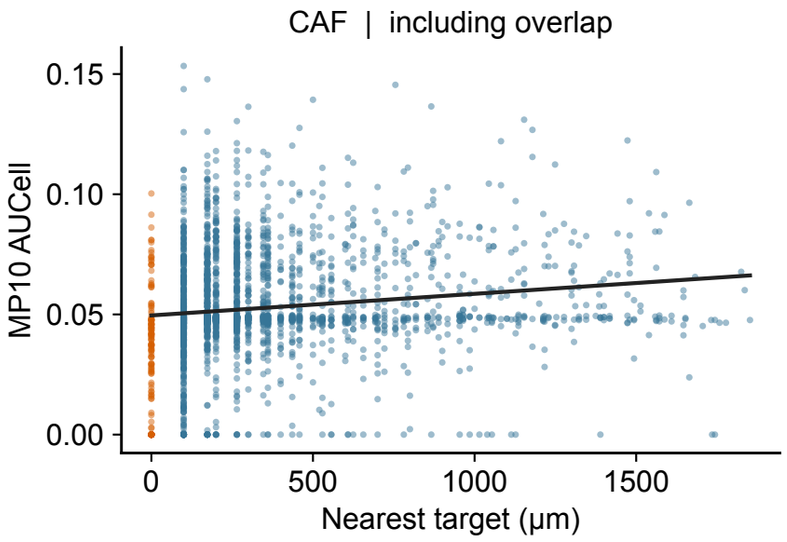


0 1.61e+03

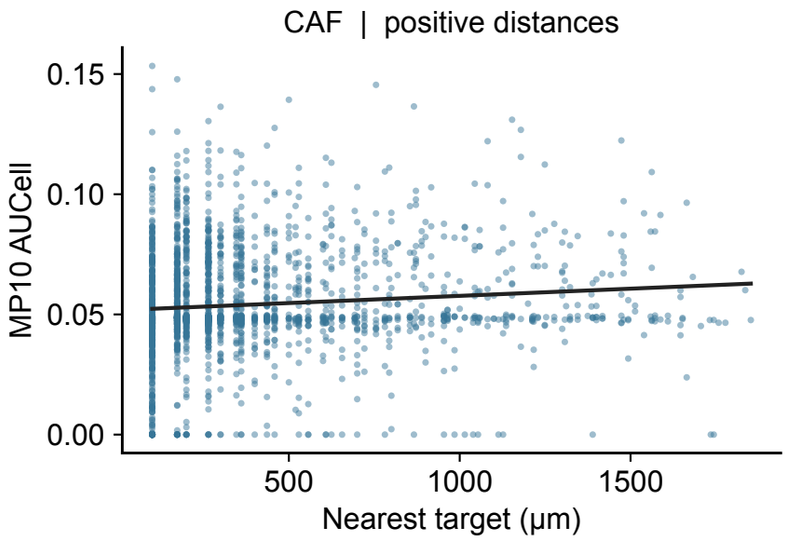
Nearest spot center (µm)

CanDis_2026_Sample9 | MP10 versus nearest target

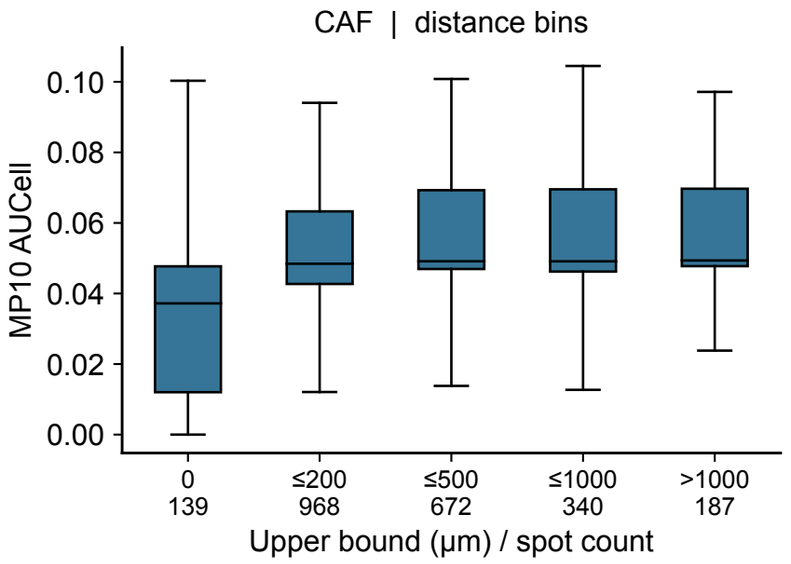
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



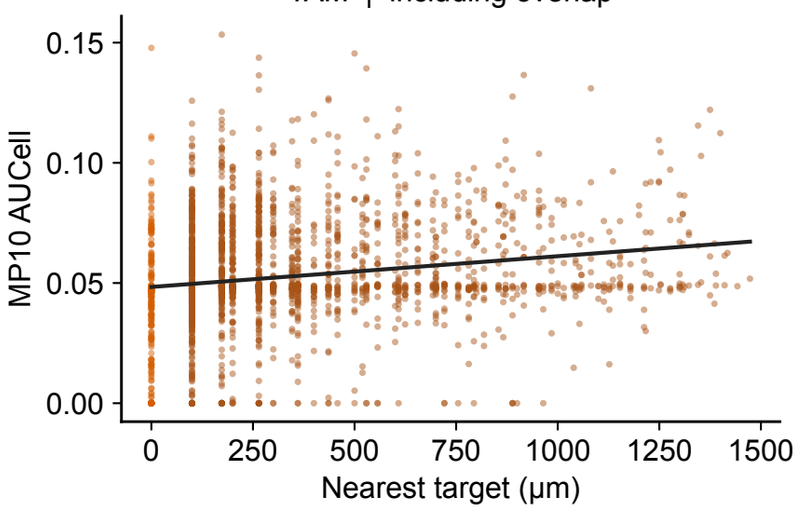
r +0.14 ρ +0.22
n 2306/2310 queries • 579 targets



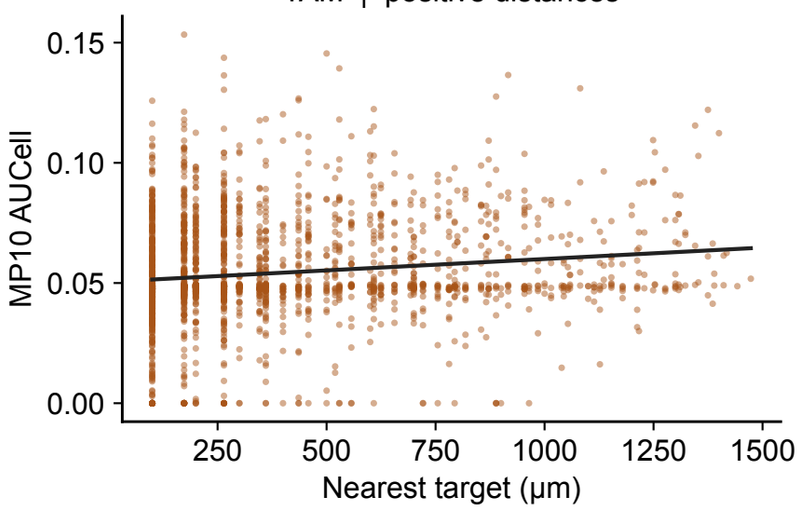
r +0.10 ρ +0.16
n 2167/2310 queries • 579 targets



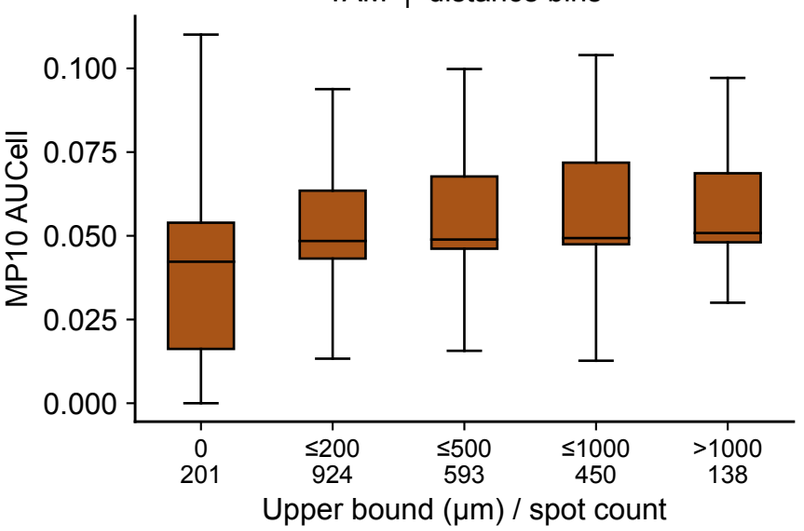
Orange = zero distance
Line = within-section linear fit



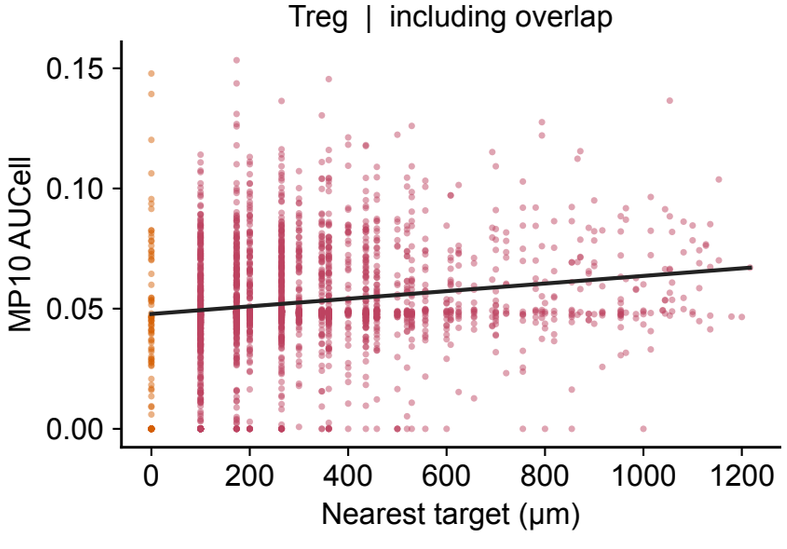
r +0.18 ρ +0.22
n 2306/2310 queries • 579 targets



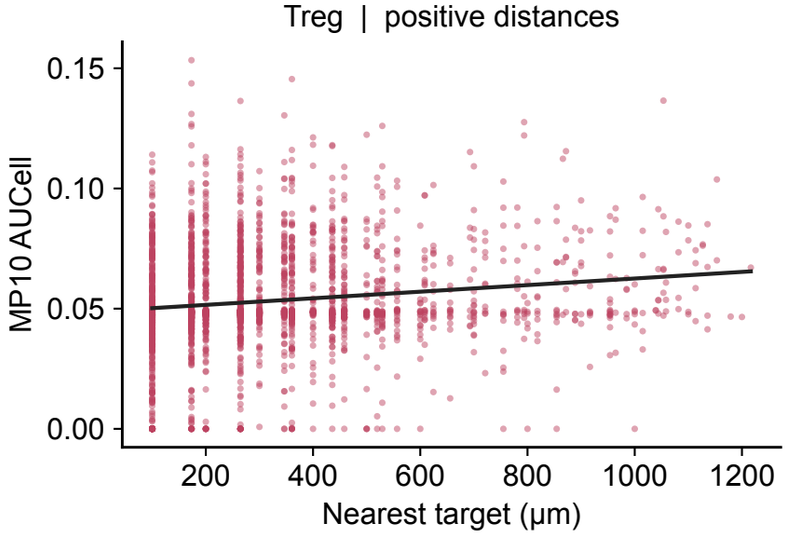
r +0.14 ρ +0.17
n 2105/2310 queries • 579 targets



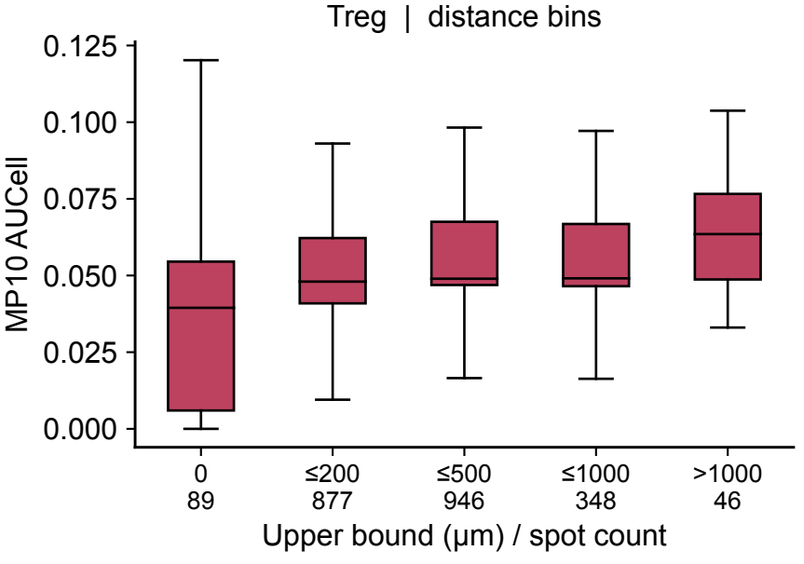
Orange = zero distance
Line = within-section linear fit



r +0.16 ρ +0.19
n 2306/2310 queries • 579 targets



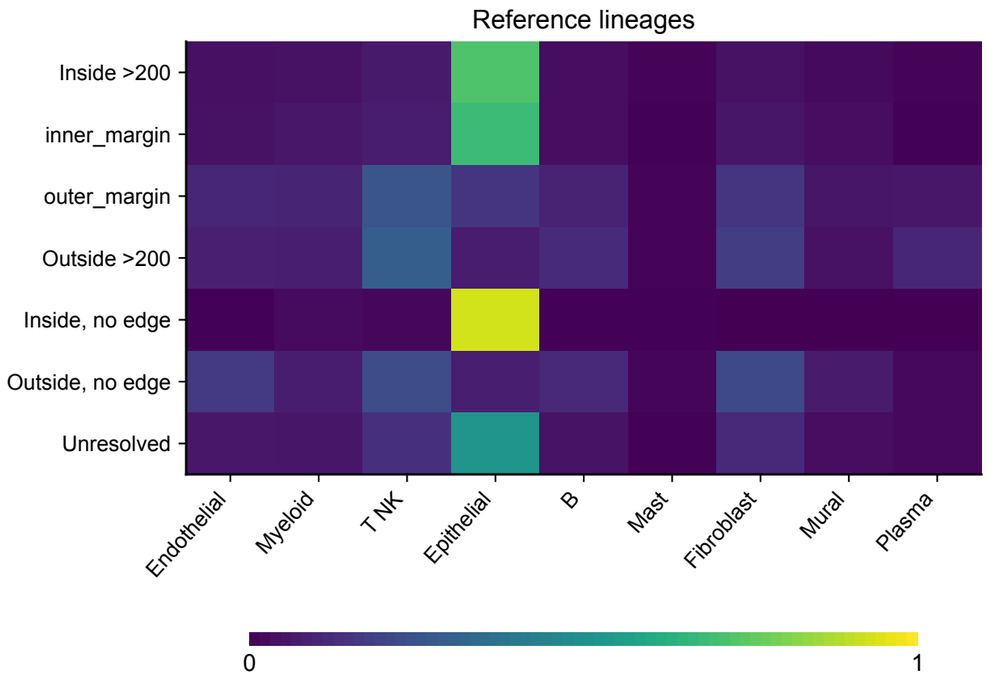
r +0.14 ρ +0.17
n 2217/2310 queries • 579 targets

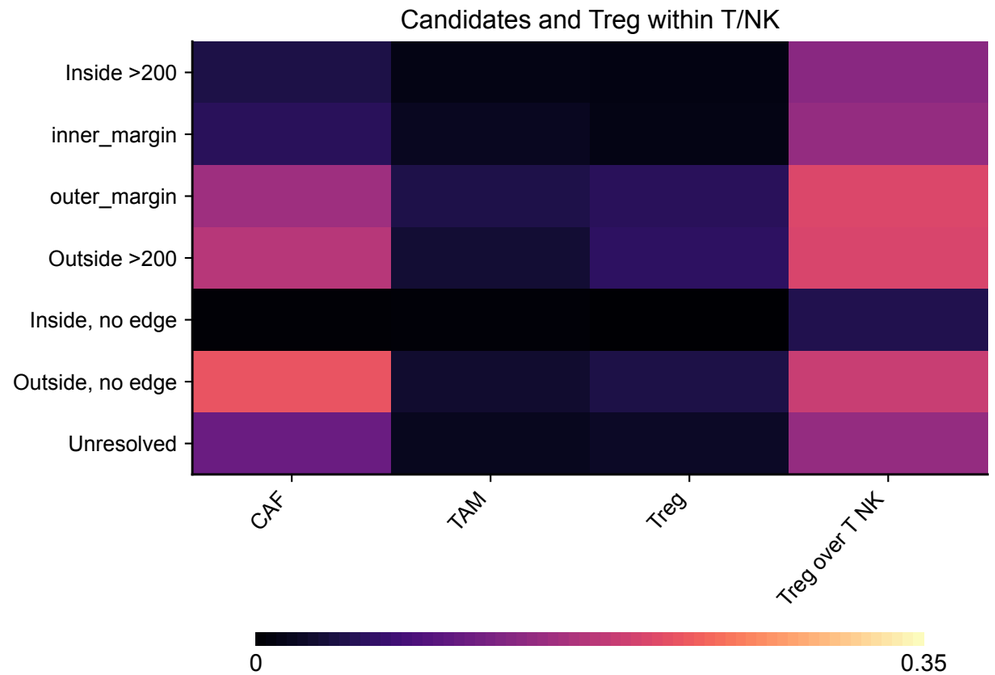


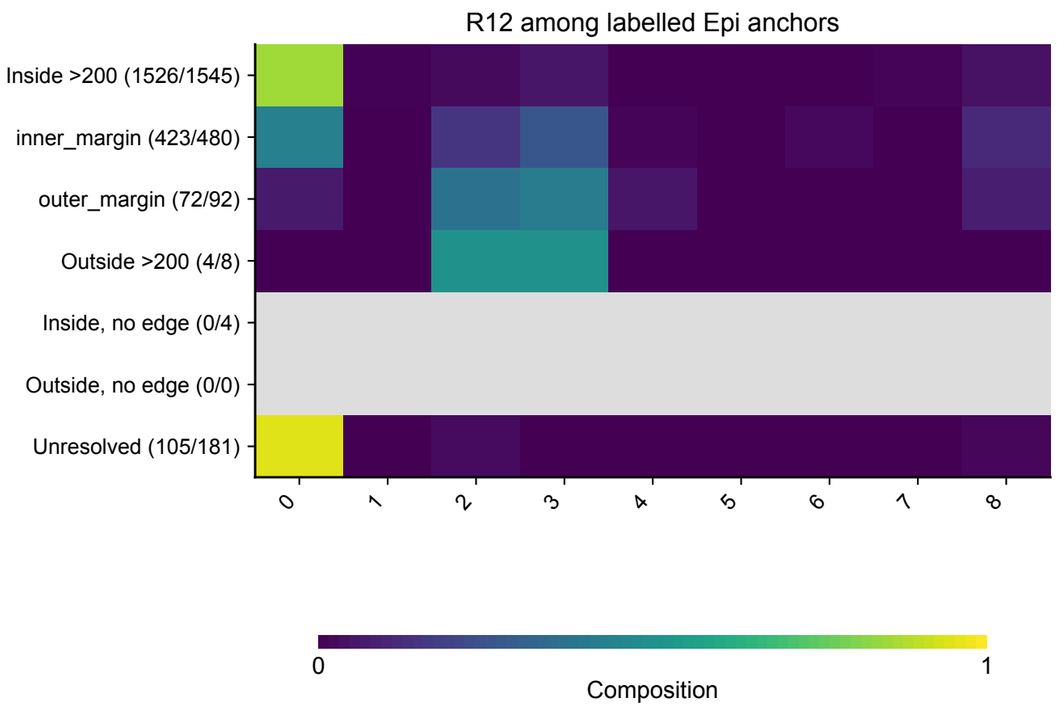
Orange = zero distance
Line = within-section linear fit

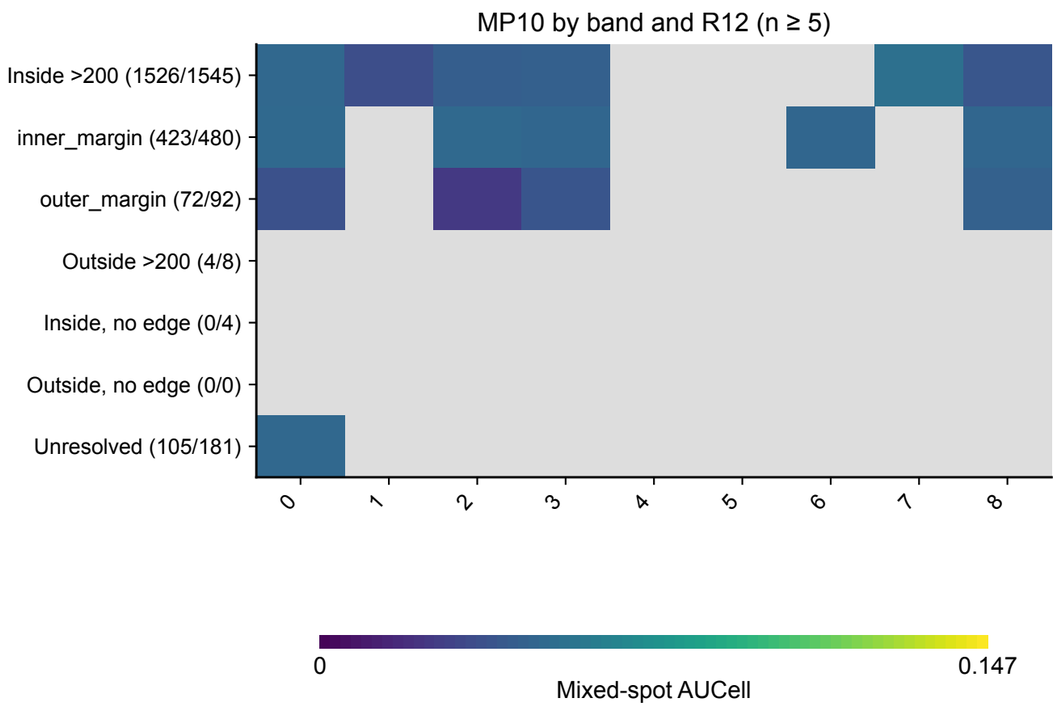
CanDis_2026_Sample9 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





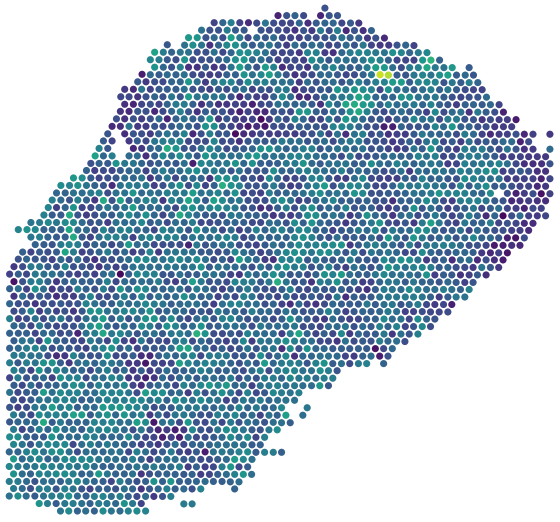




CanDis_2026_Sample9 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

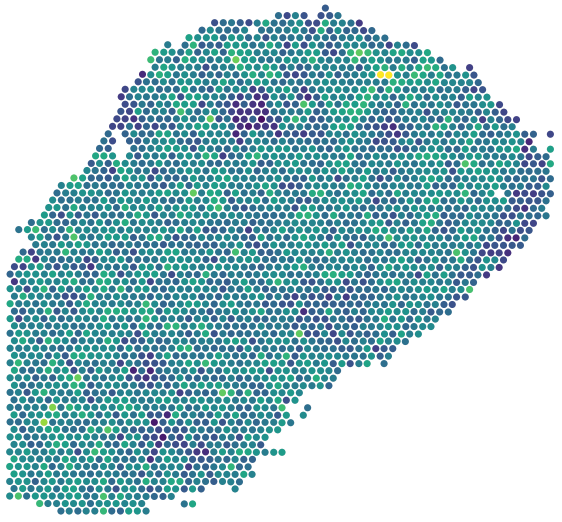
E2F Targets



0 0.152

UCell score

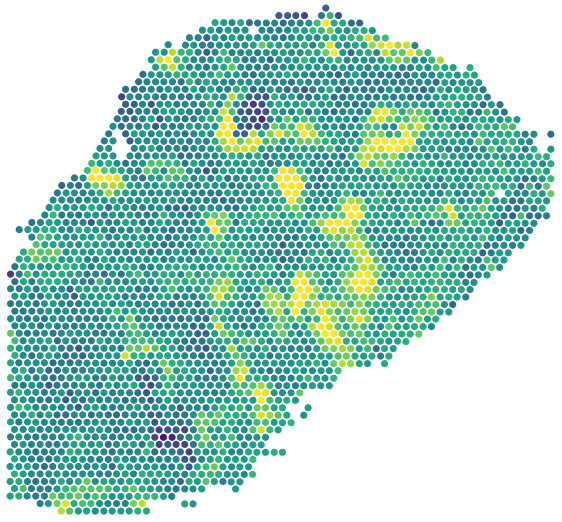
G2M Checkpoint



0 0.134

UCell score

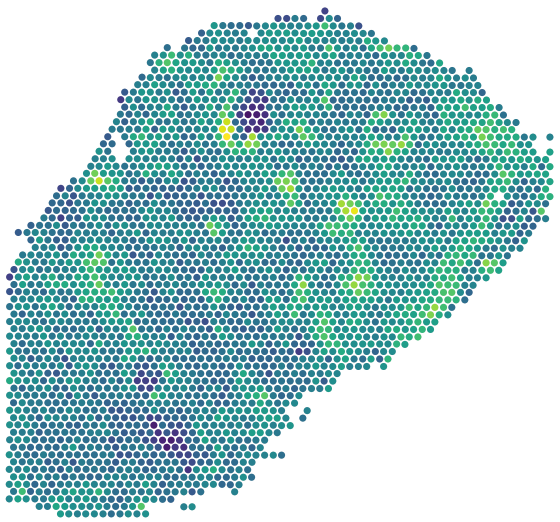
Hypoxia



0 0.211

UCell score

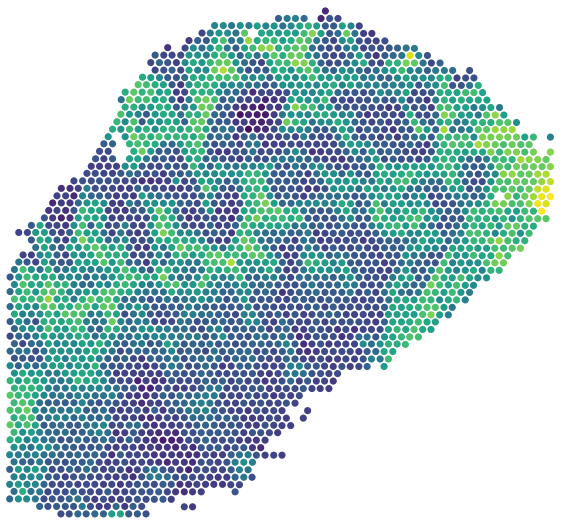
Tnfa Signaling Via Nfkb



0 0.275

UCell score

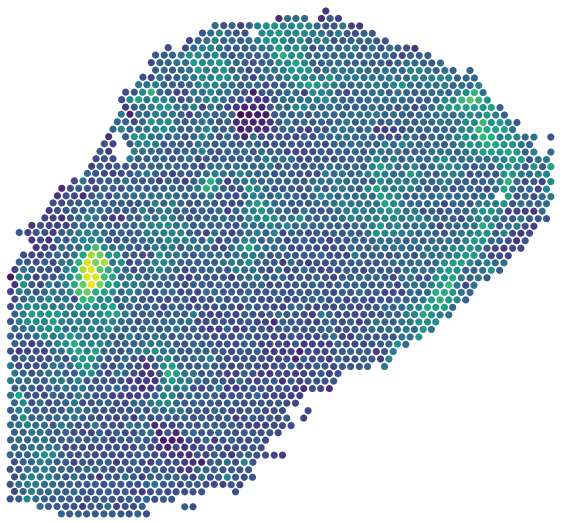
Emt



0 0.386

UCell score

Interferon Gamma Response



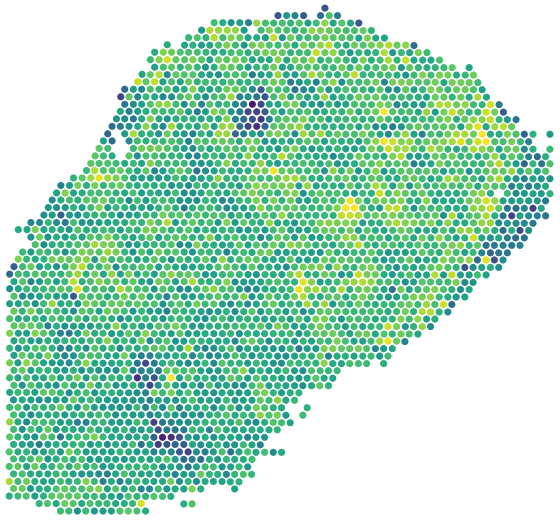
0 0.275

UCell score

CanDis_2026_Sample9 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

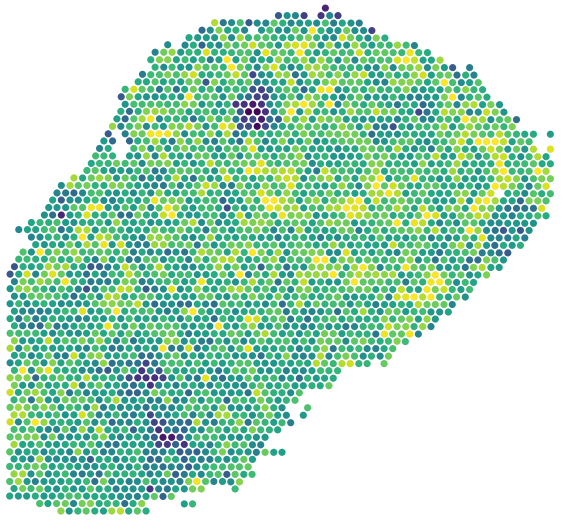
Il2 Stat5 Signaling



0 0.144

UCell score

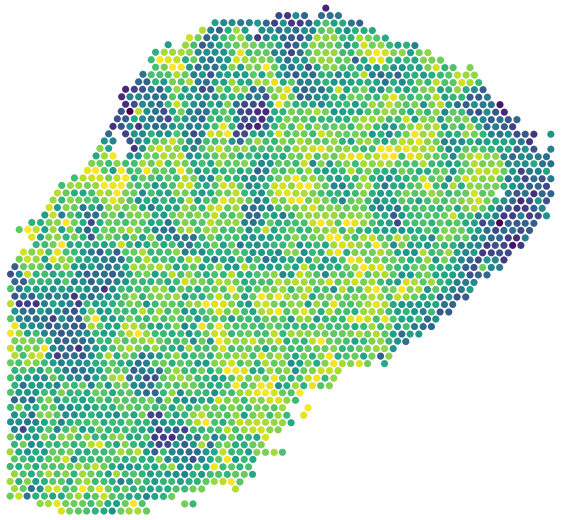
Tgf Beta Signaling



0 0.232

UCell score

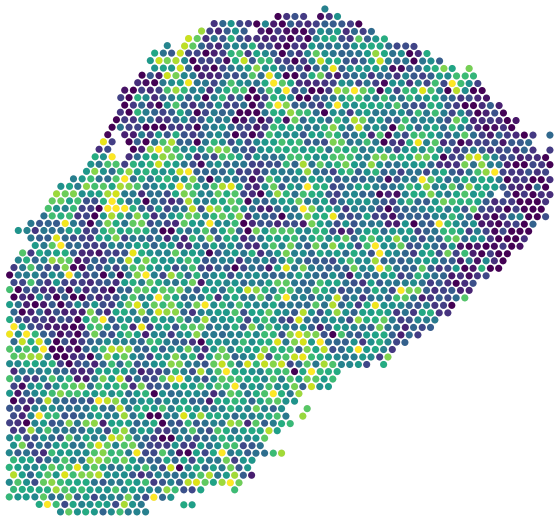
Cholesterol Homeostasis



0 0.21

UCell score

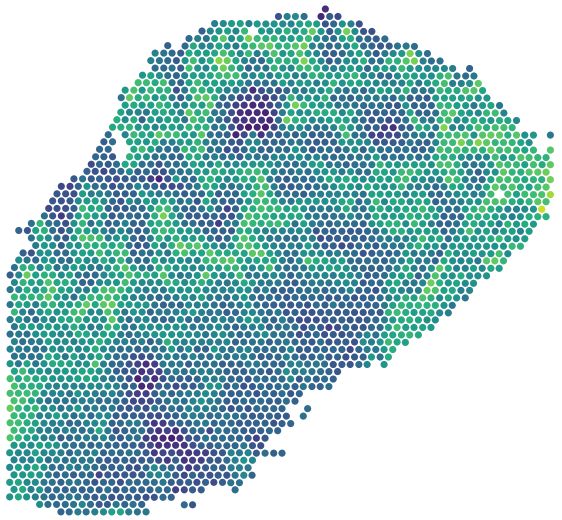
Cholesterol Biosynthesis



0 0.166

UCell score

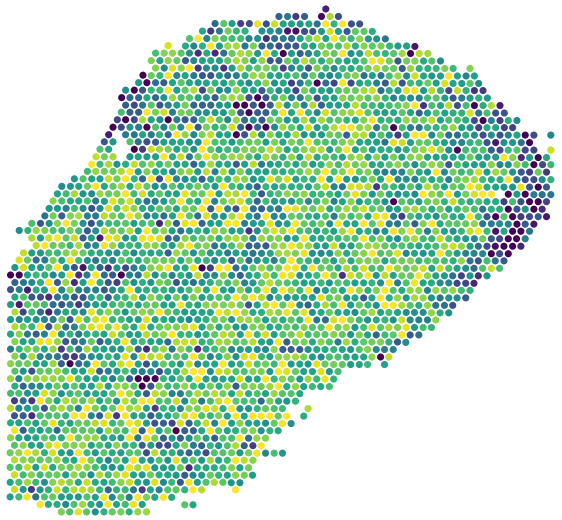
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



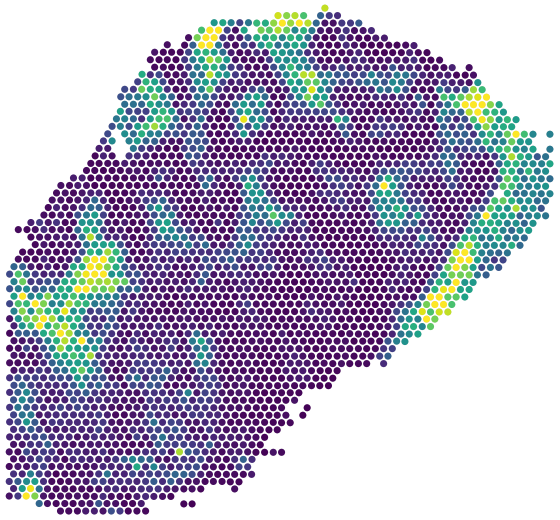
0 0.151

UCell score

CanDis_2026_Sample9 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

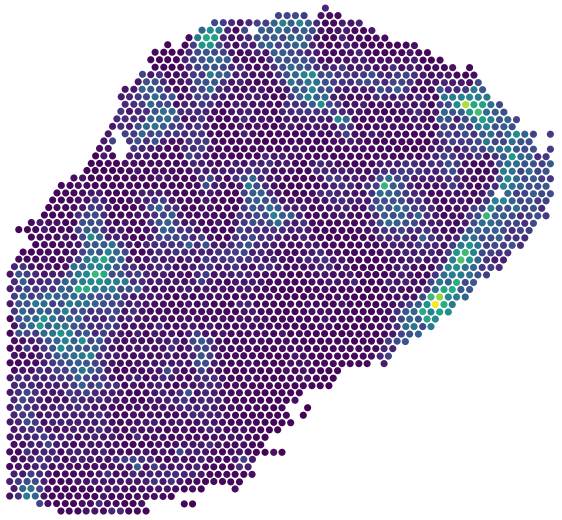
Treg composition



0 0.102

Fraction of all 30

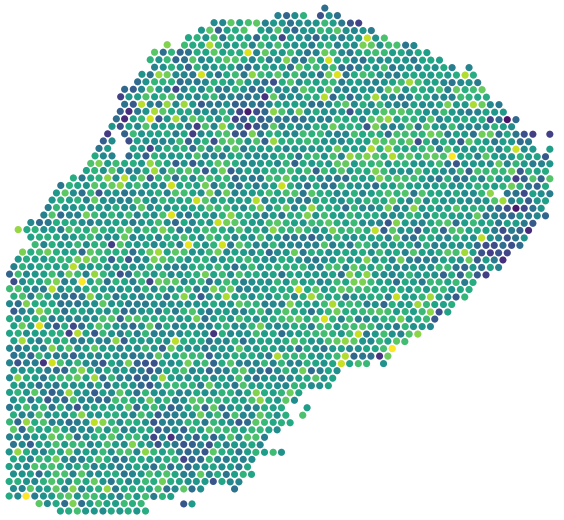
Treg estimated abundance



0 2.85

Posterior mean estimate

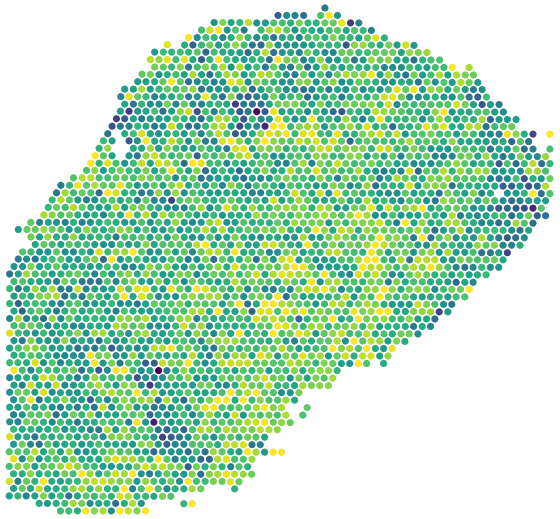
Treg Resting Vs Activated Dn



0 0.0778

UCell score

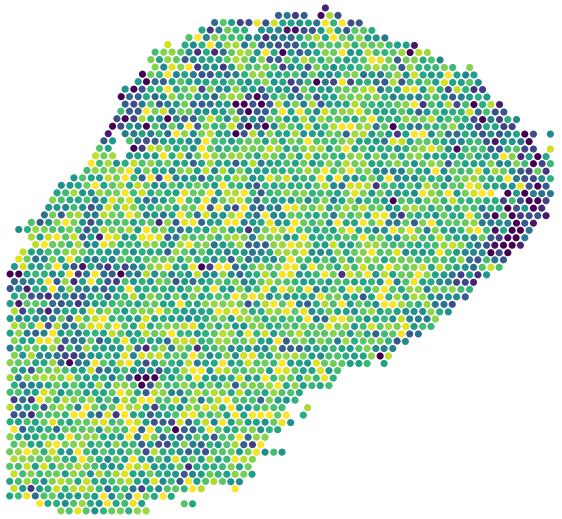
Treg Resting Vs Activated Up



0 0.0609

UCell score

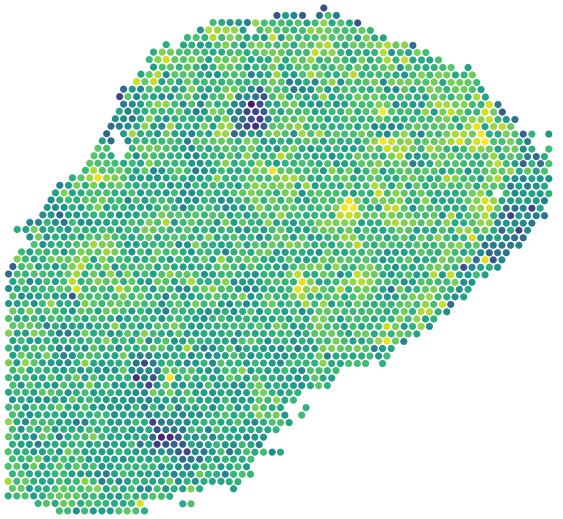
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

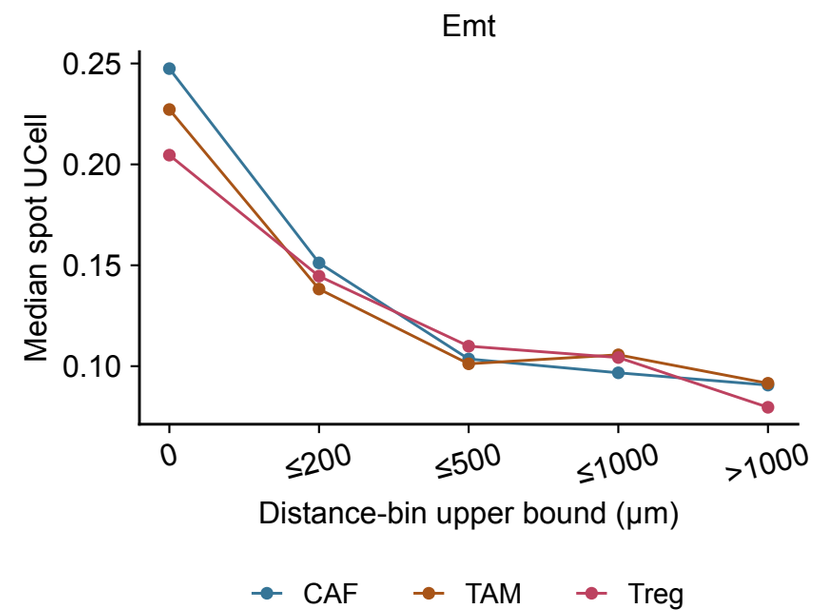


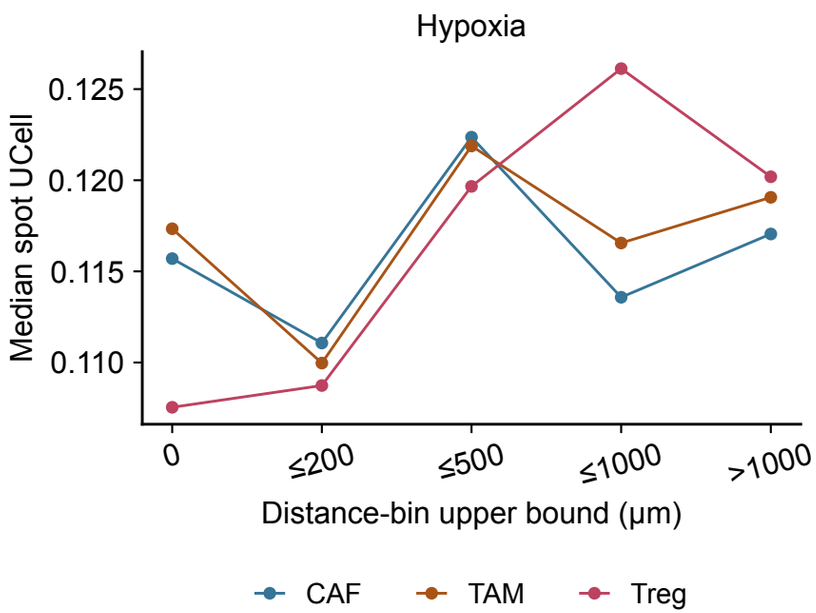
0 0.144

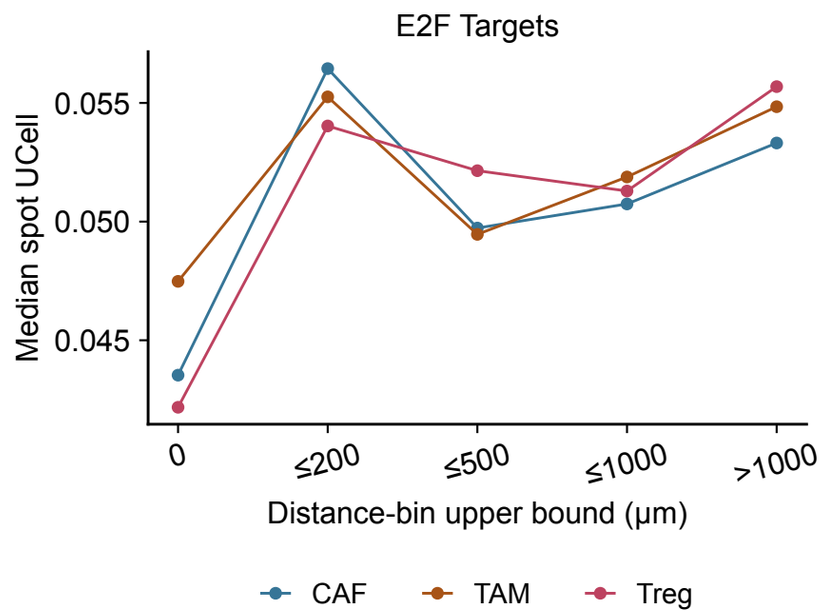
UCell score

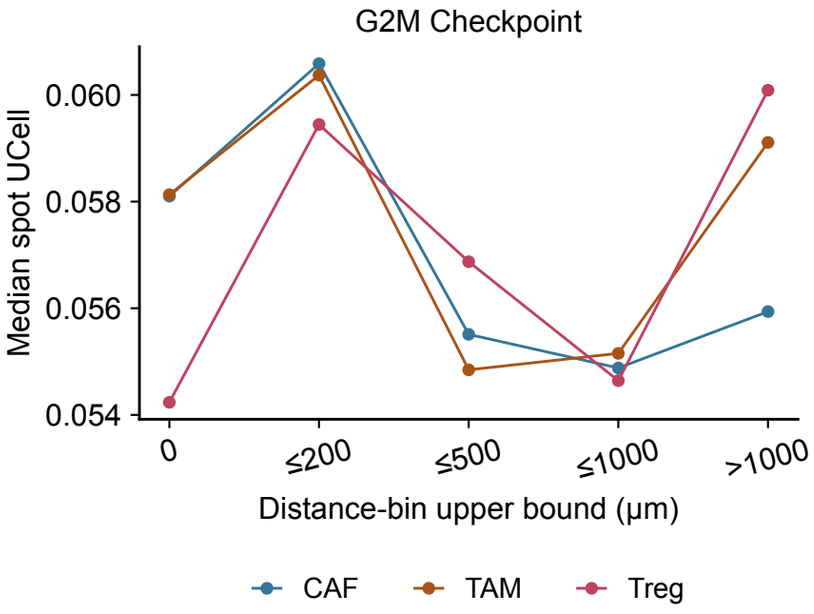
CanDis_2026_Sample9 | Programs by target distance

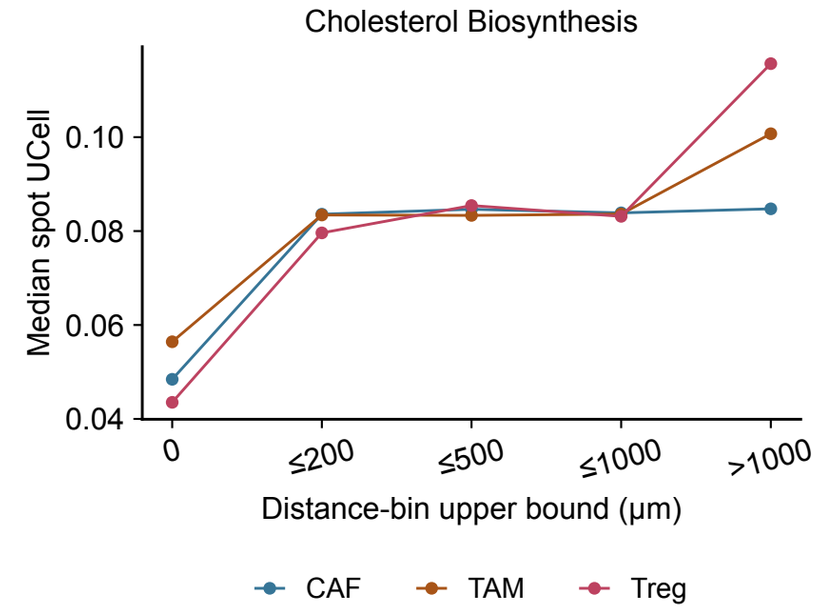
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

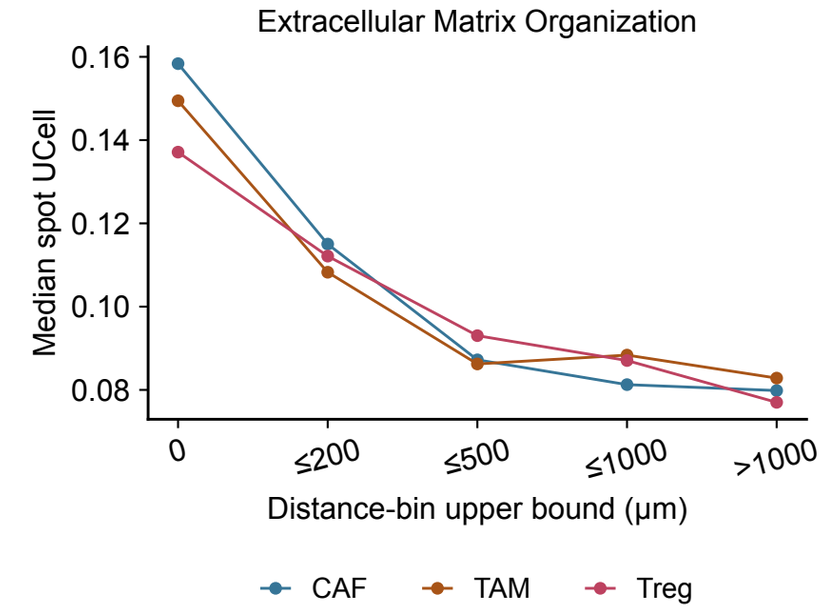












CanDis_2026_Sample9 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6